

COMPASS_Synoptic_MacrophyteFluxes_2025_CO2

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2025-11-26

Set Up

Pull in Raw Datafiles

```
## [1] "C:\\Users\\readz\\AppData\\Local\\Temp\\RtmpaUw4A6\\Test_FW_Autochamber_Data\\Test_FW_Autochamber_
```

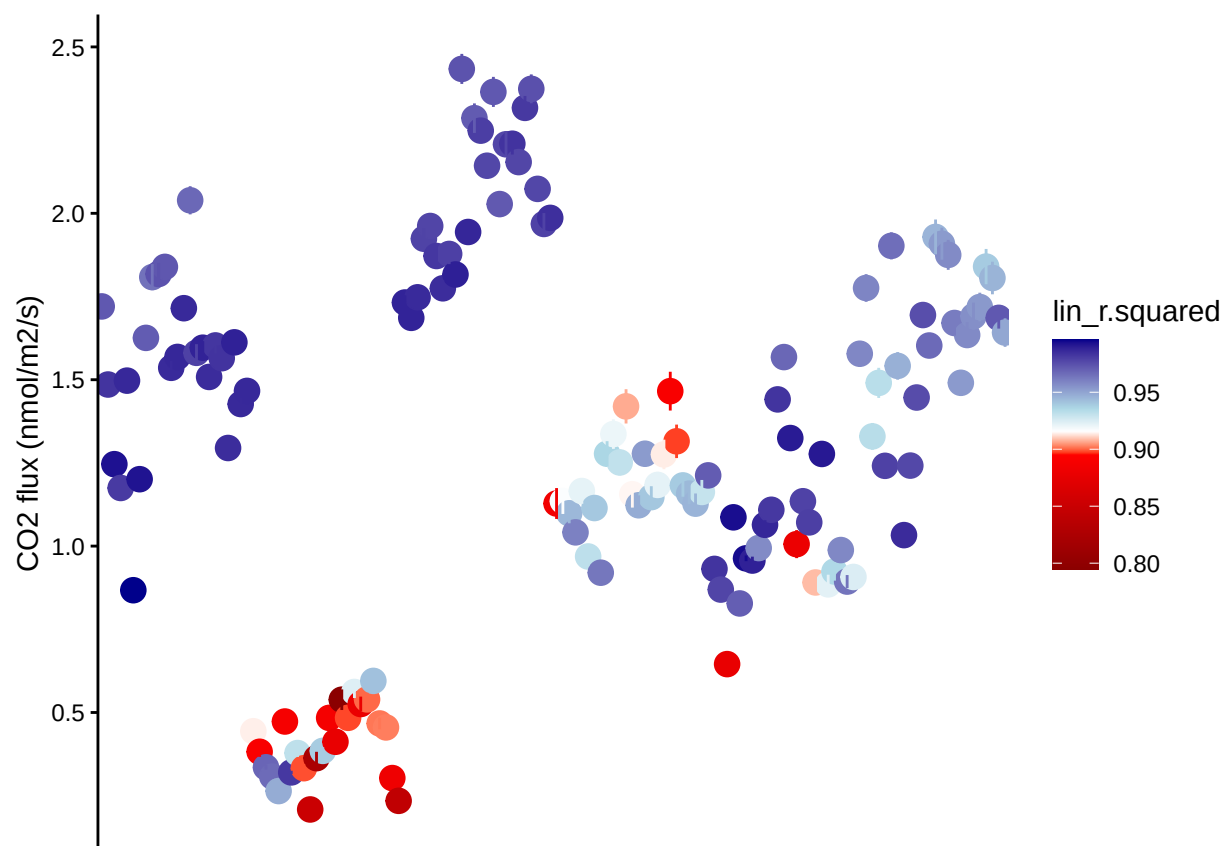
```
## TG10-01725-2026-05-12T090000.data: read 422623 rows of TG10-01725 data, 2026-05-13 11:08:32 to 2026-
```

Read in metadata file and match to raw data

Match metadata to raw data

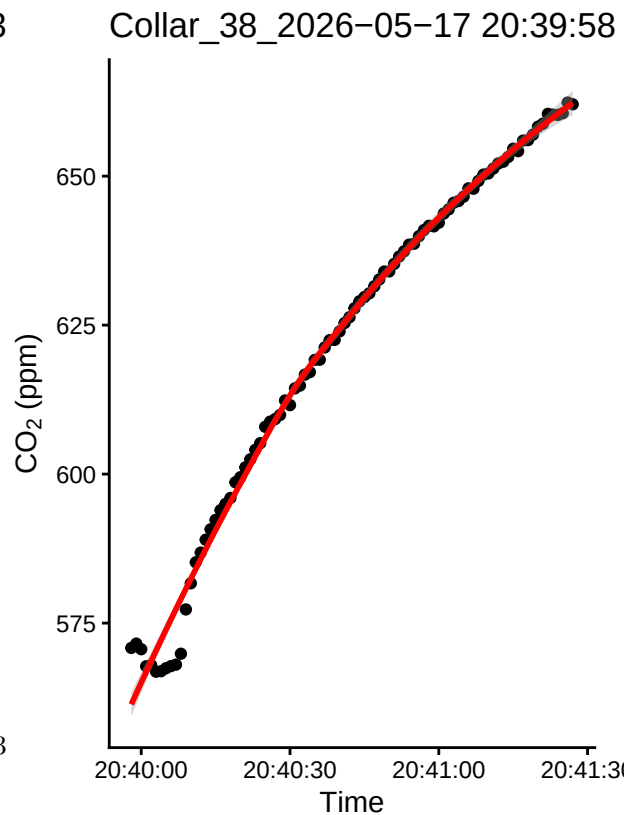
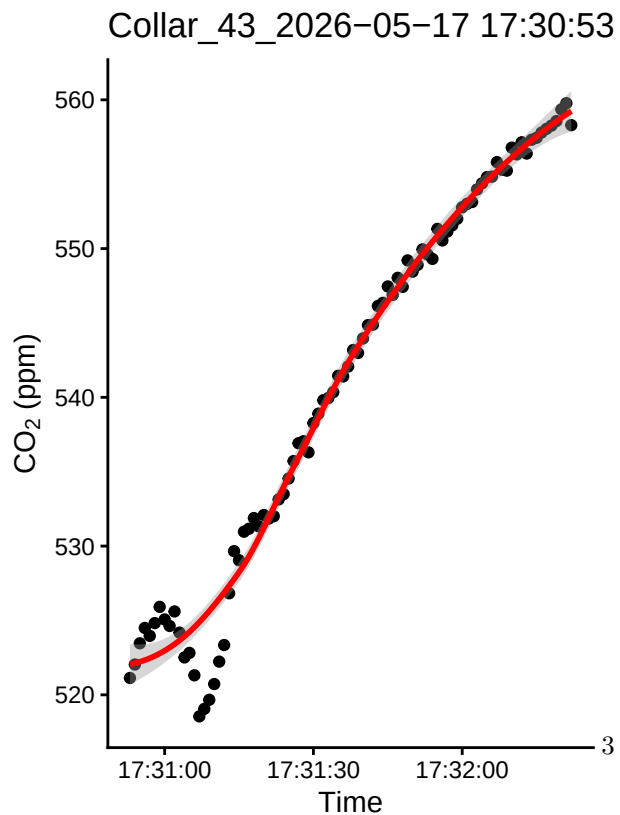
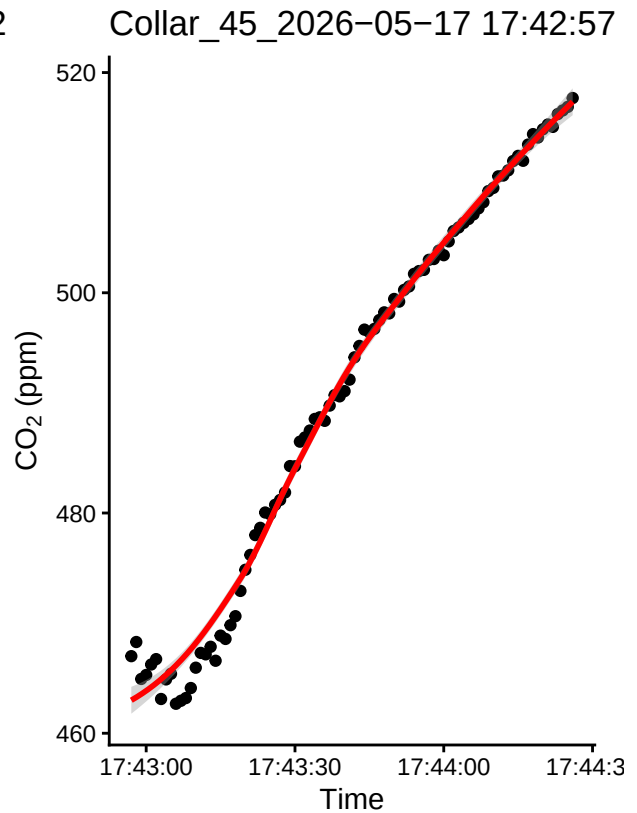
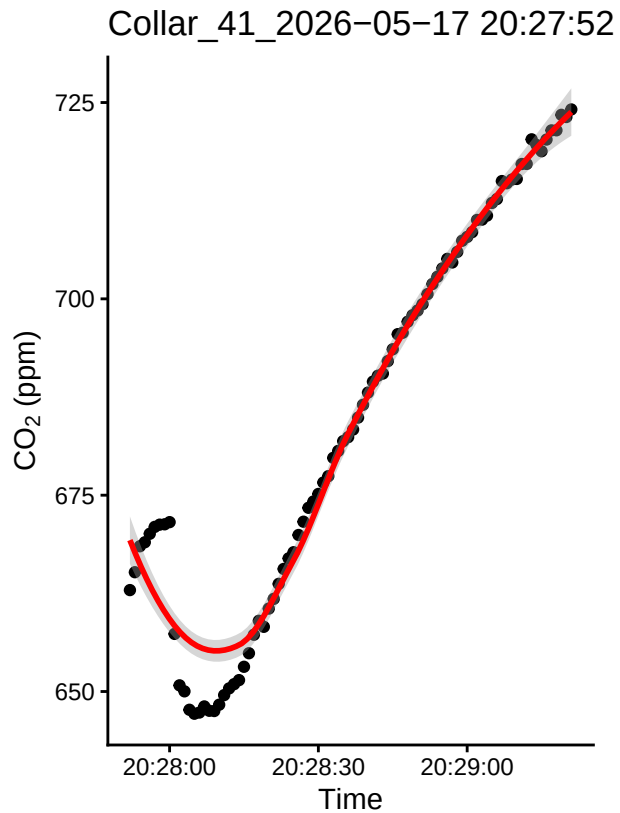
```
## 8 entries had no timestamp matches!
```

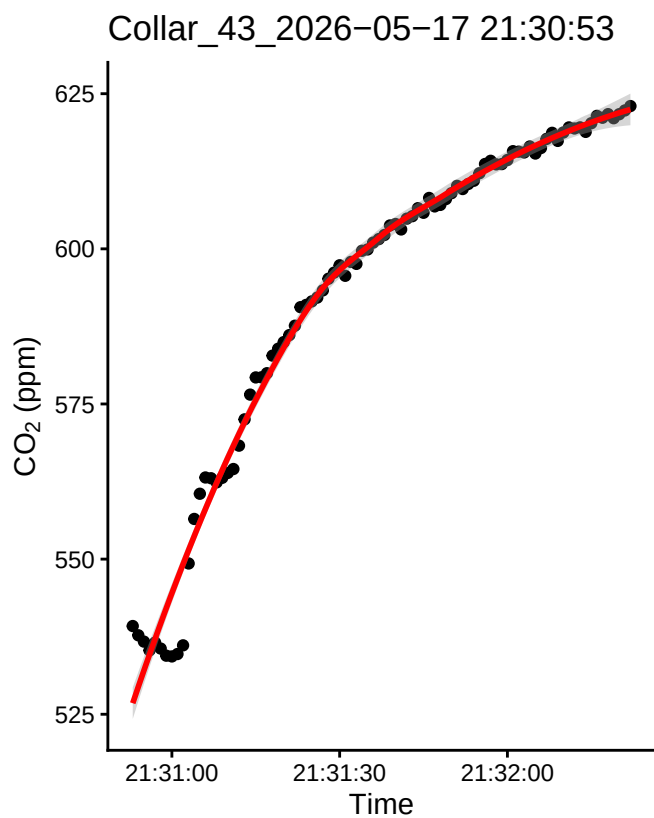
Calculate fluxes



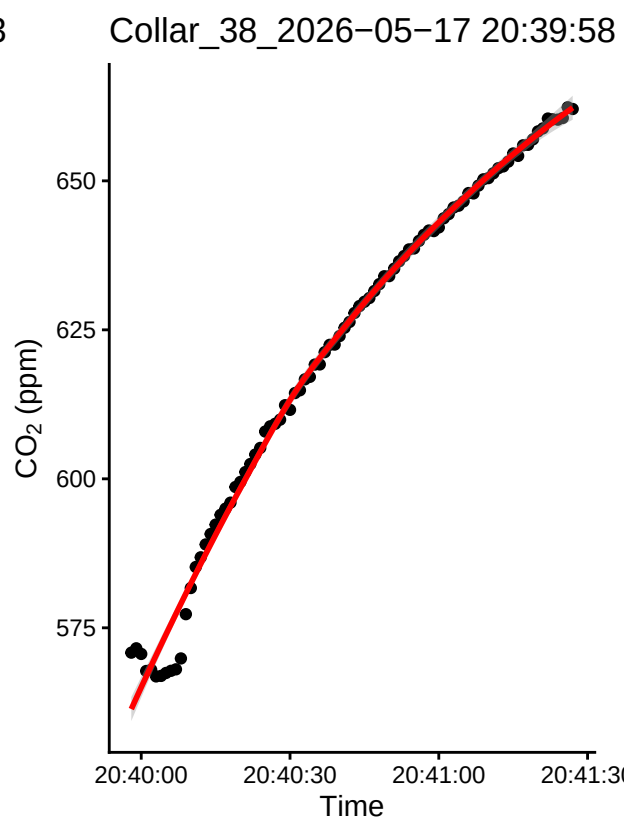
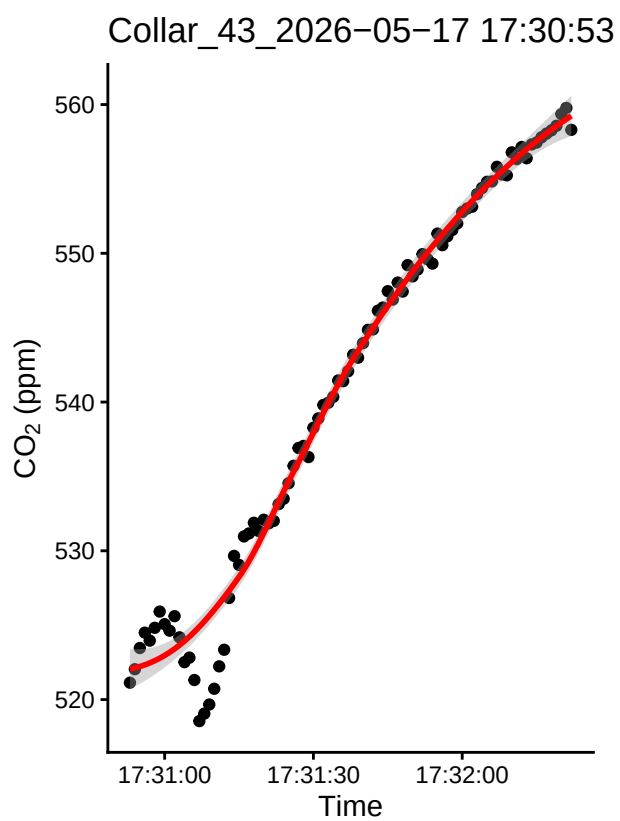
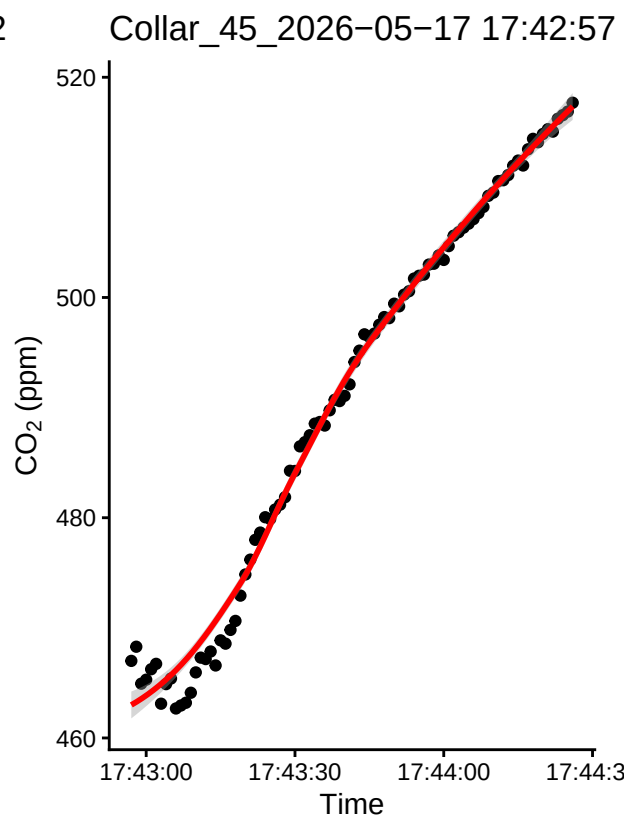
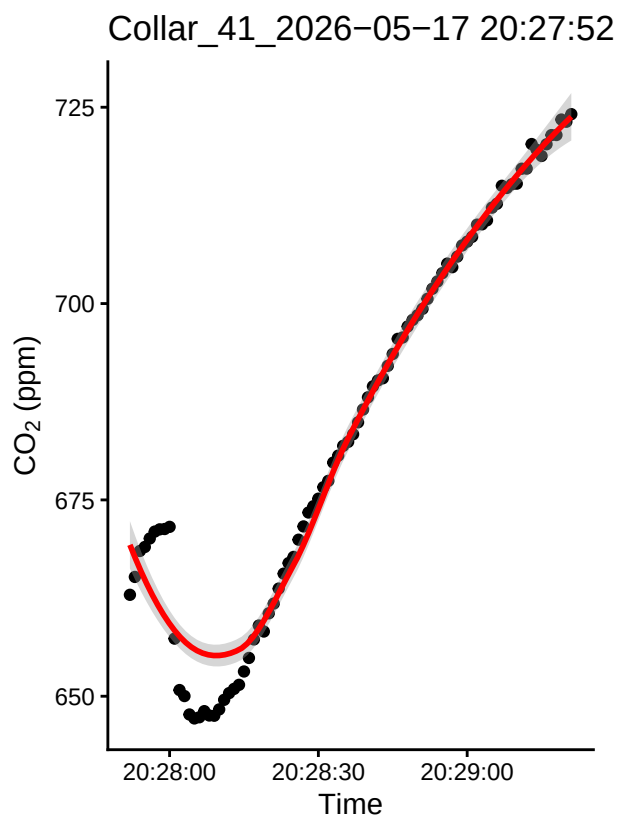
Check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

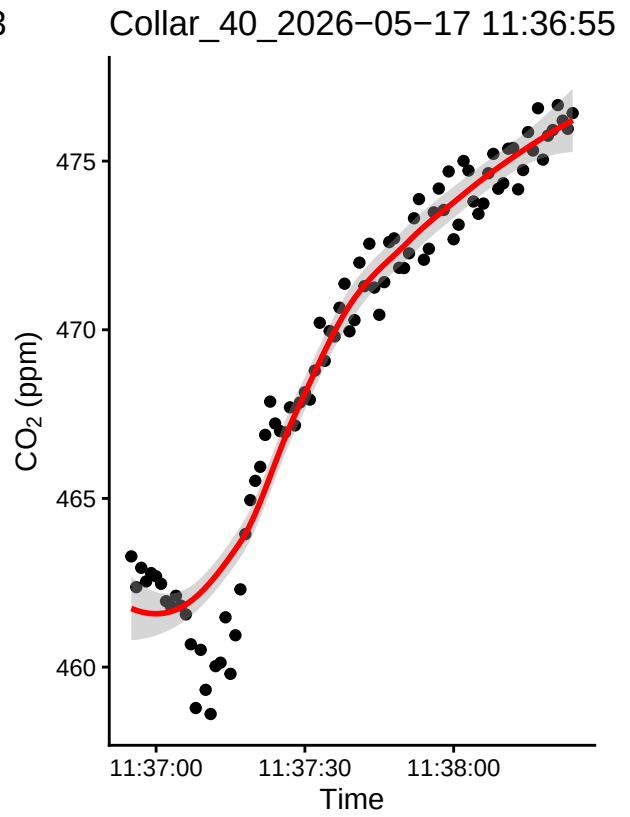
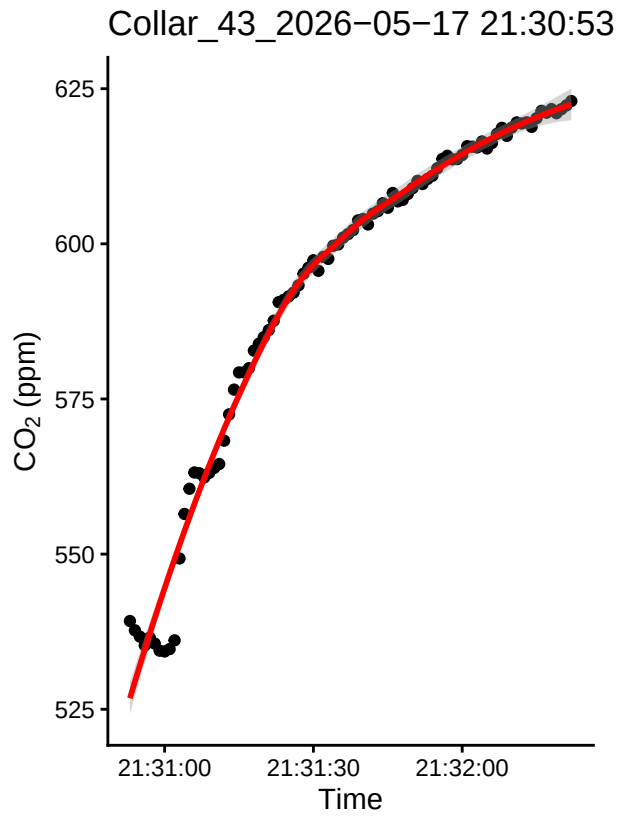
Spot check a few fluxes with $R^2 > 0.9$

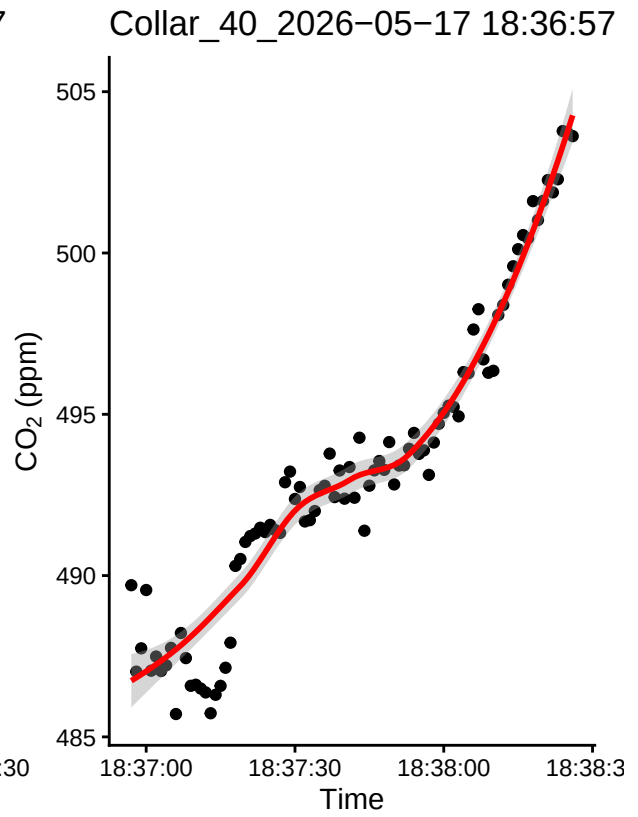
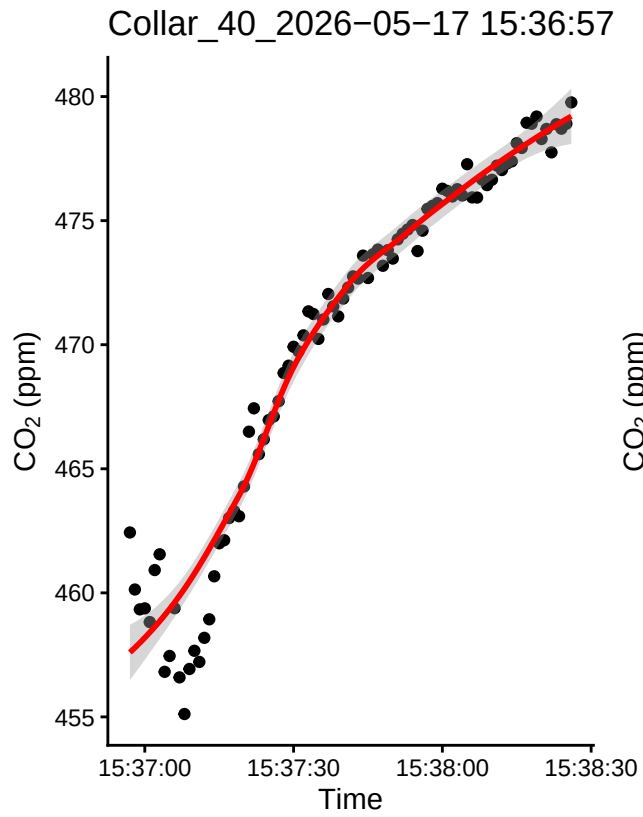


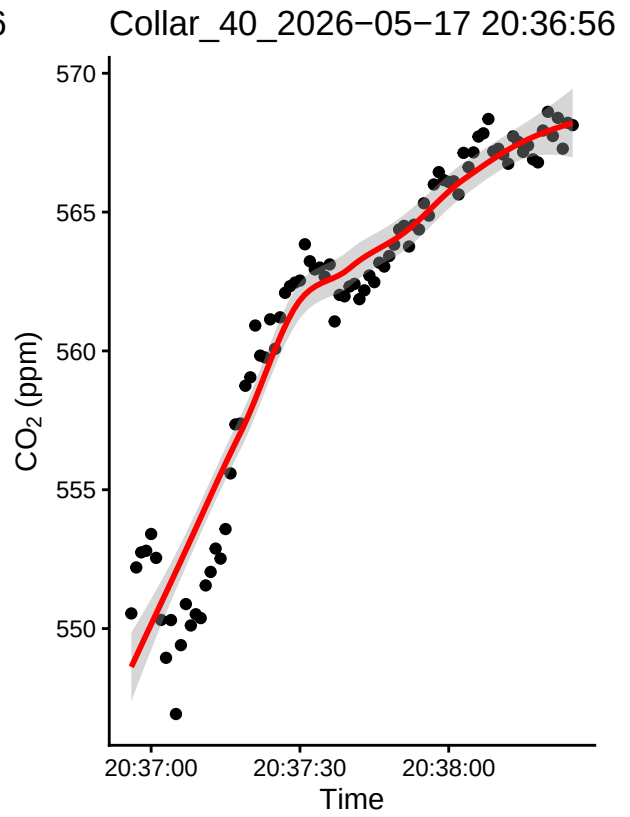
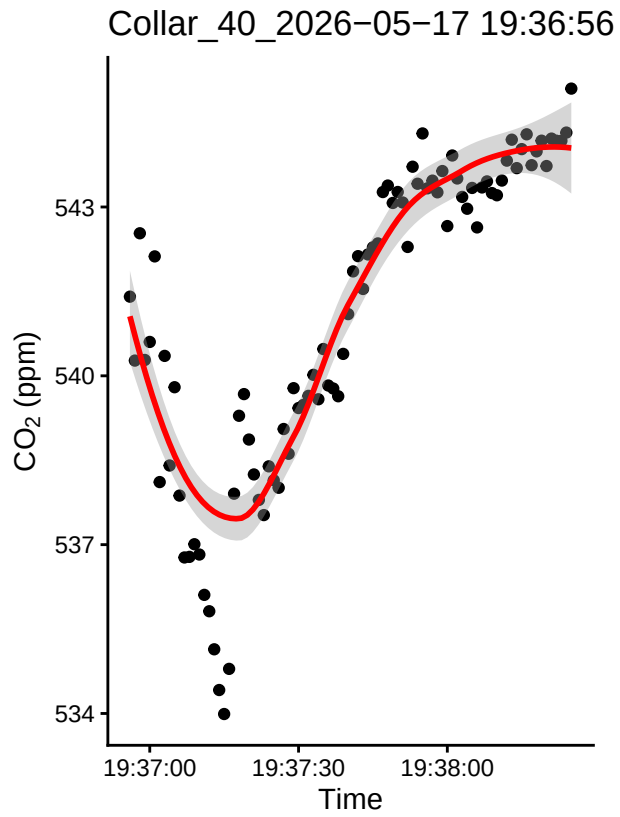


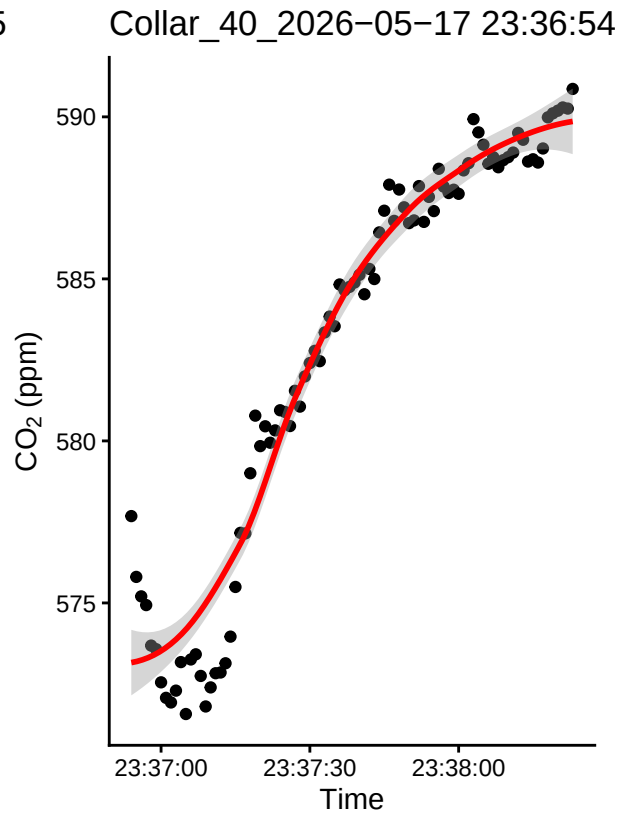
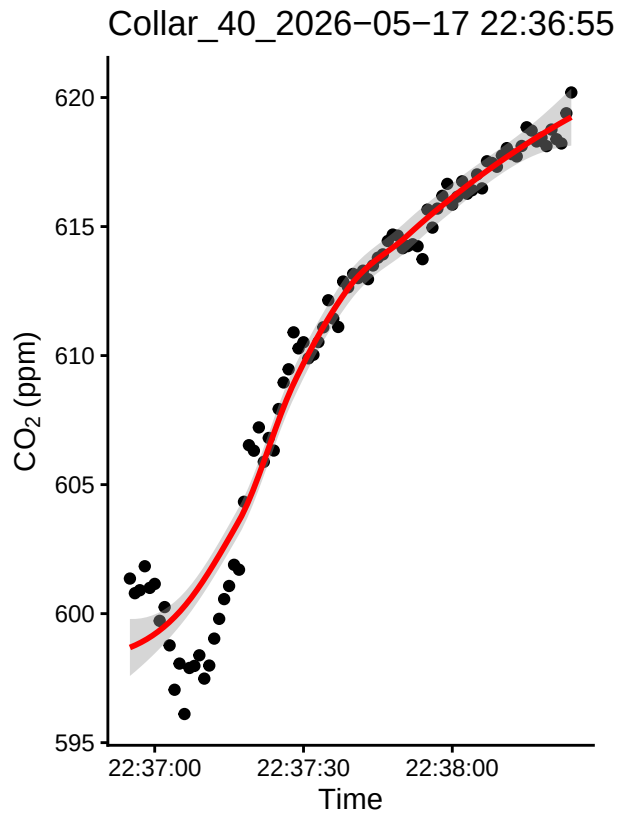
Take a look at the bad fluxes <0.9

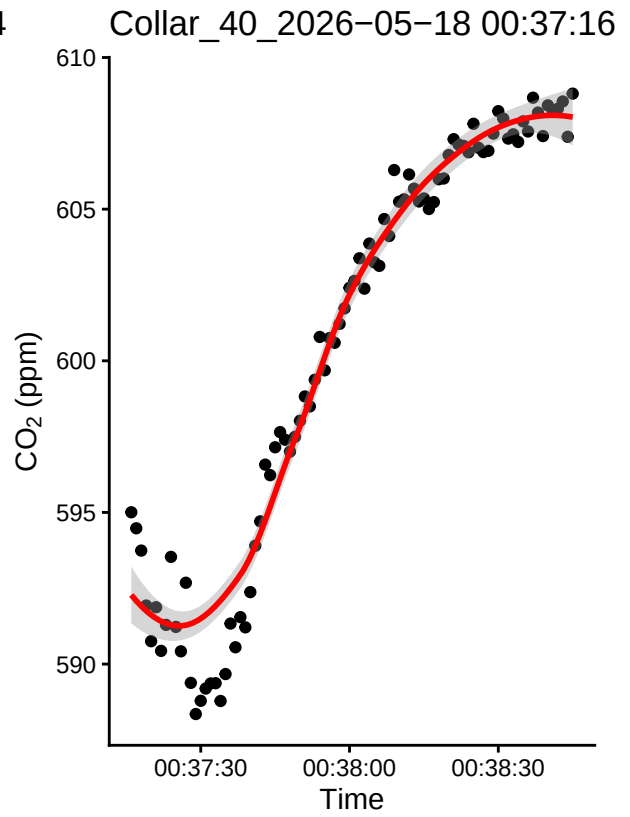
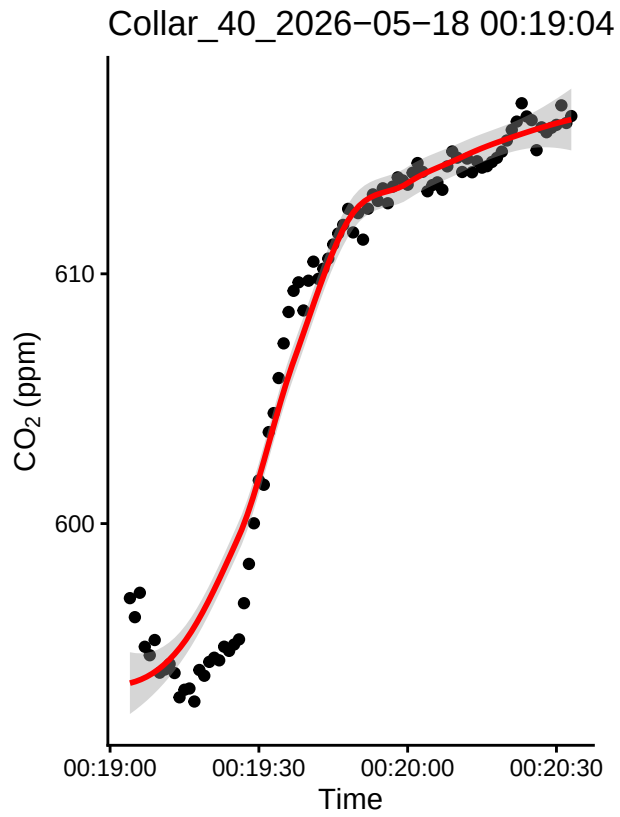


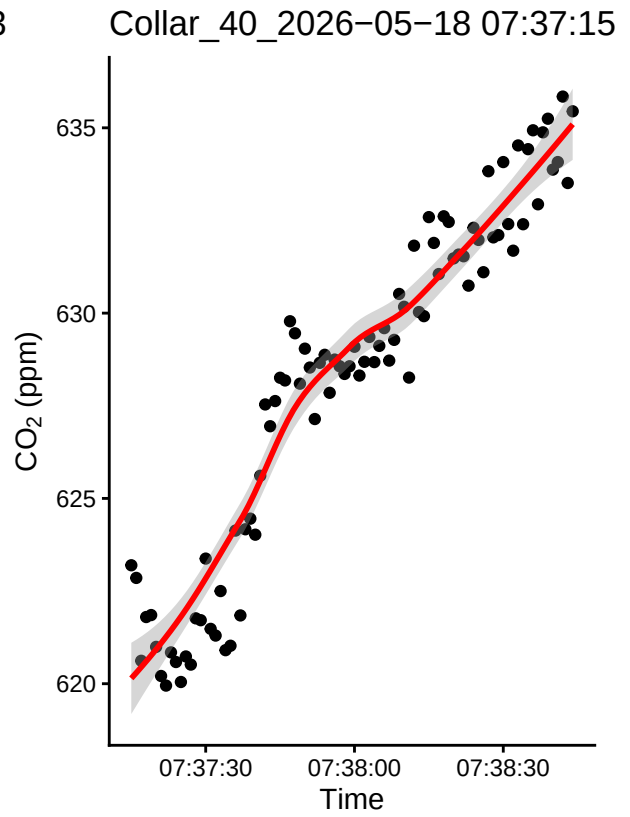
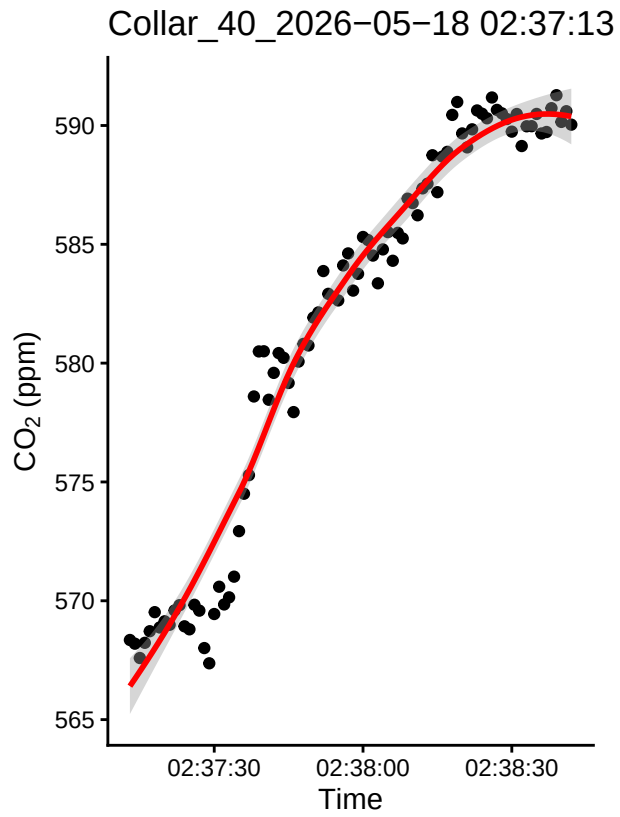


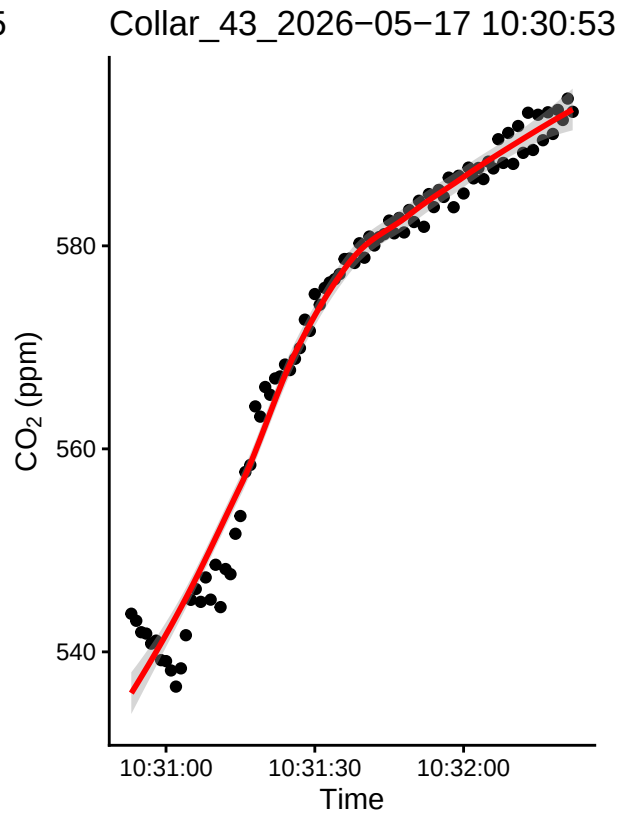
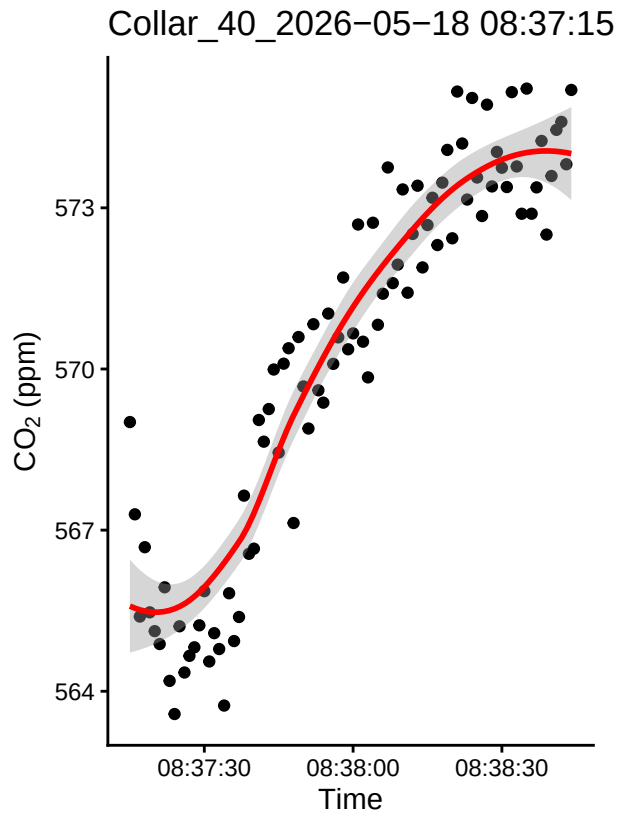


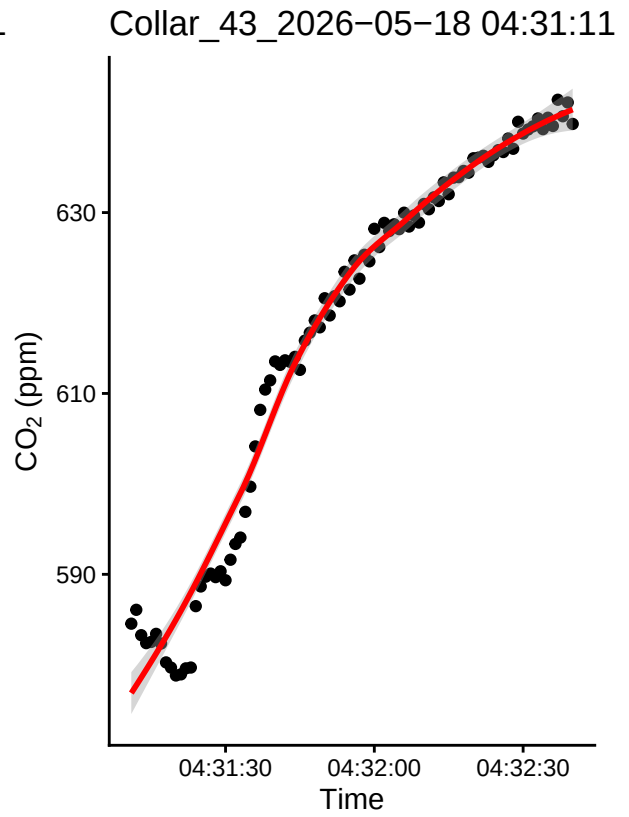
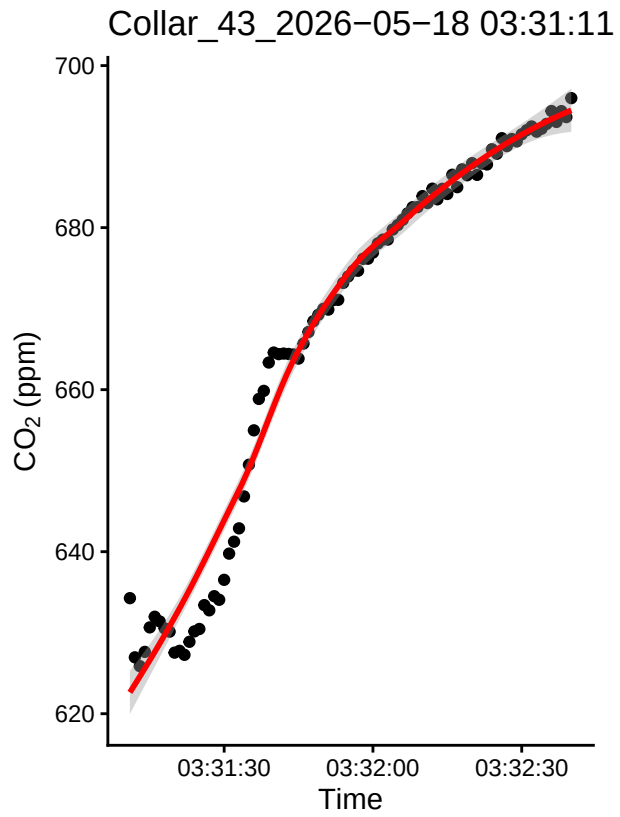


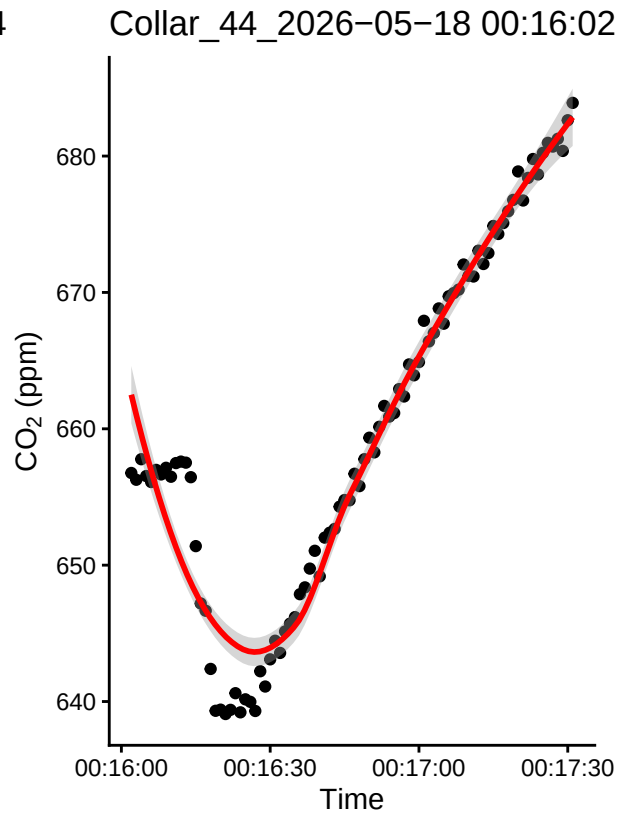
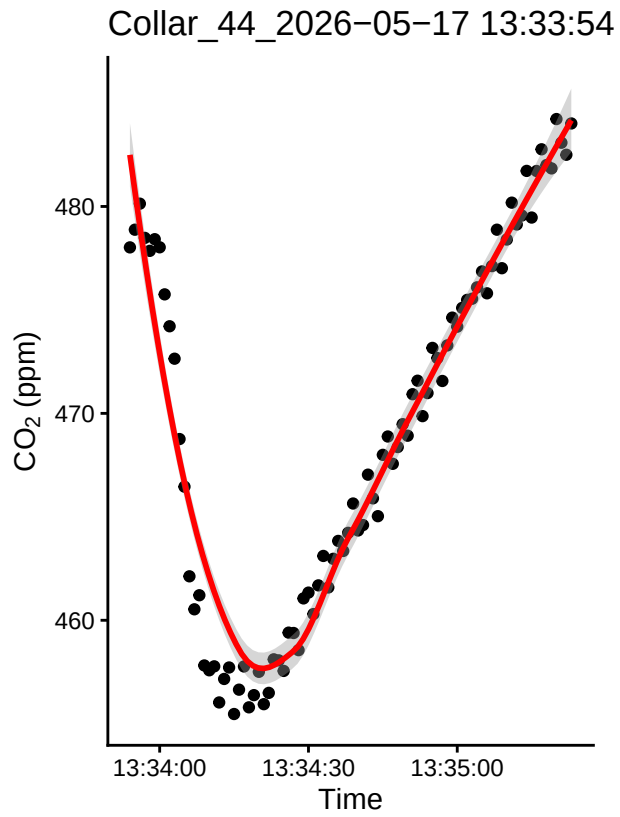




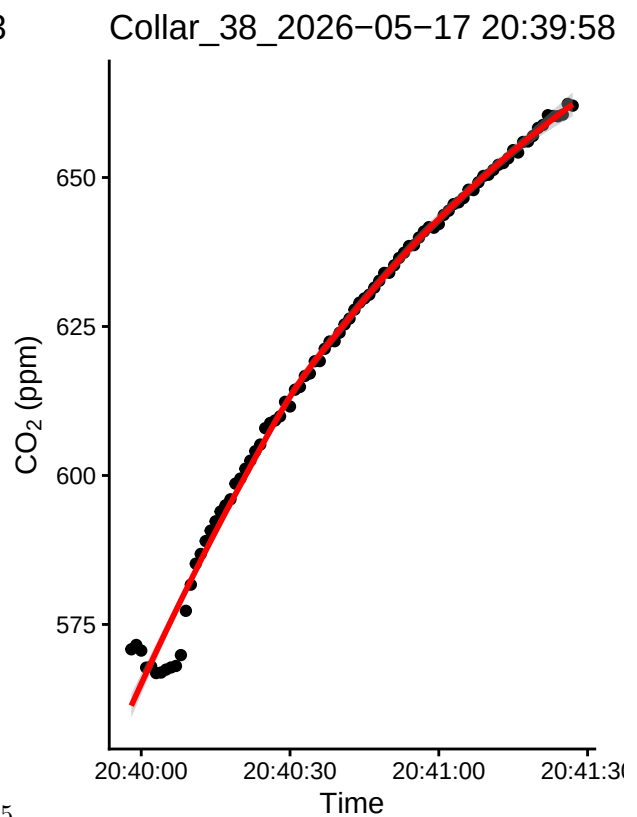
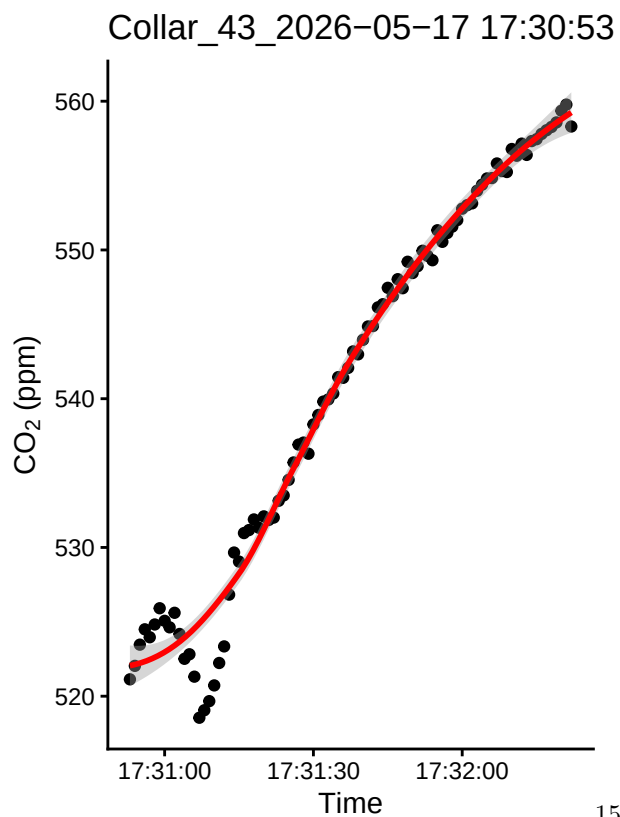
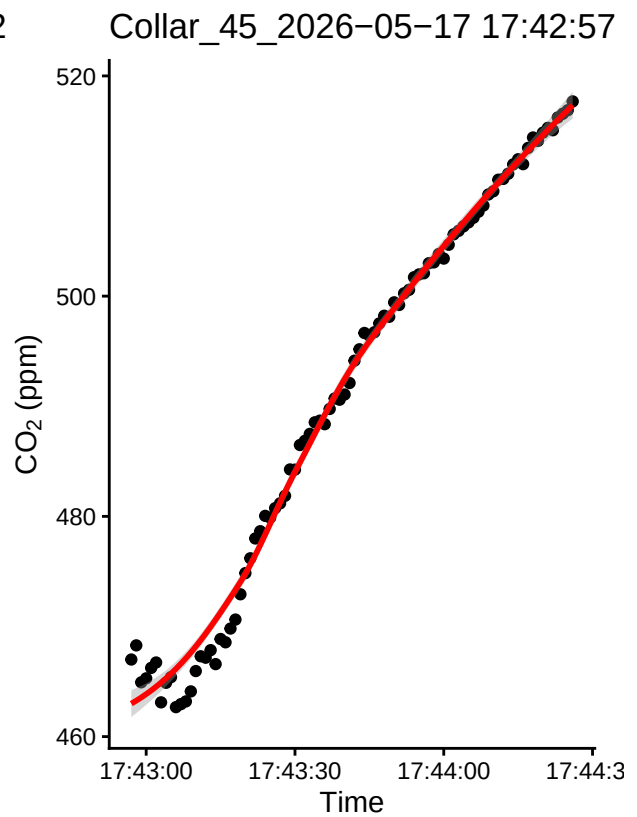
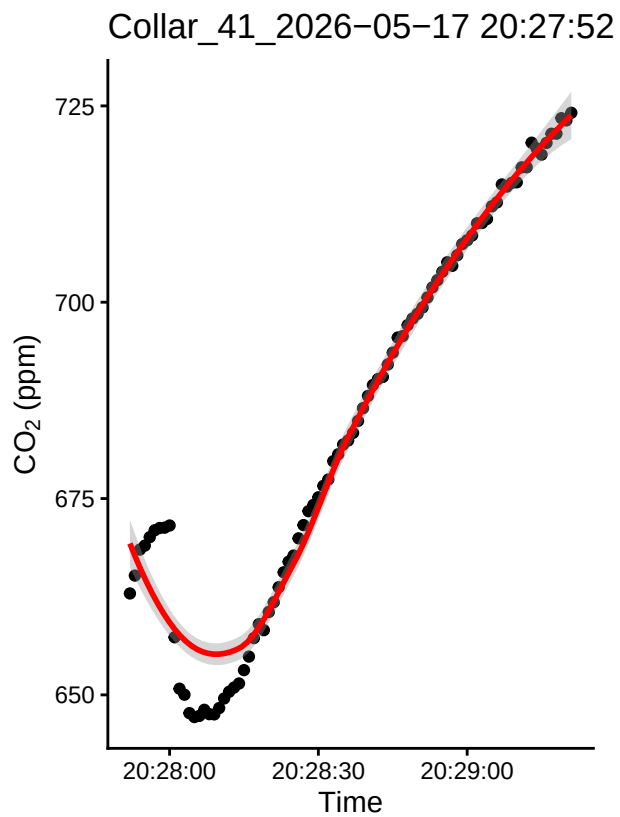


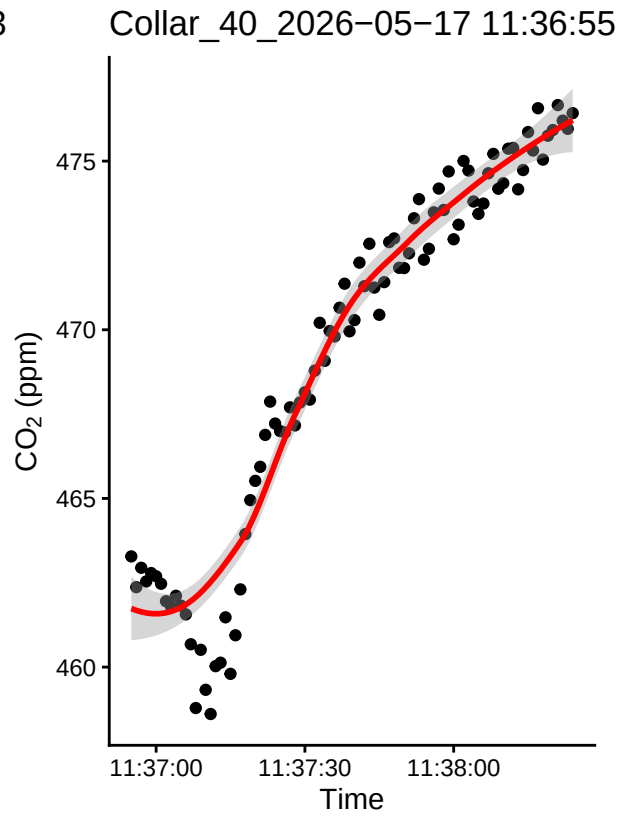
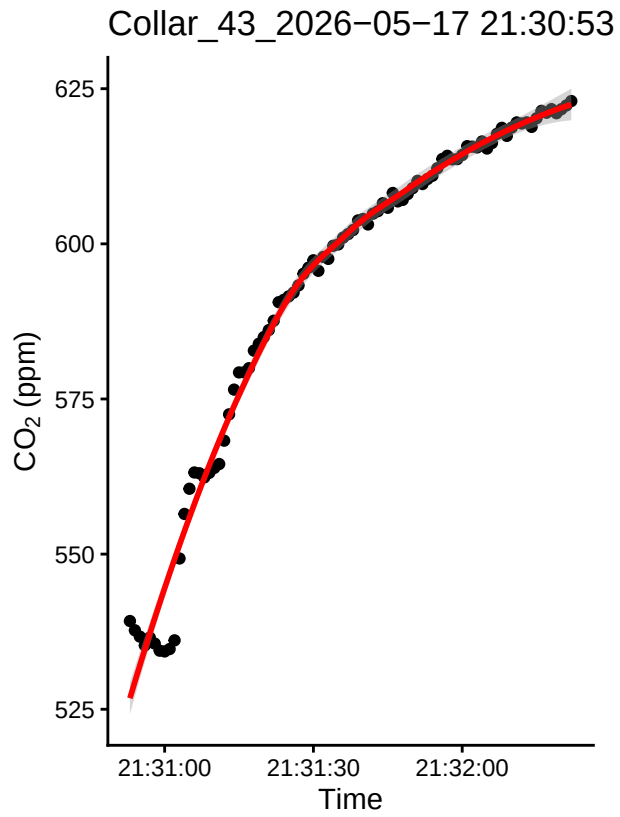


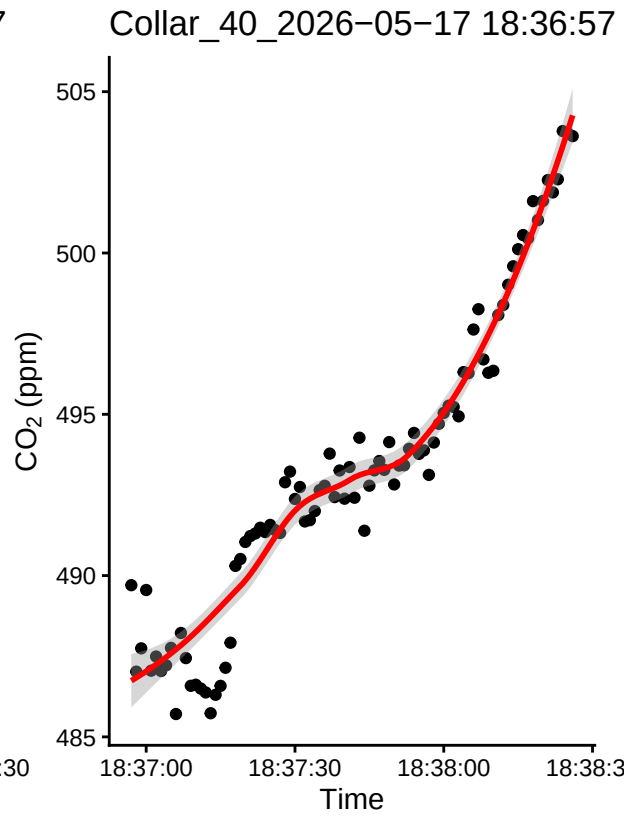
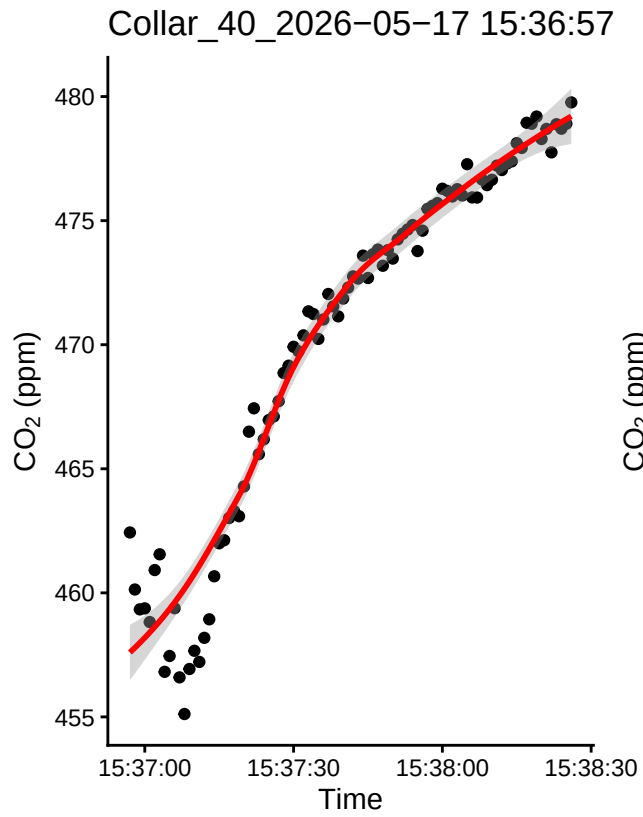


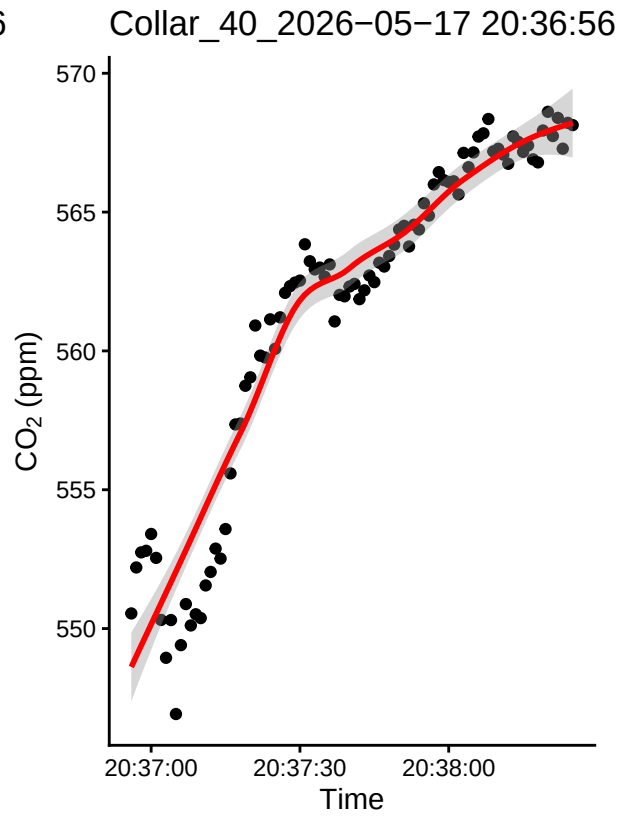
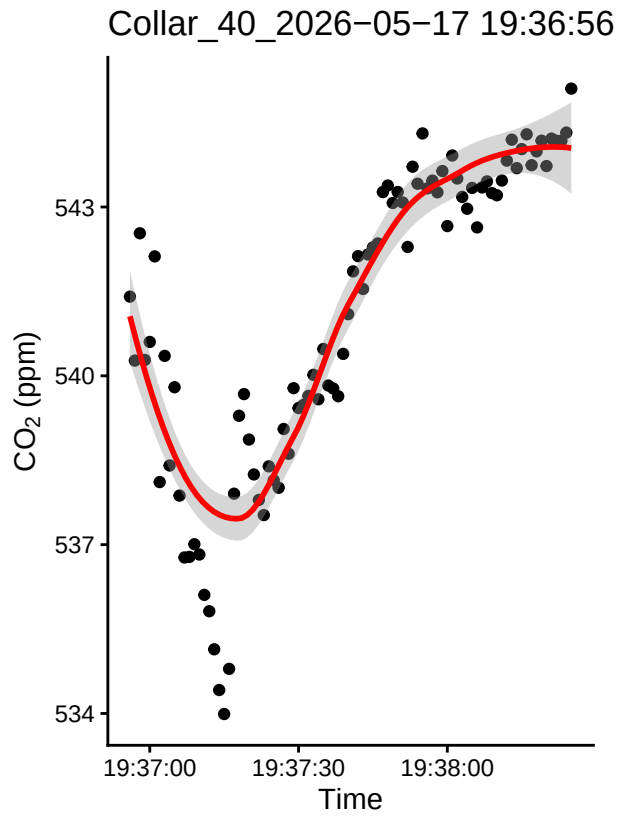


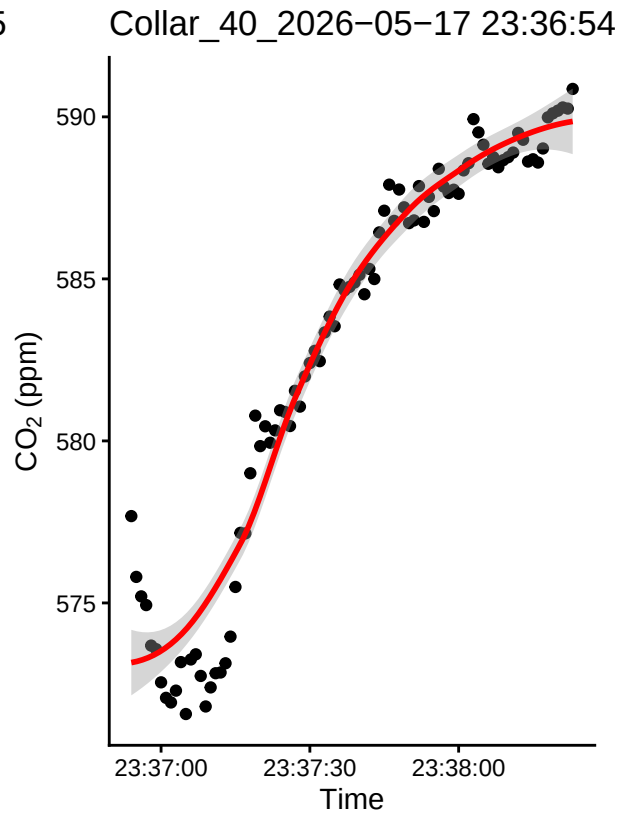
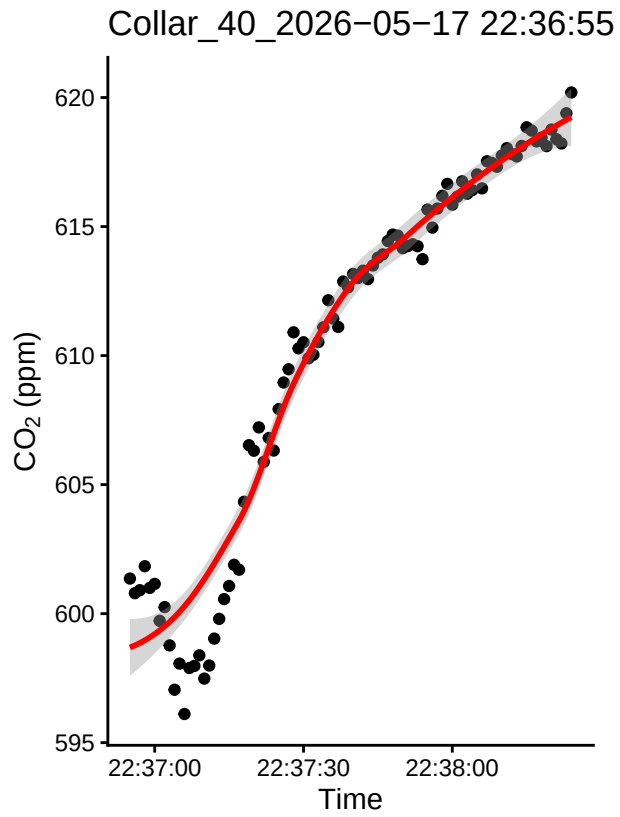
Look at the really high or really low ones to see if there is a timing issue or something

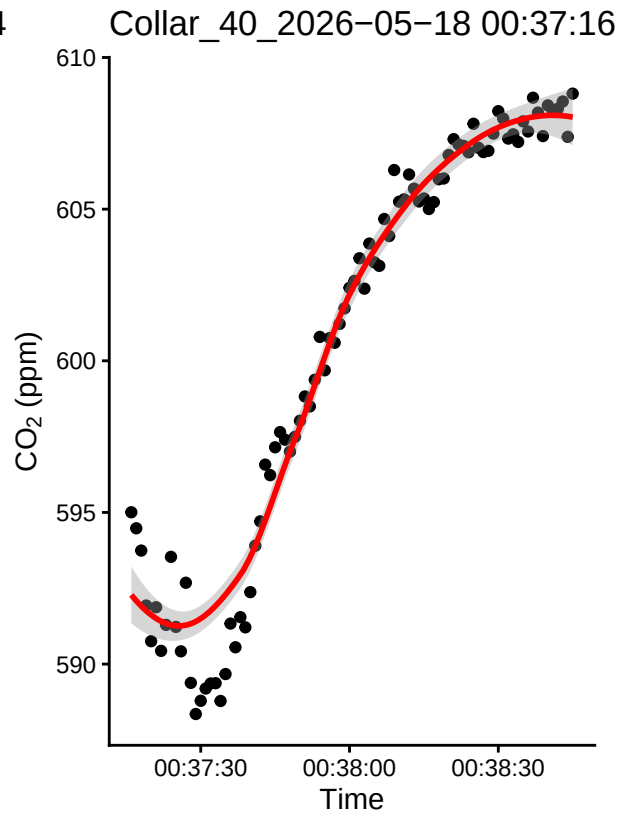
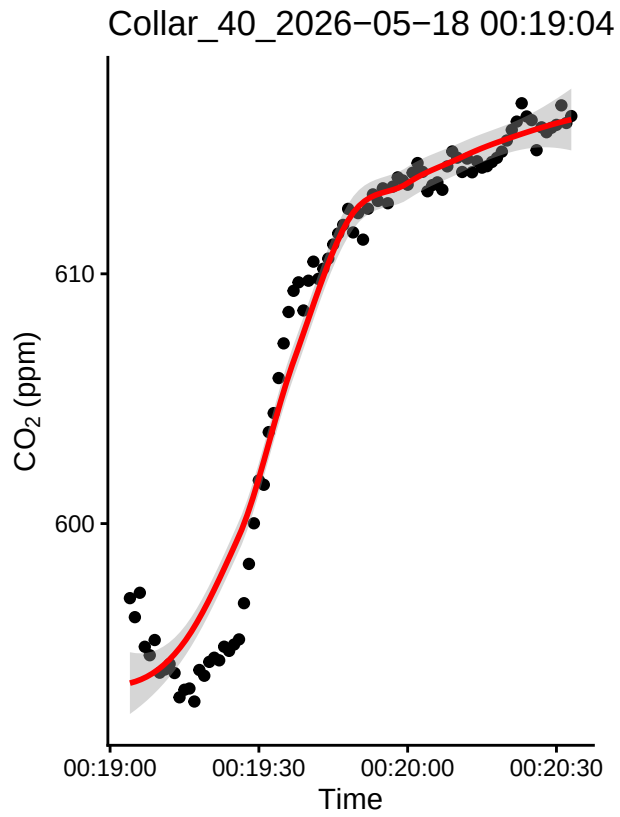


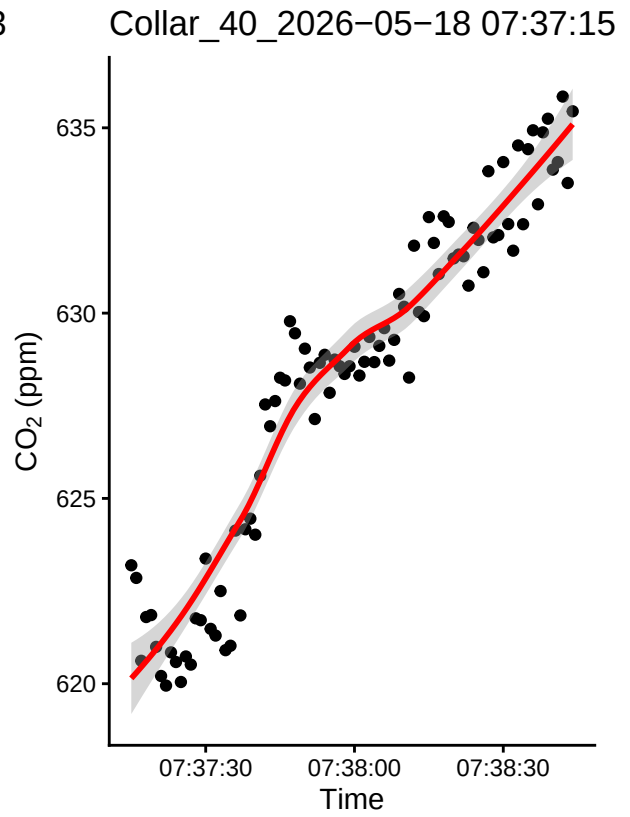
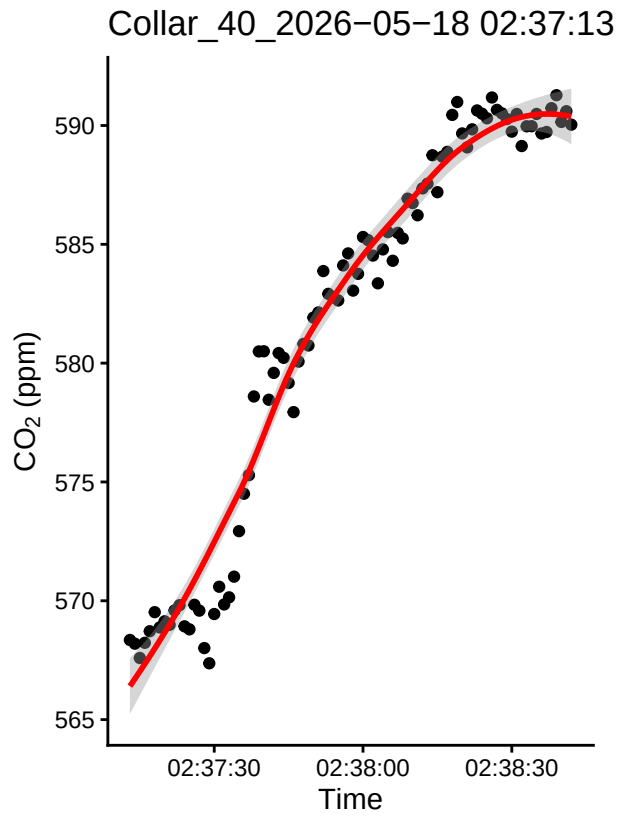


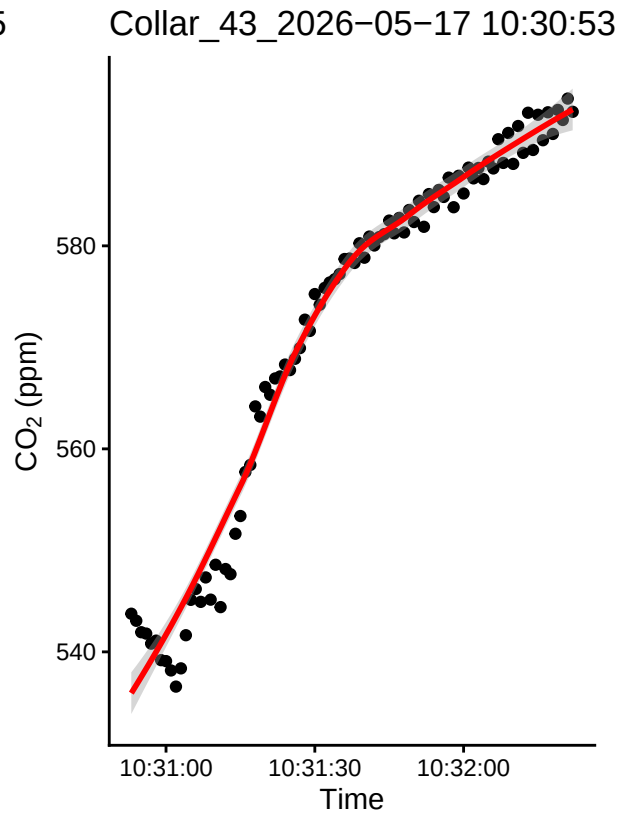
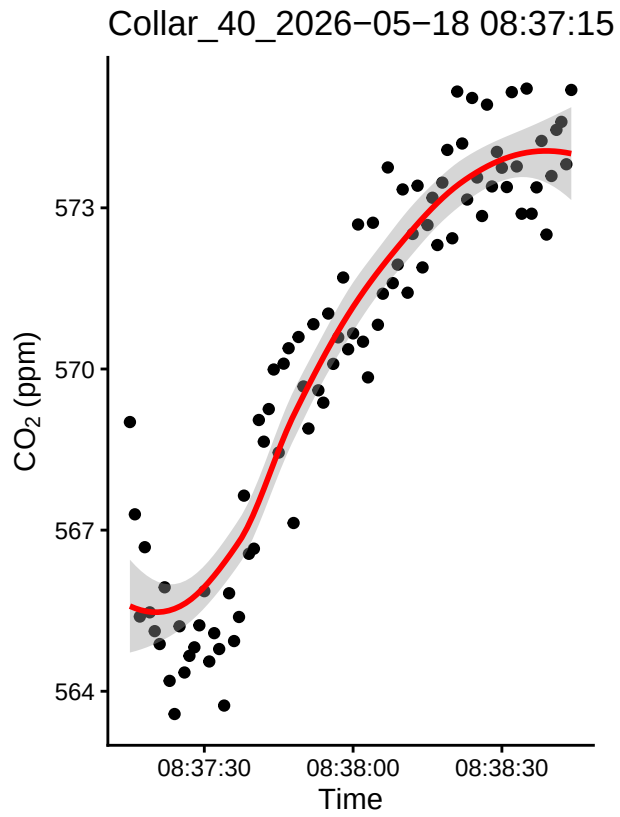


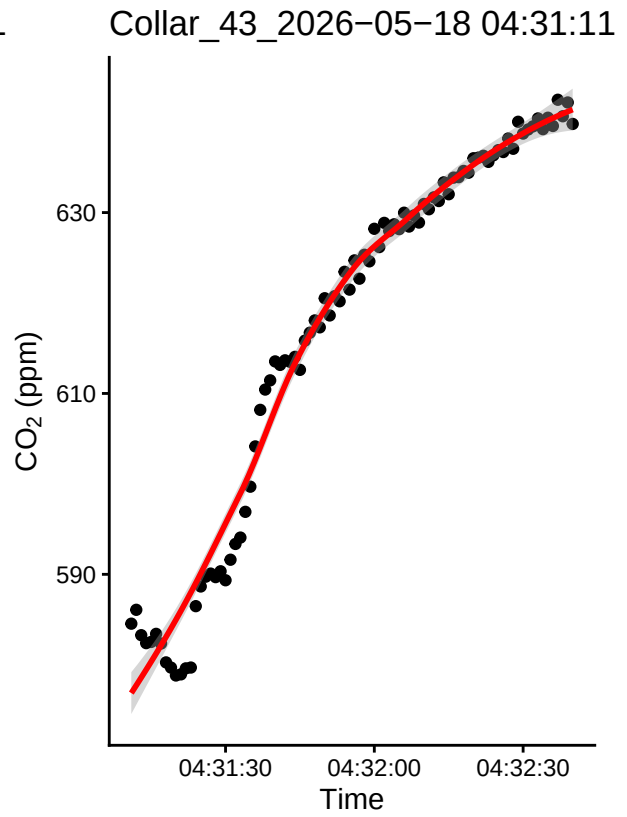
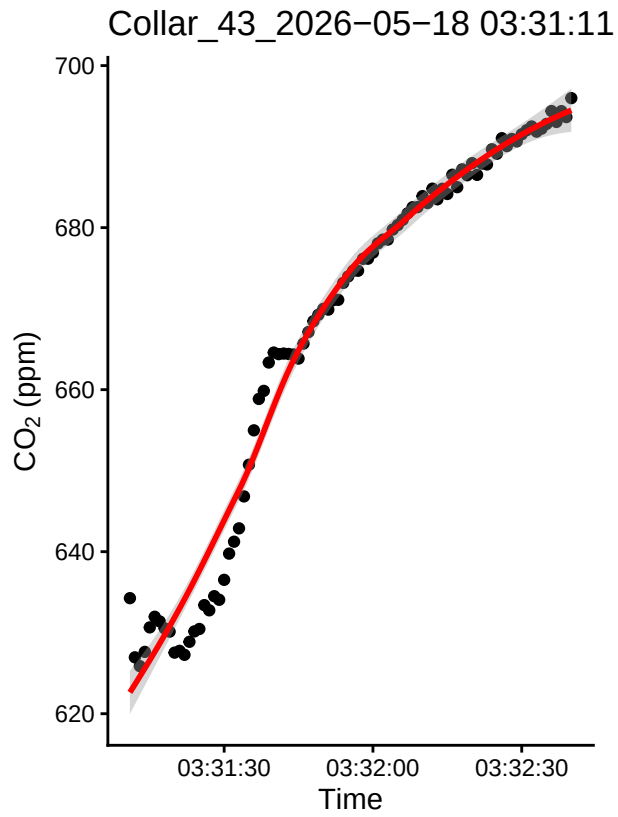




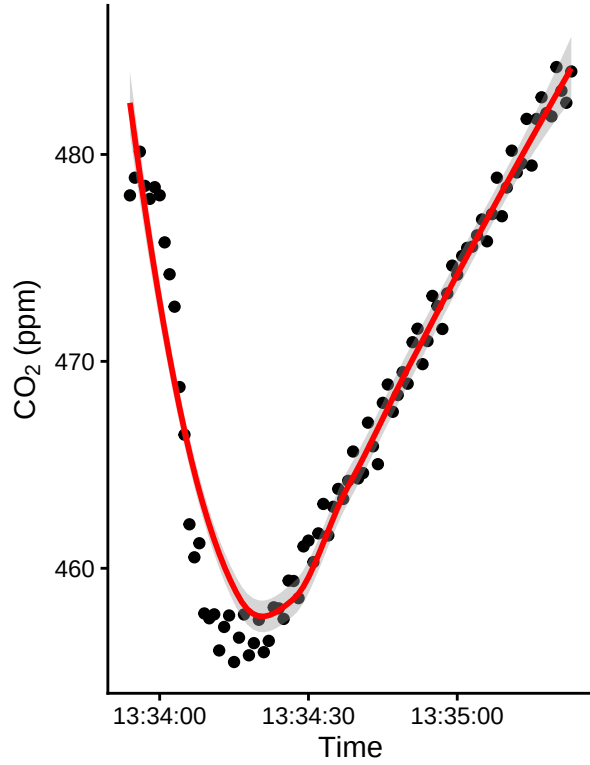




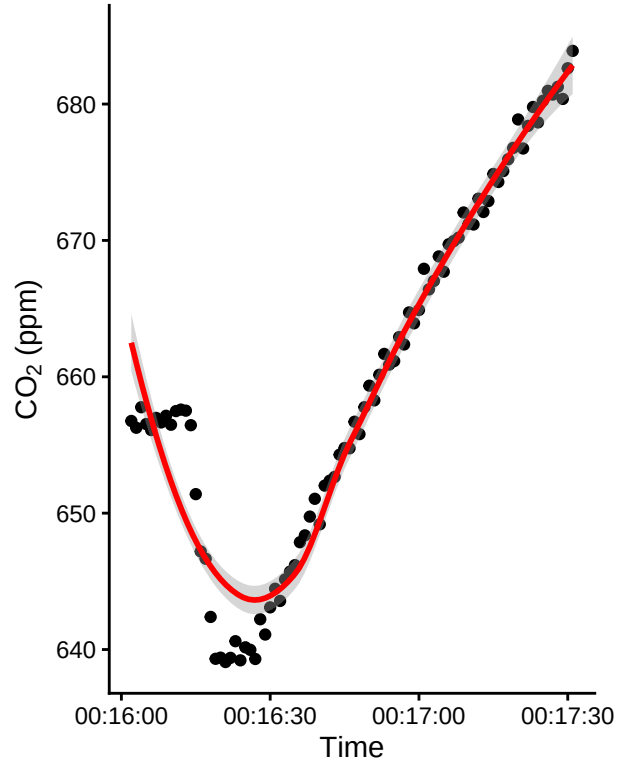


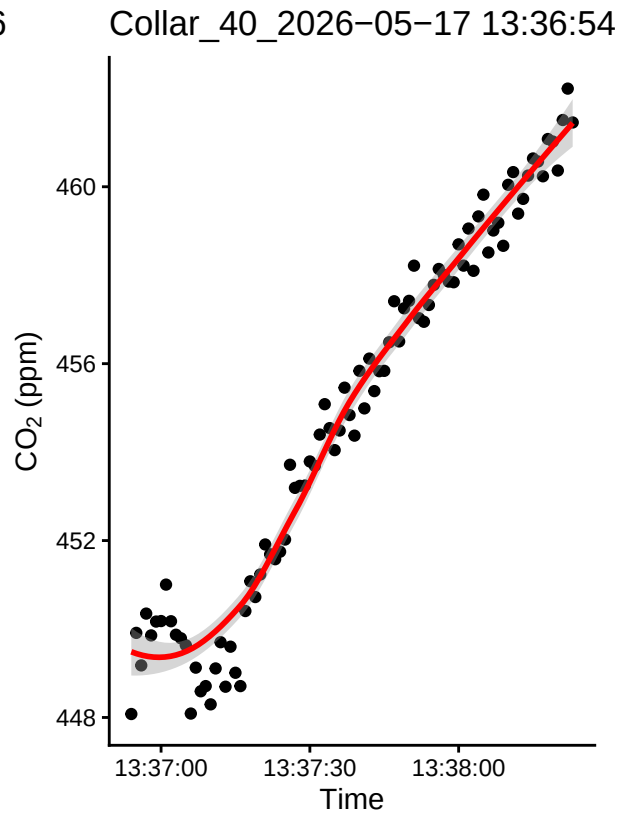
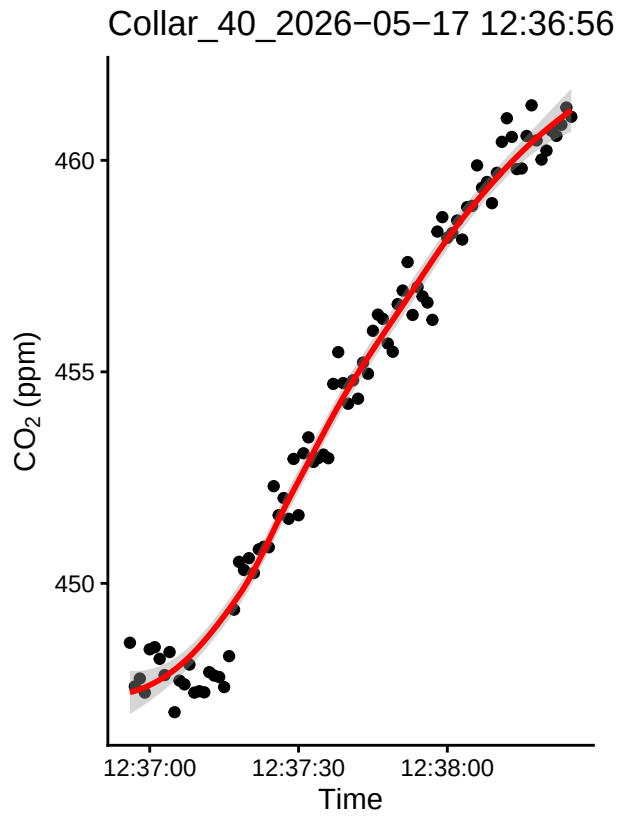


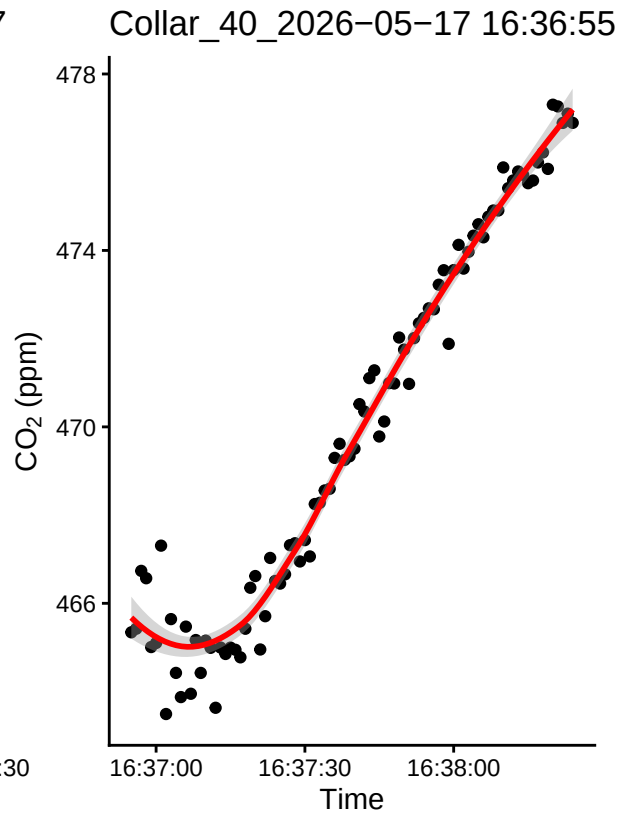
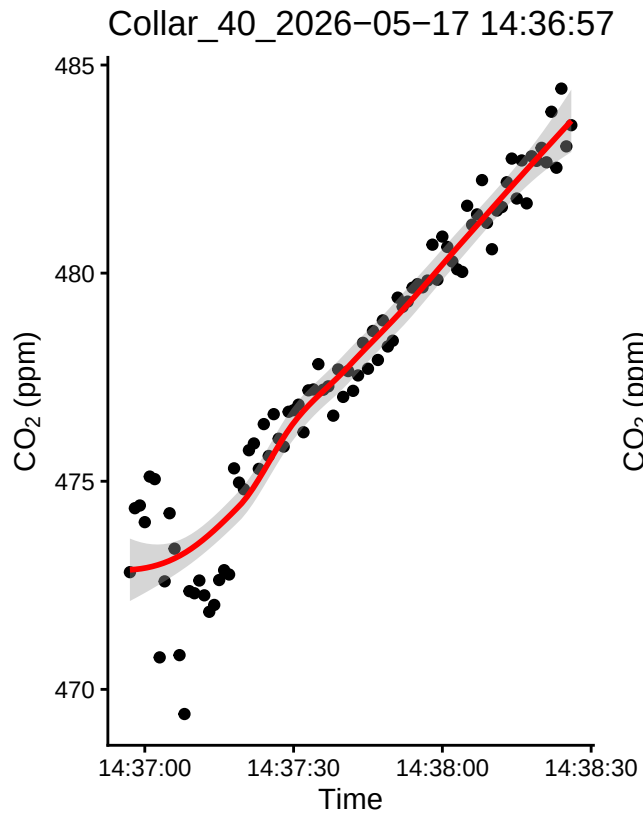
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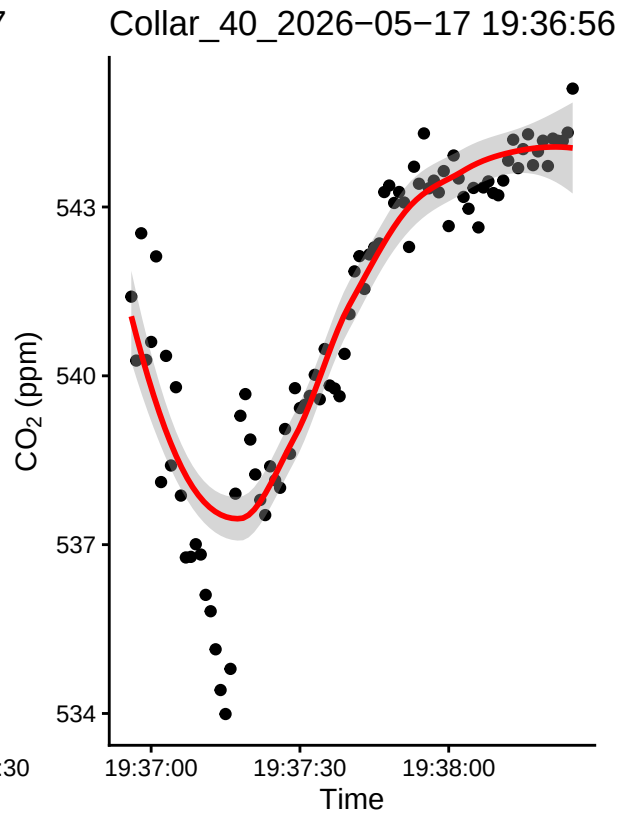
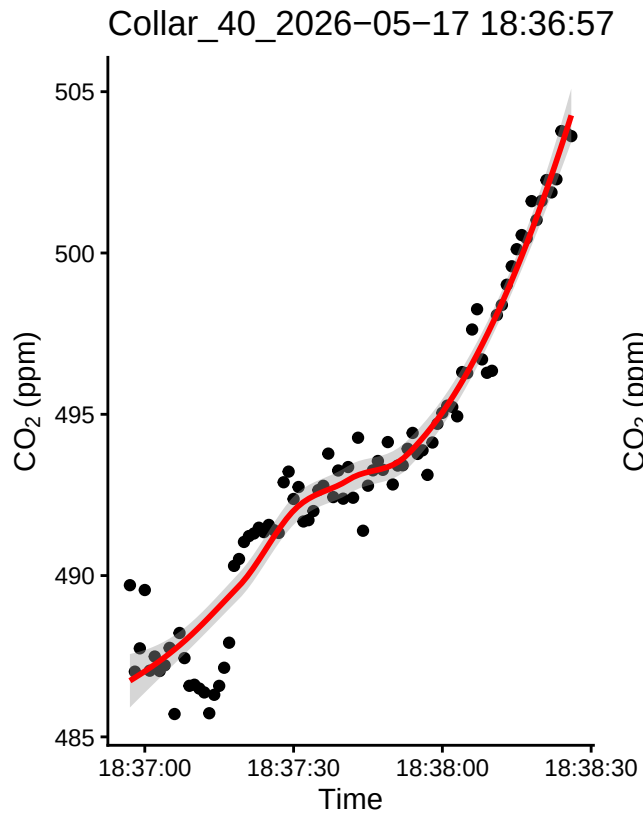


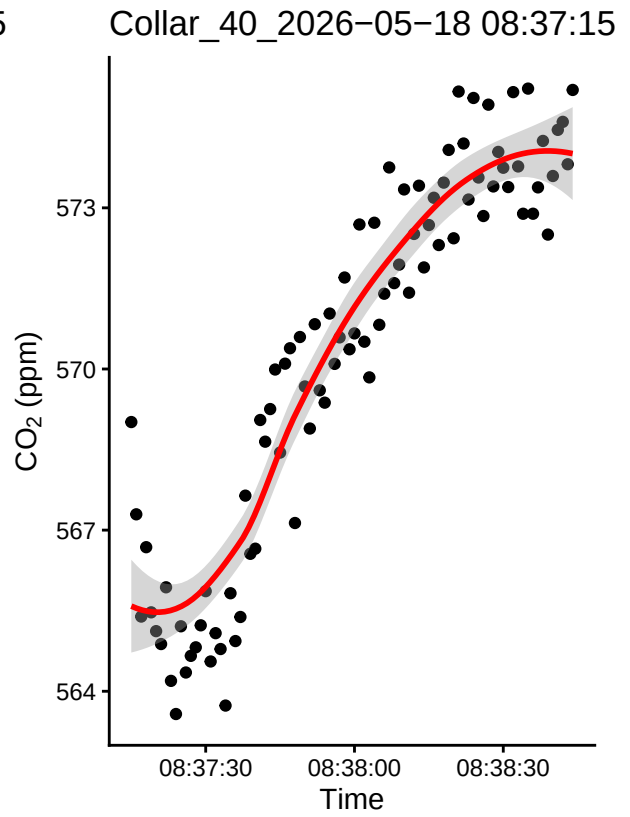
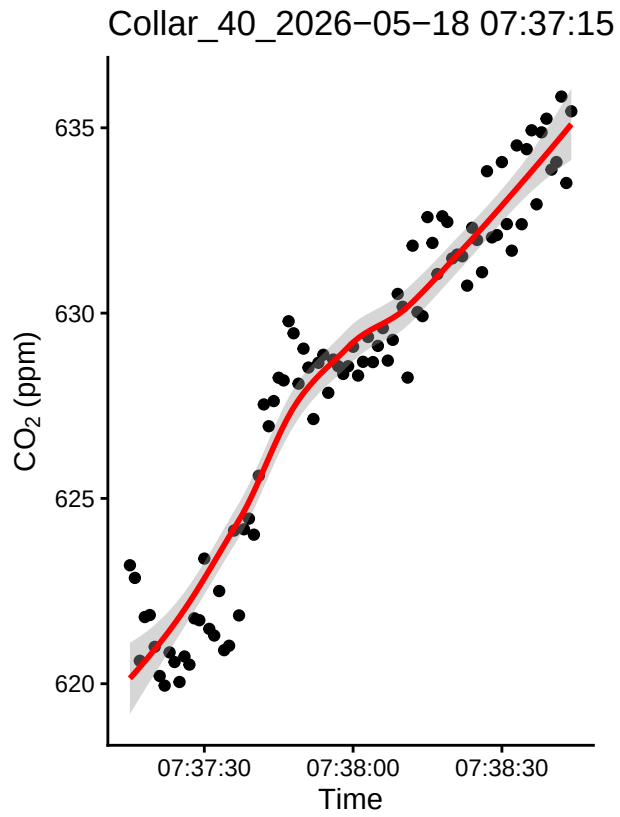
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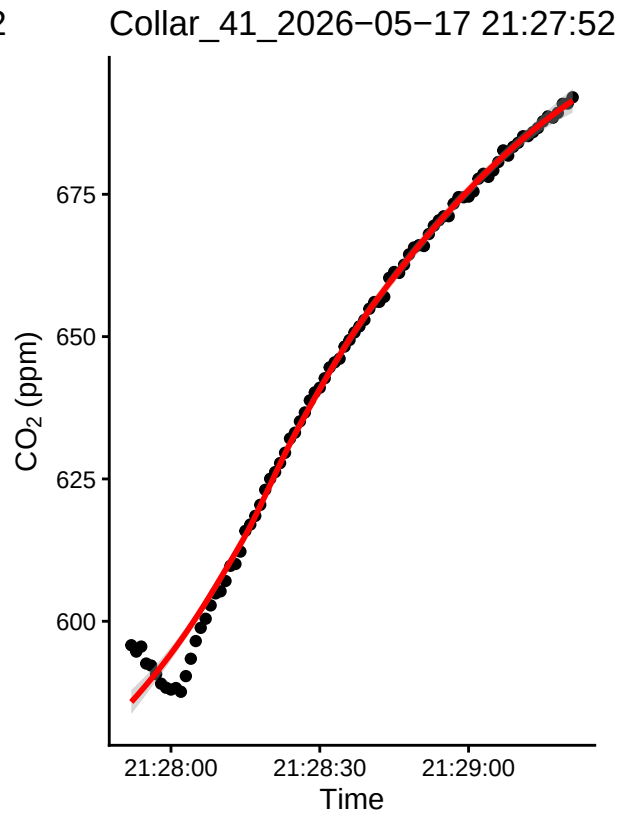
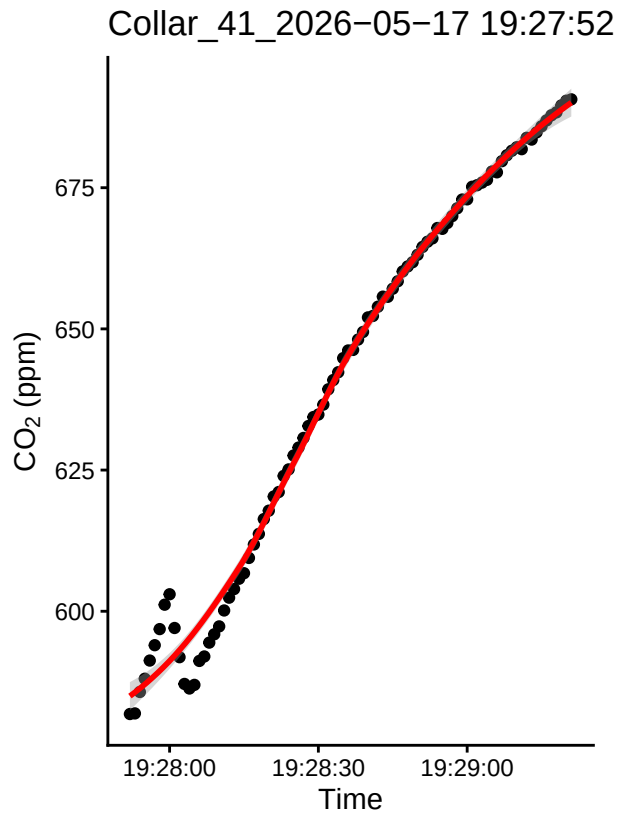


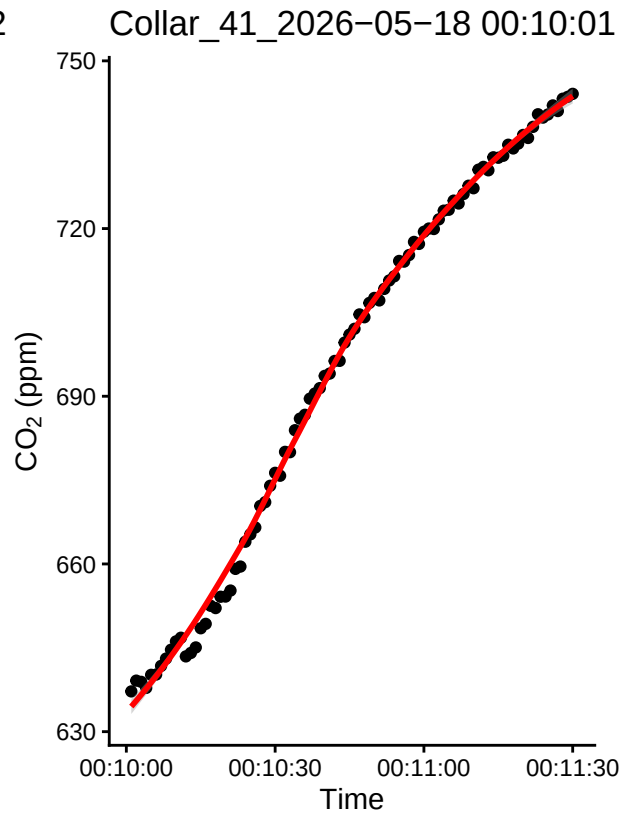
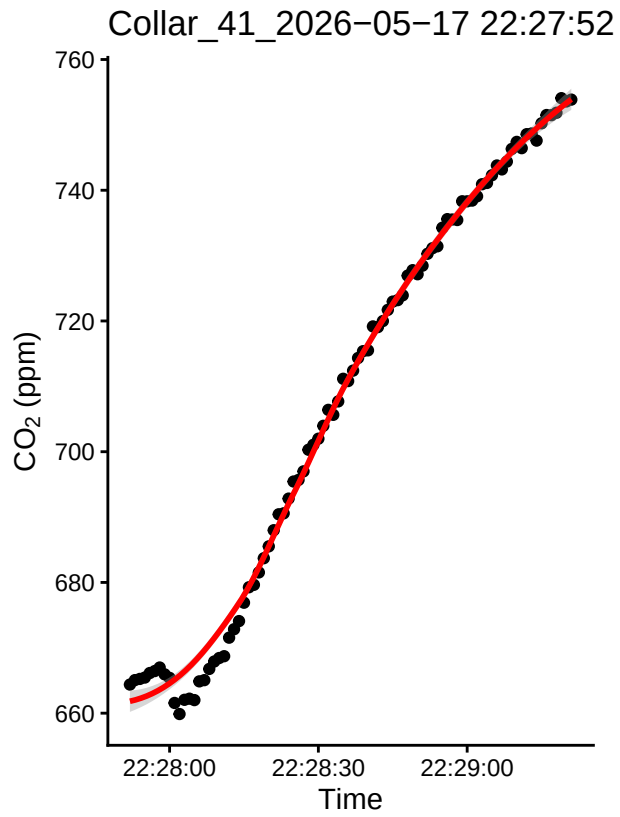


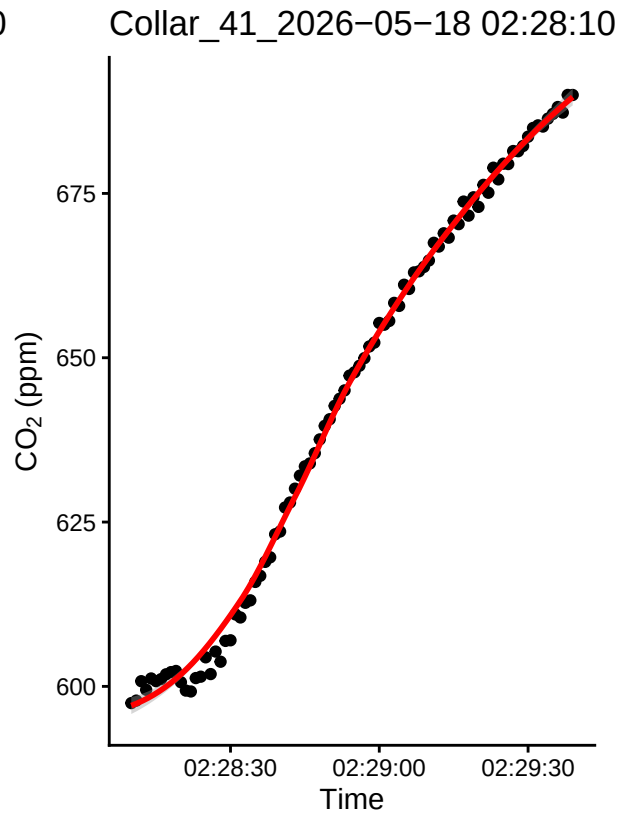
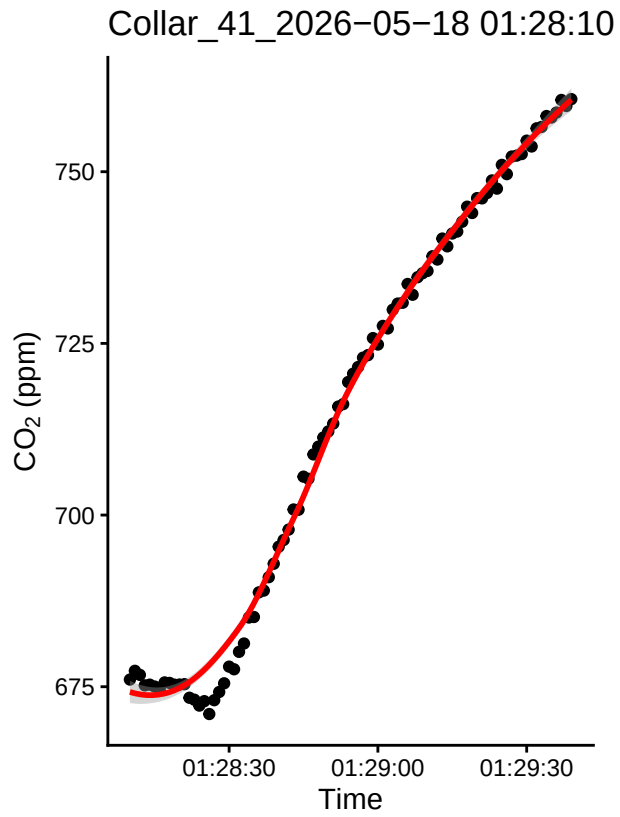


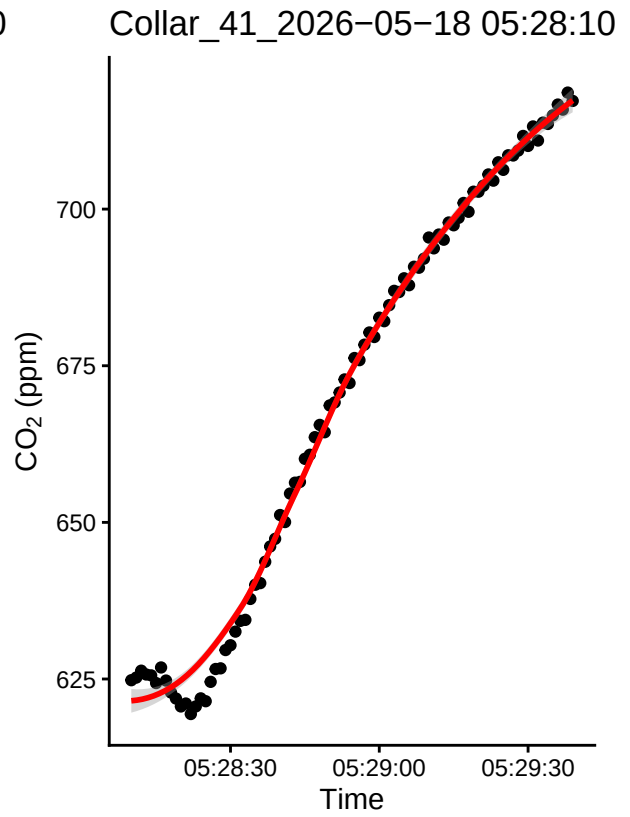
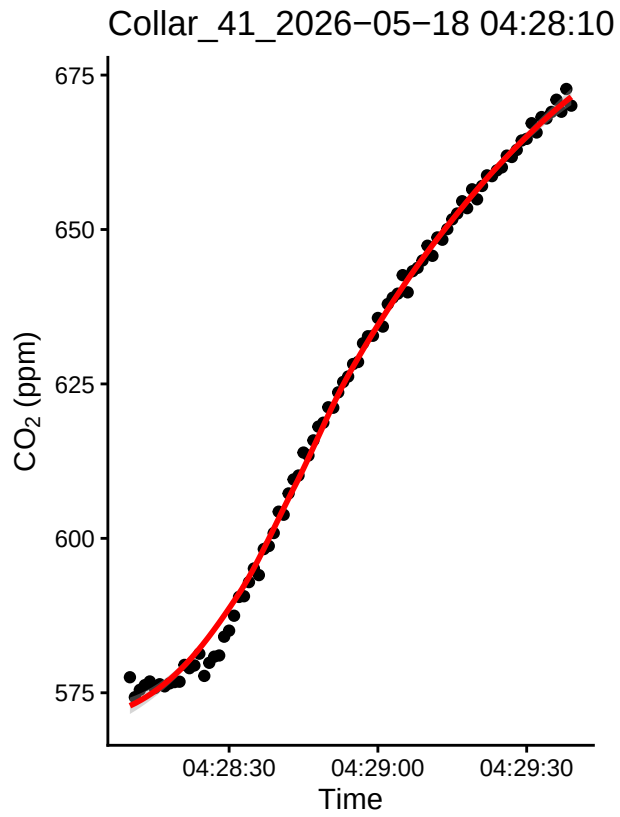




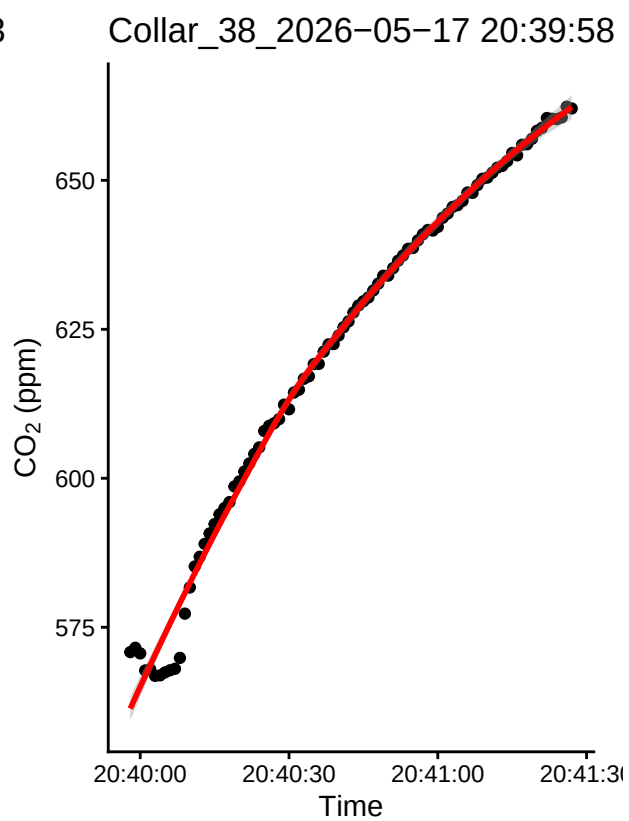
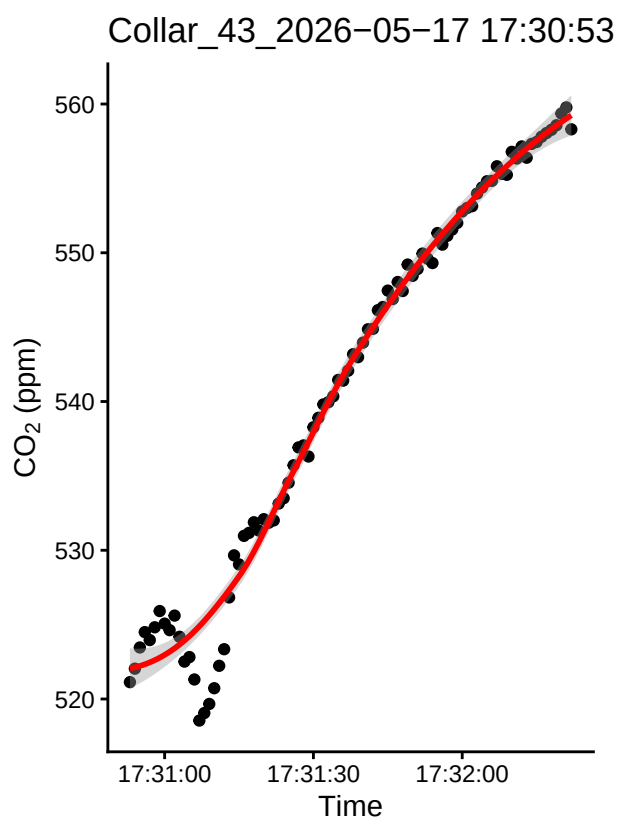
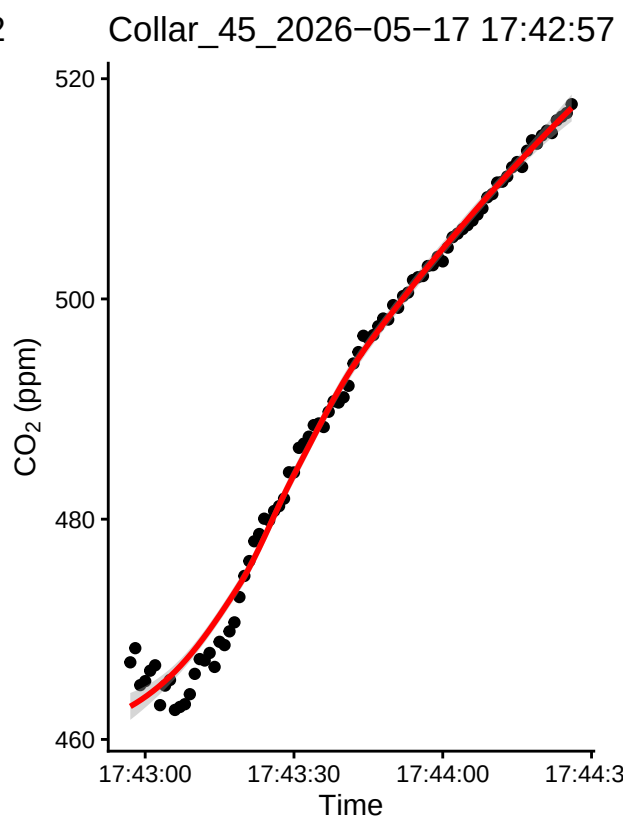
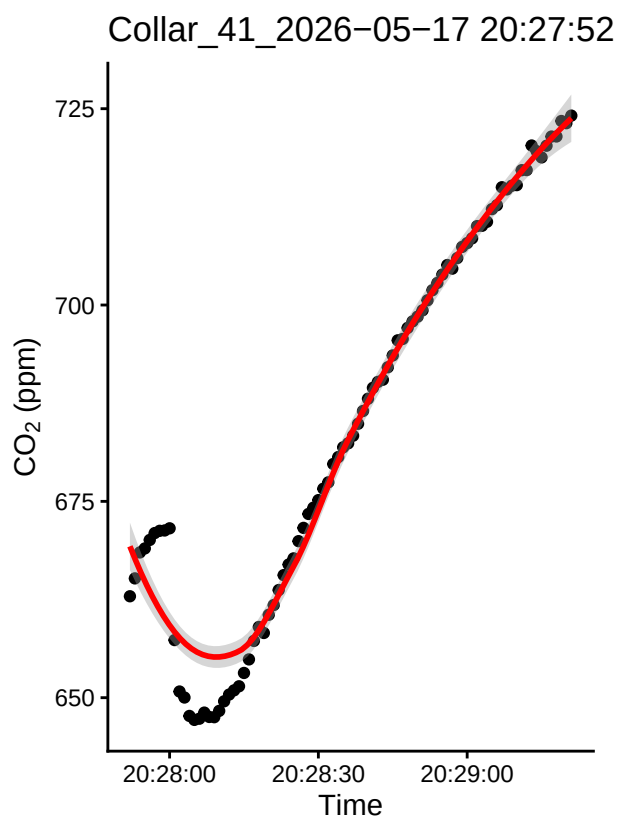


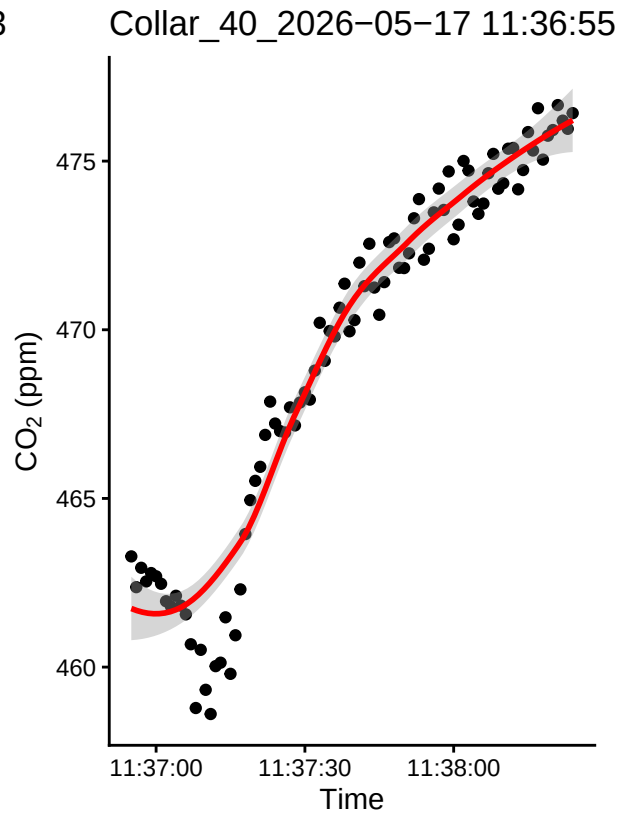
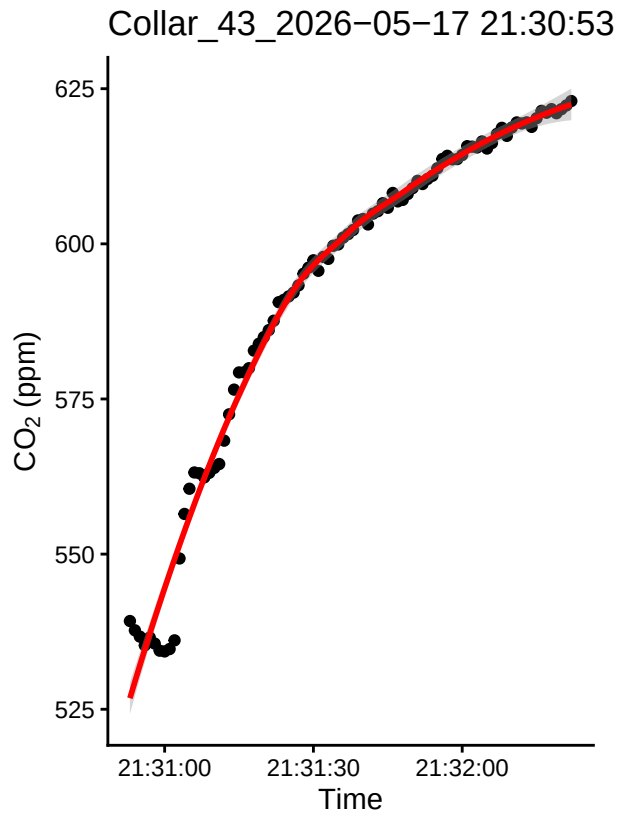


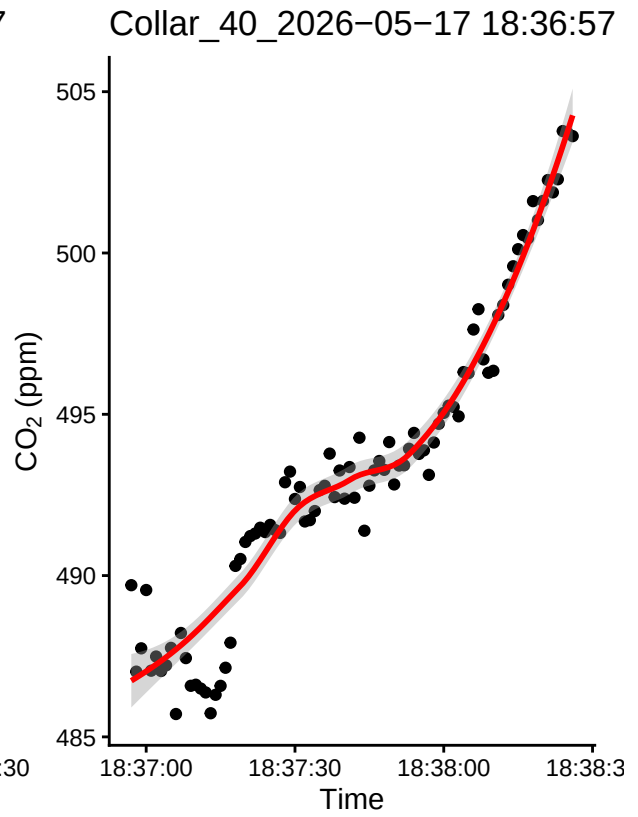
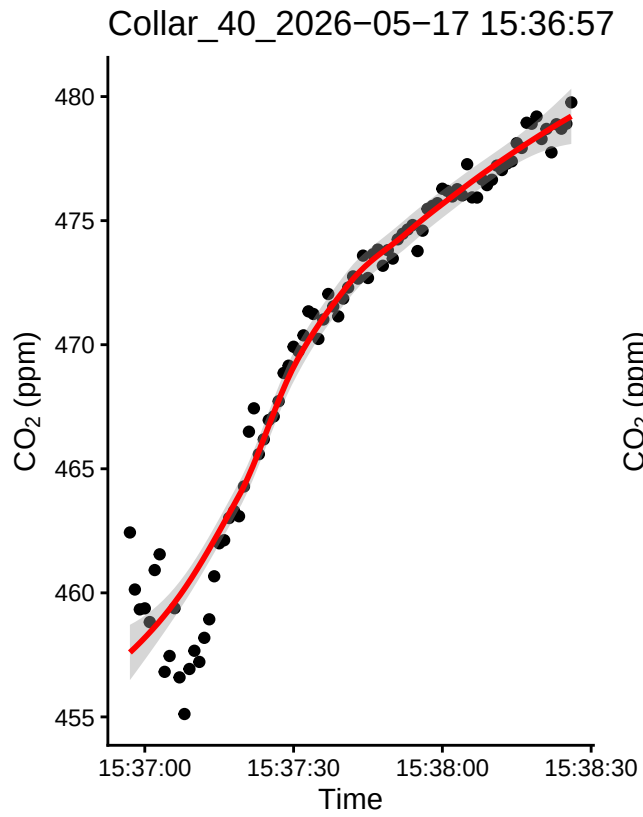


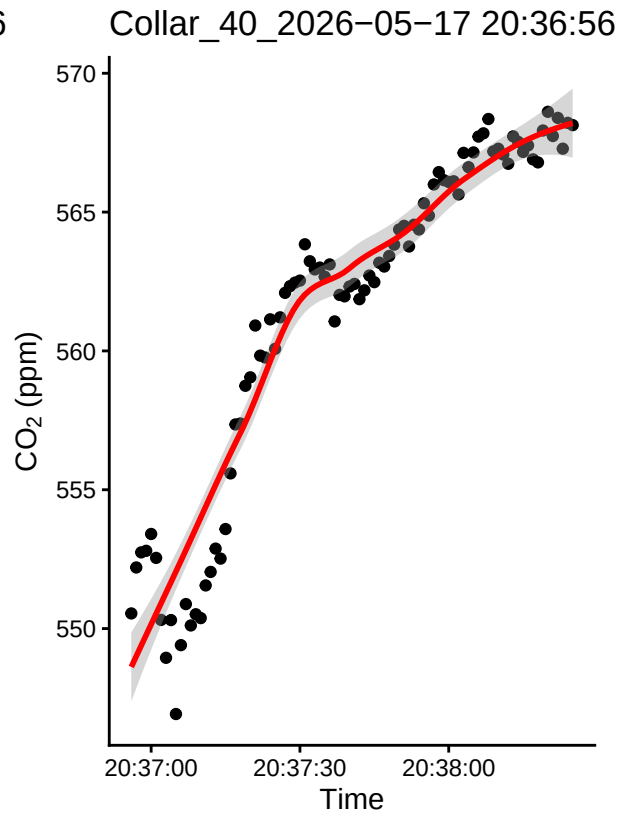
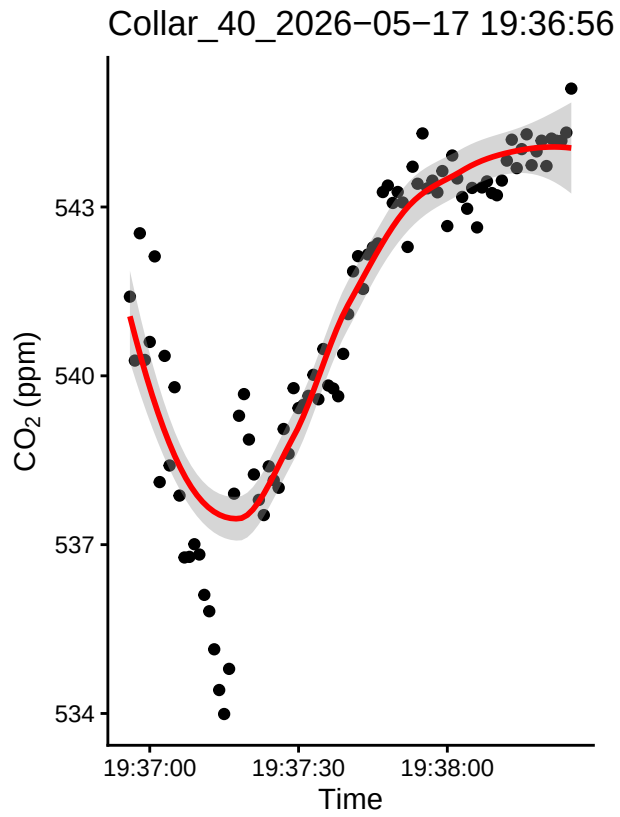


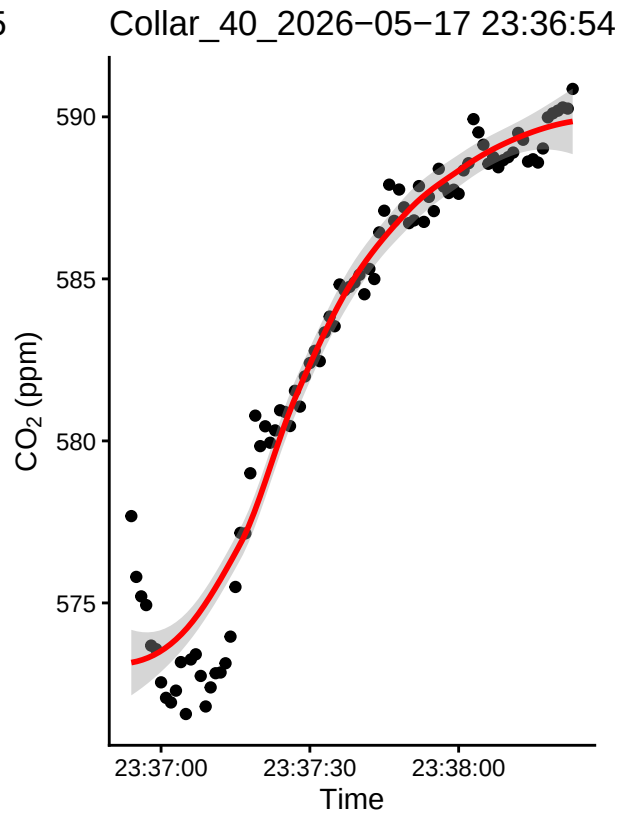
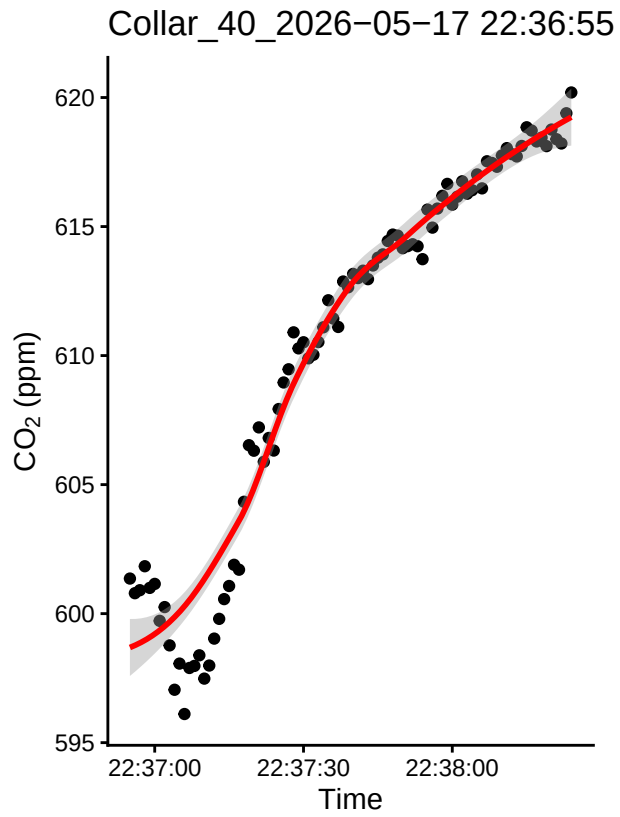
Look at the negative CO₂ fluxes

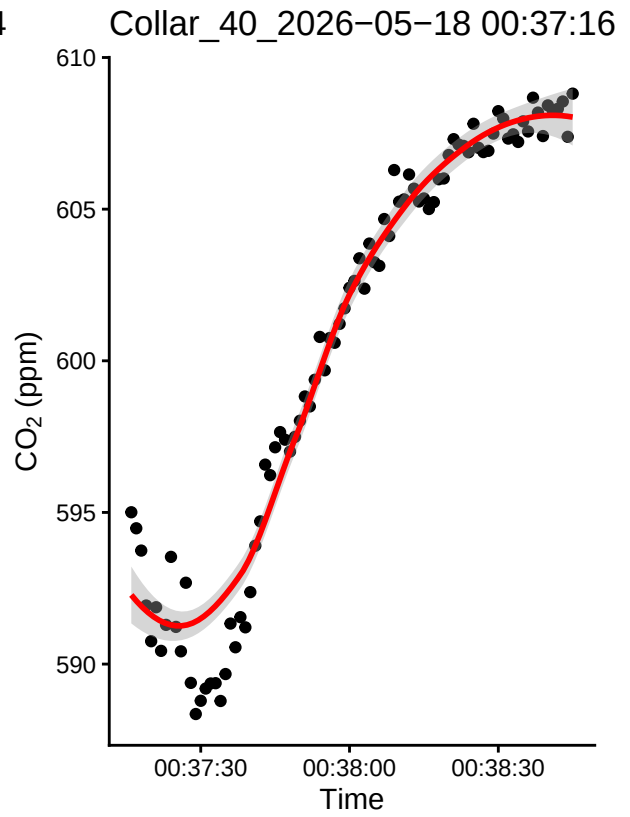
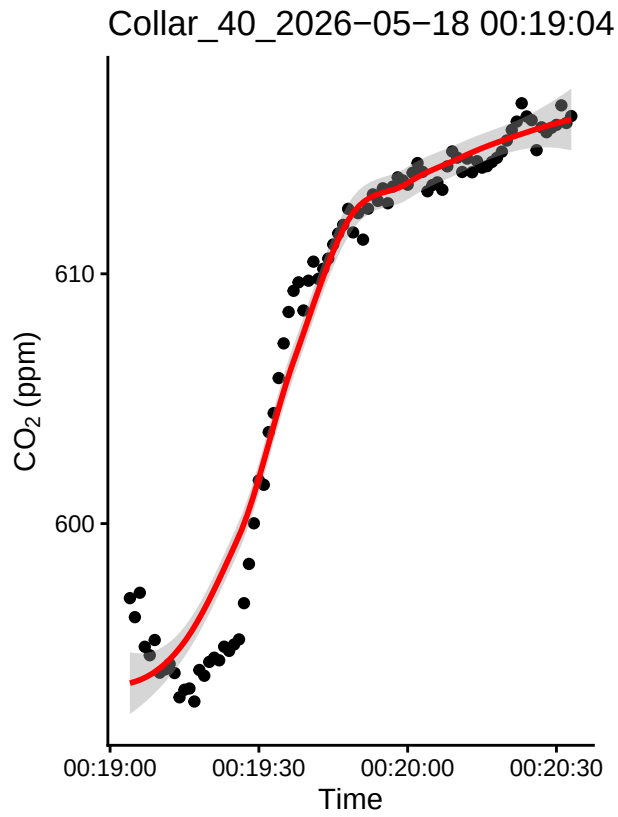


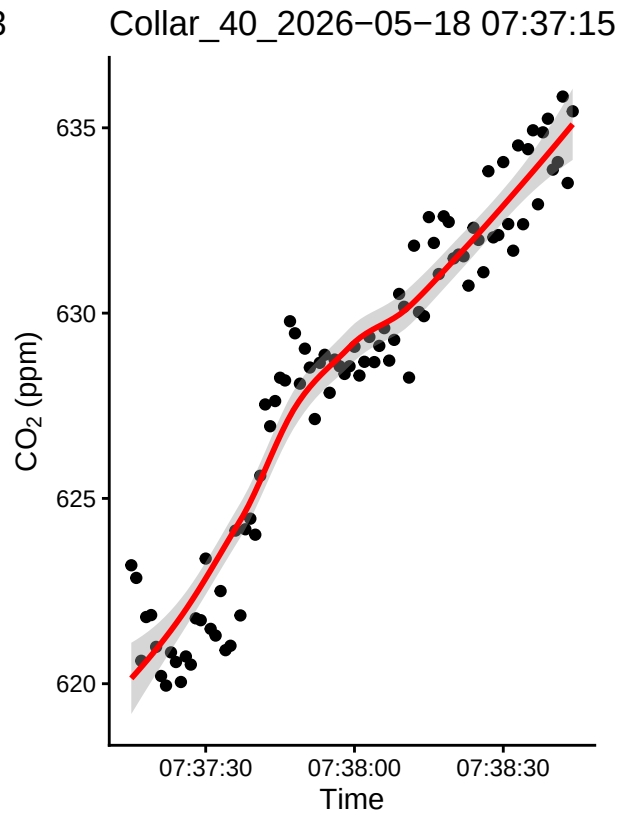
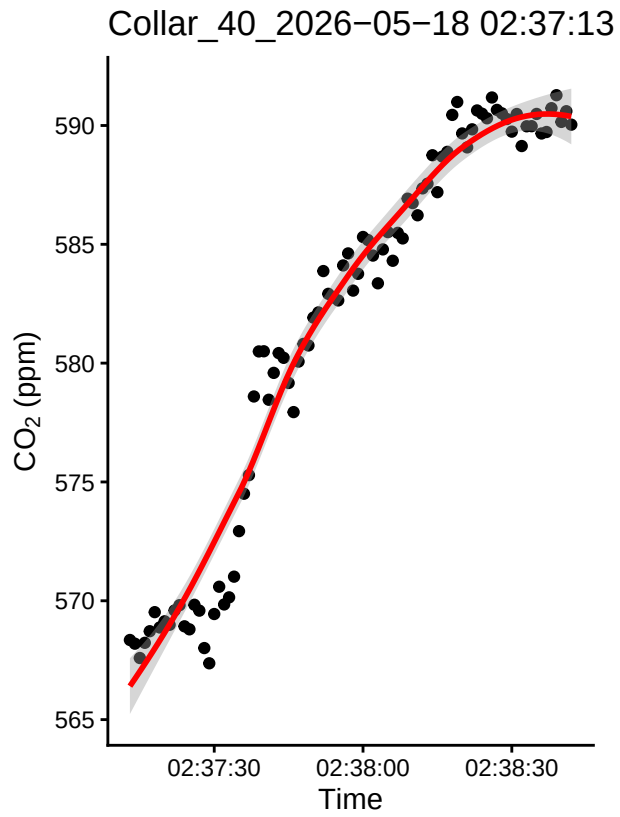


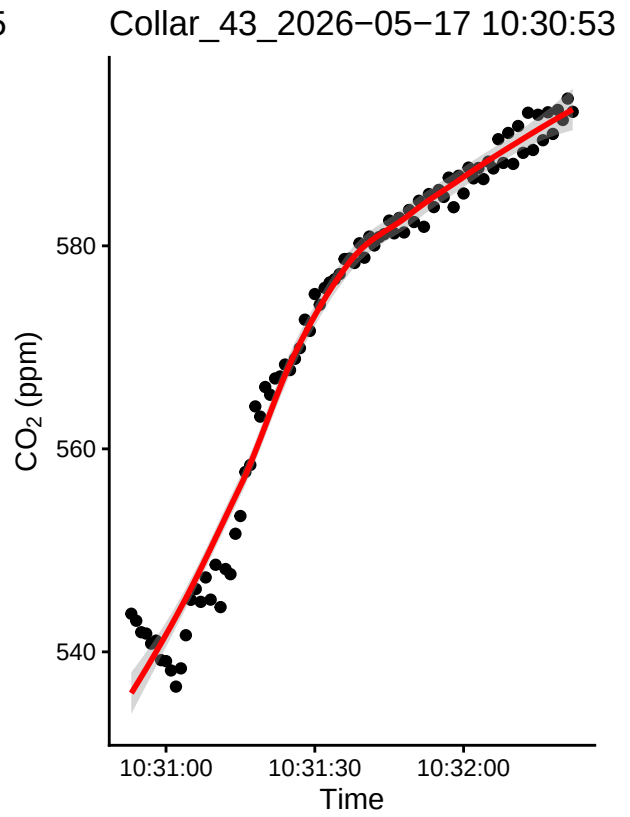
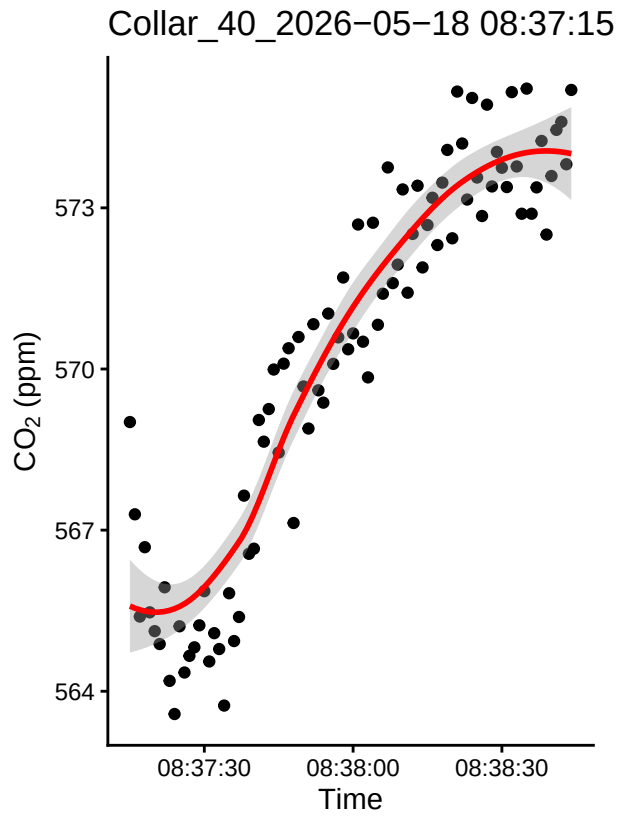


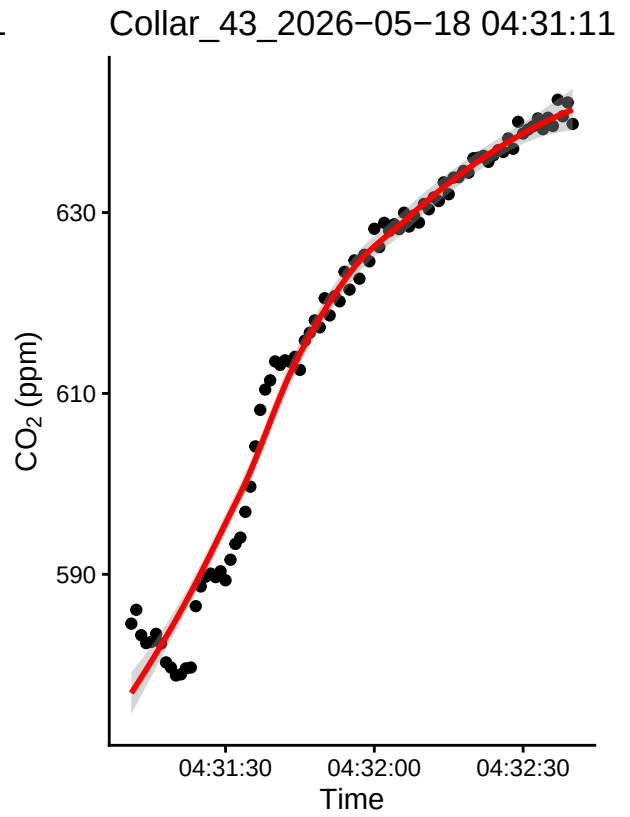
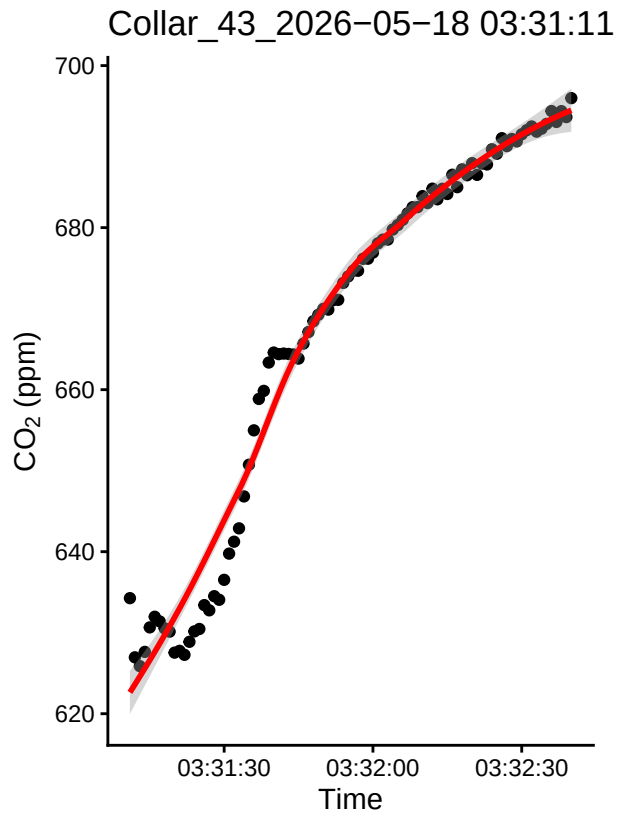




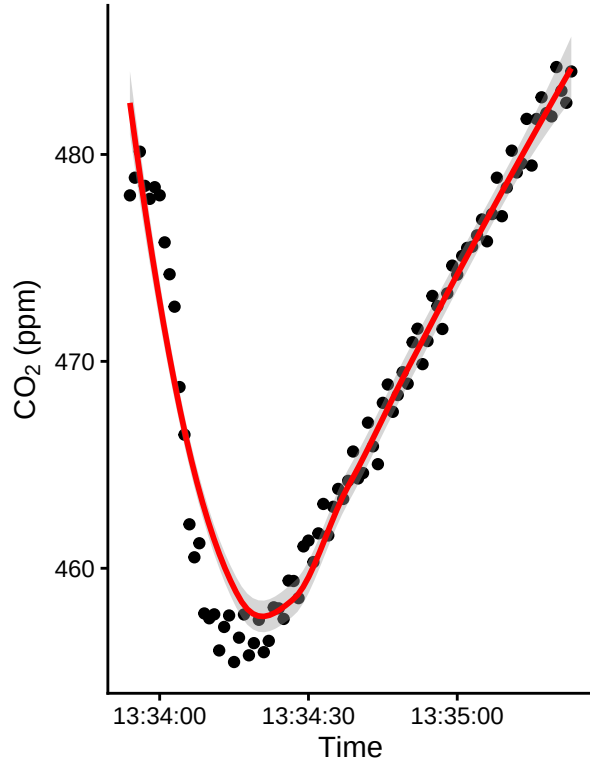




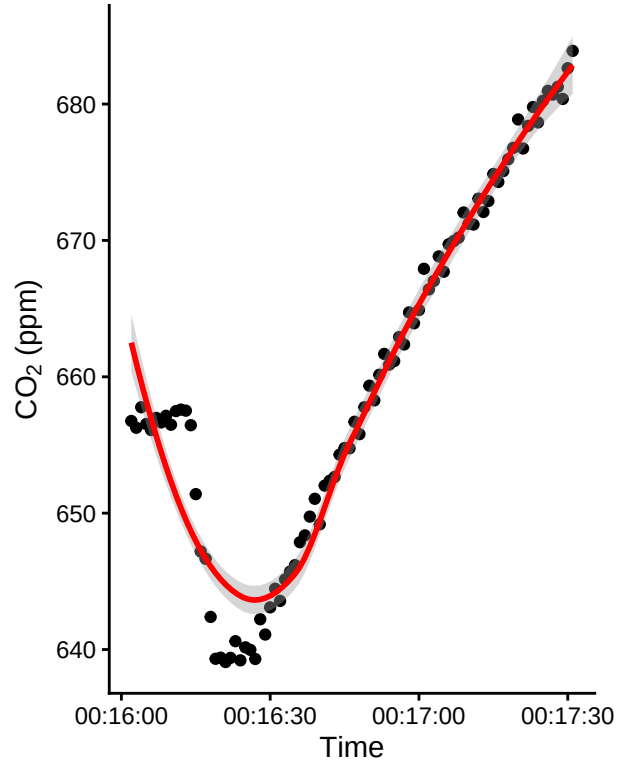


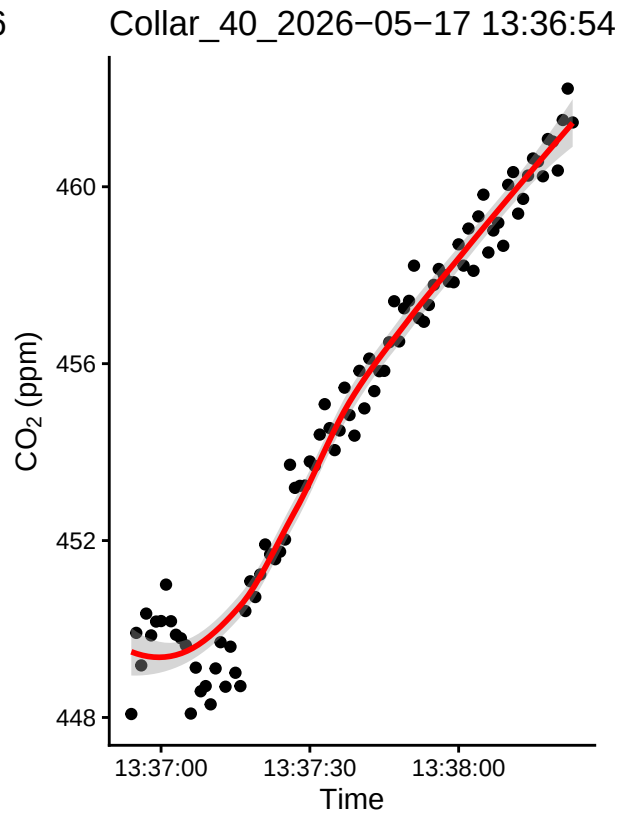
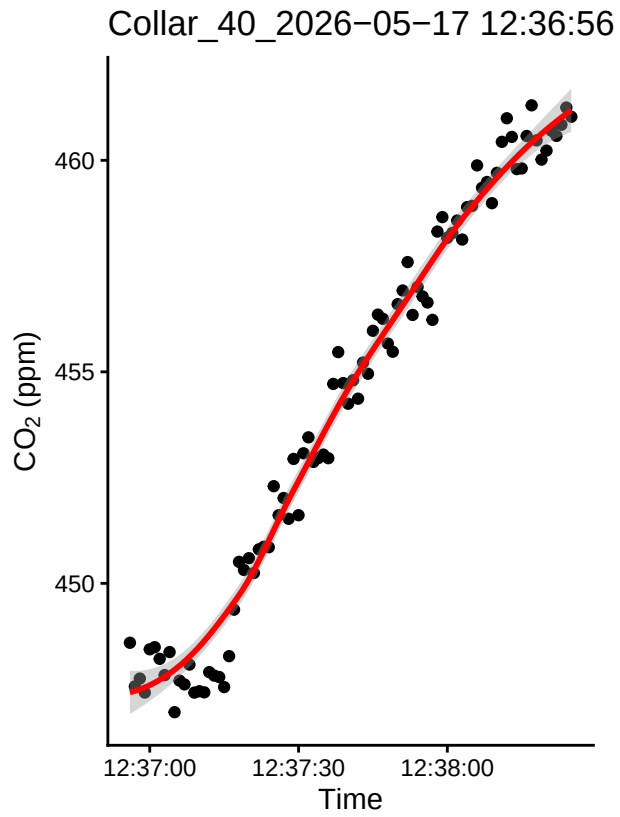


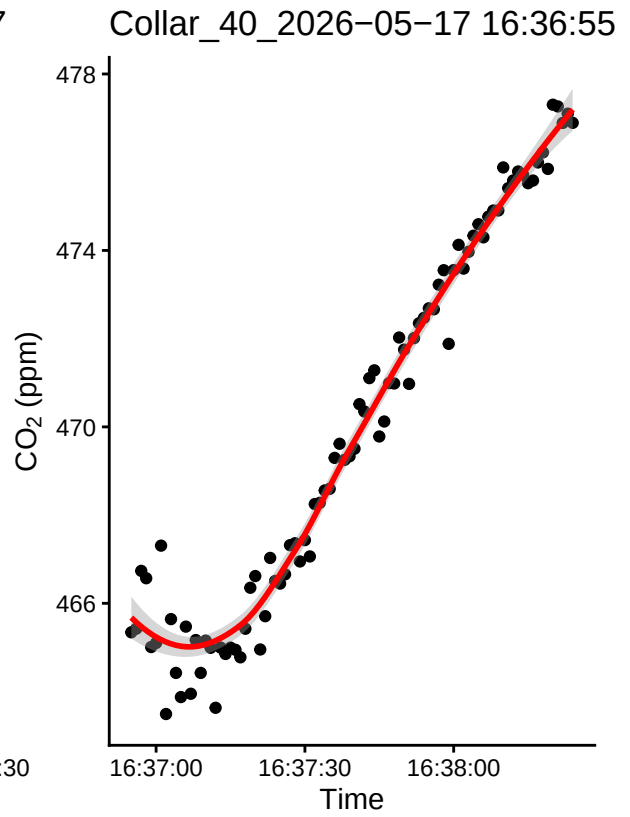
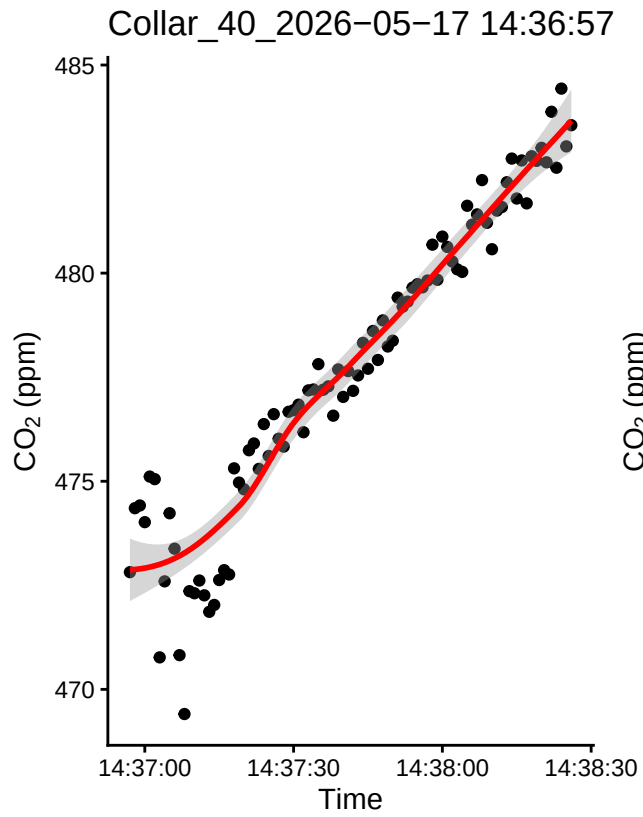
Collar_44_2026-05-17 13:33:54

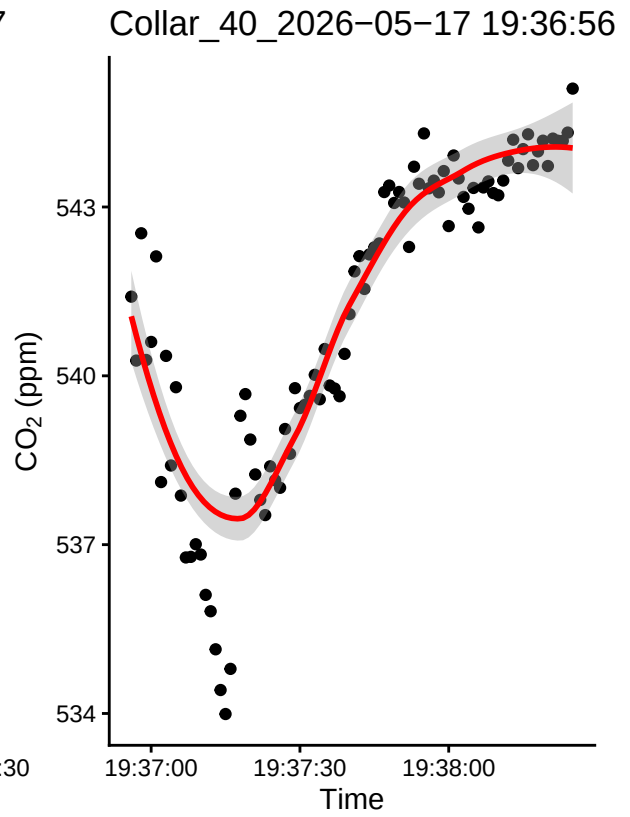
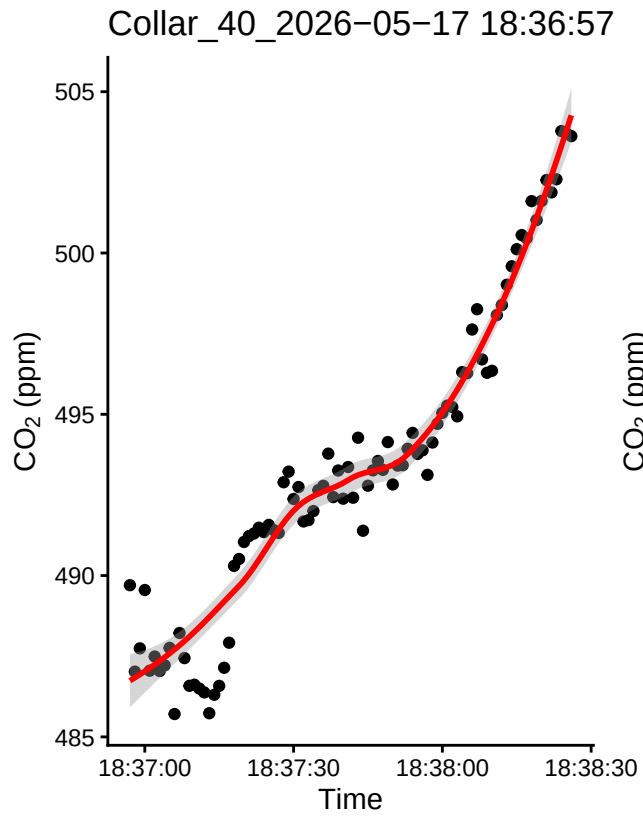


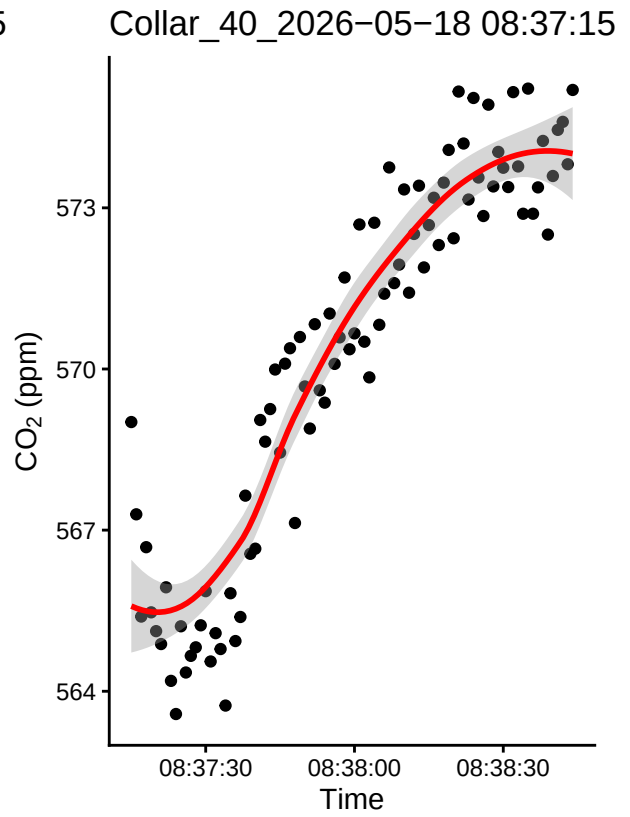
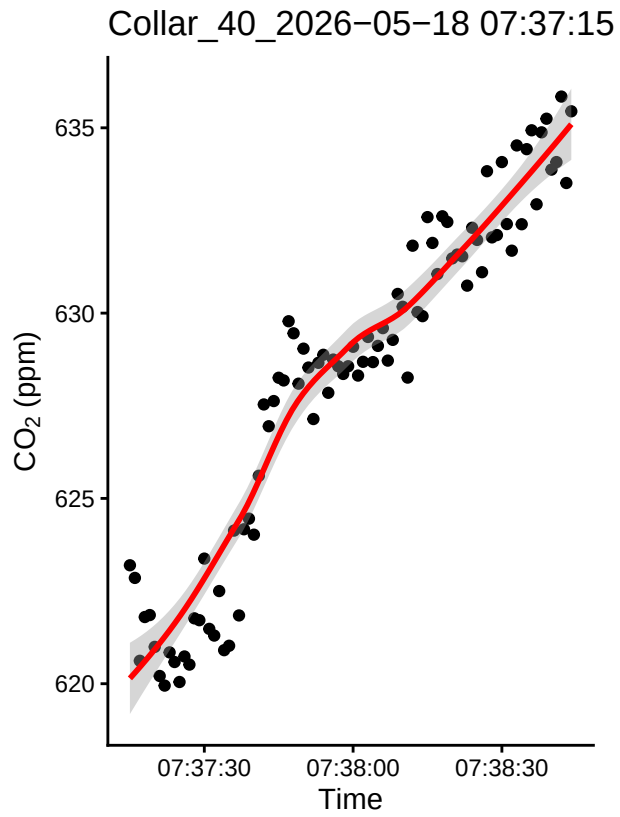
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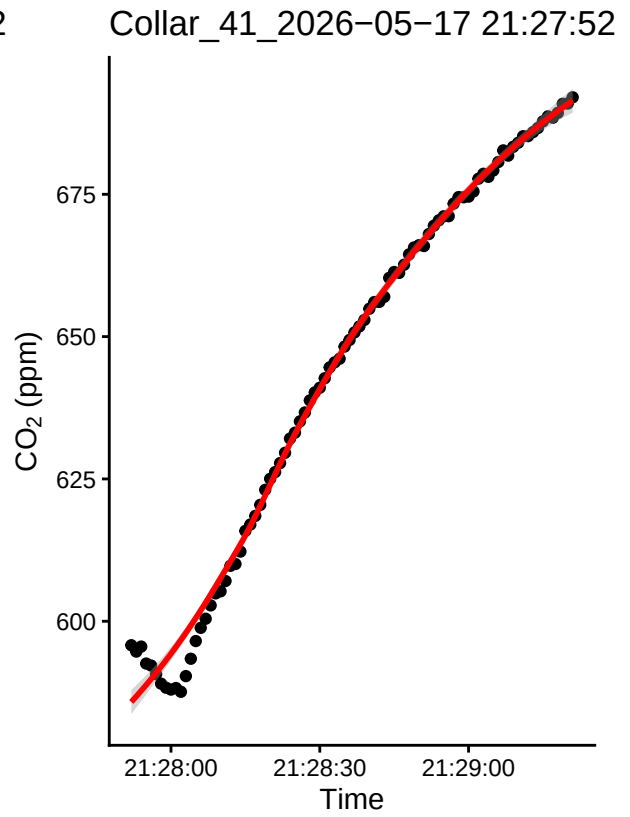
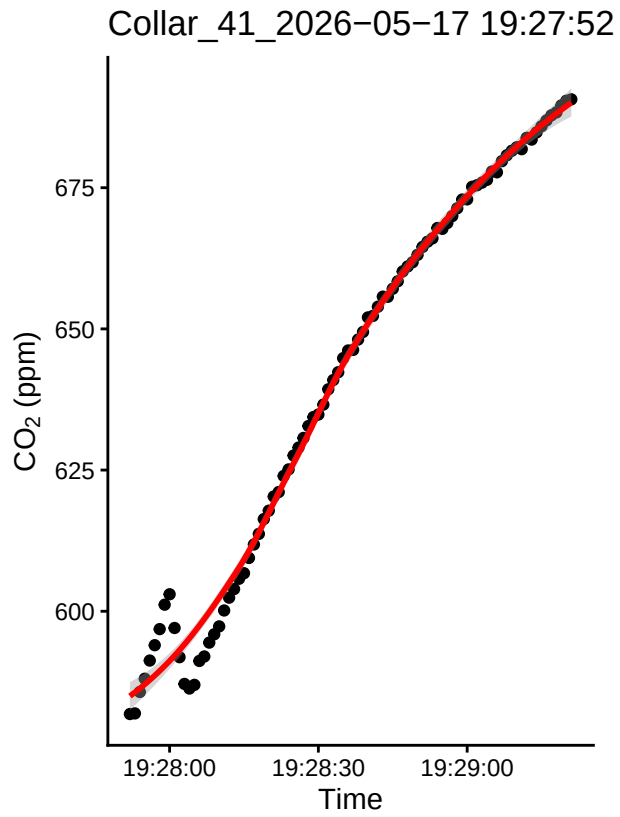


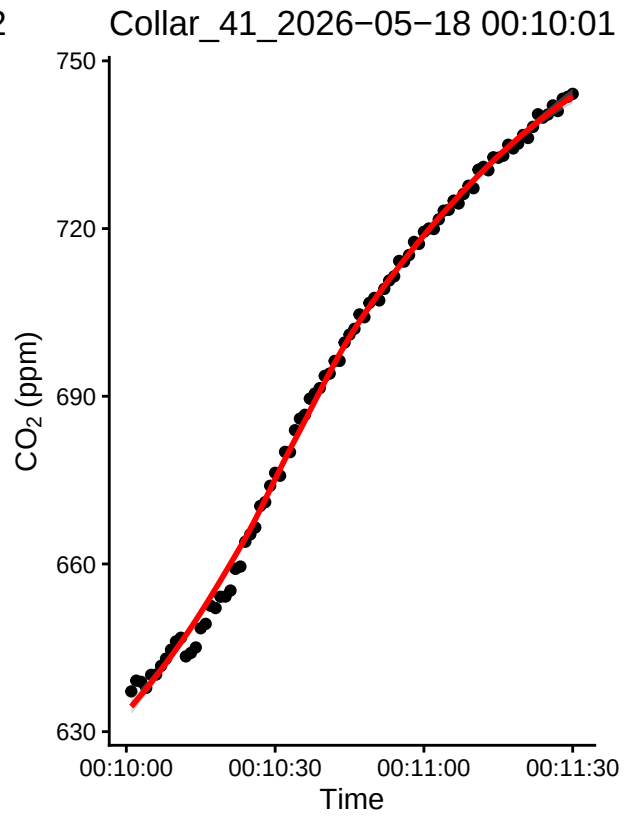
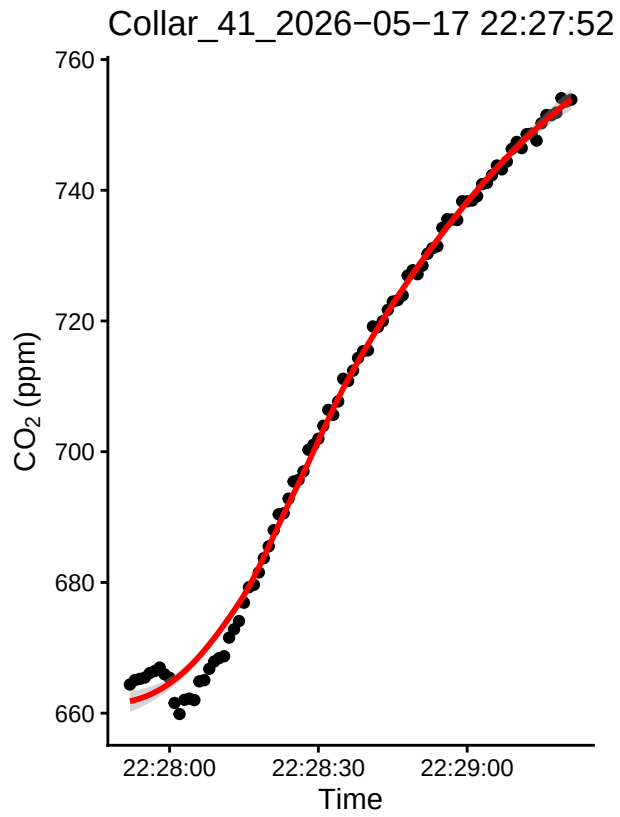


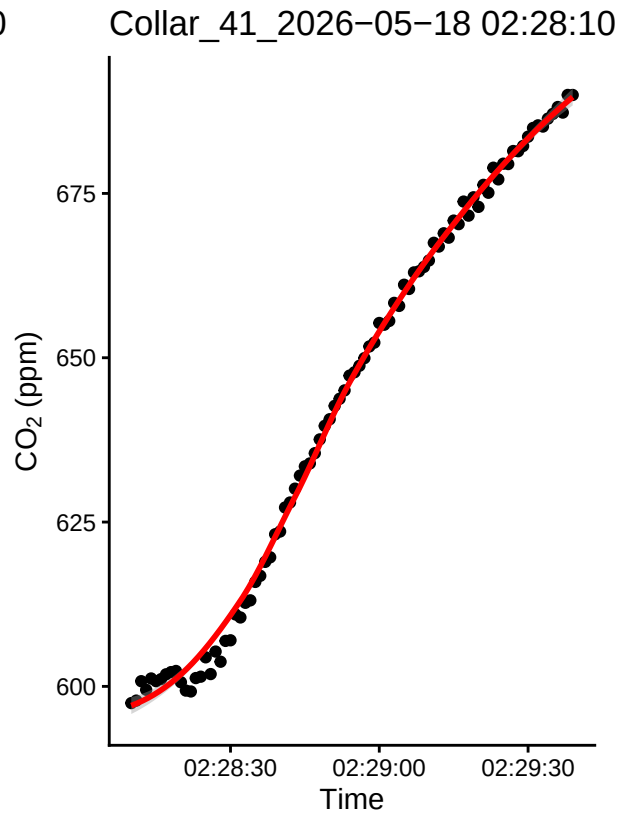
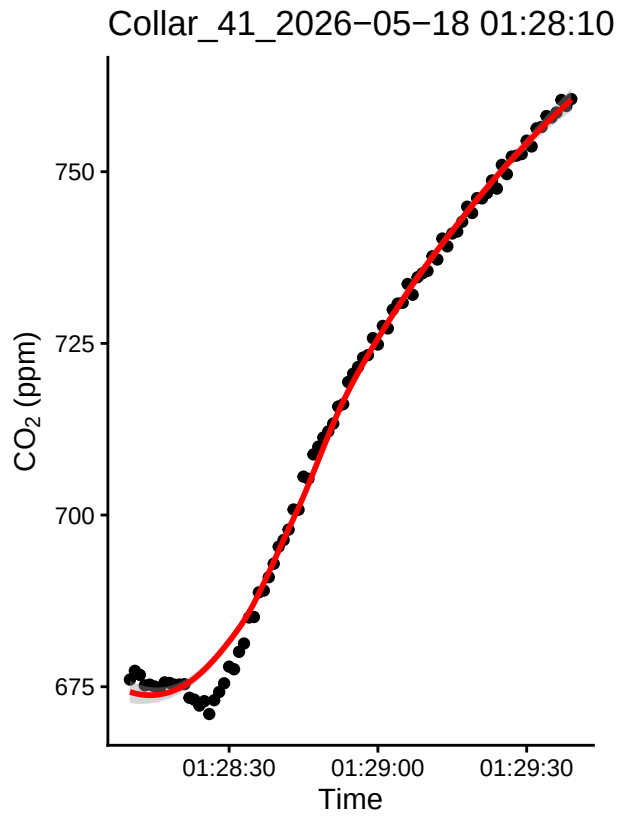


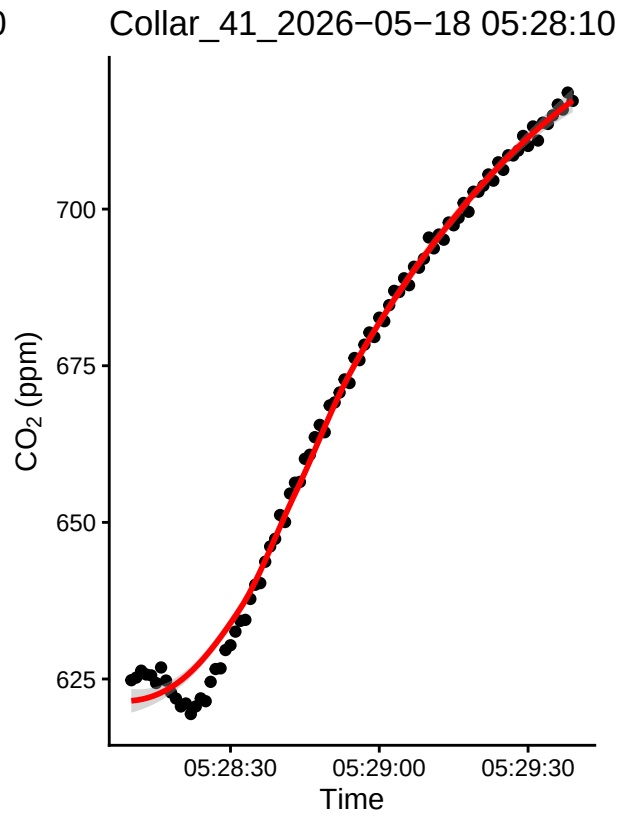
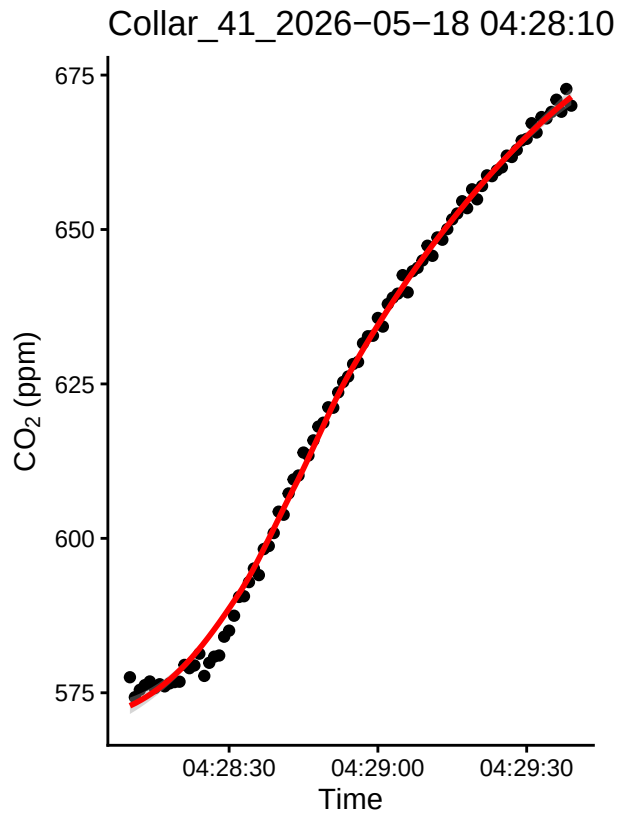




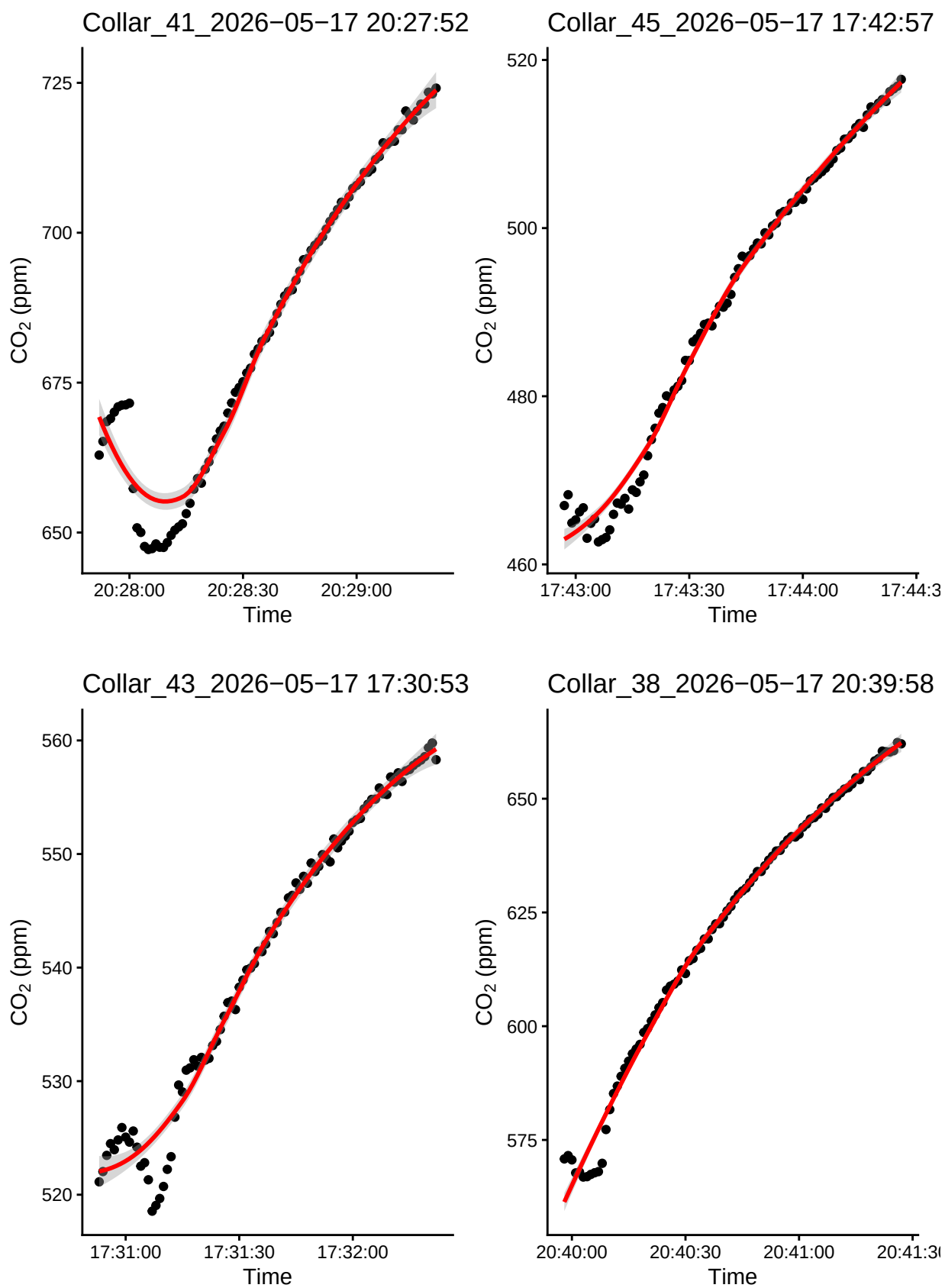


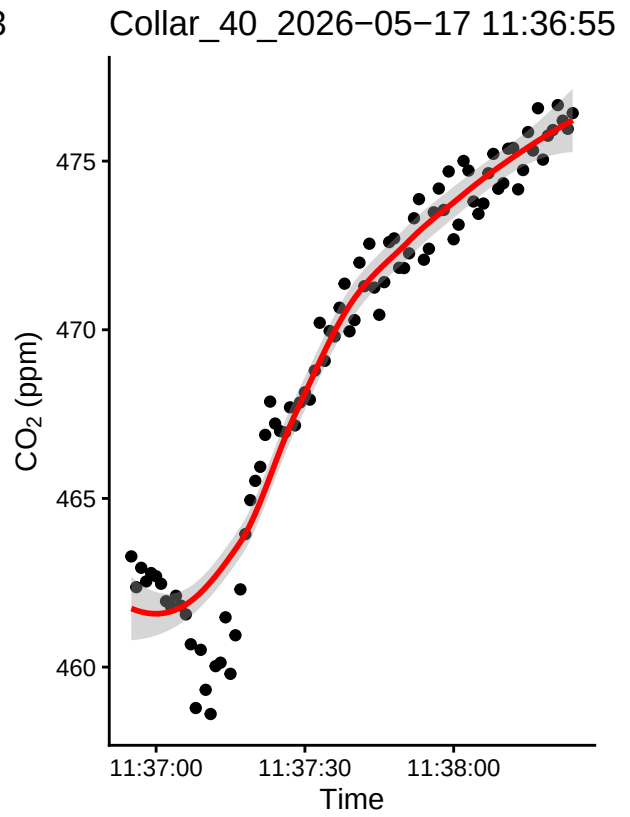
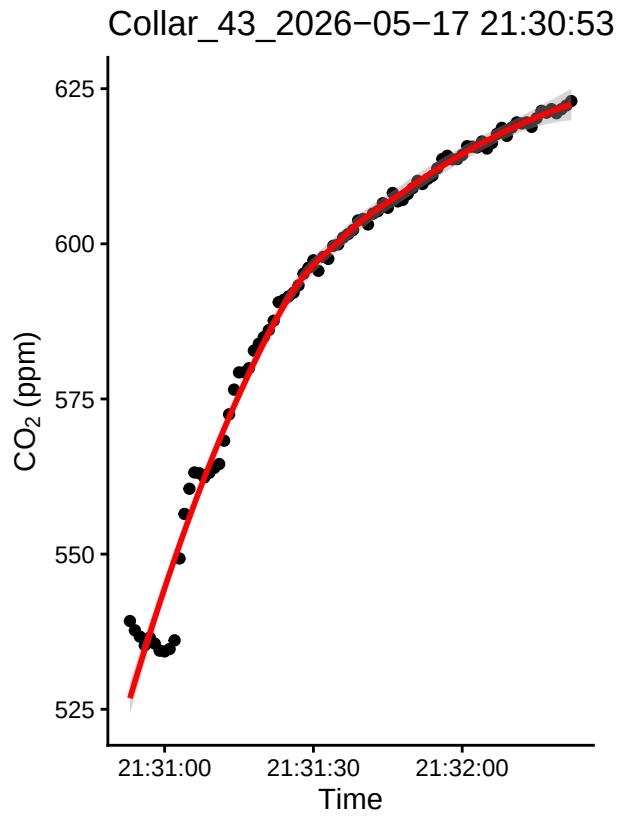


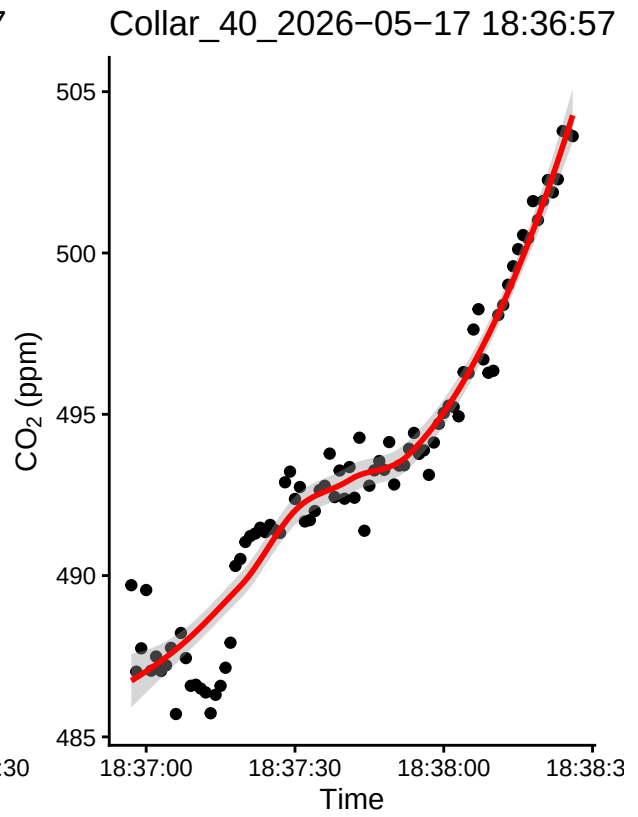
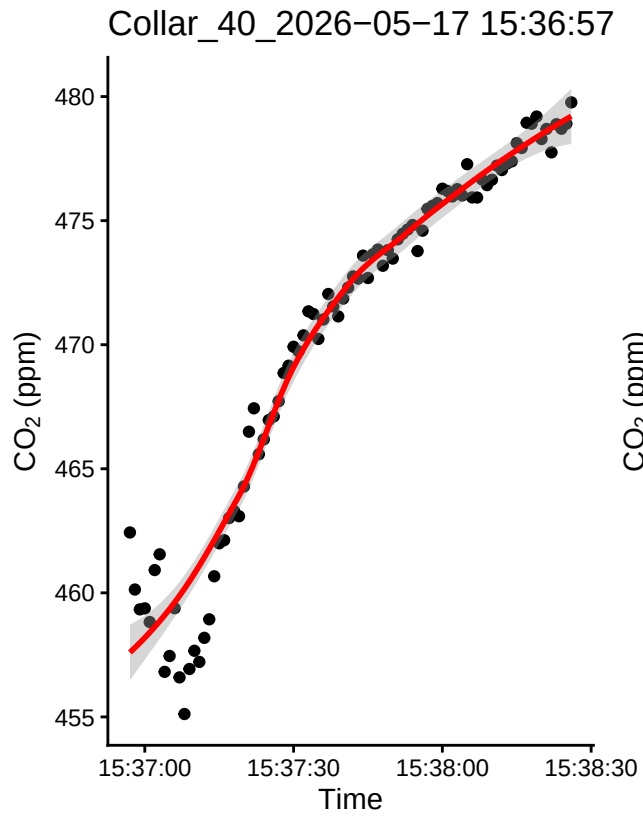


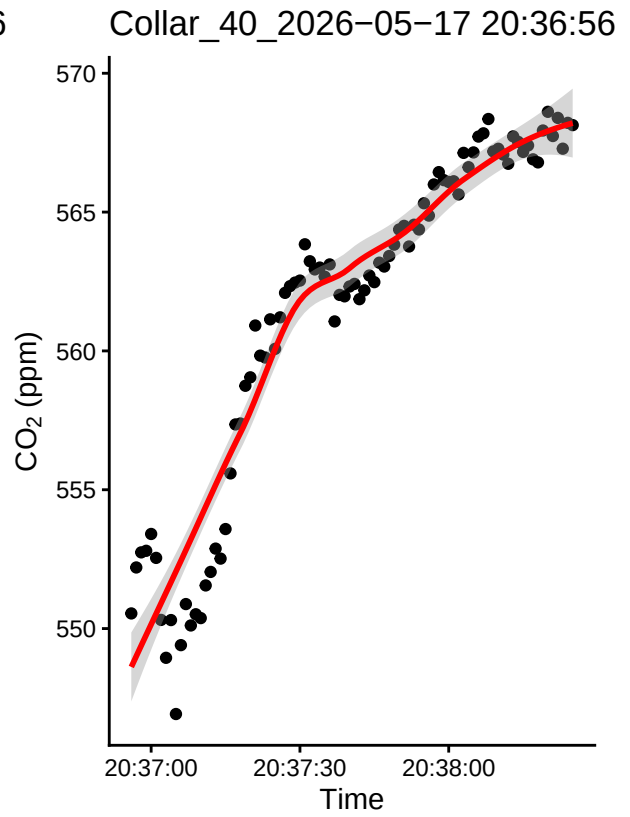
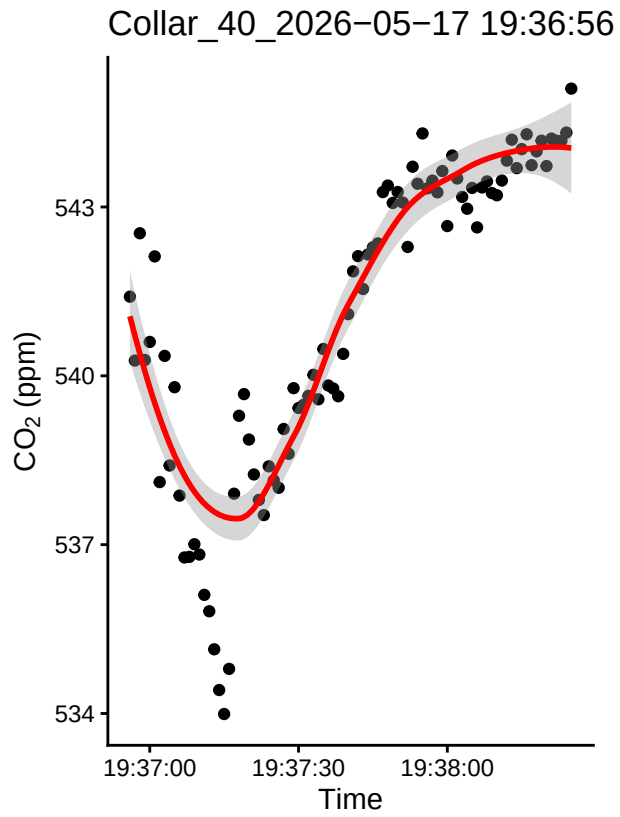


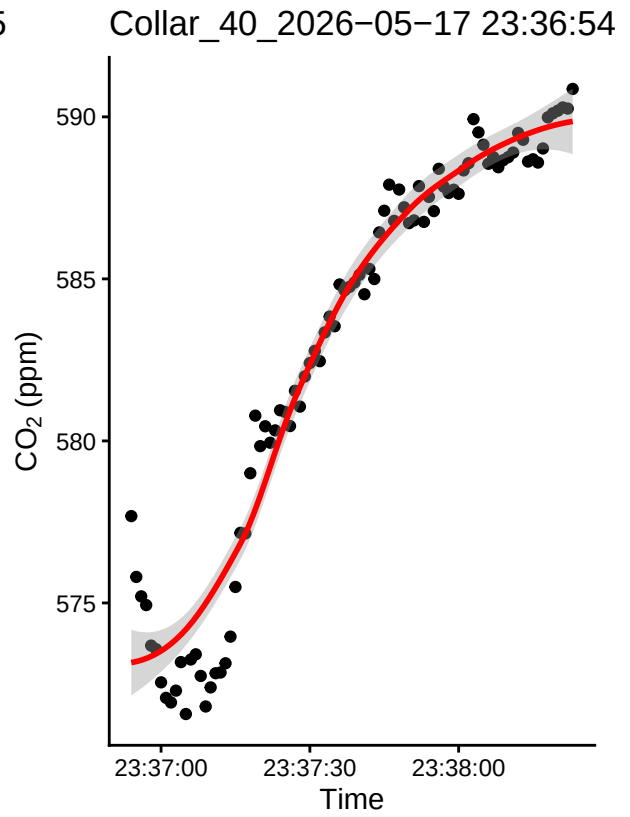
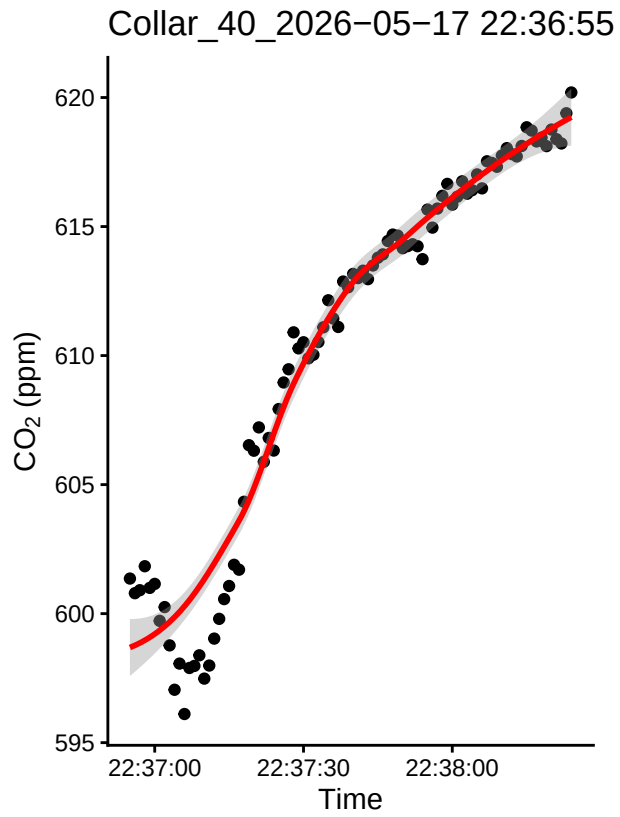
Plot all fluxes

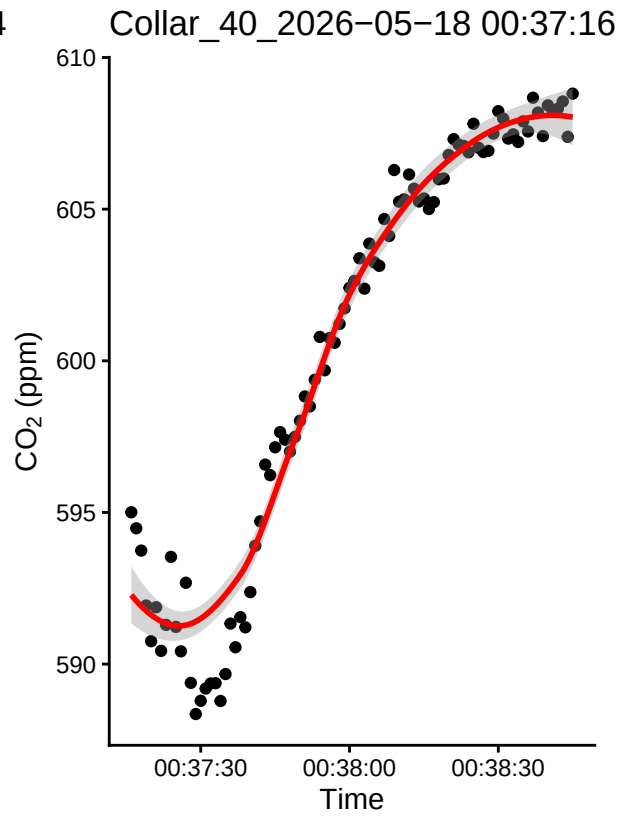
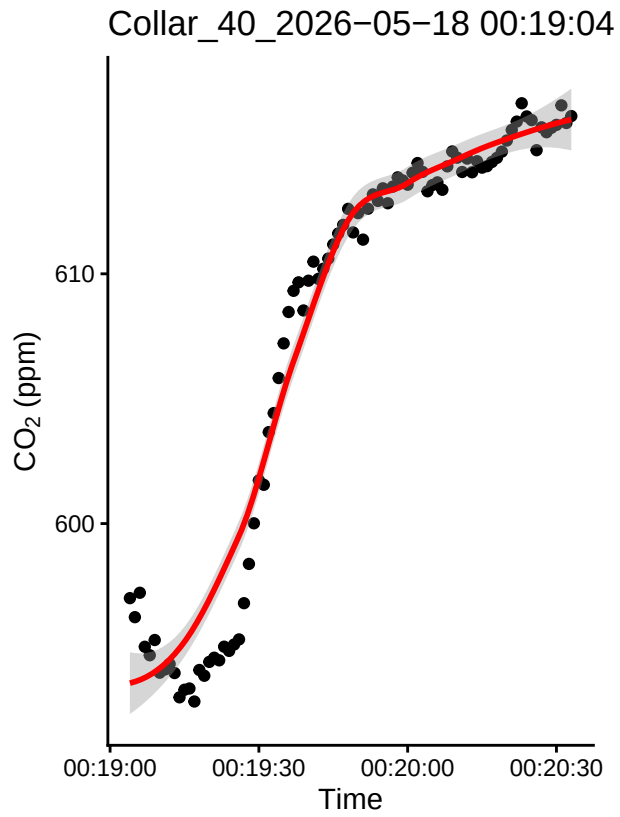


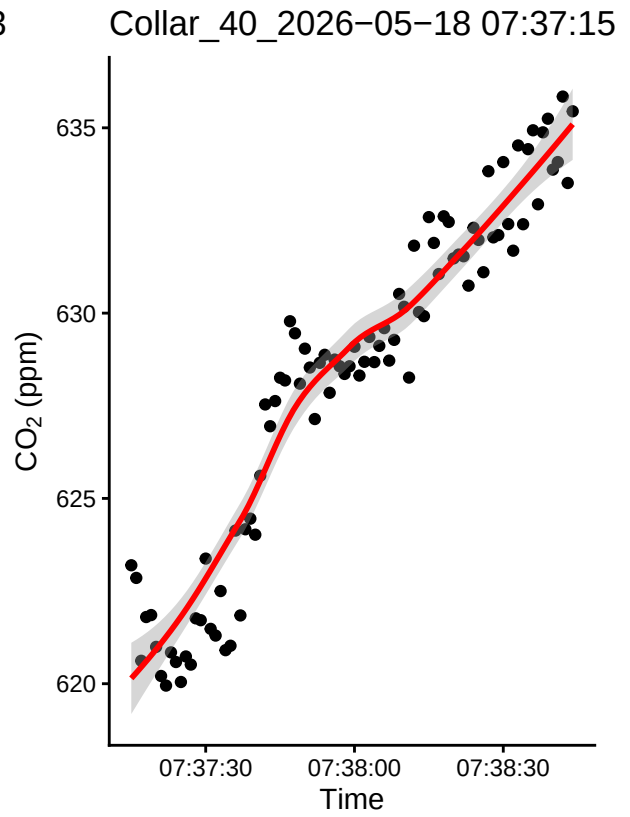
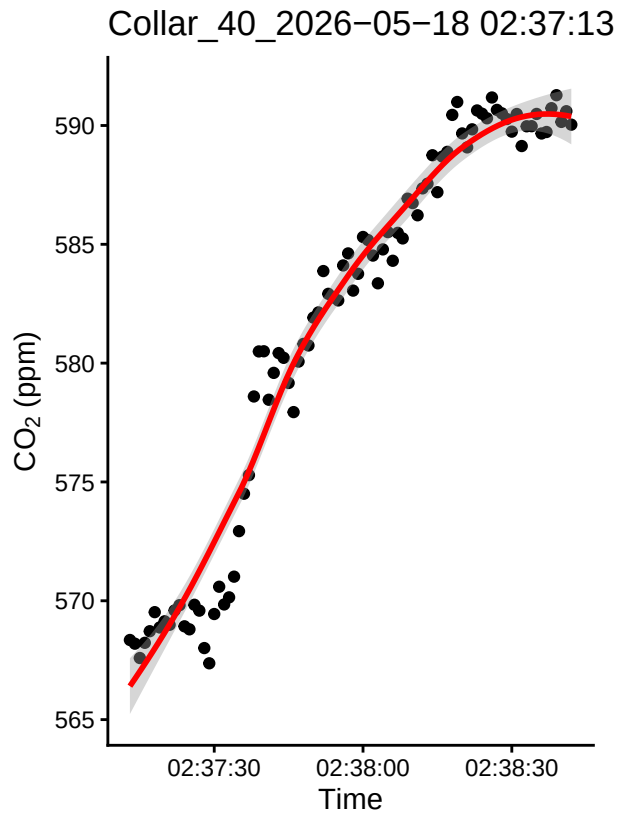


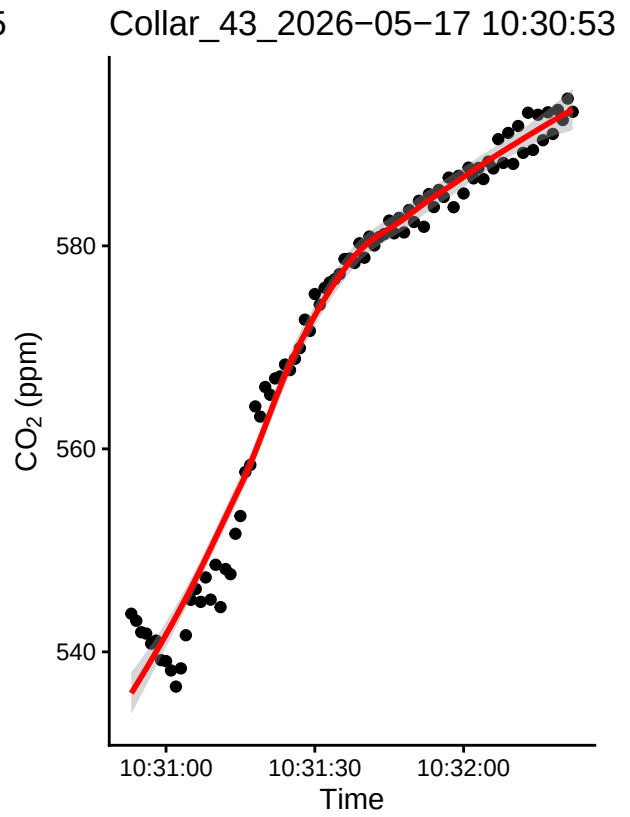
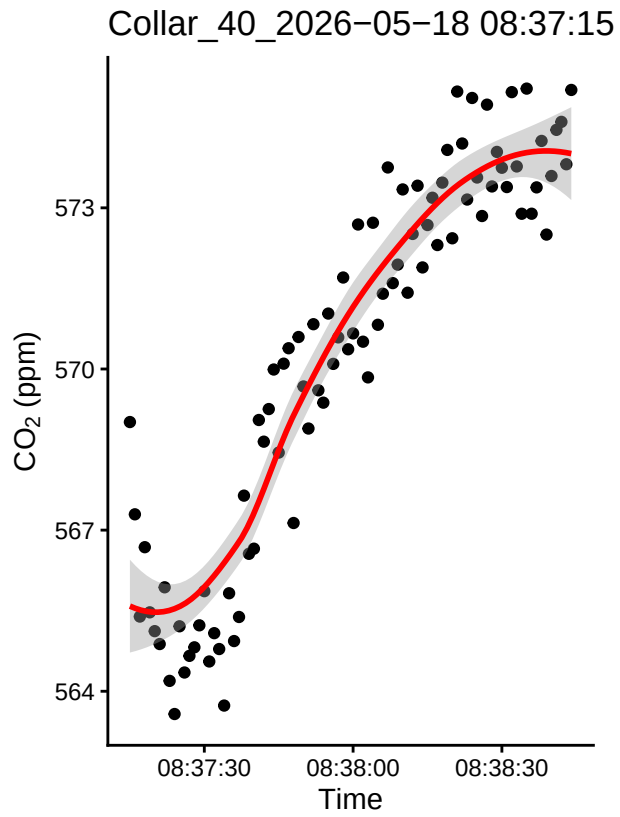


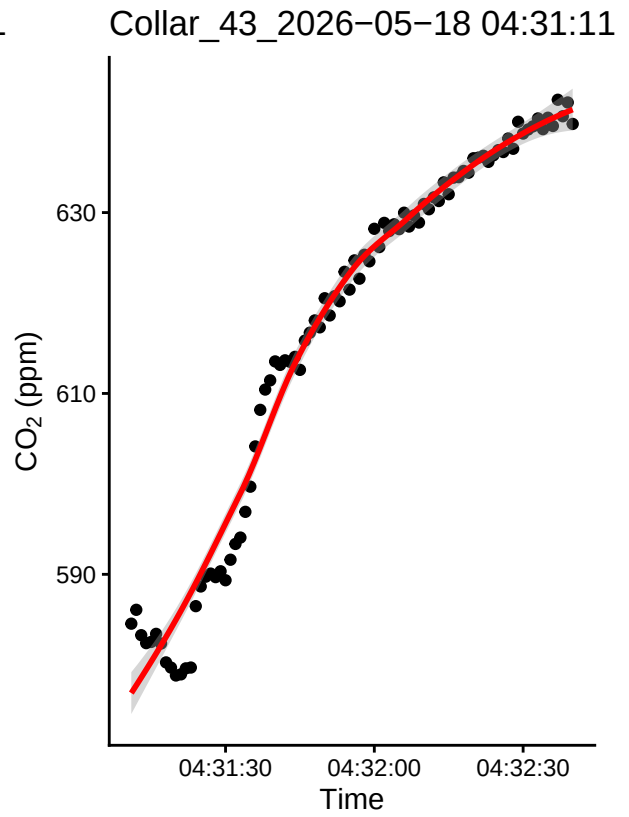
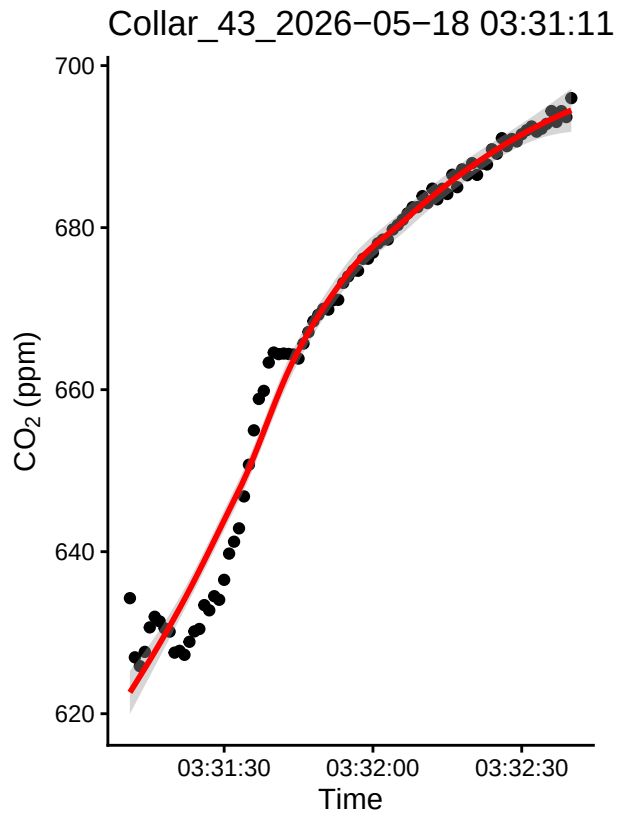


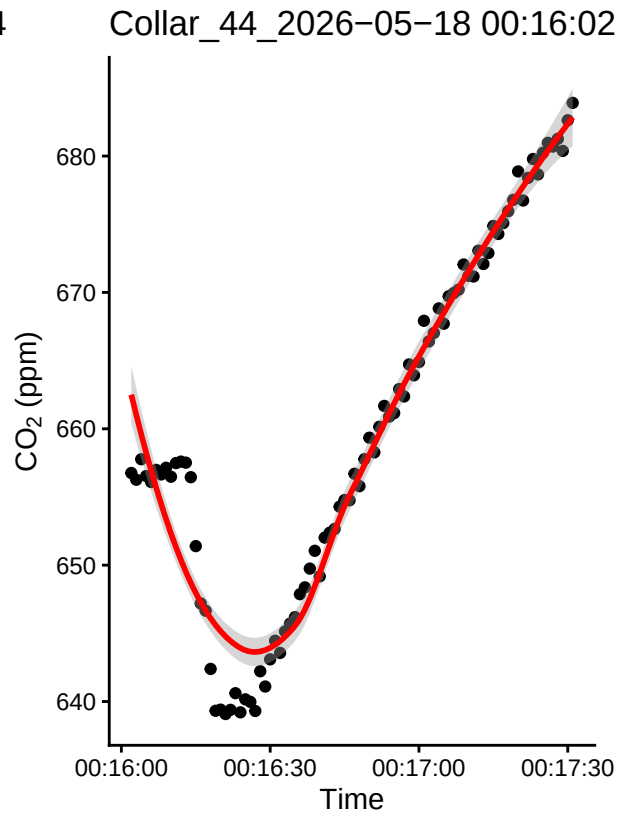
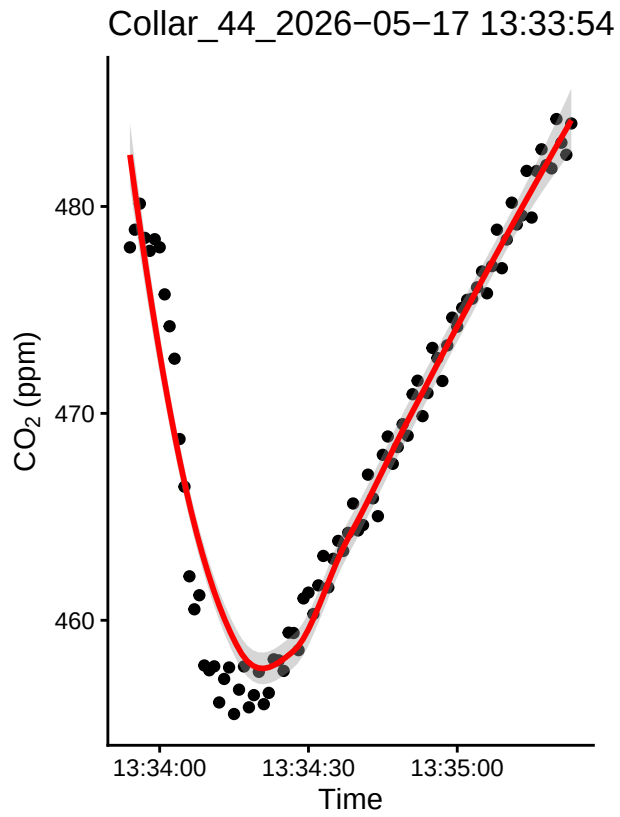


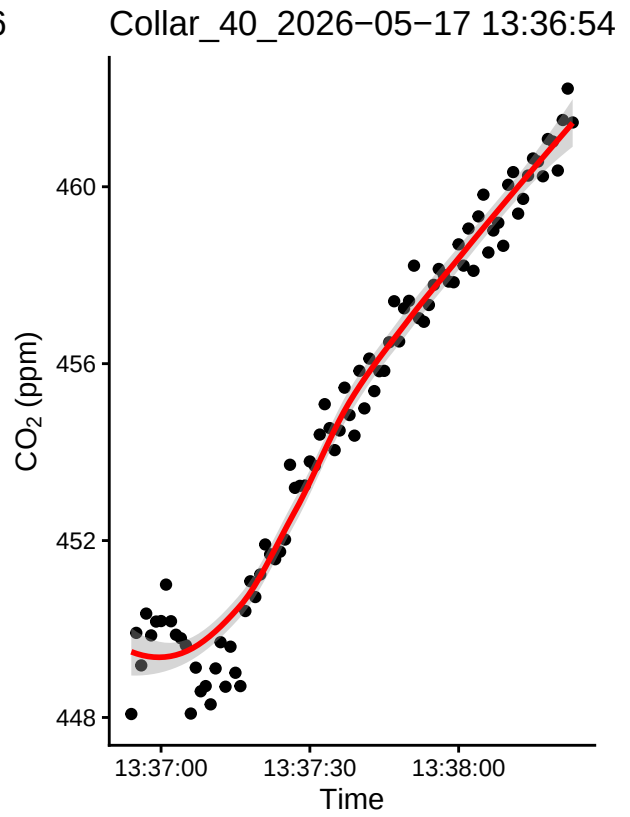
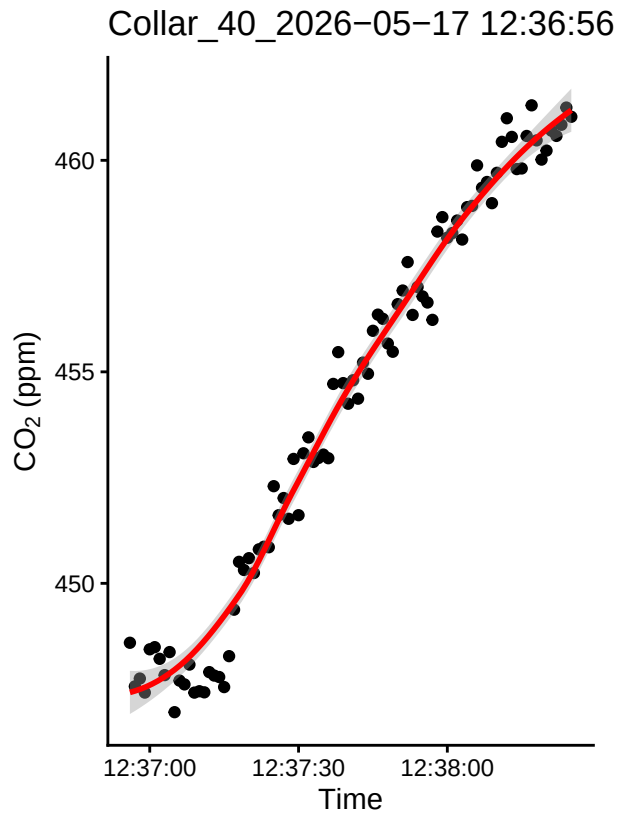


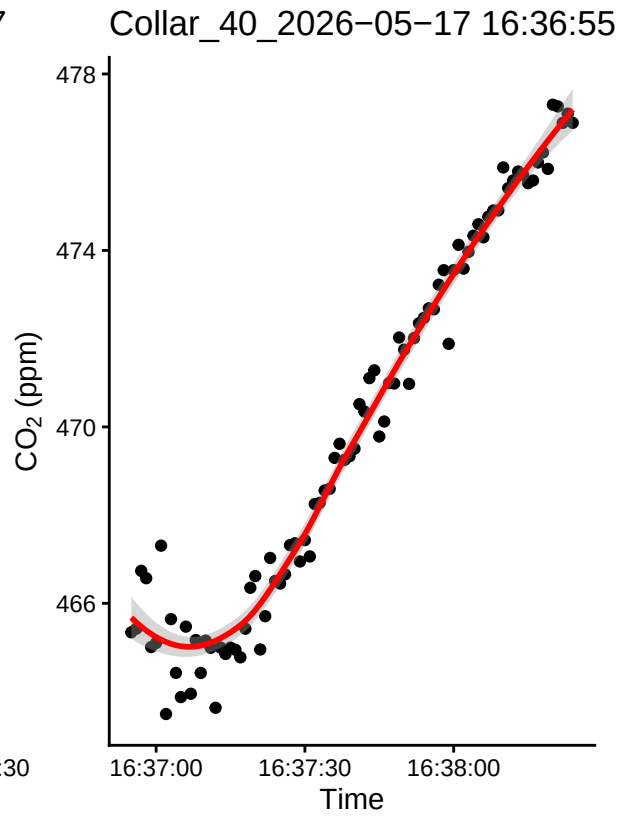
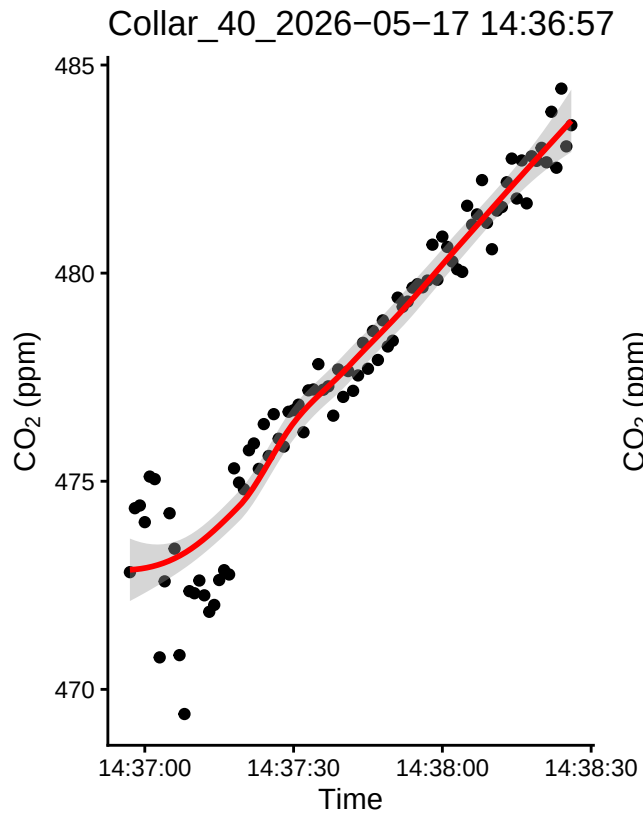


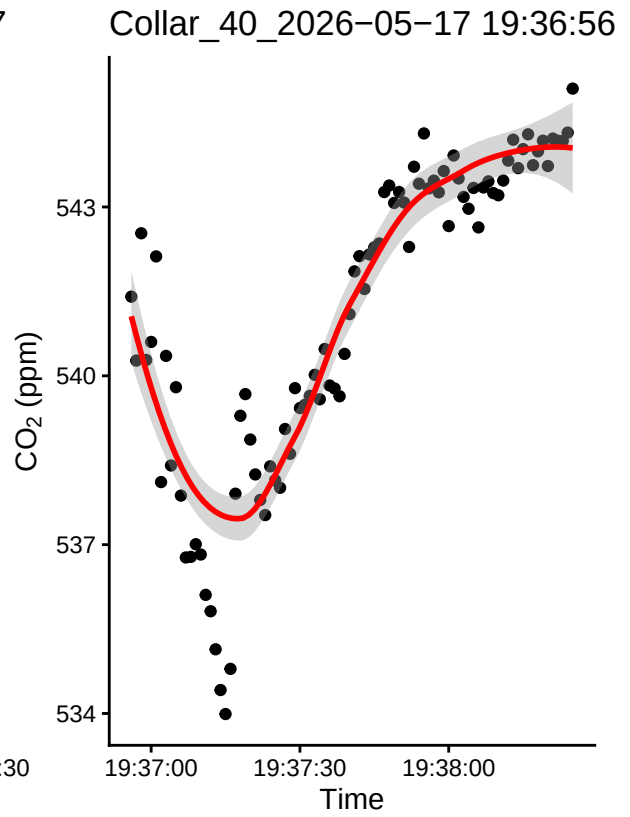
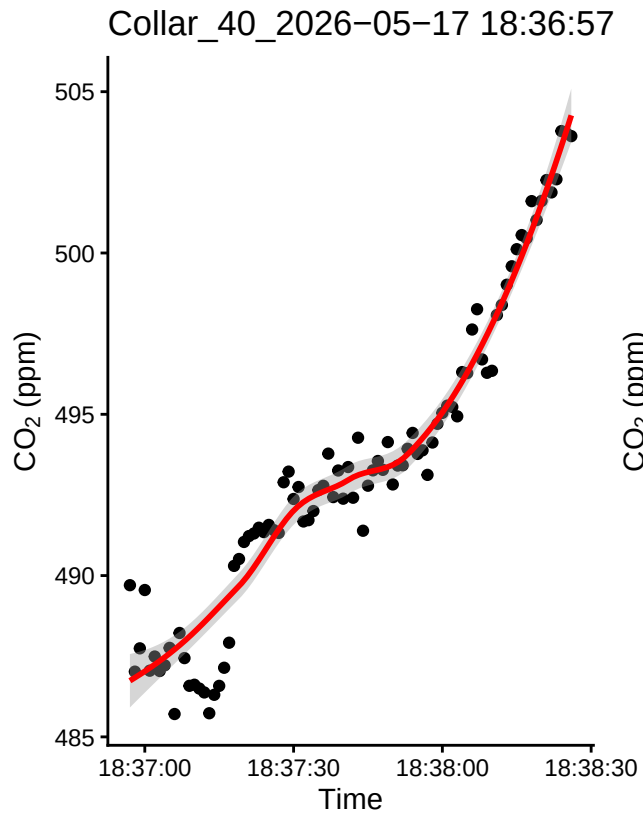


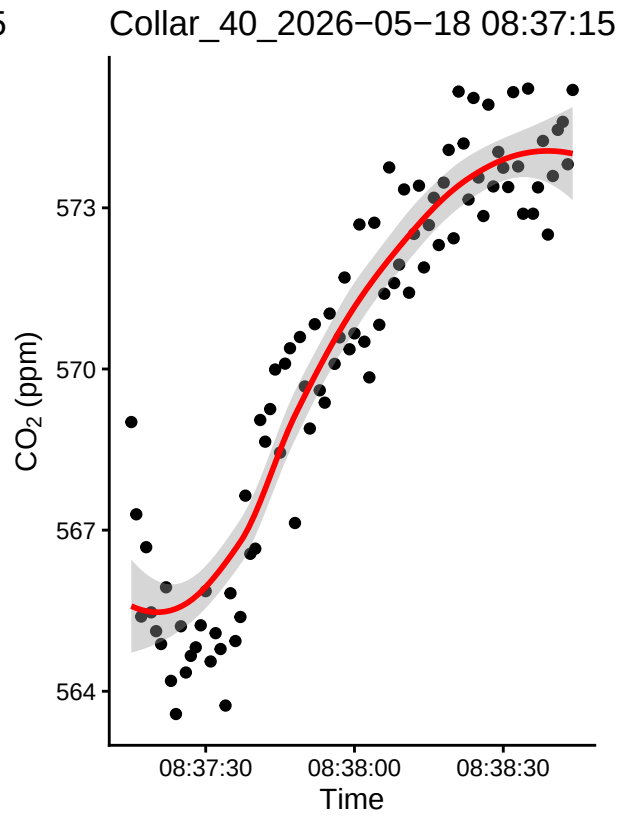
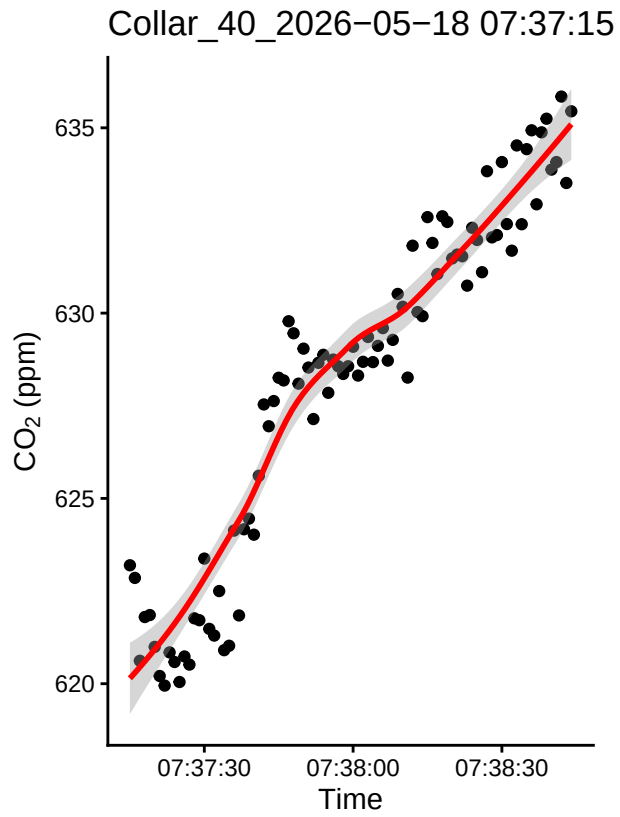


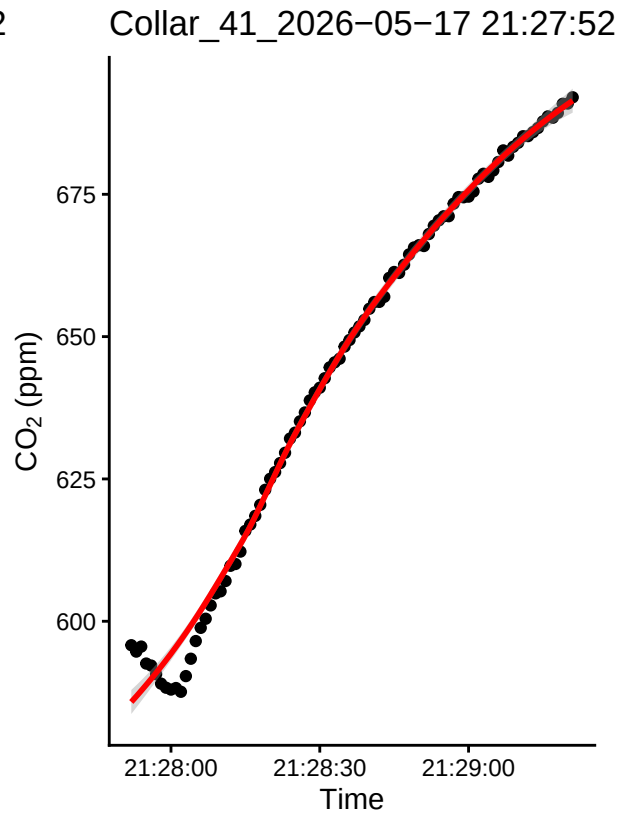
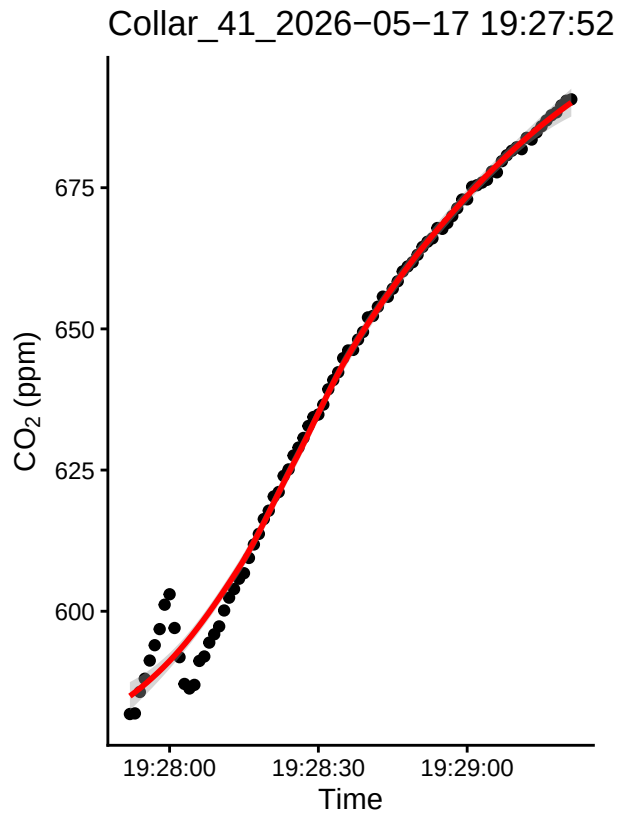


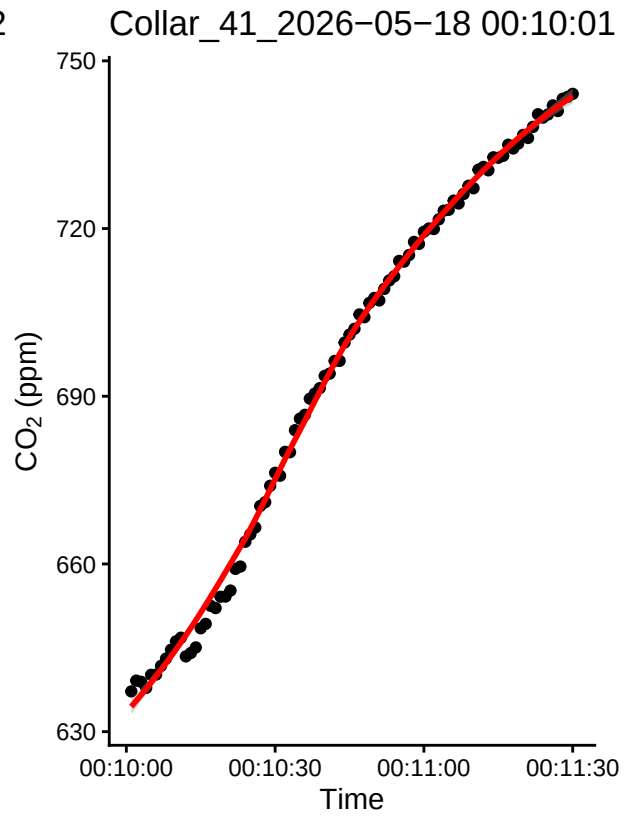
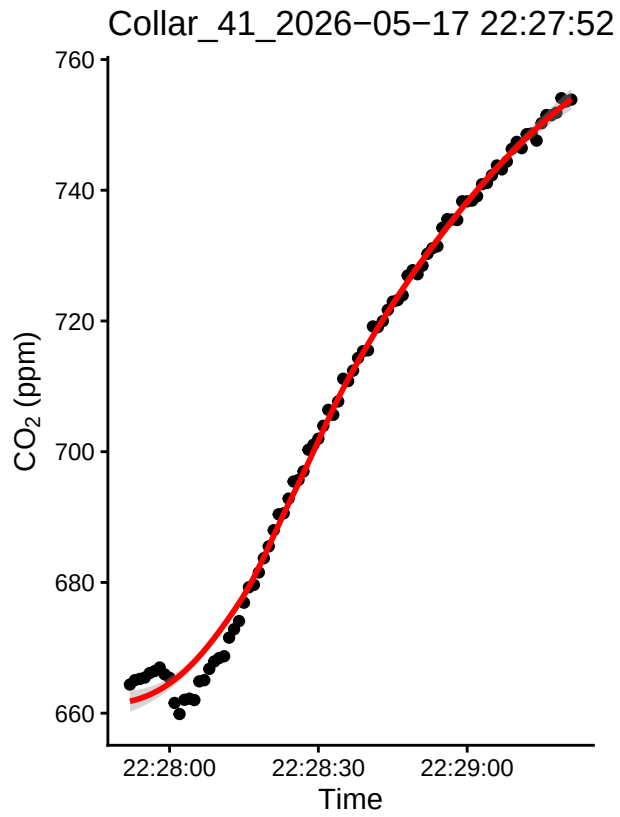


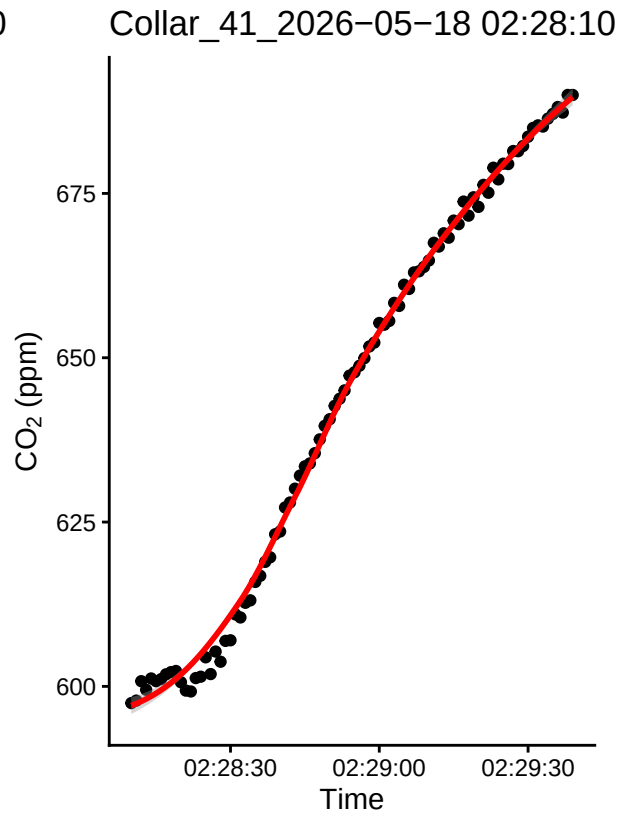
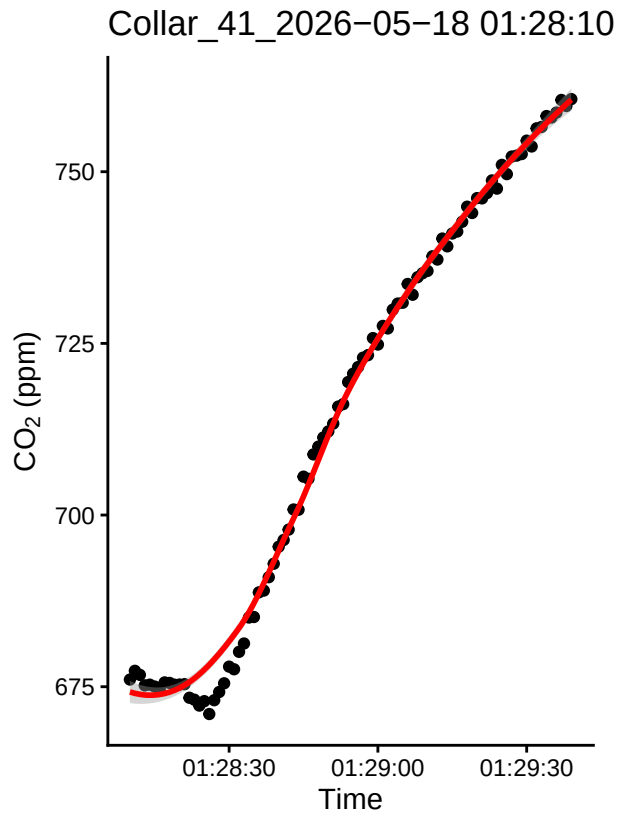


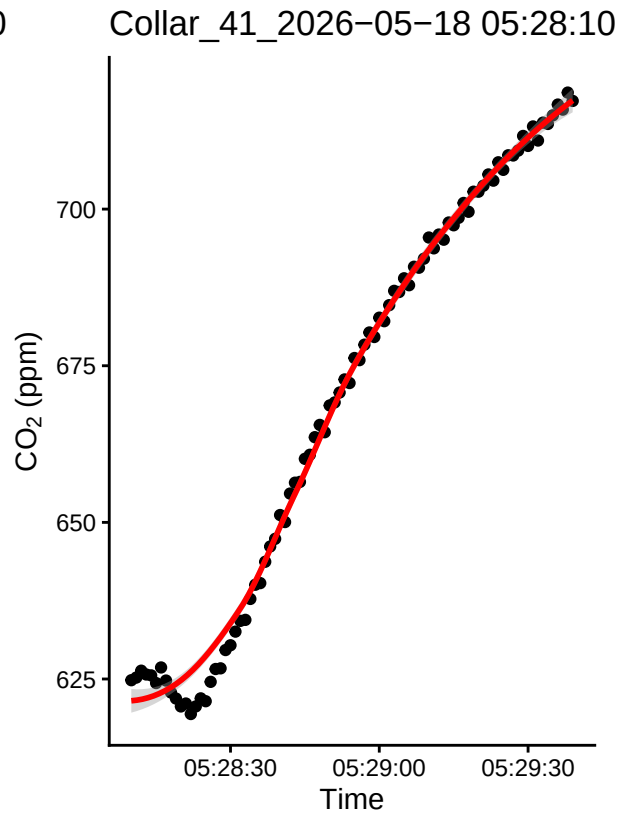
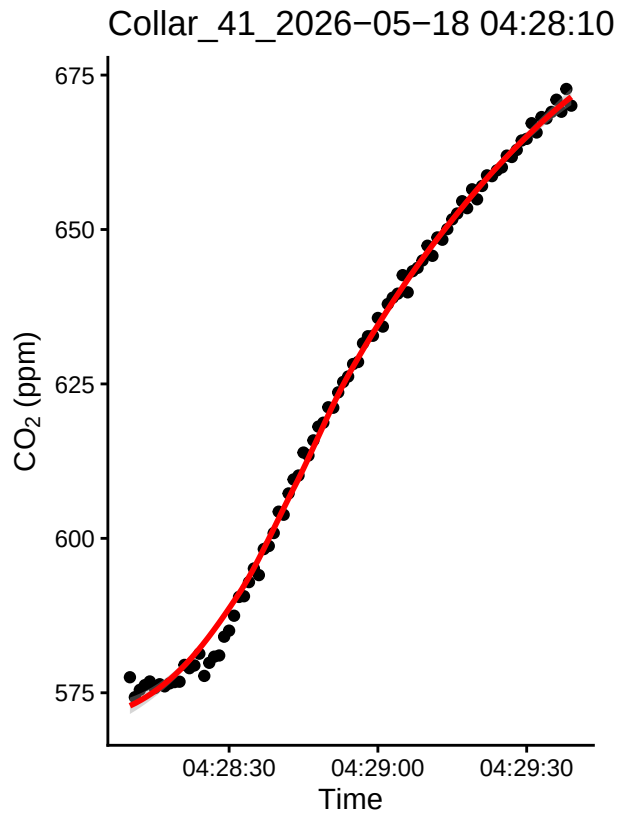


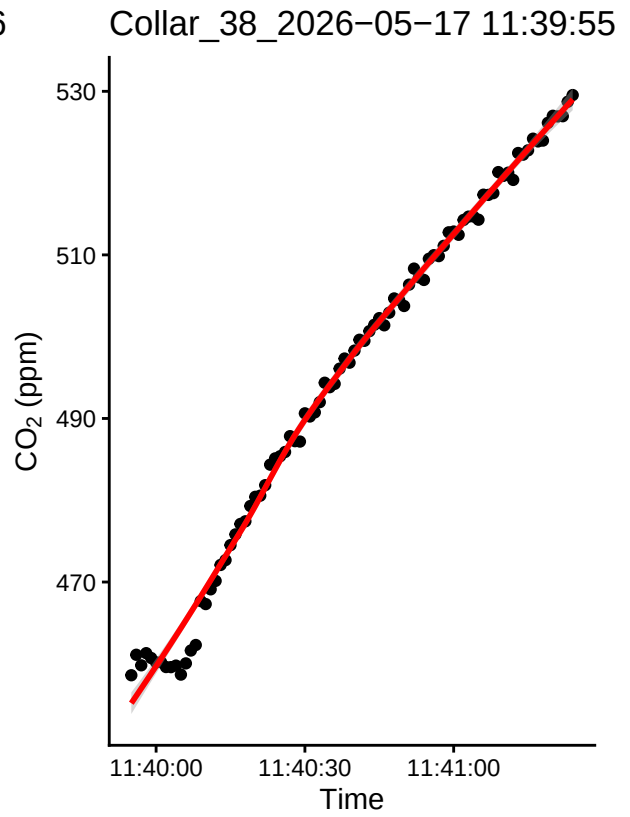
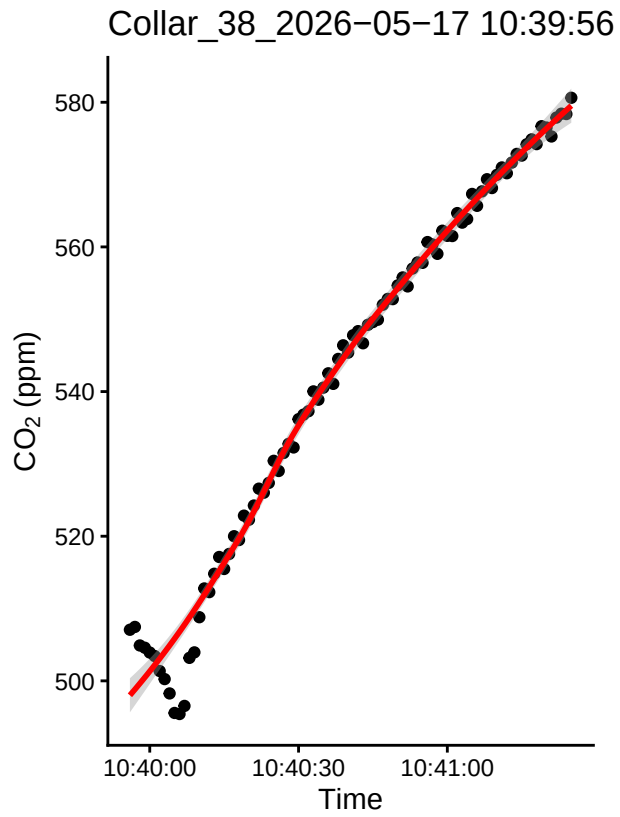


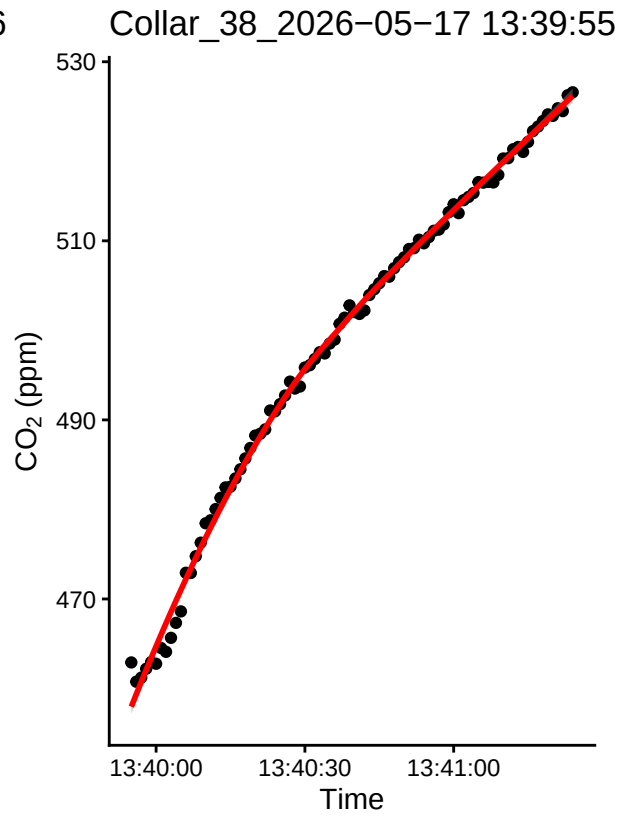
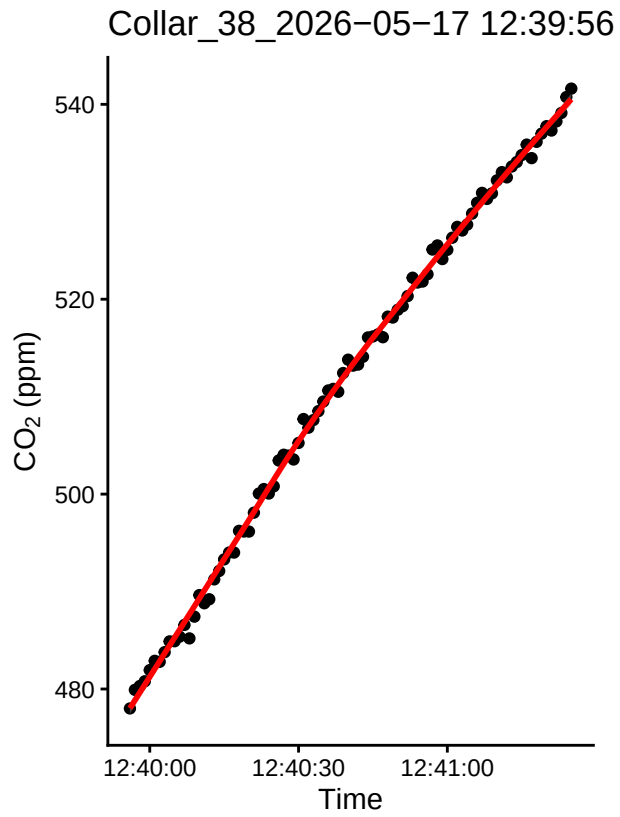


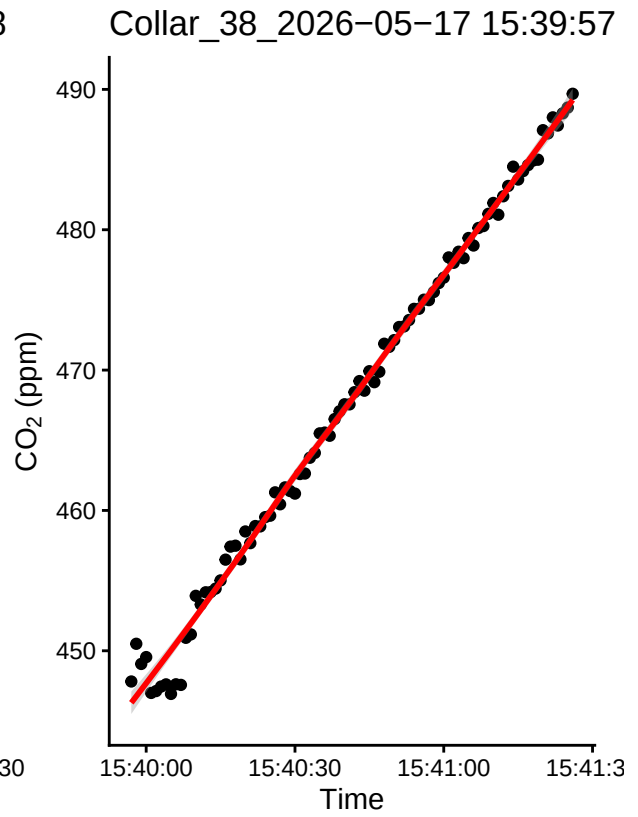
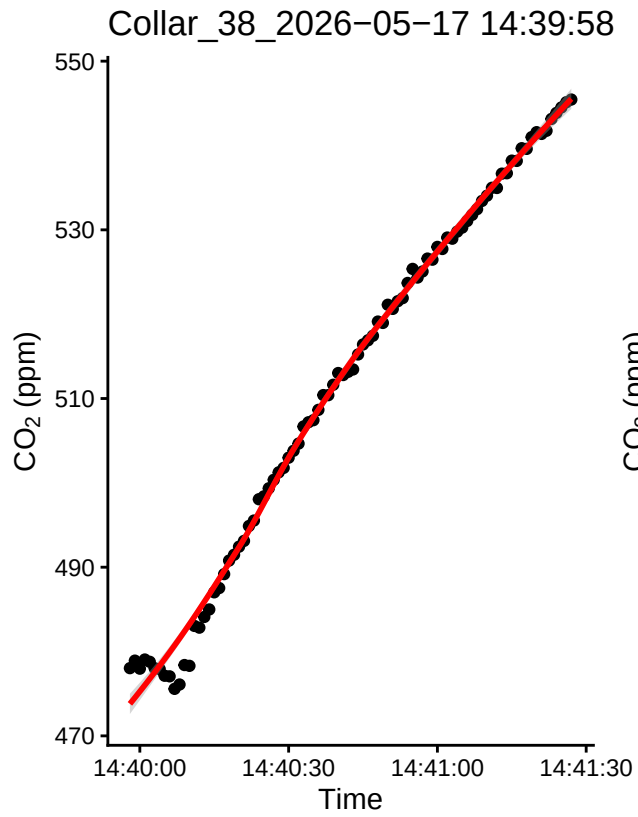


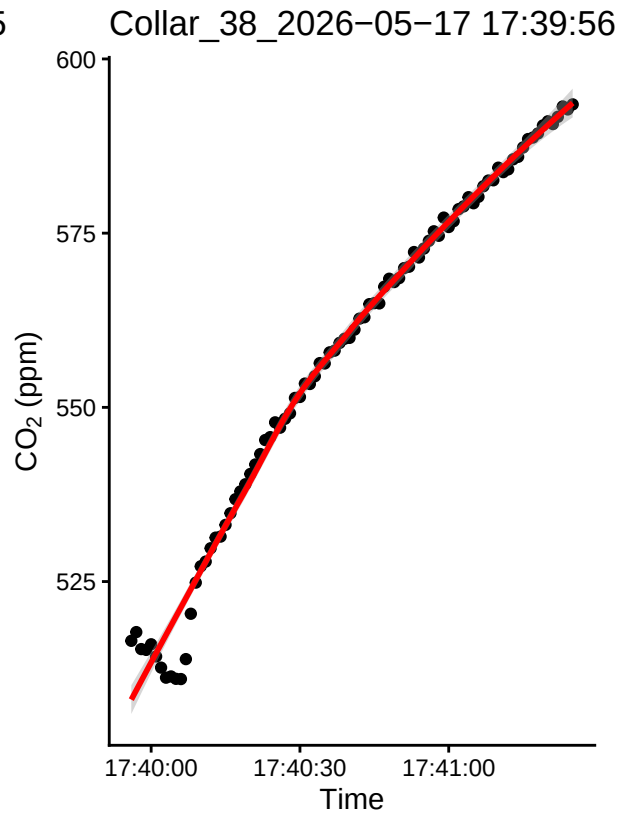
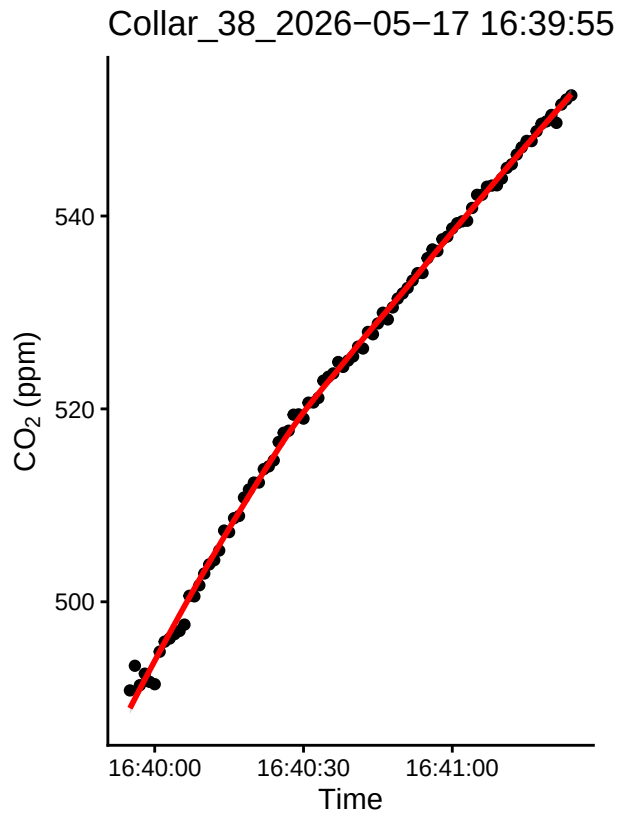


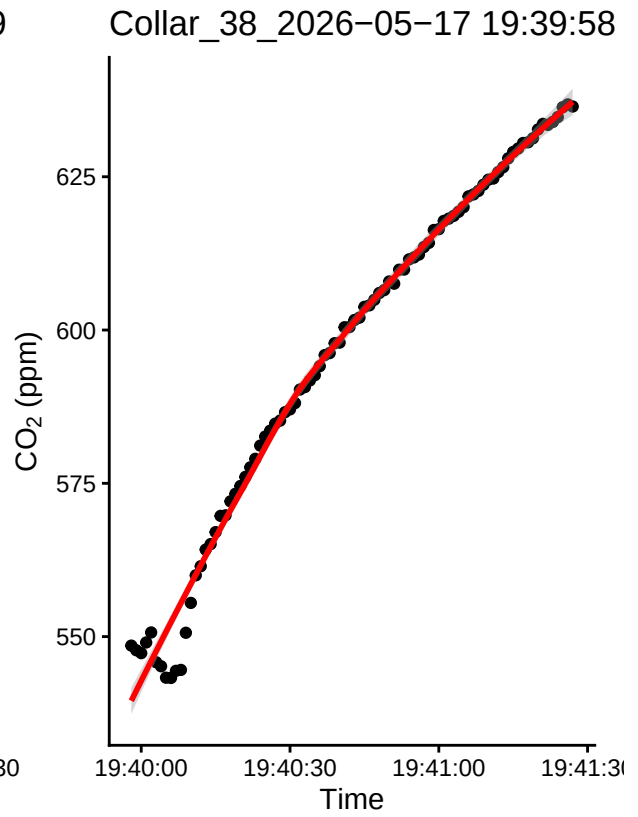
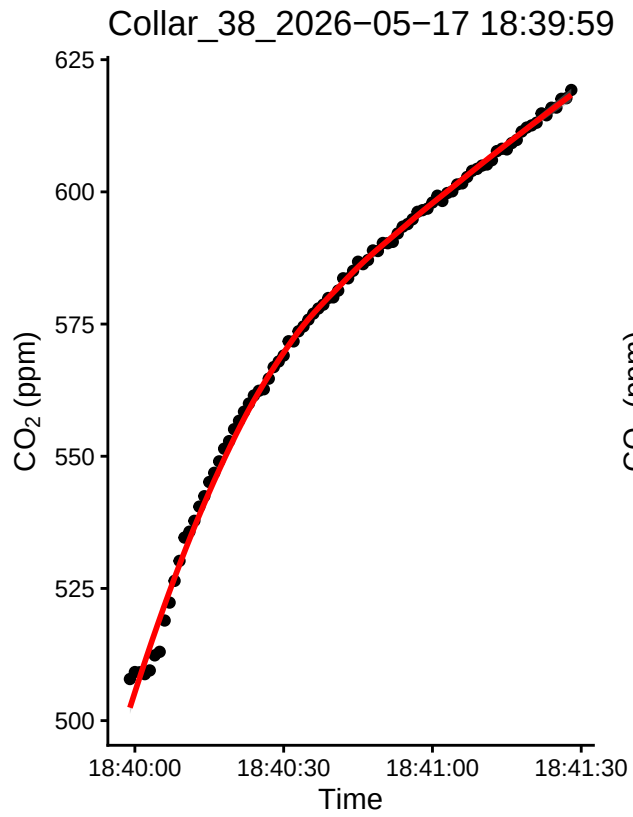


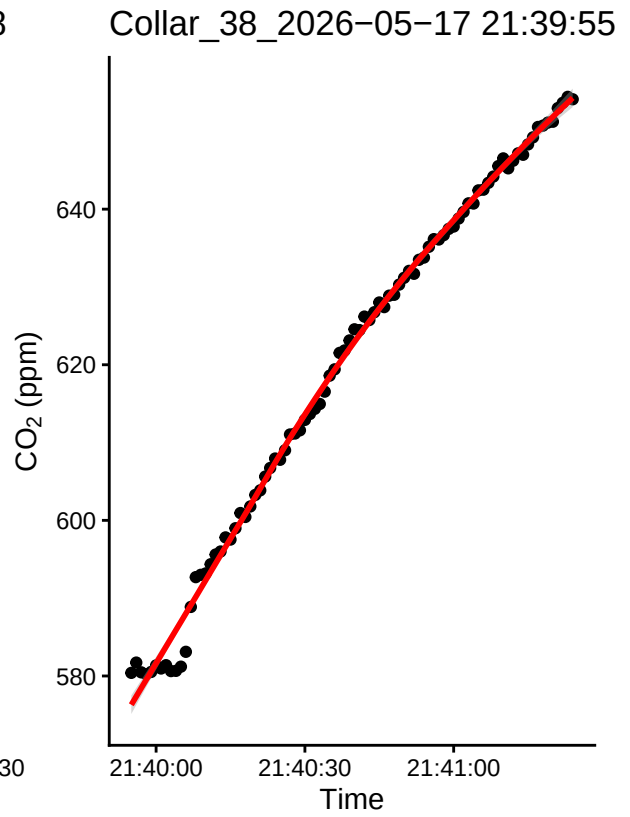
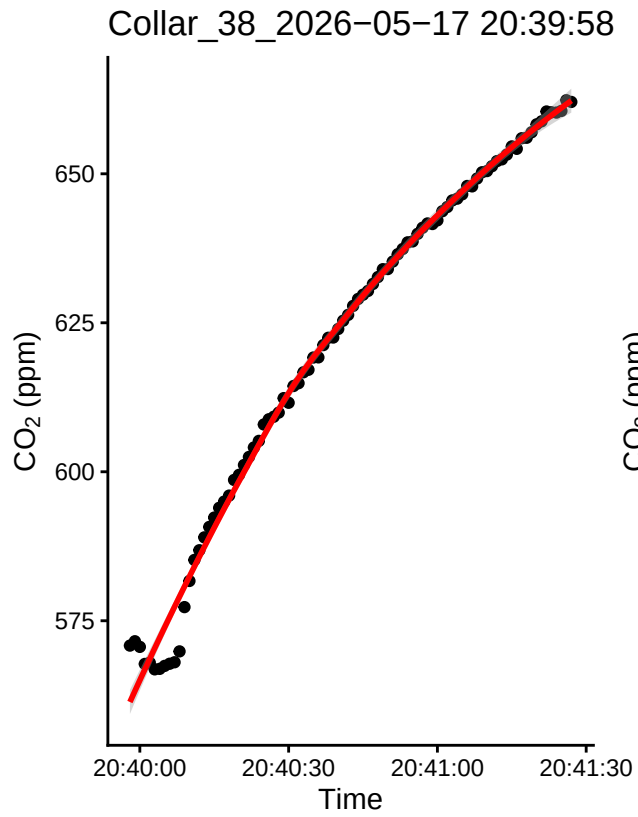


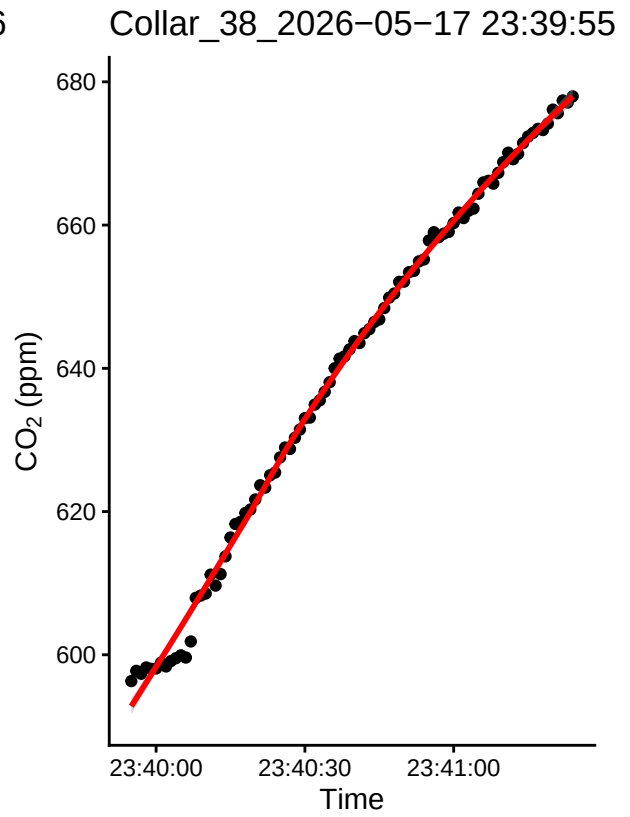
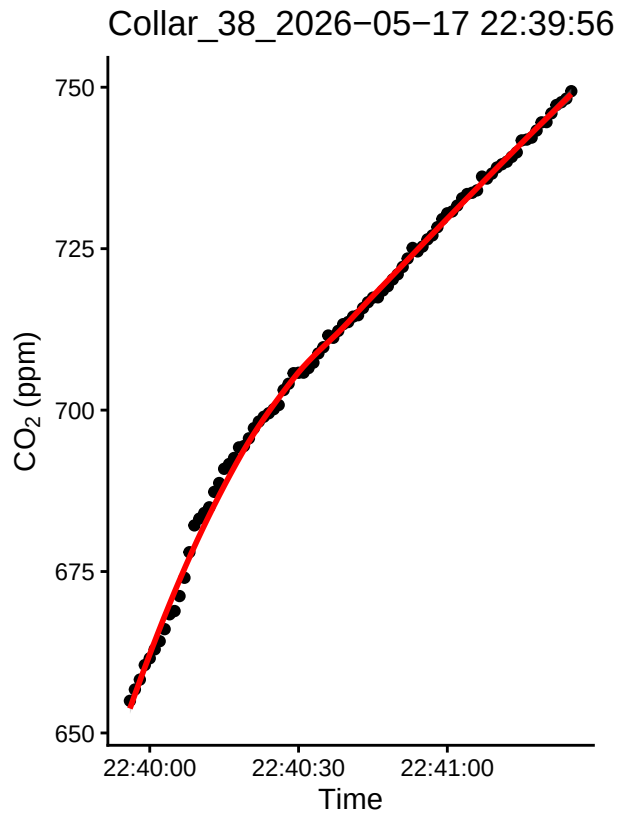


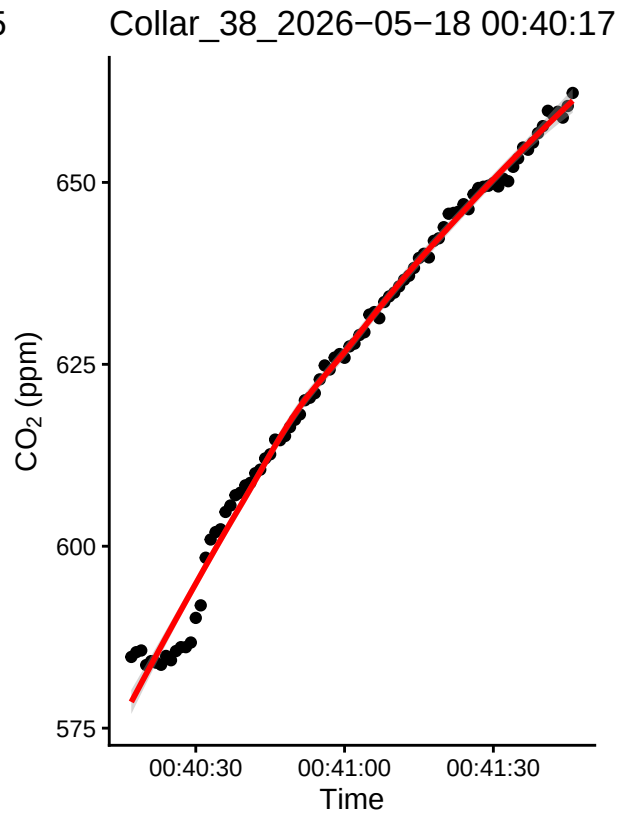
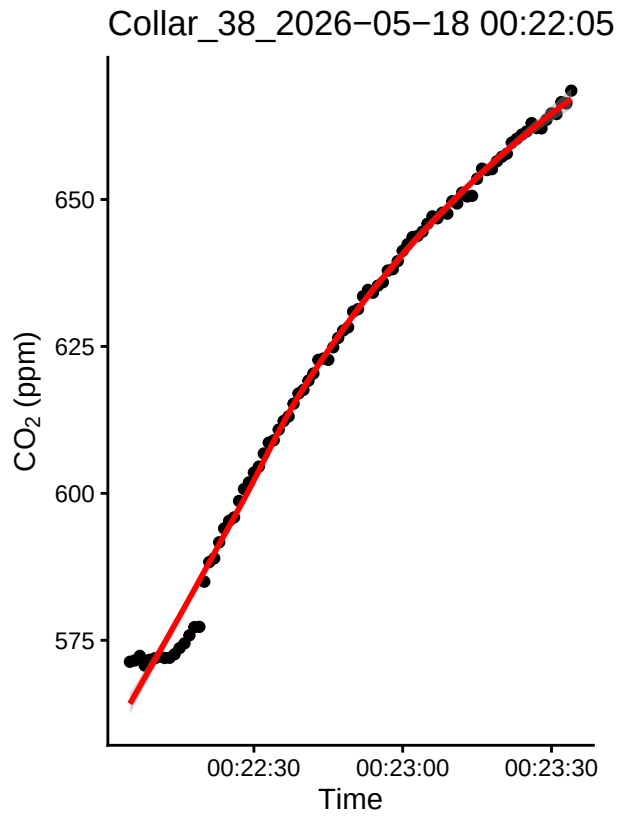


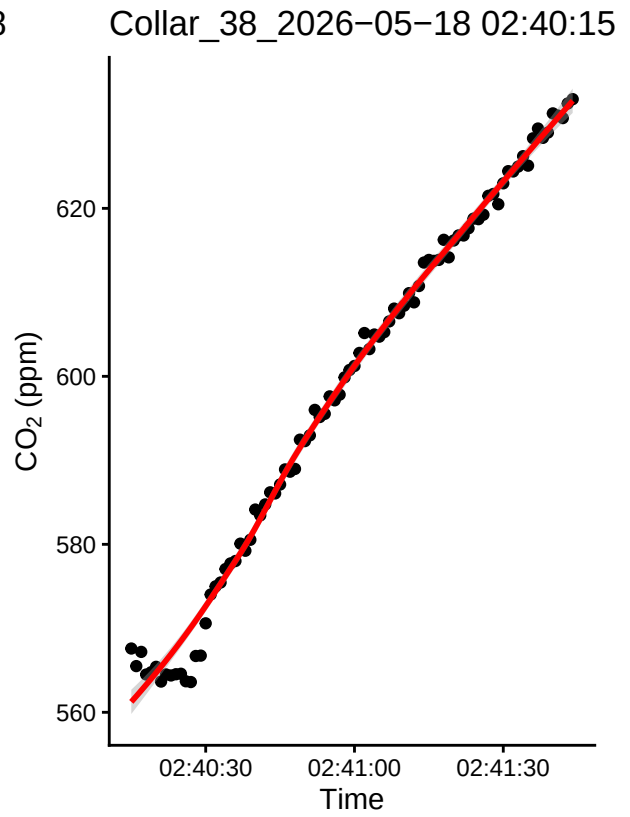
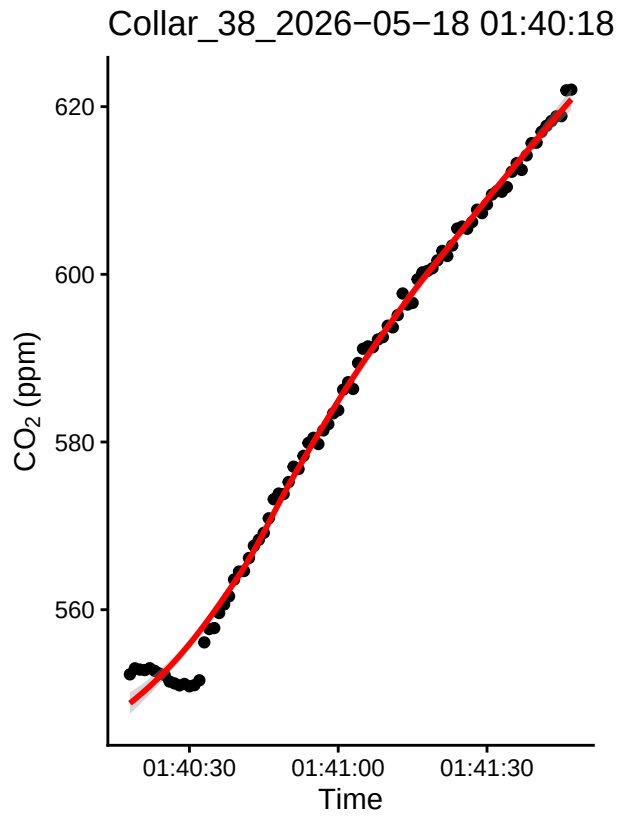


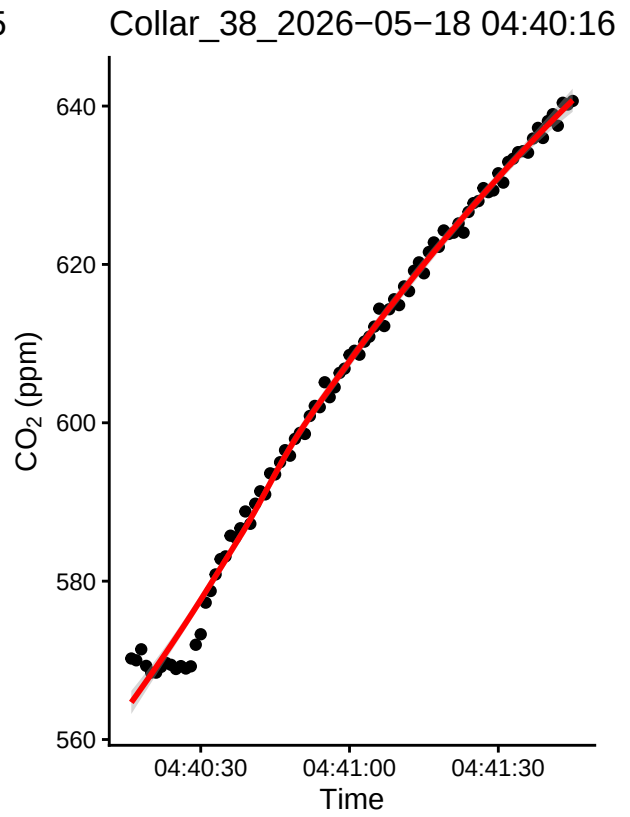
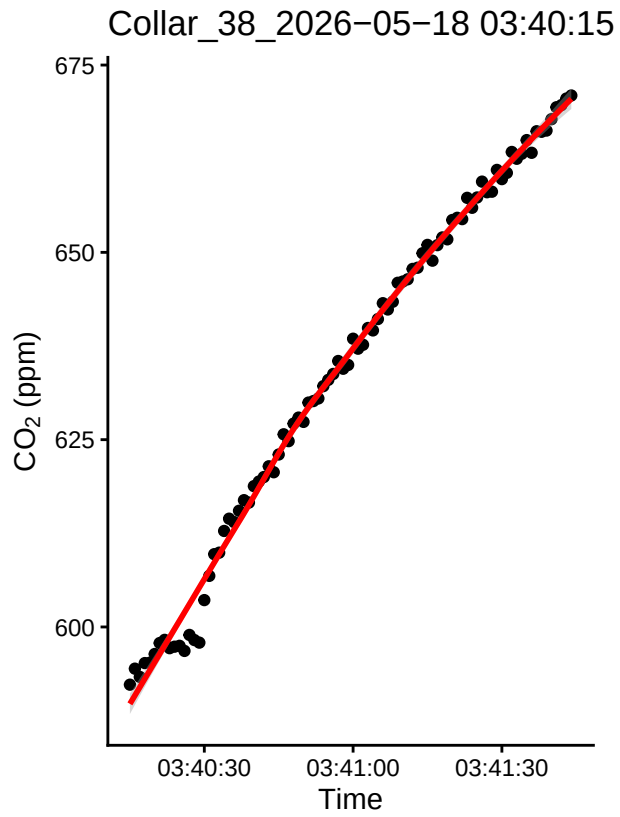


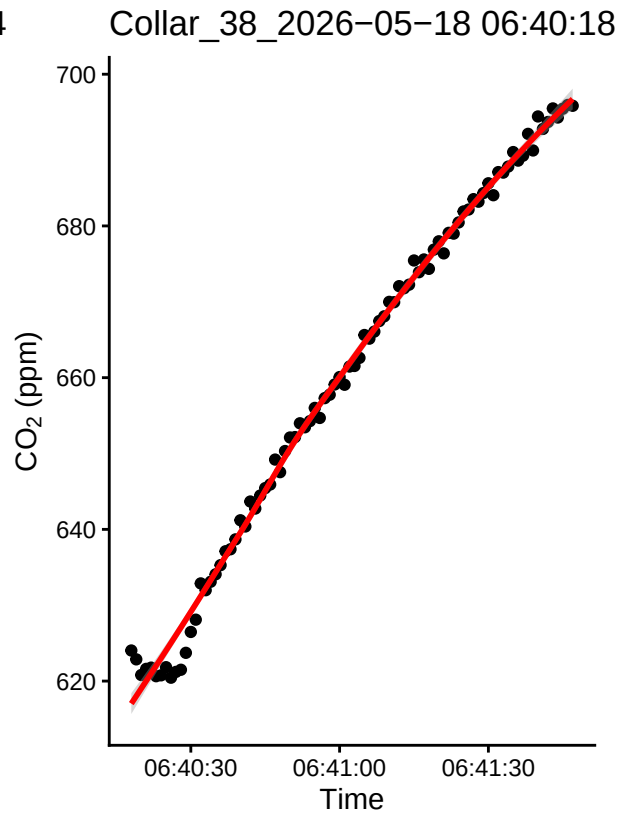
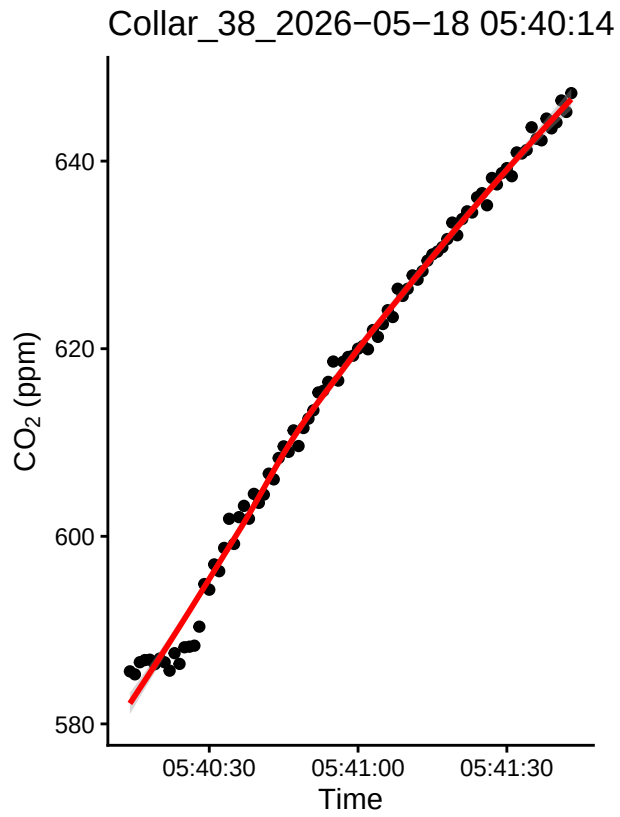


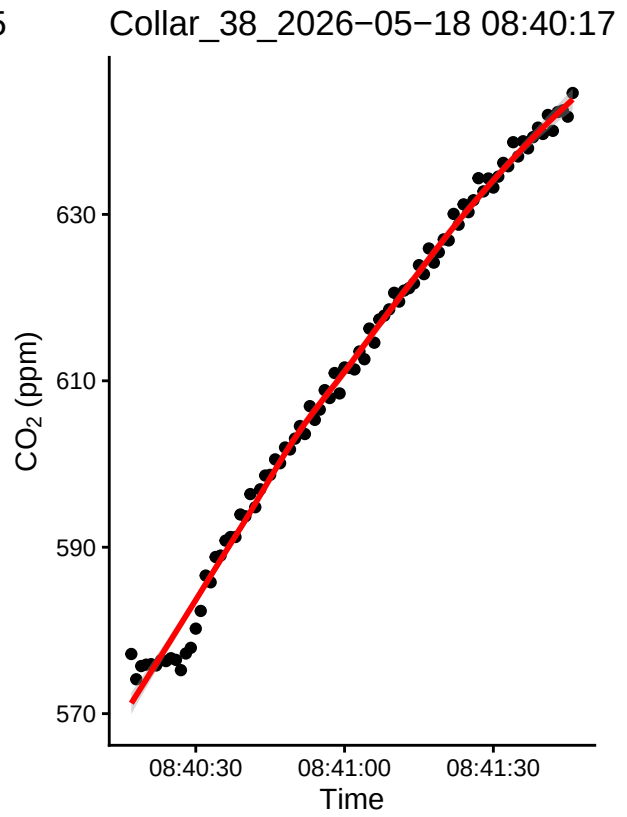
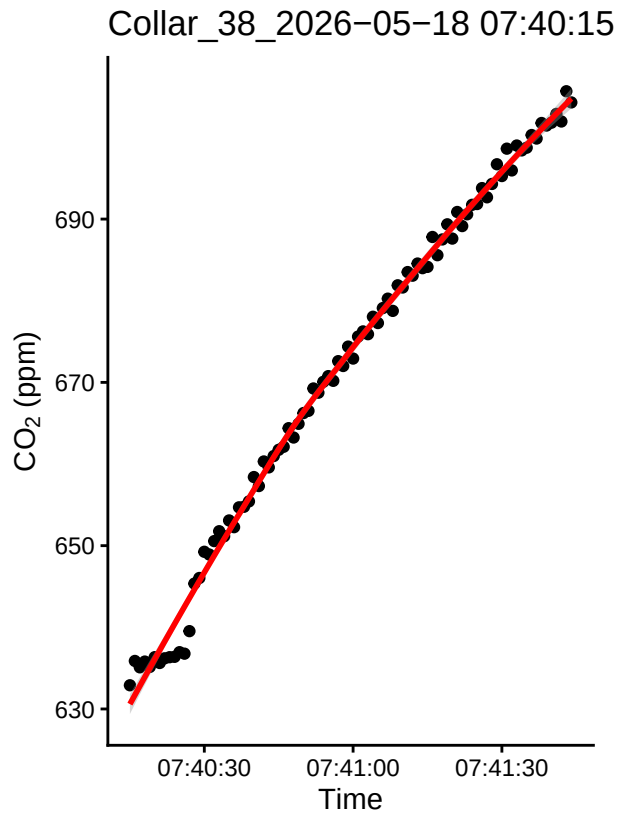


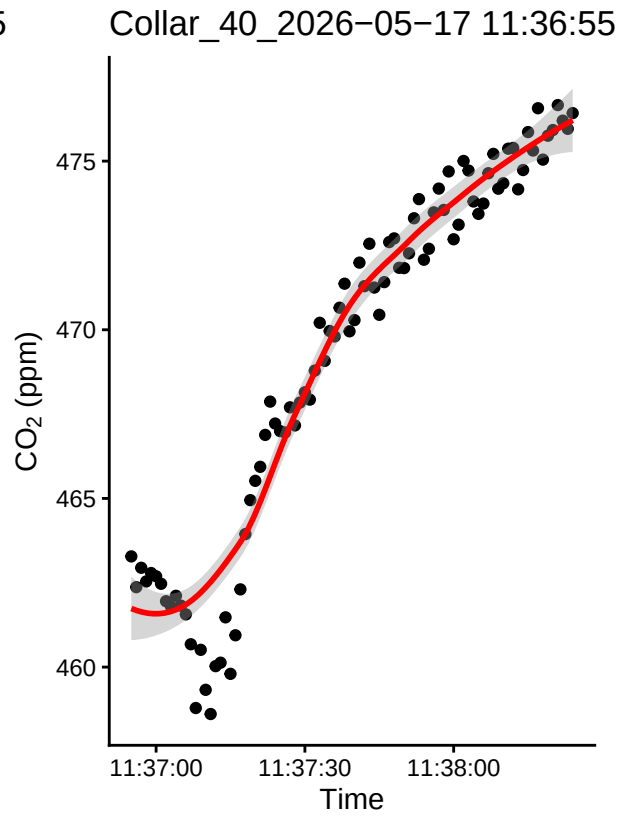
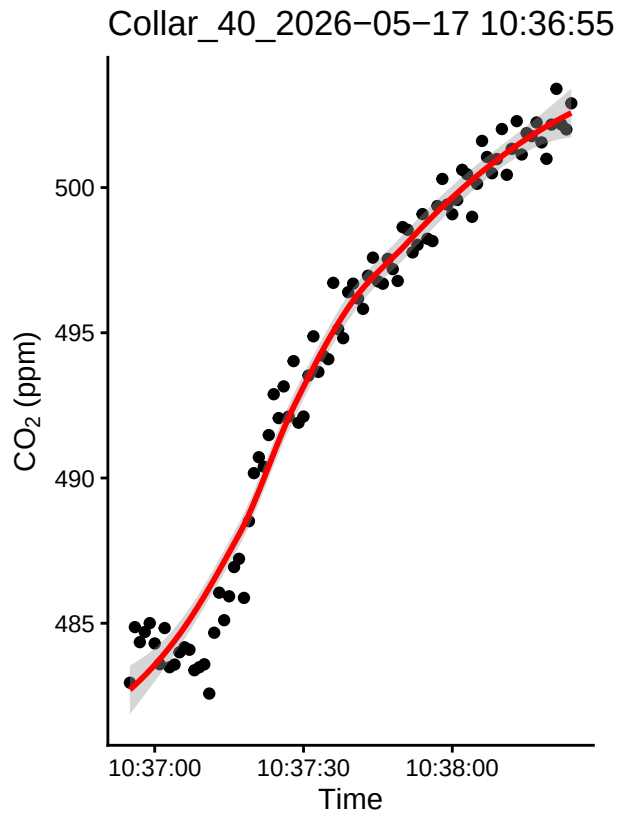


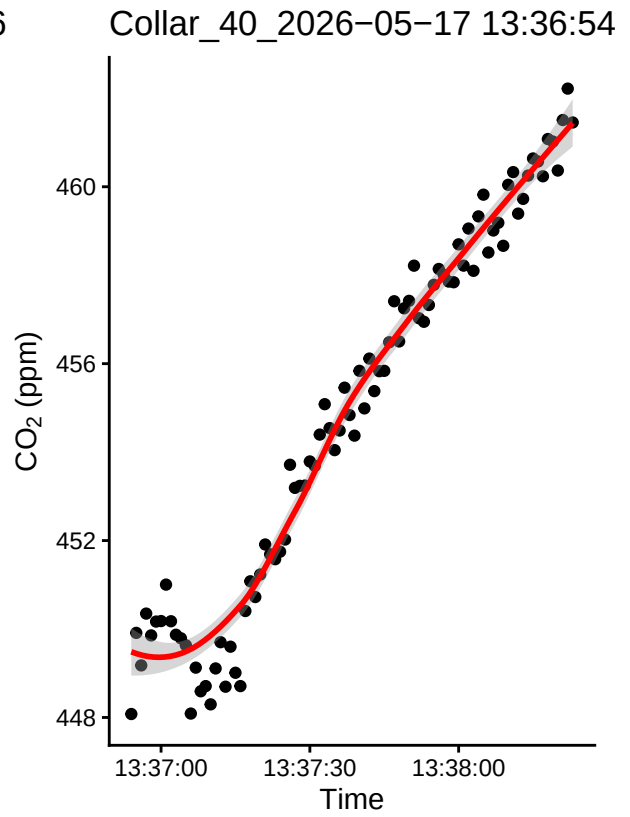
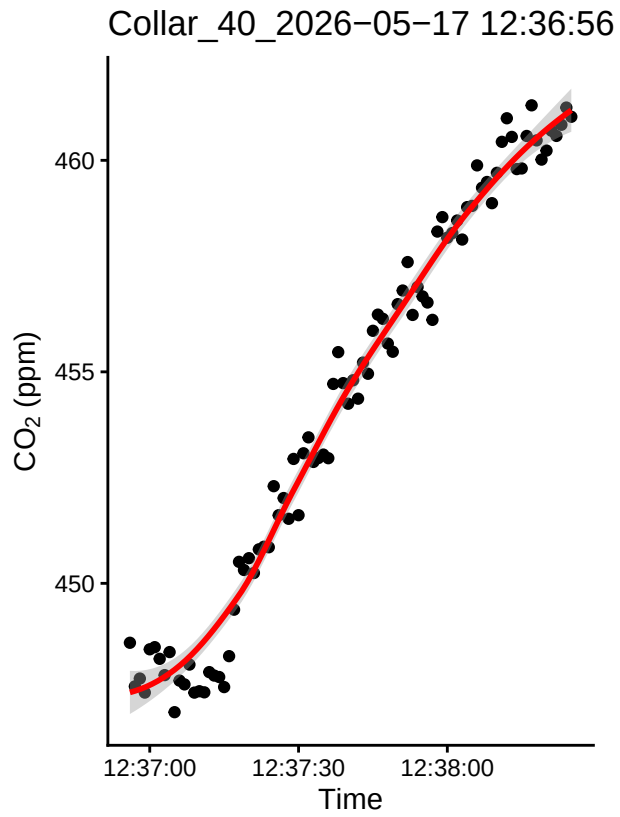


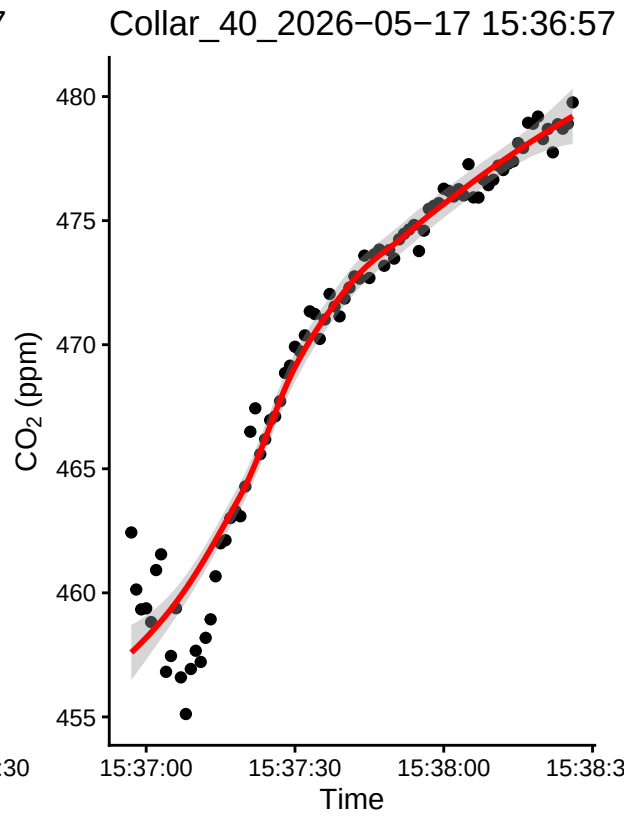
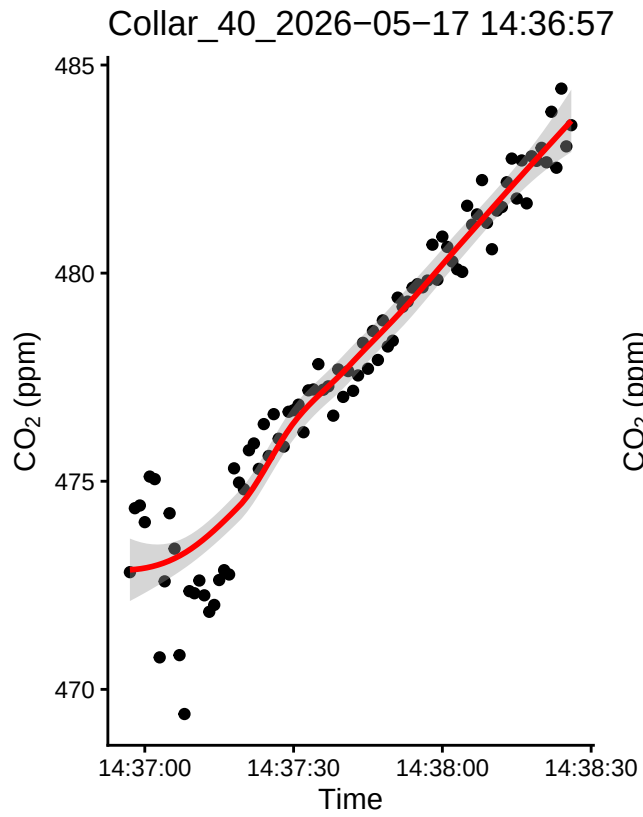


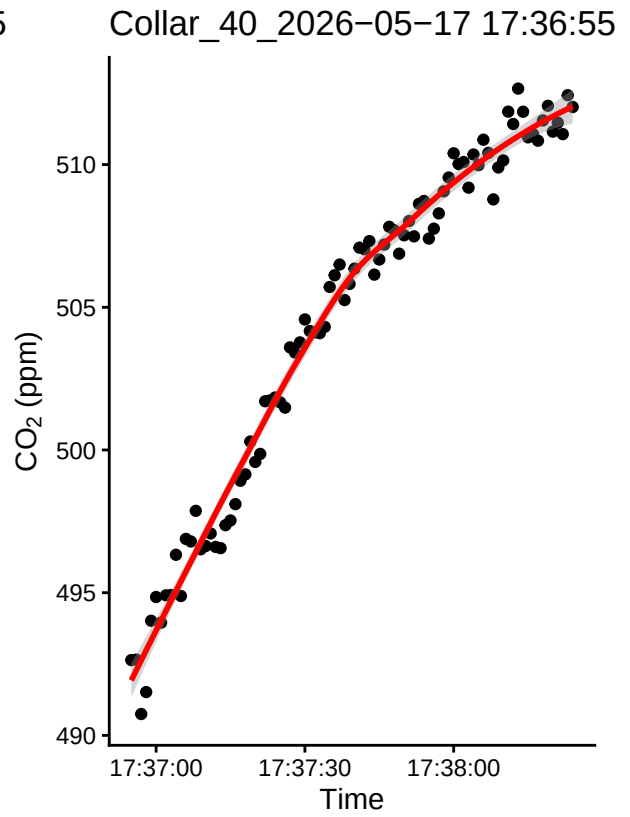
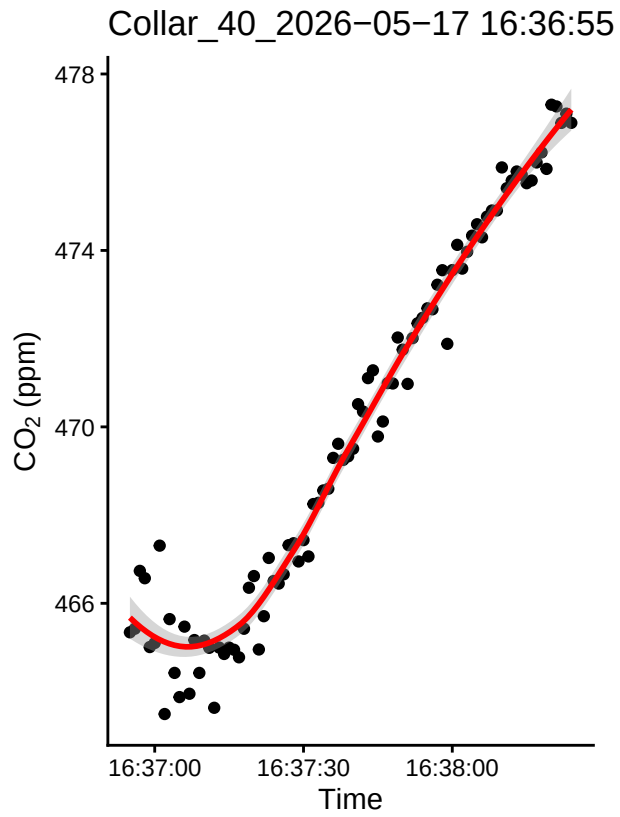


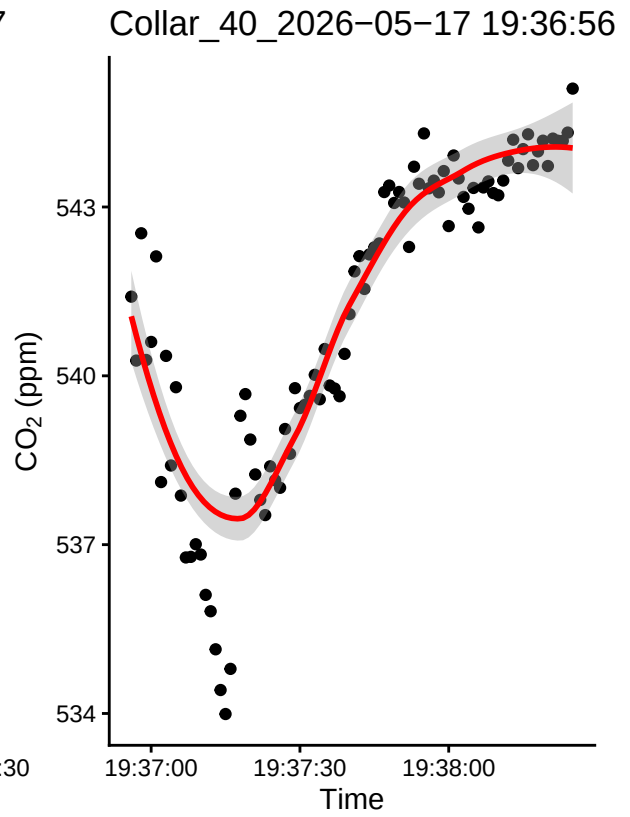
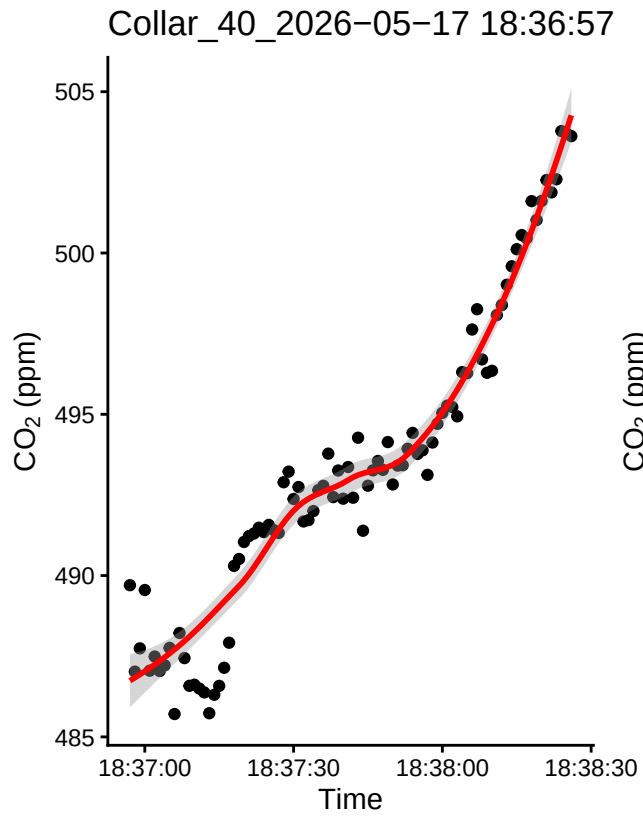


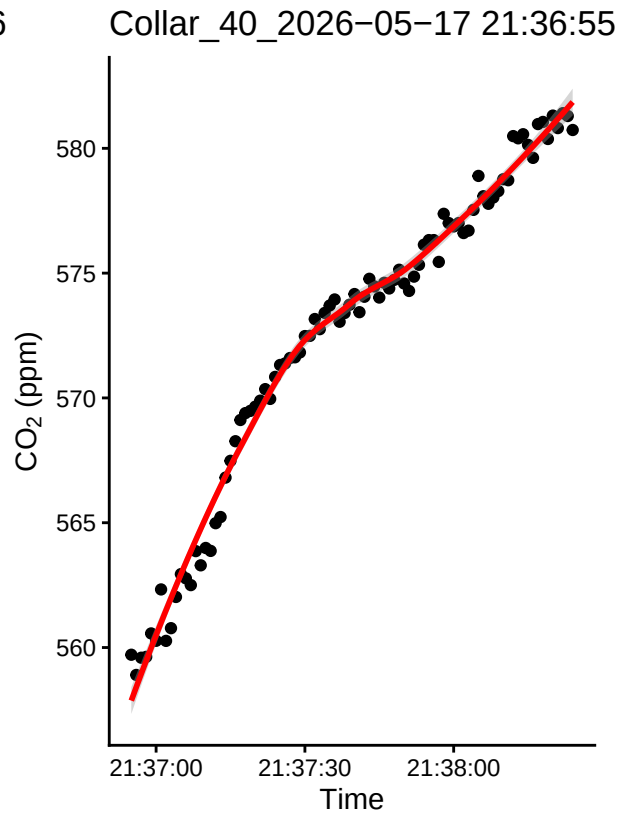
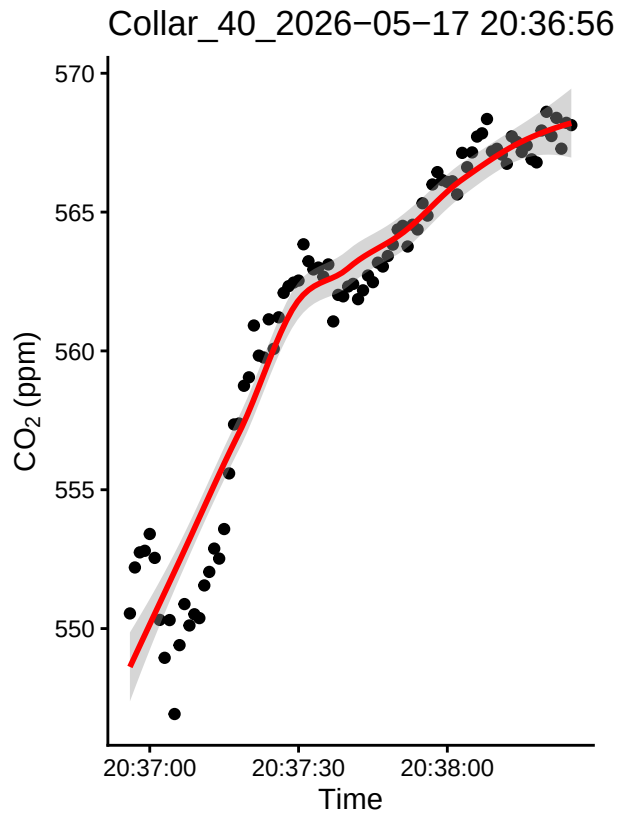


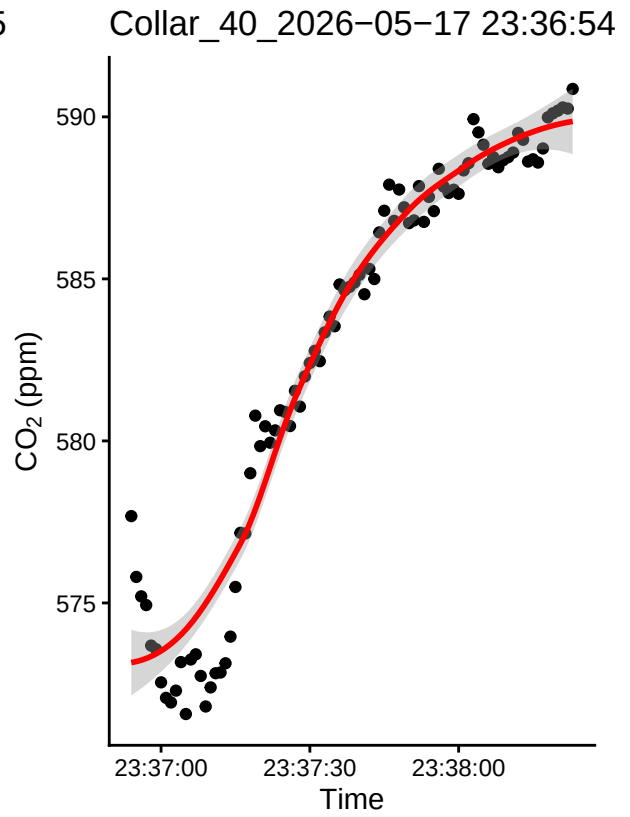
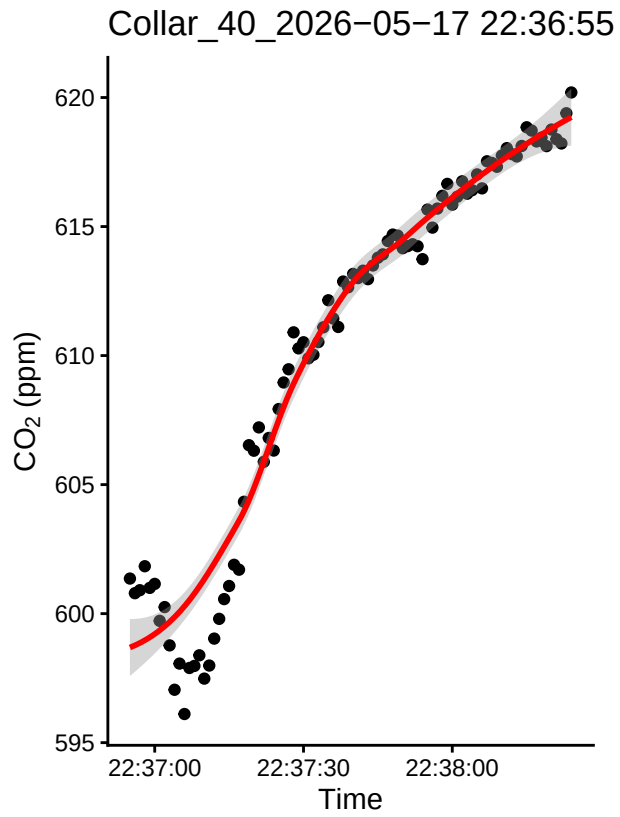


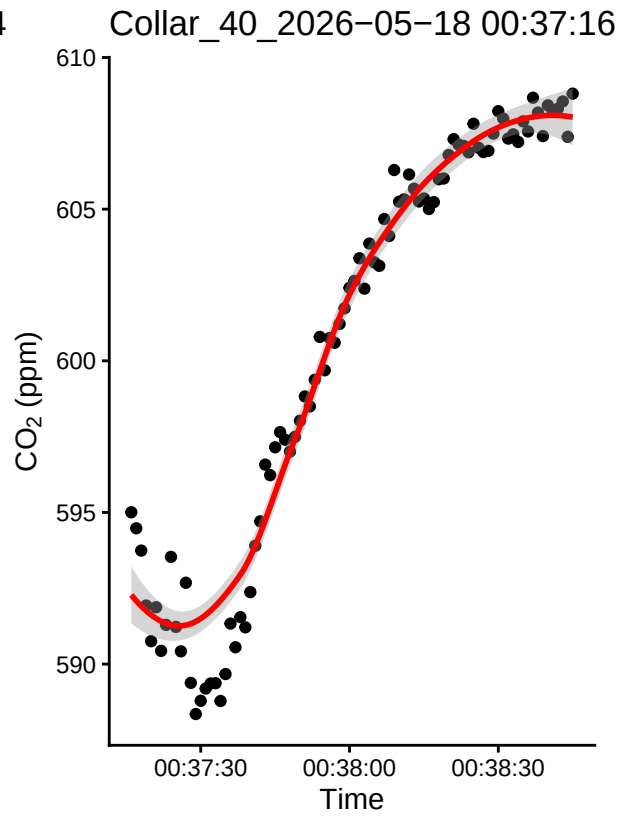
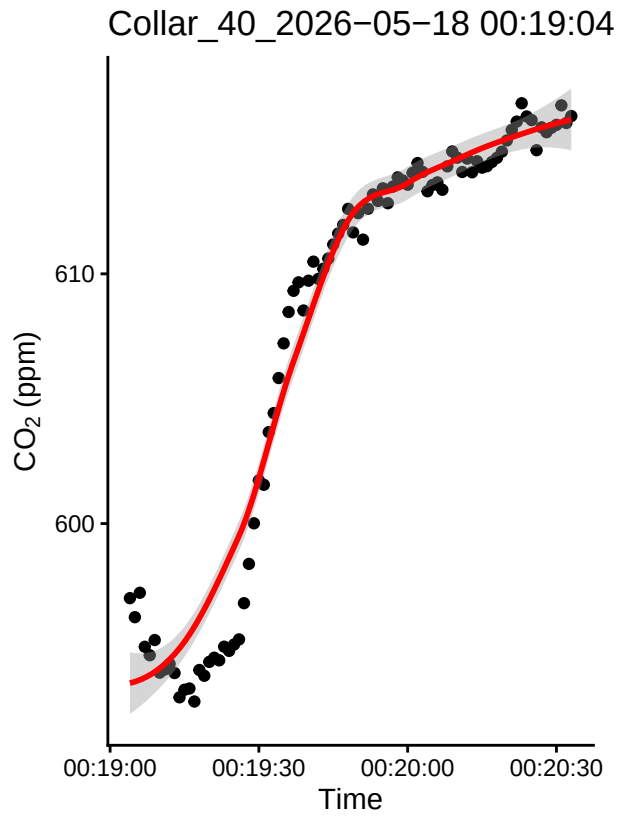


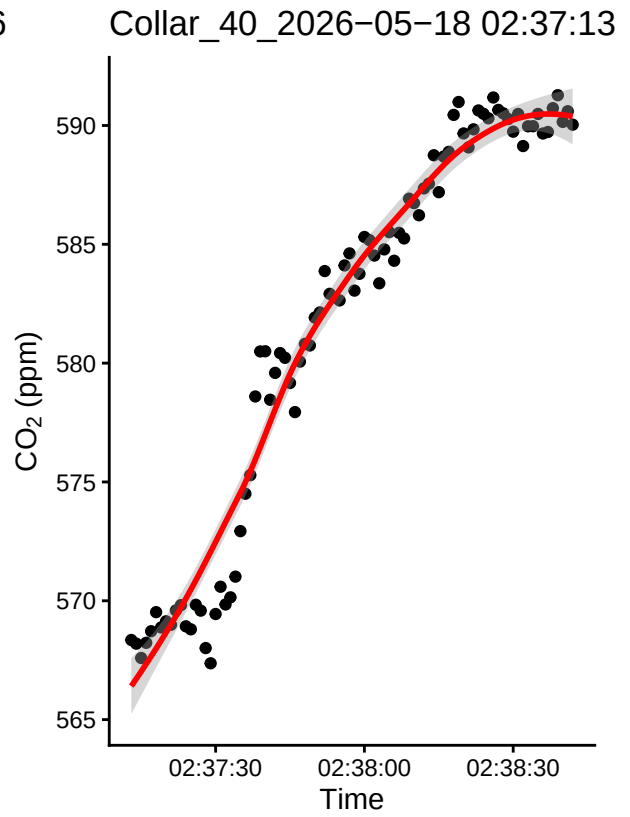
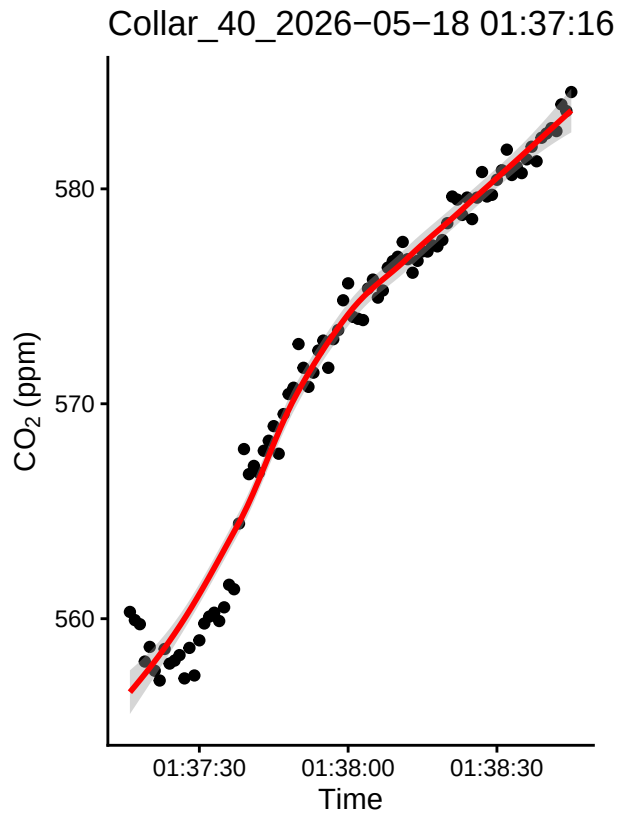


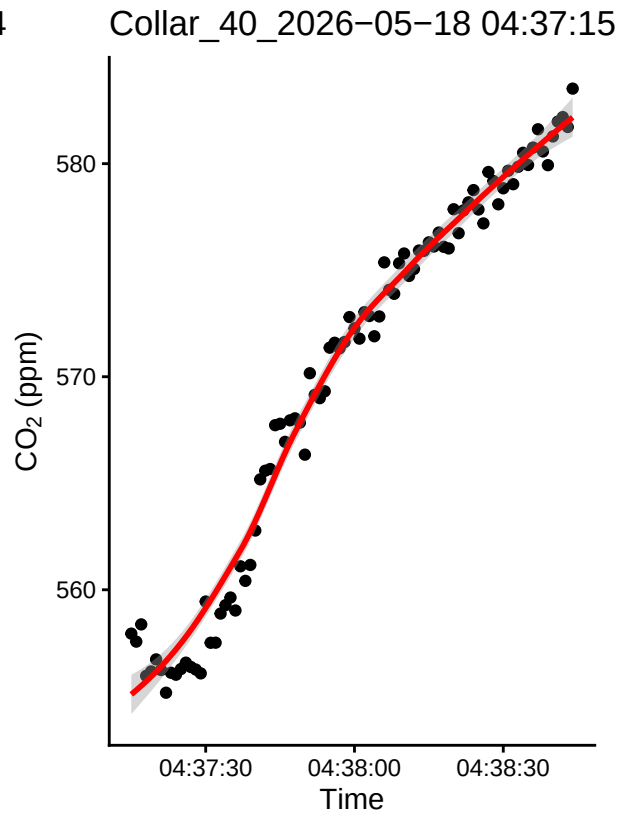
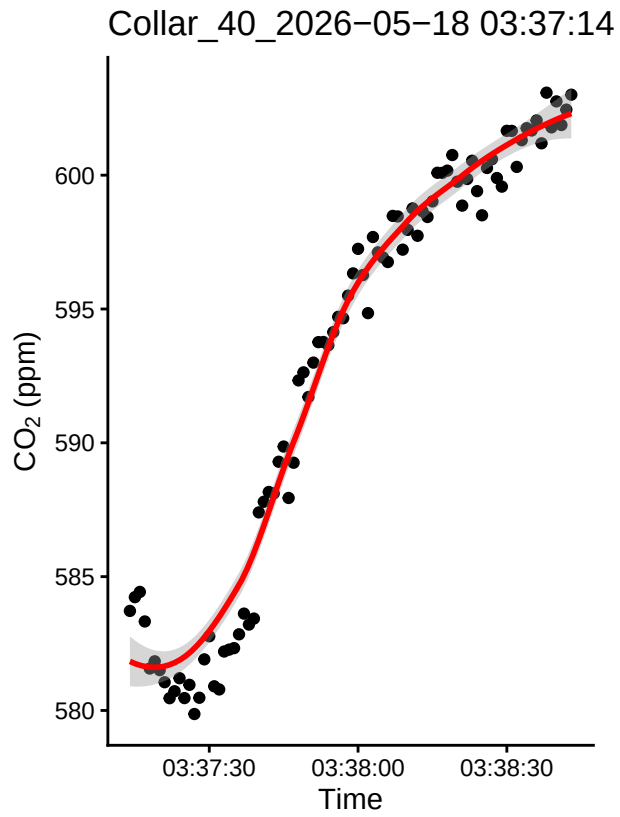


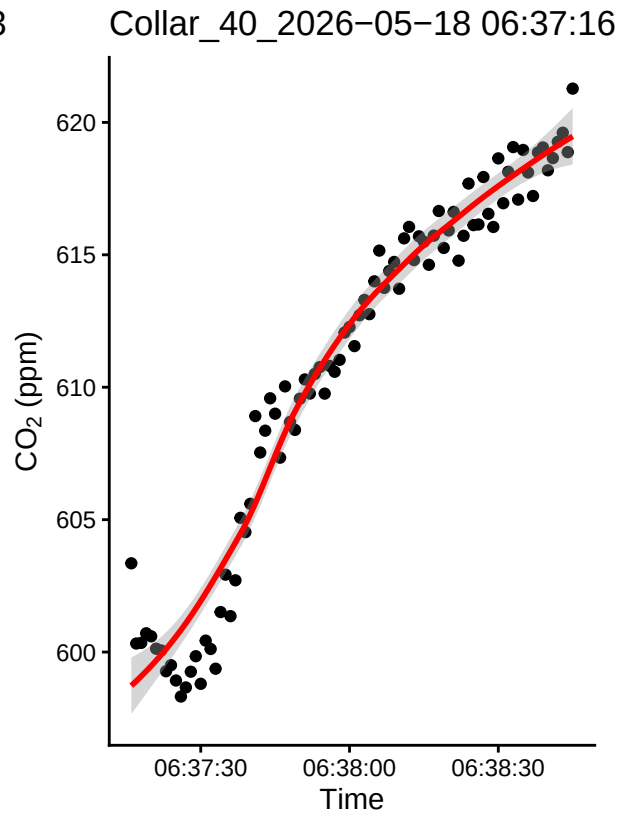
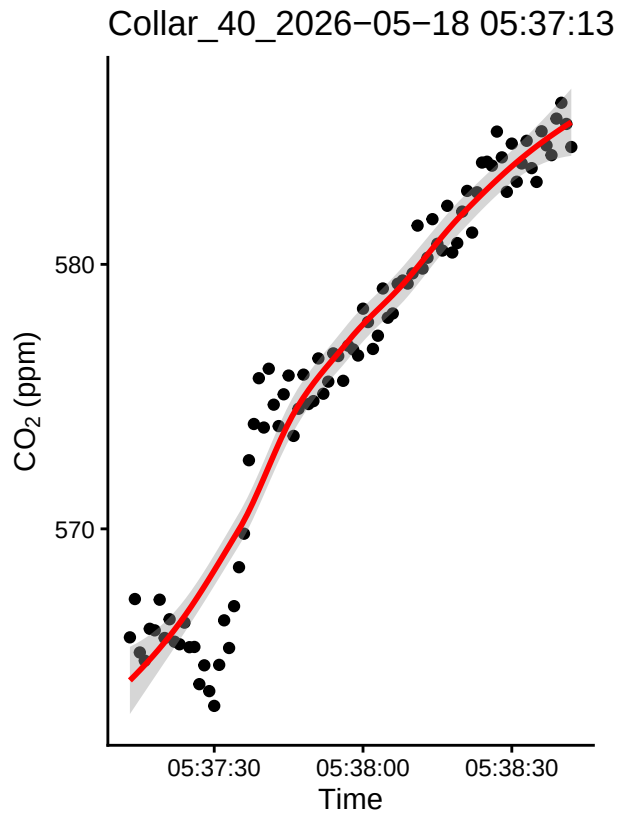


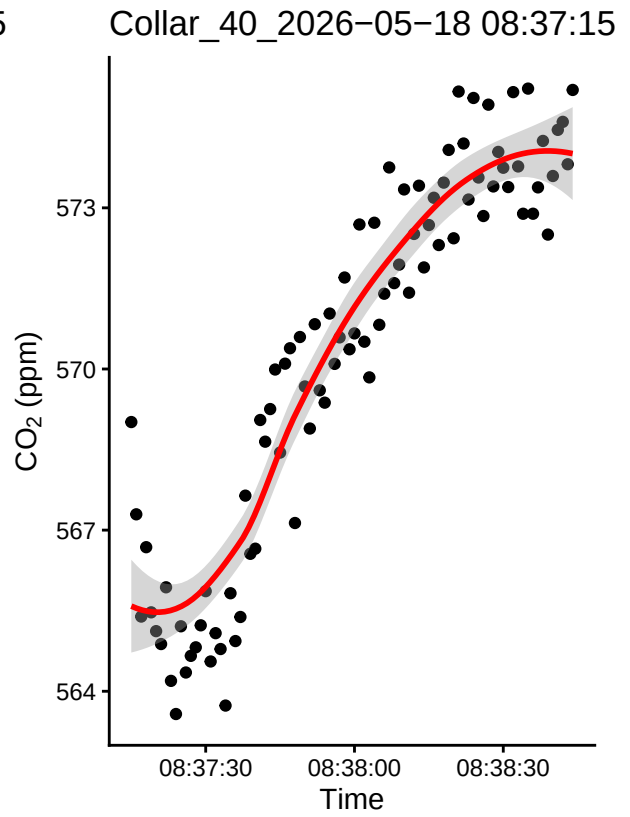
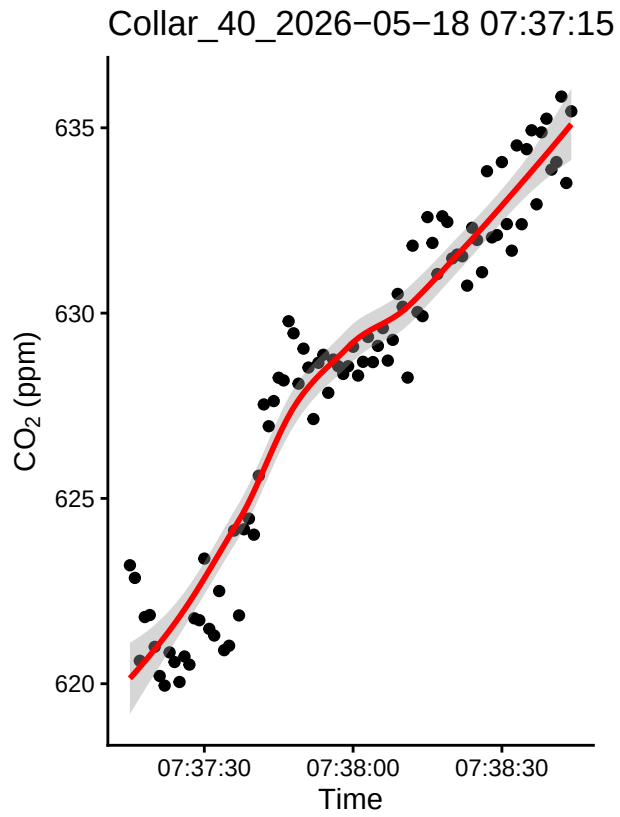


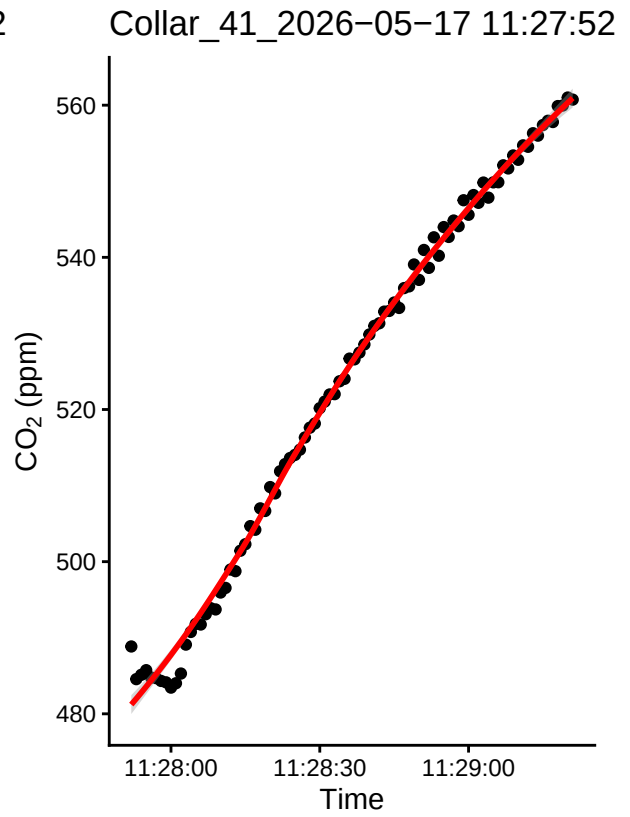
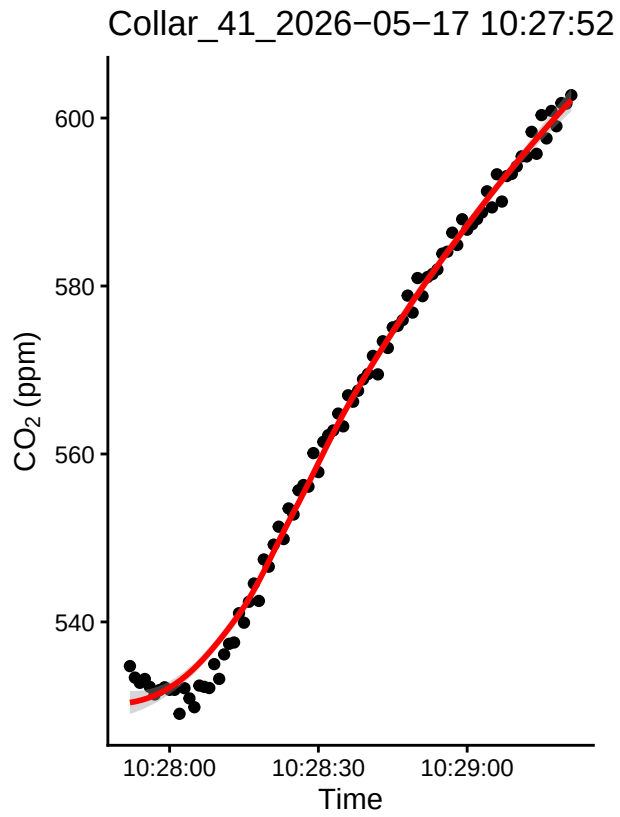


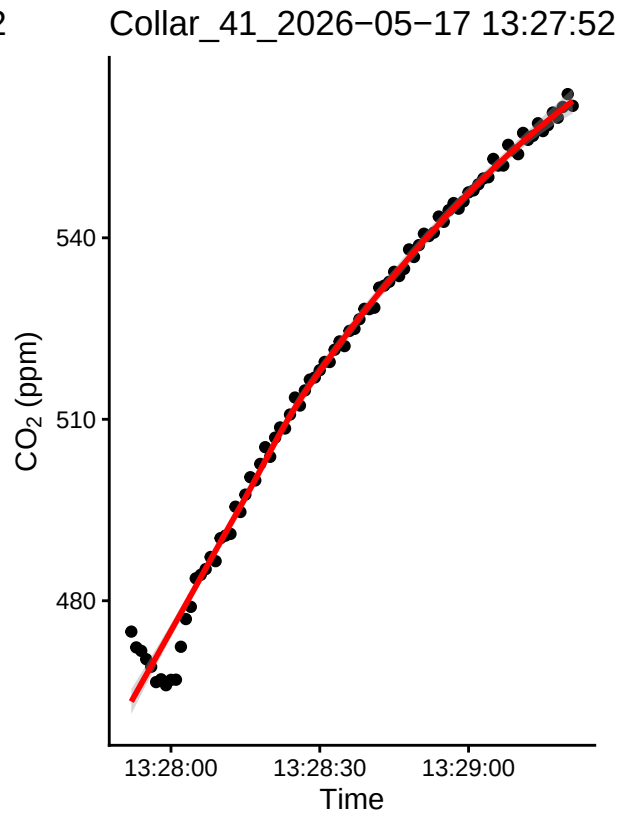
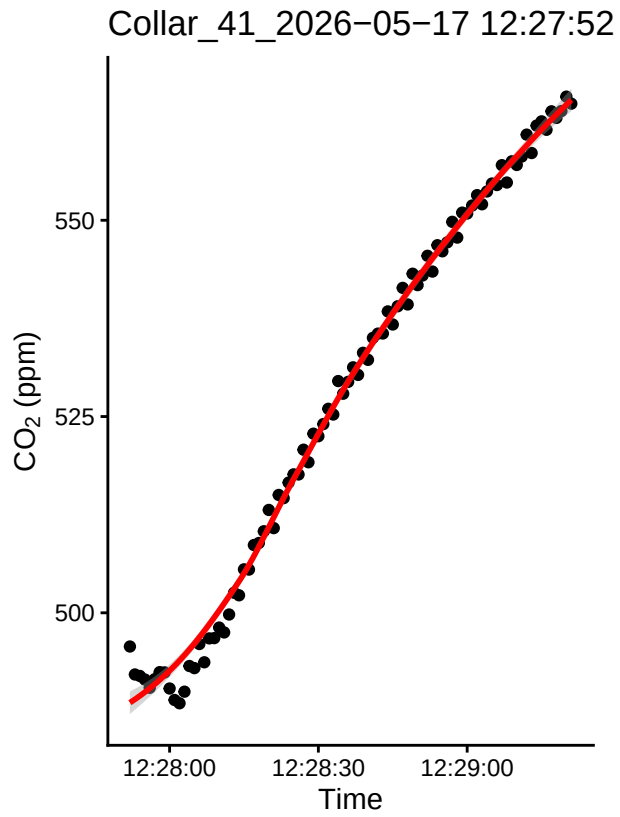


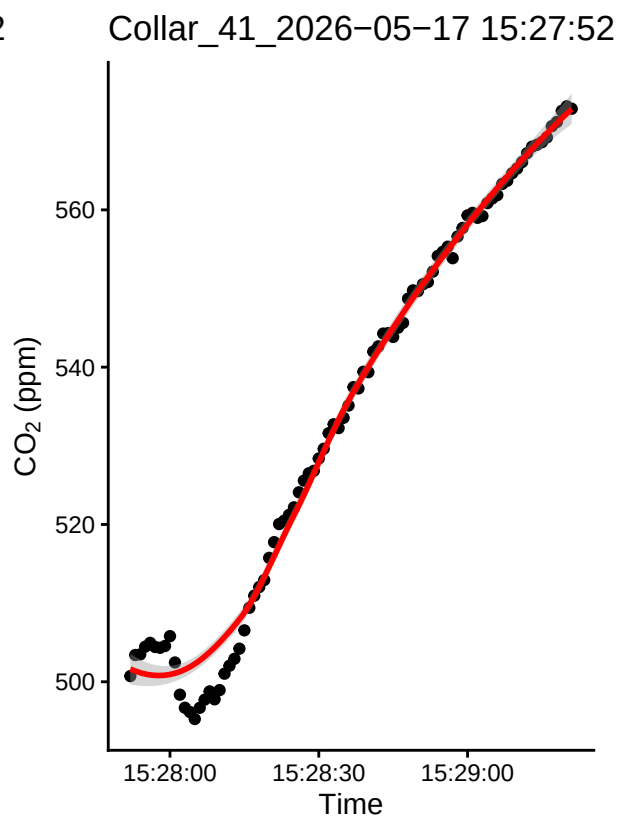
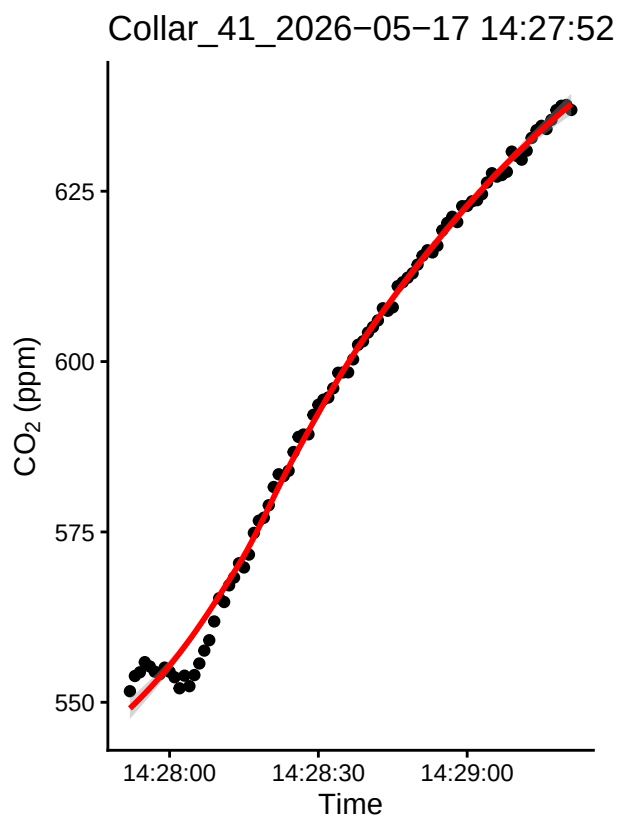


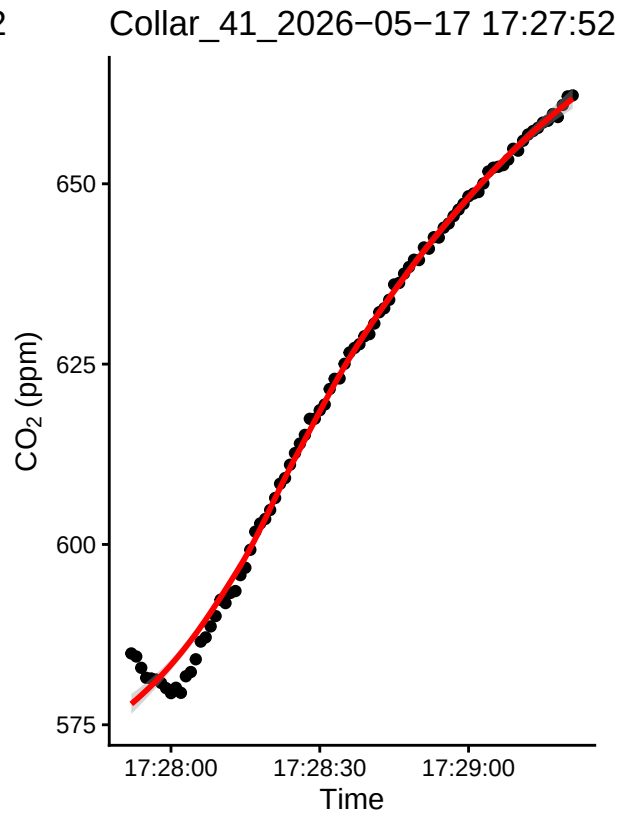
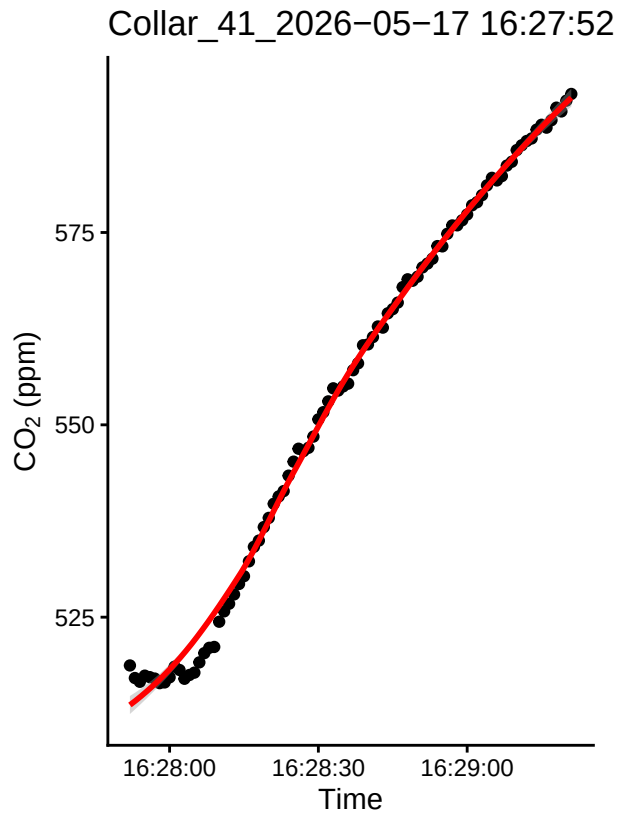


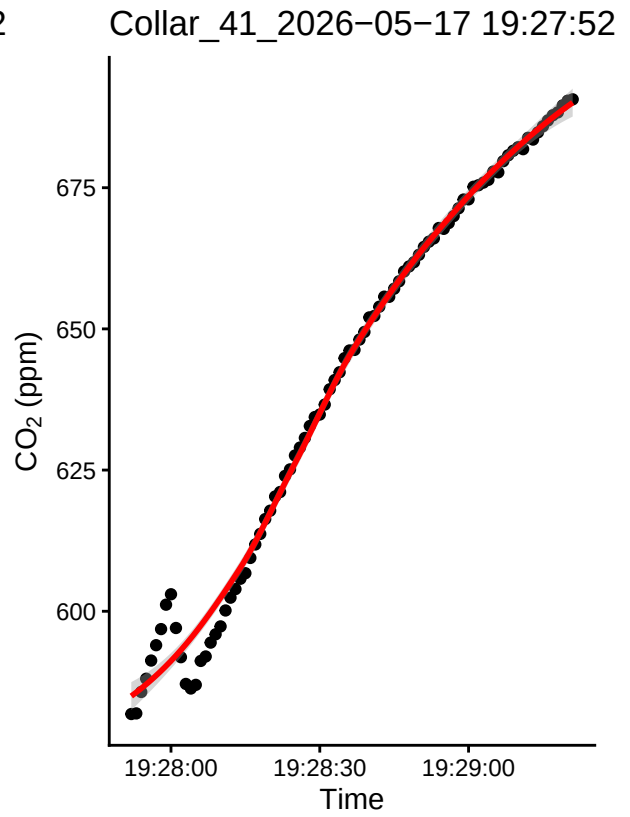
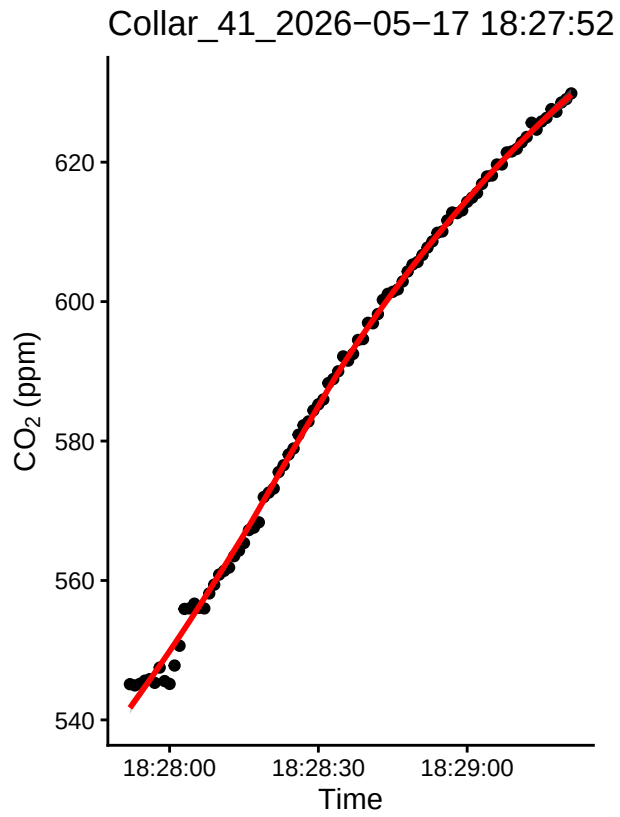


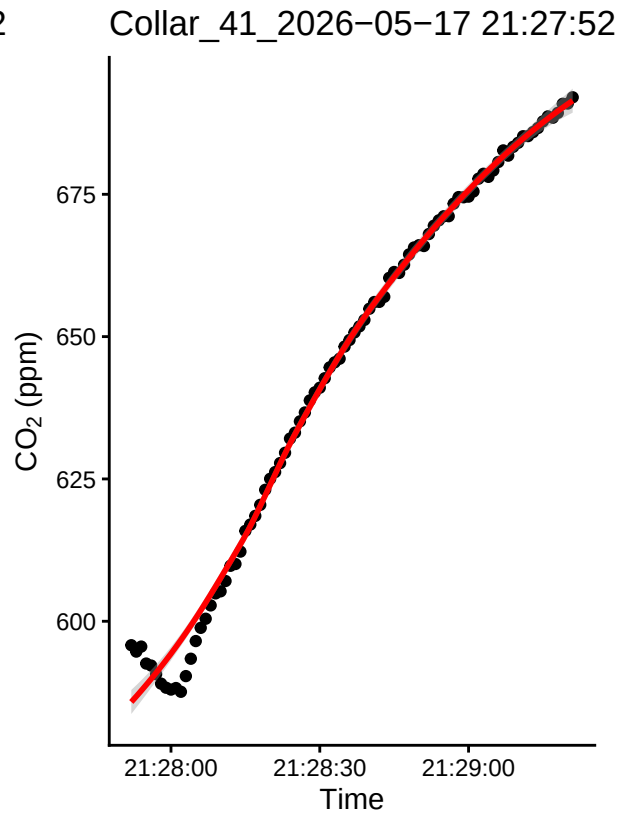
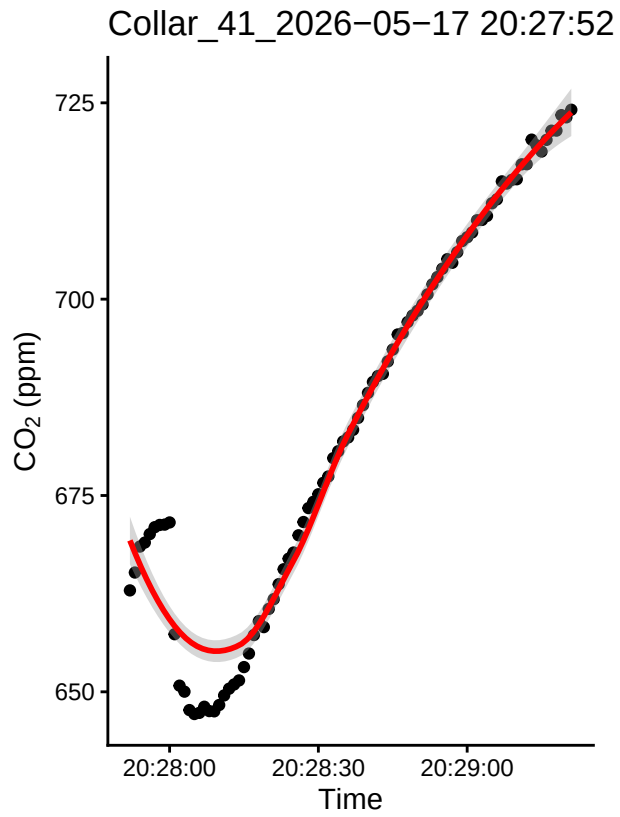


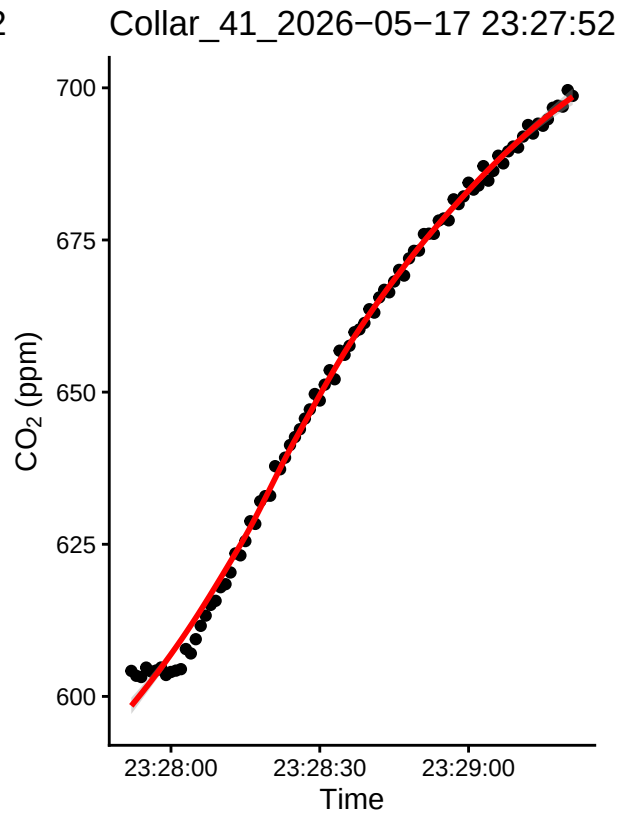
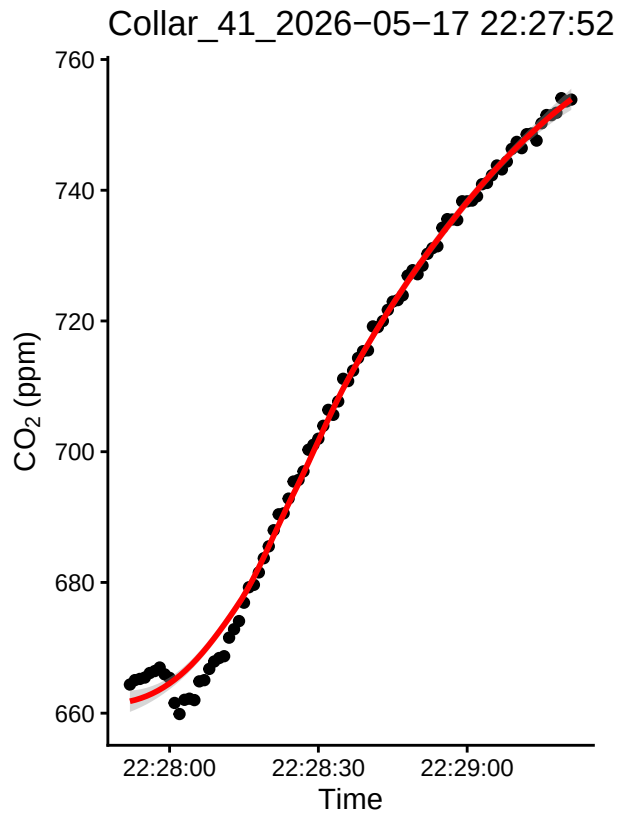


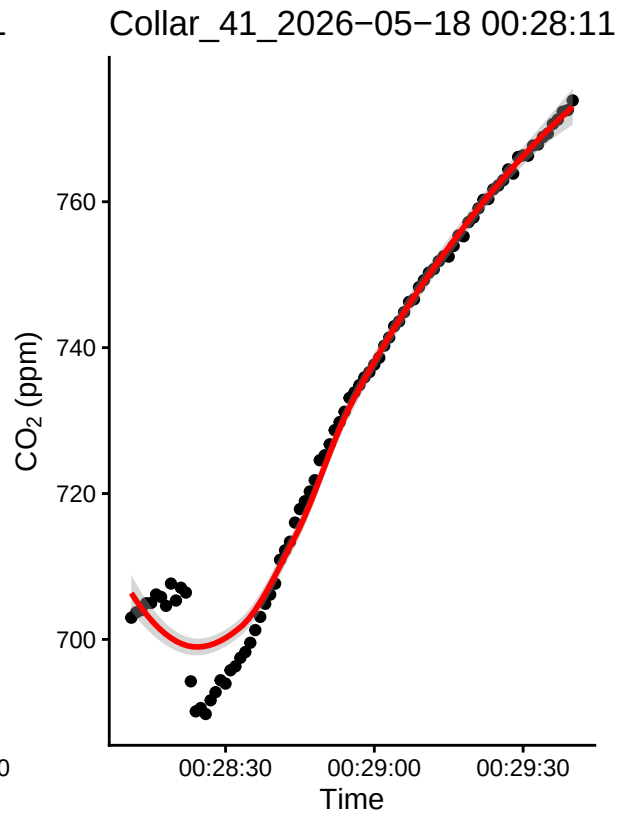
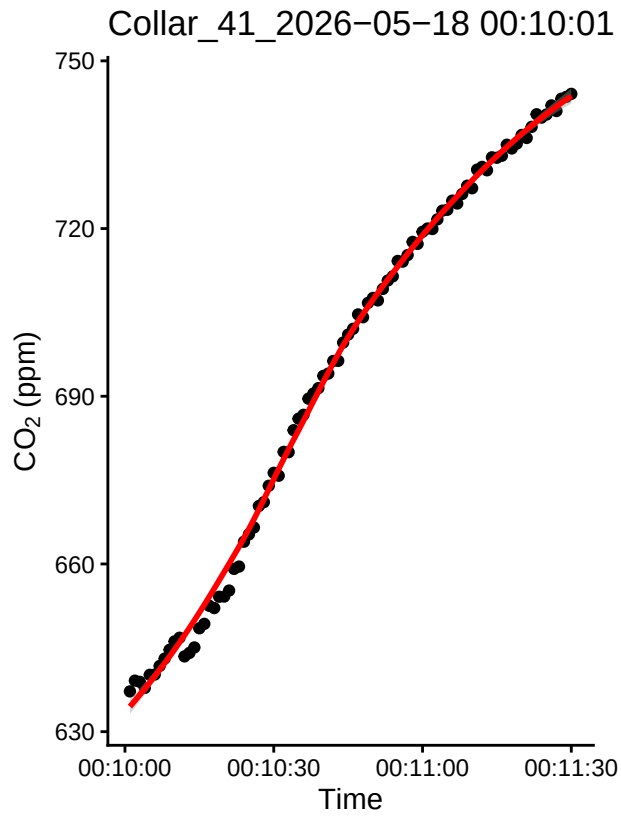


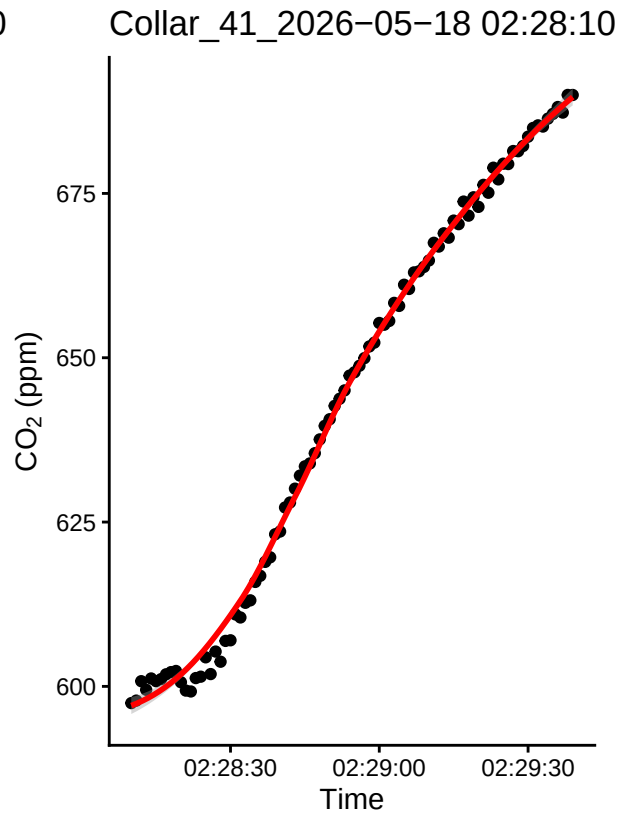
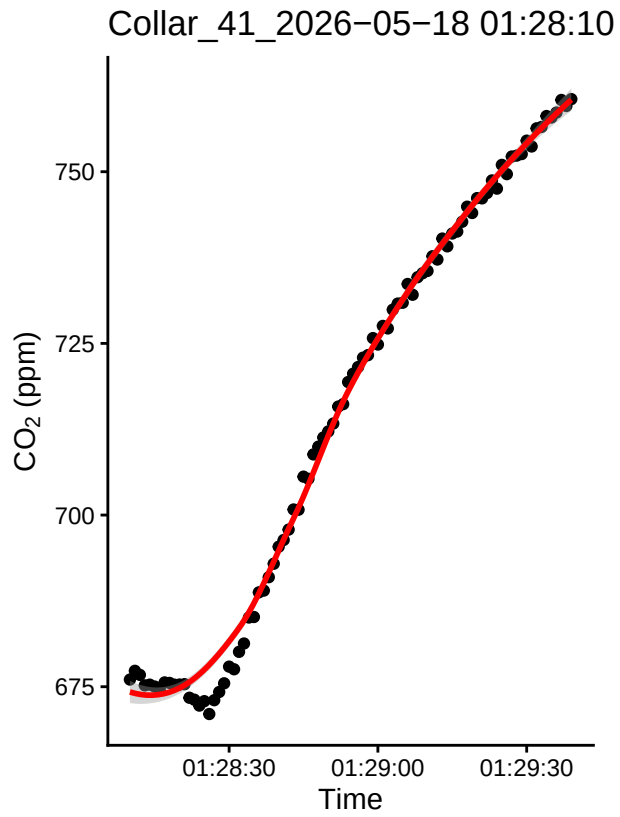


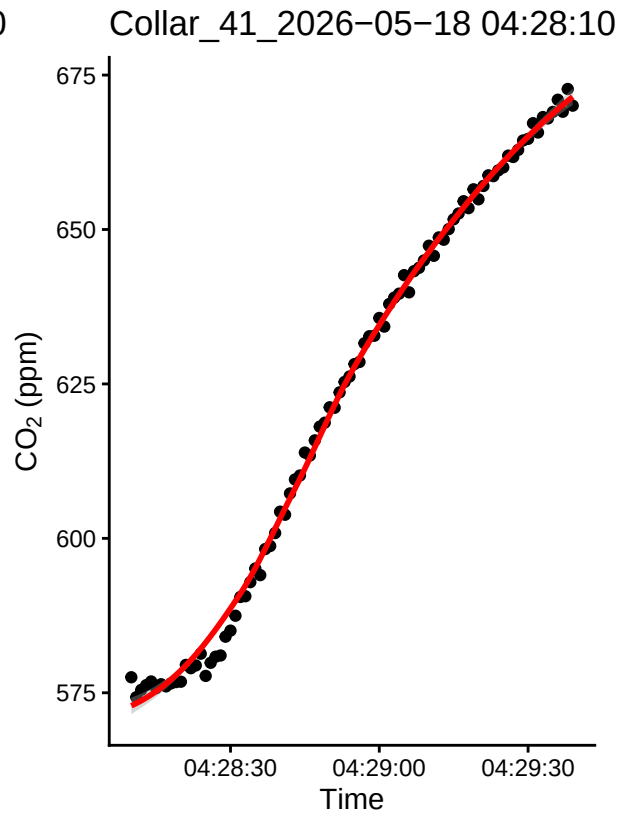
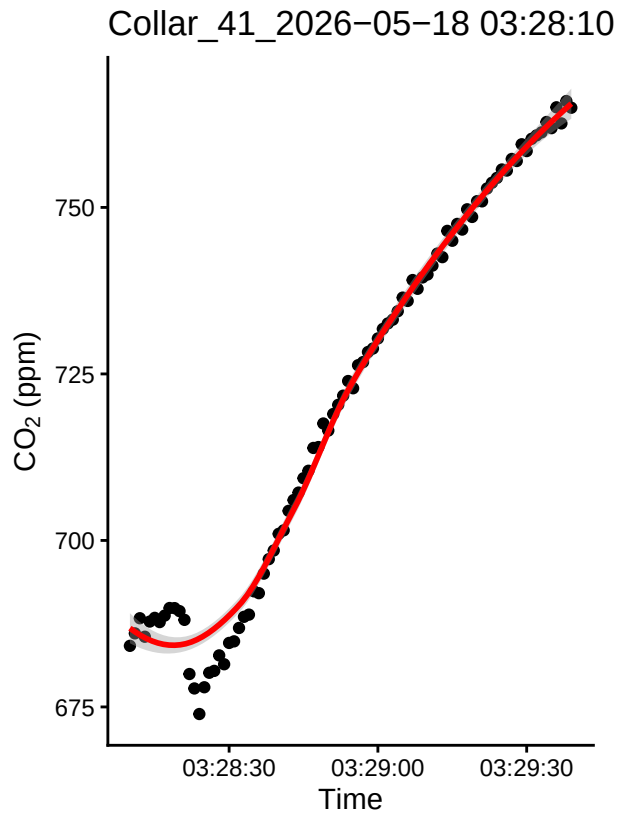


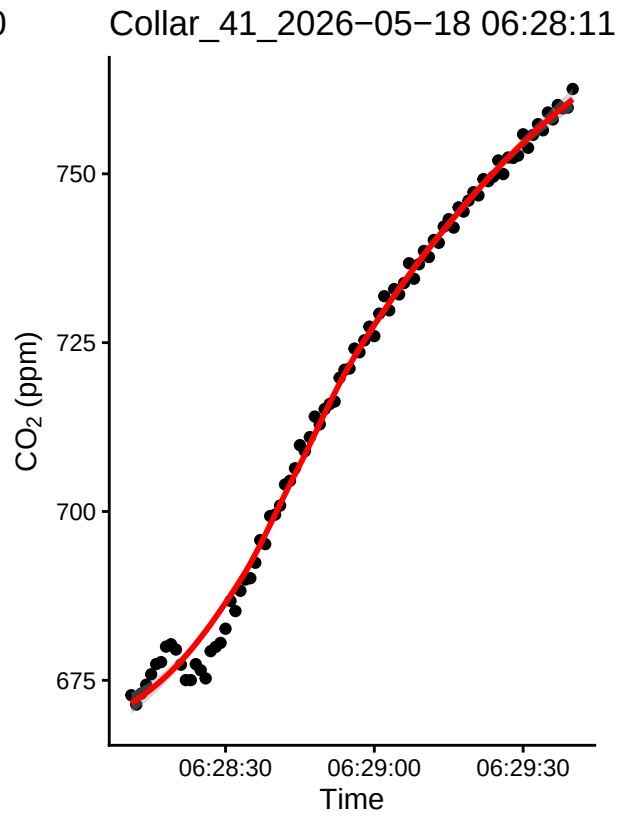
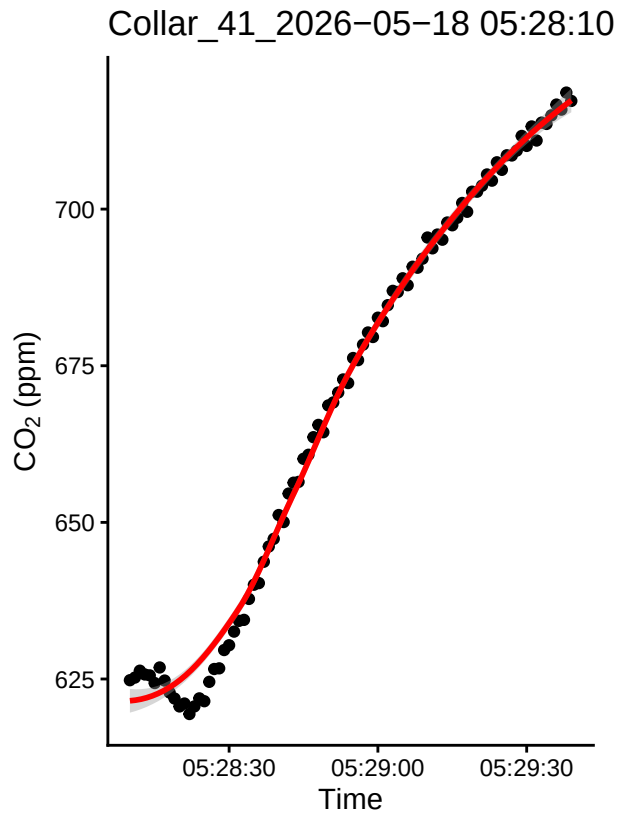


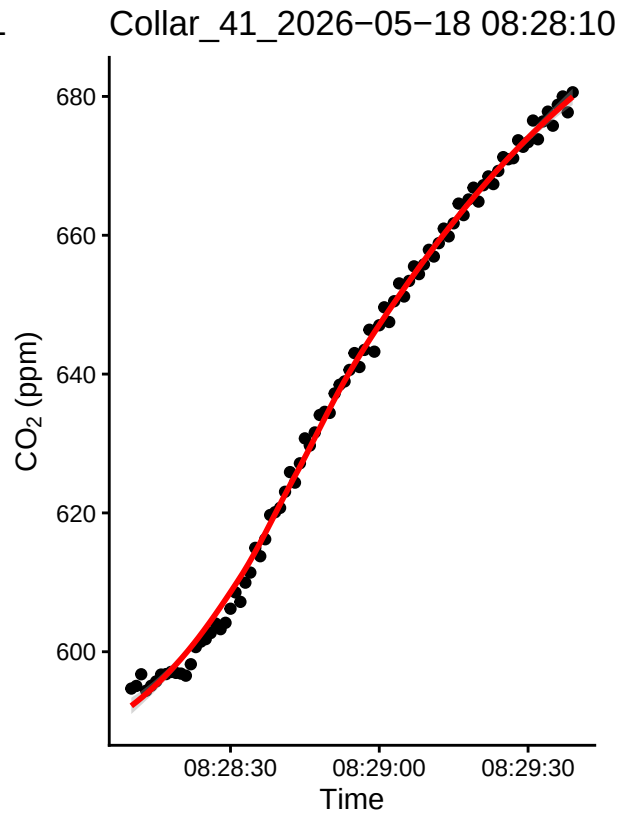
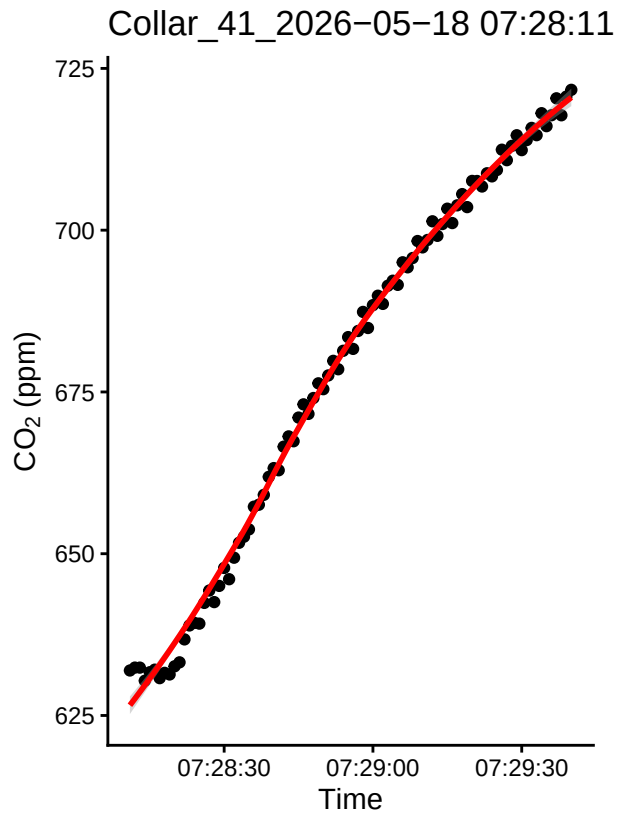


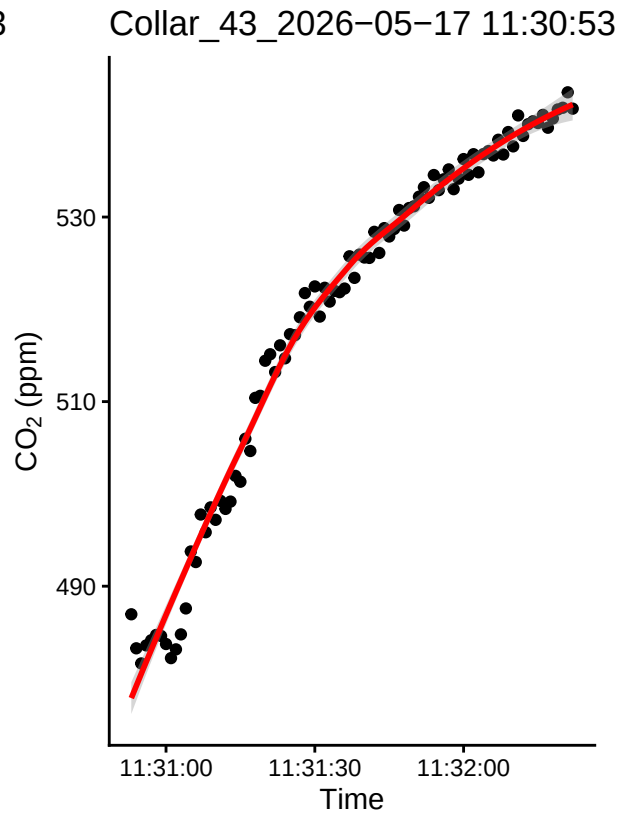
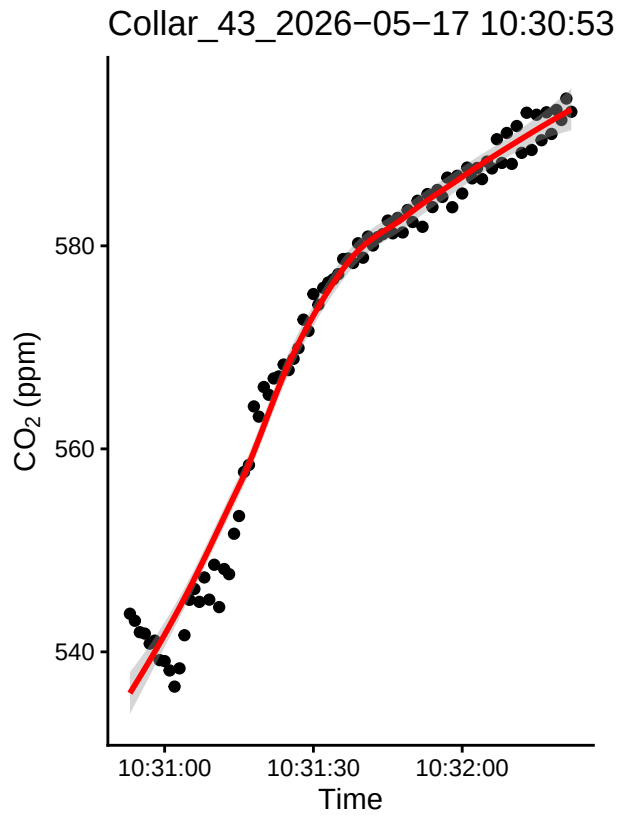


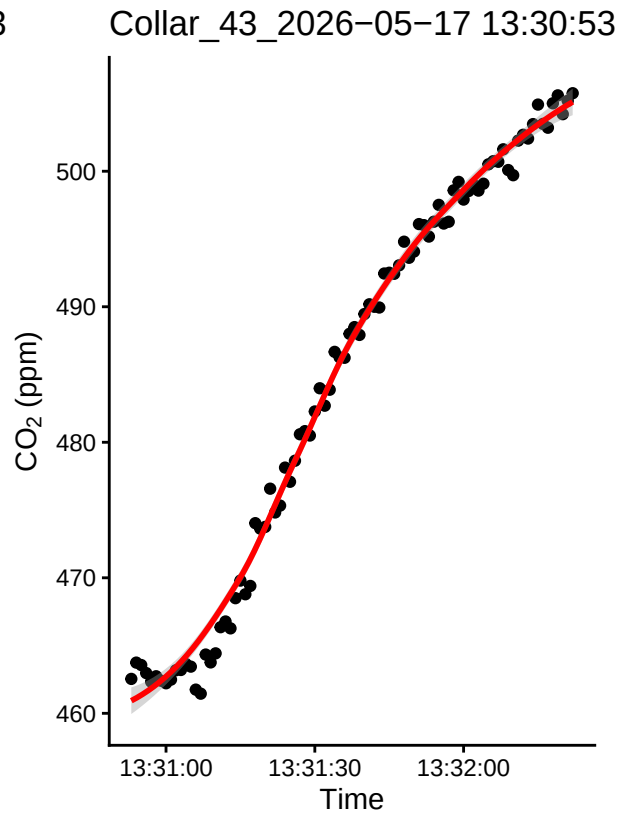
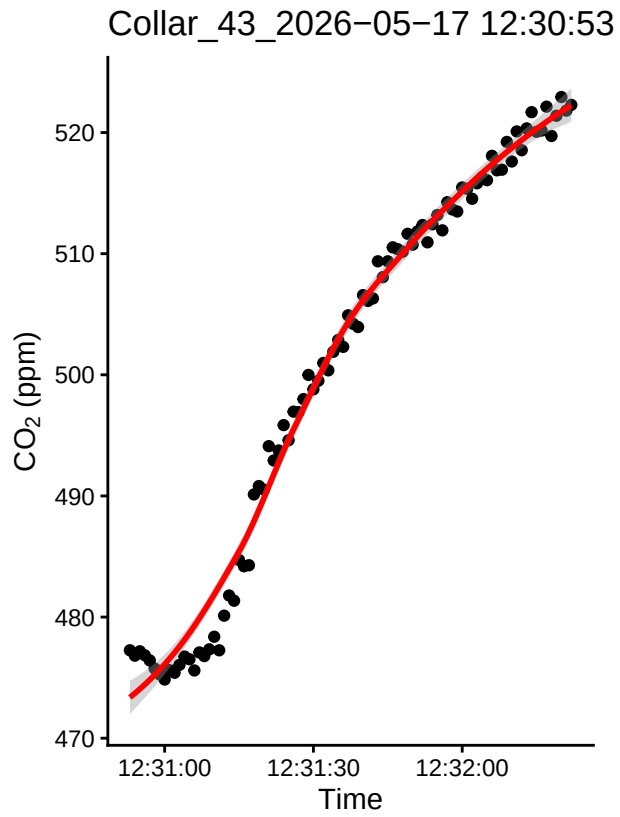


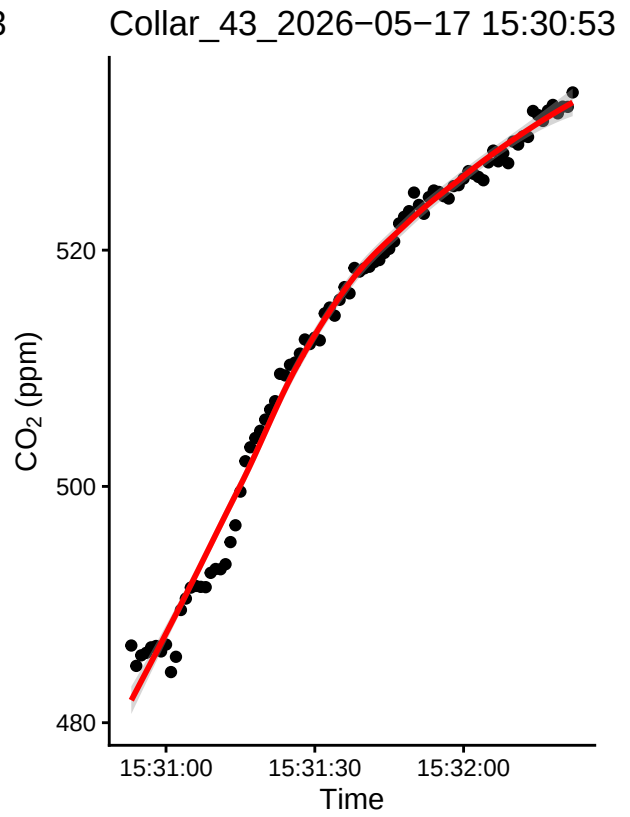
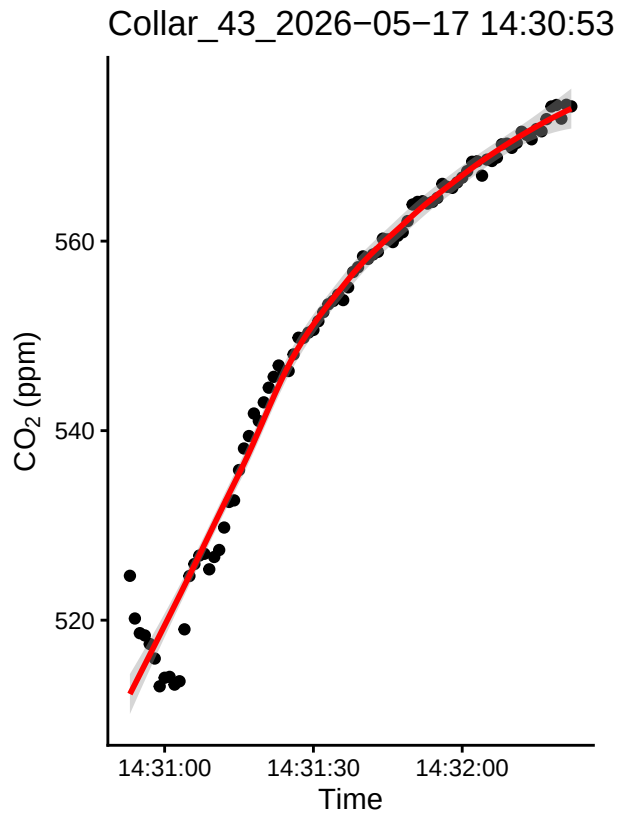


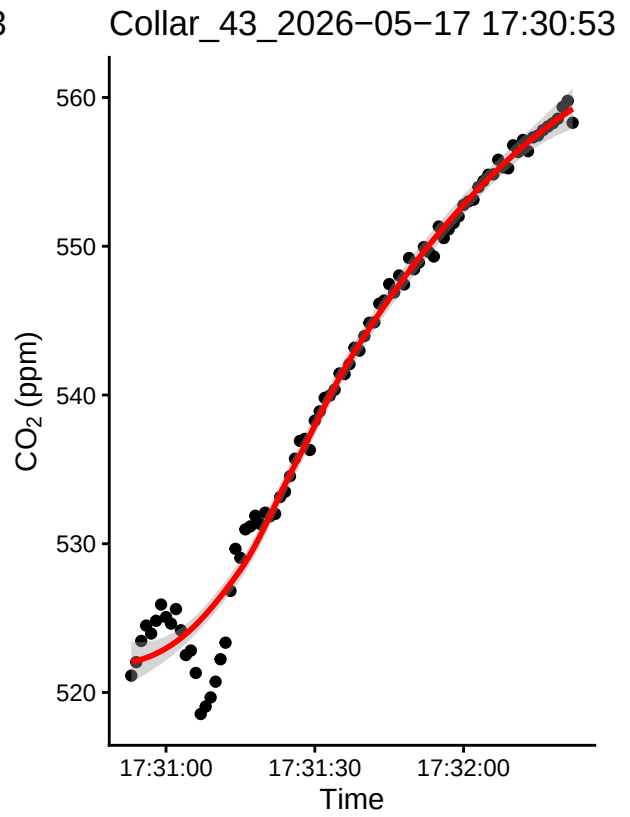
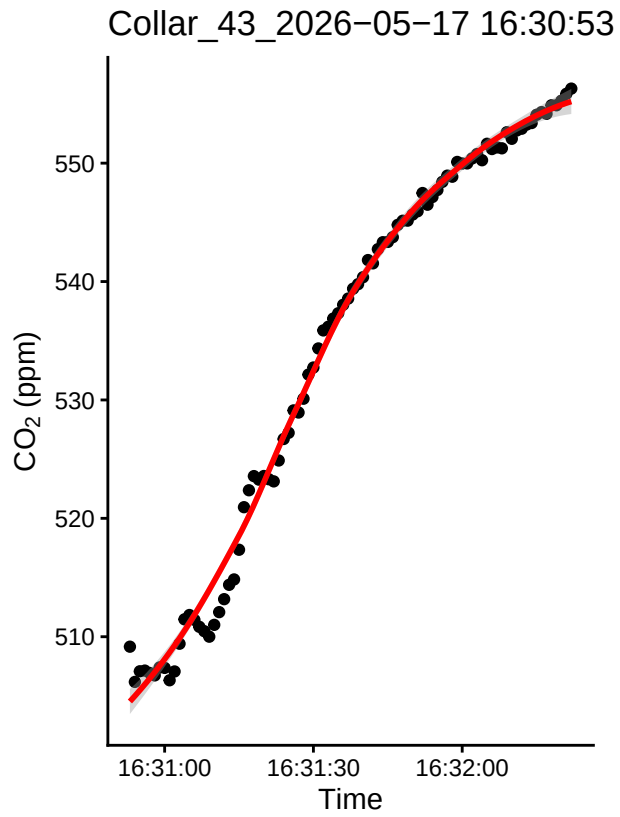


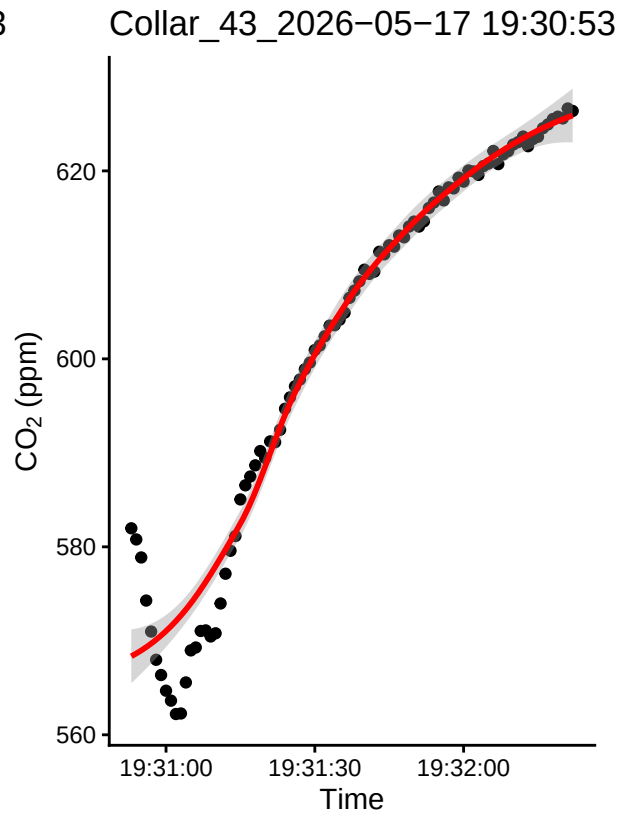
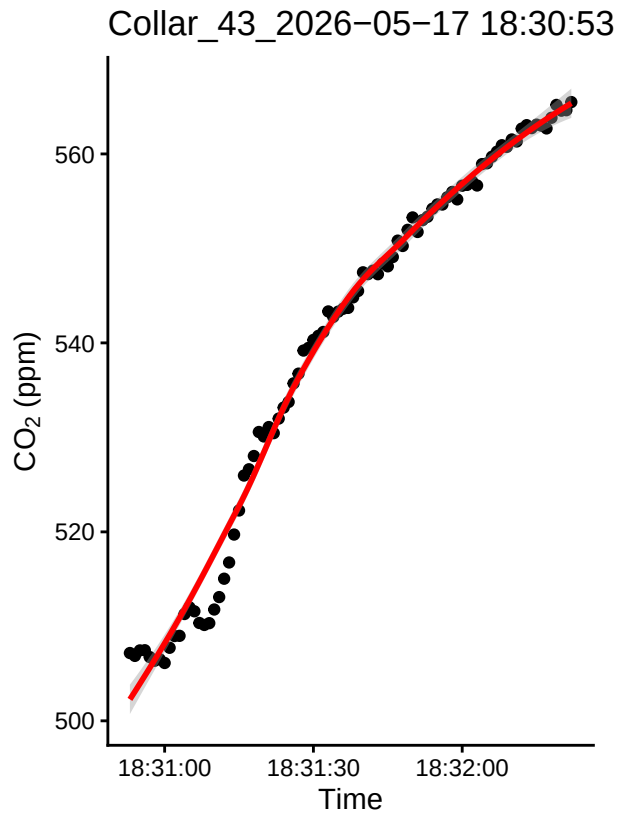


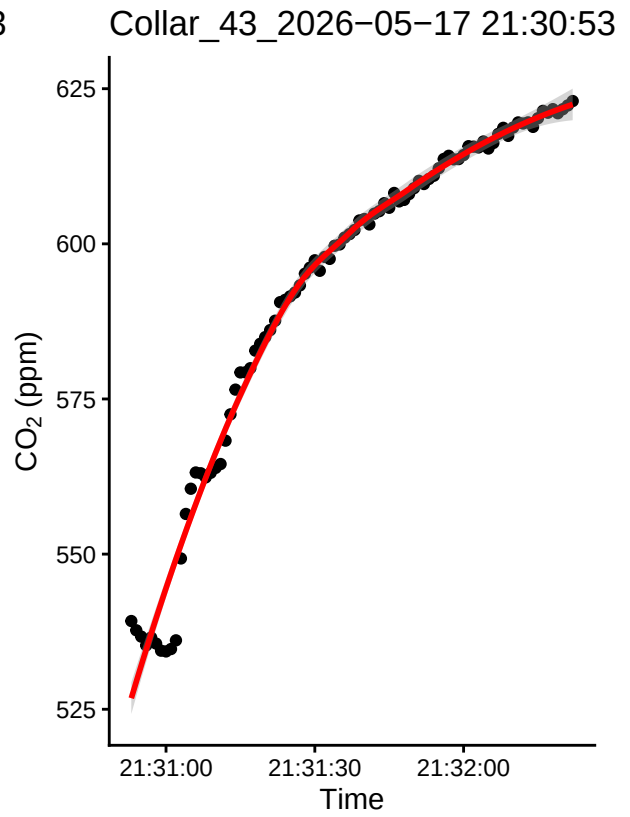
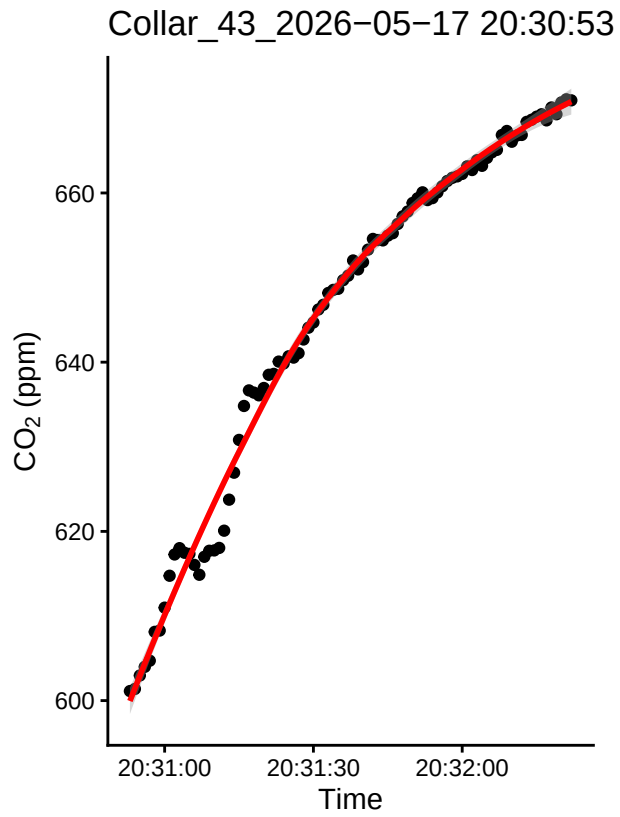


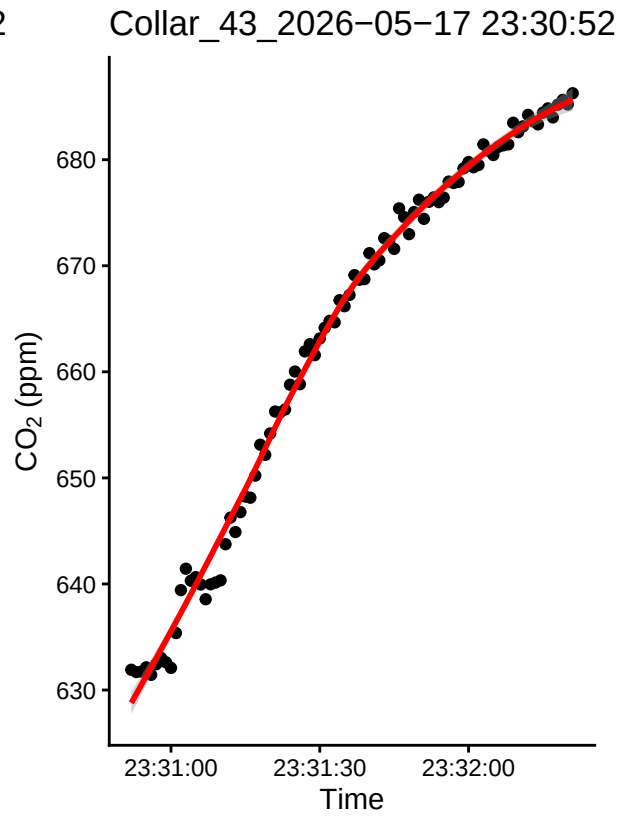
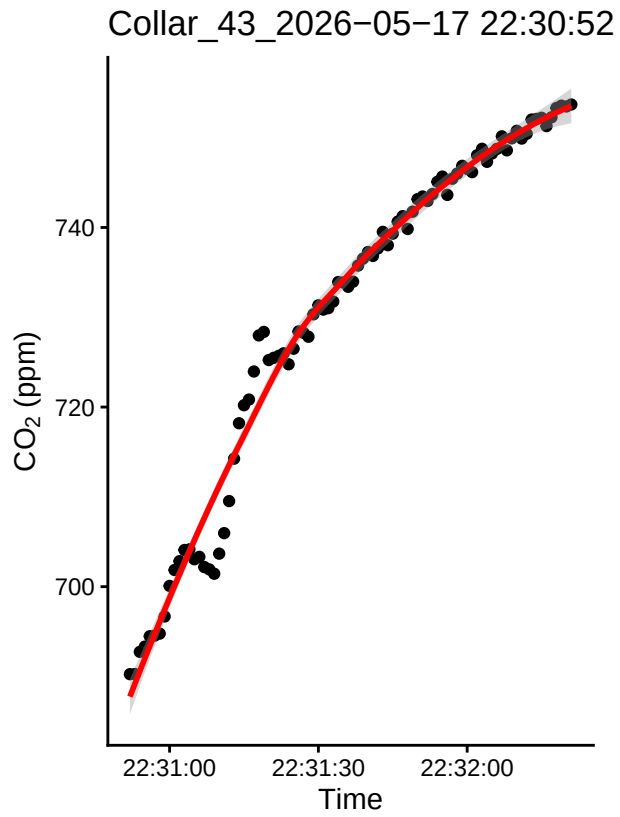


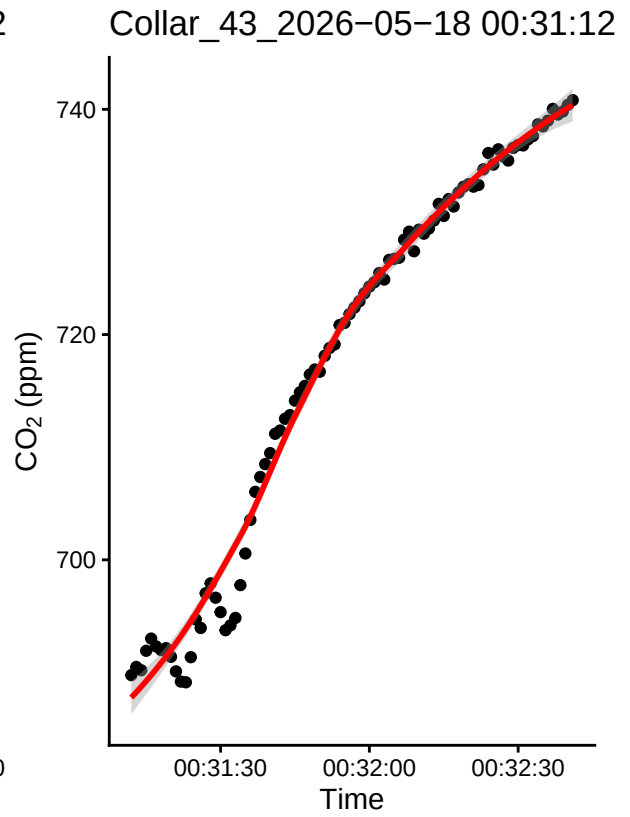
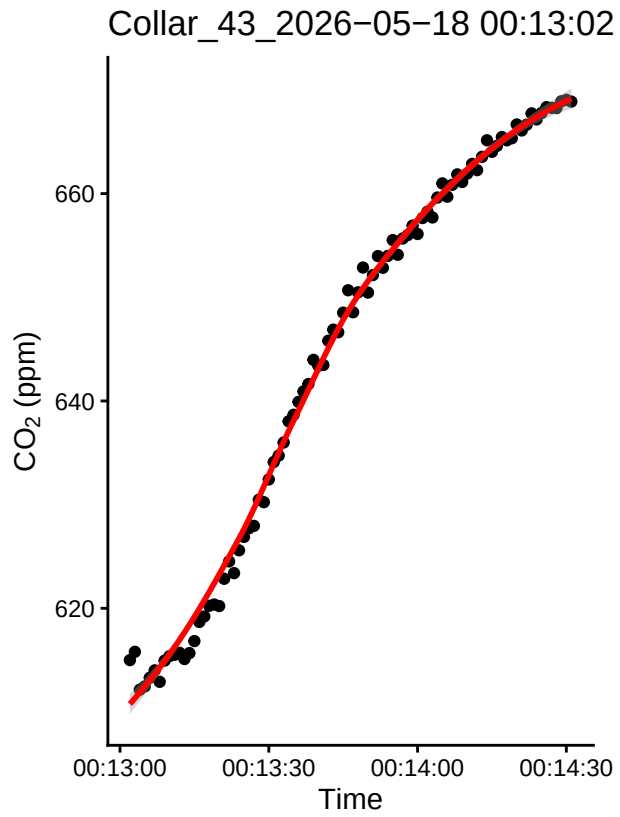


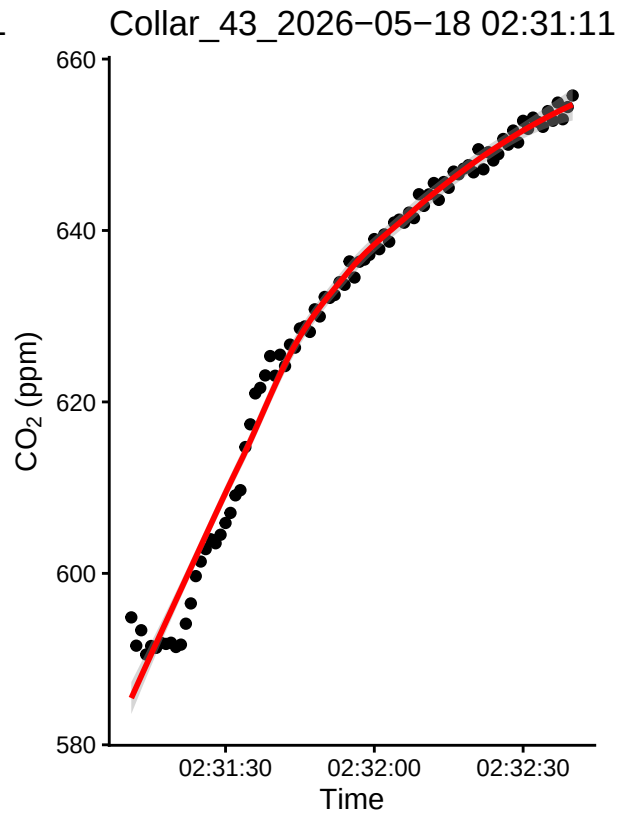
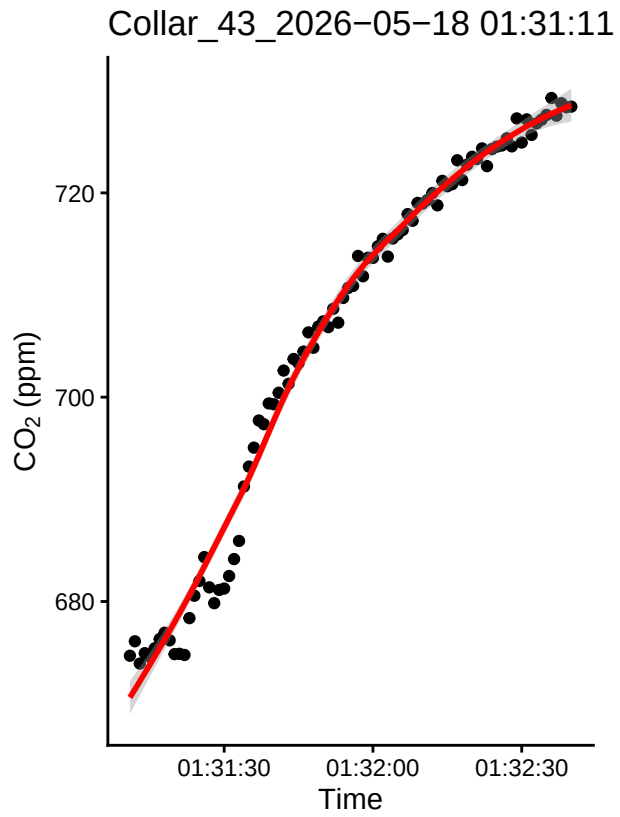


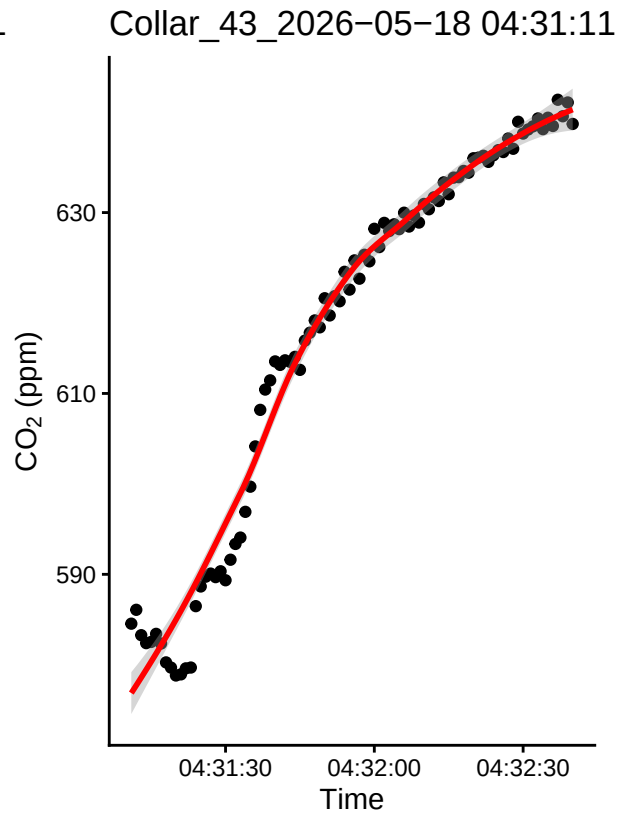
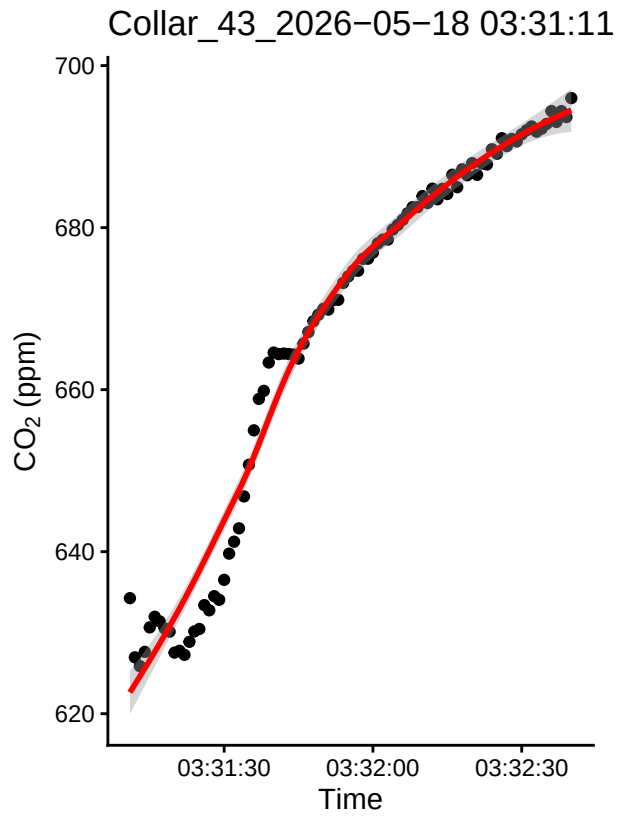


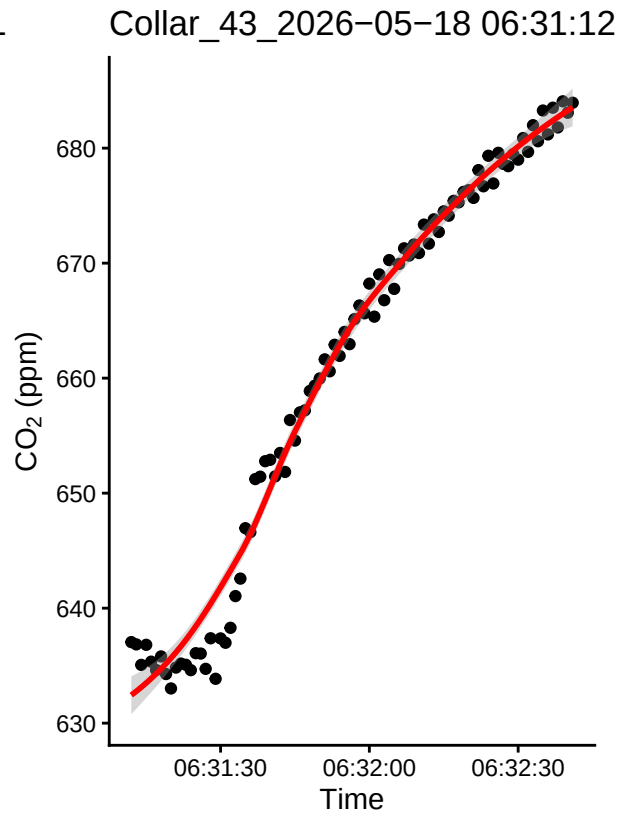
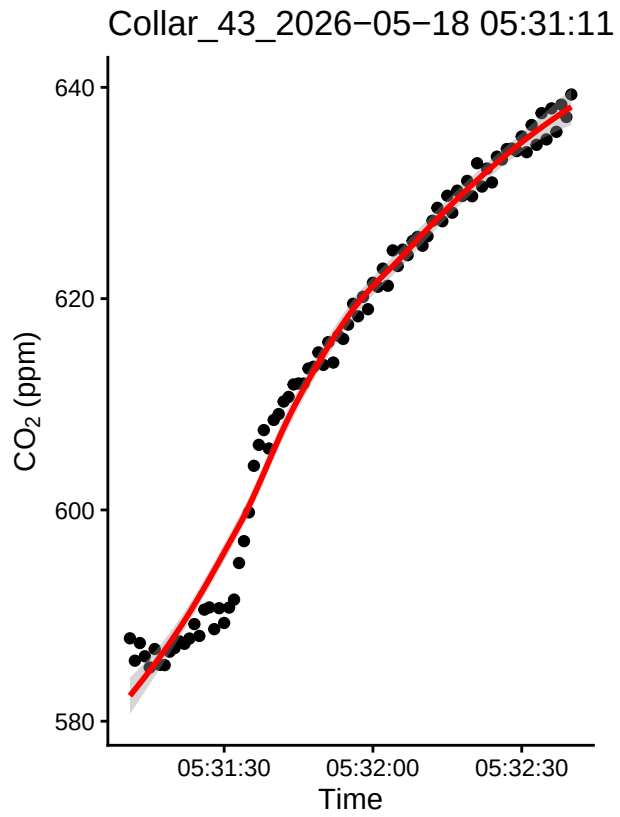


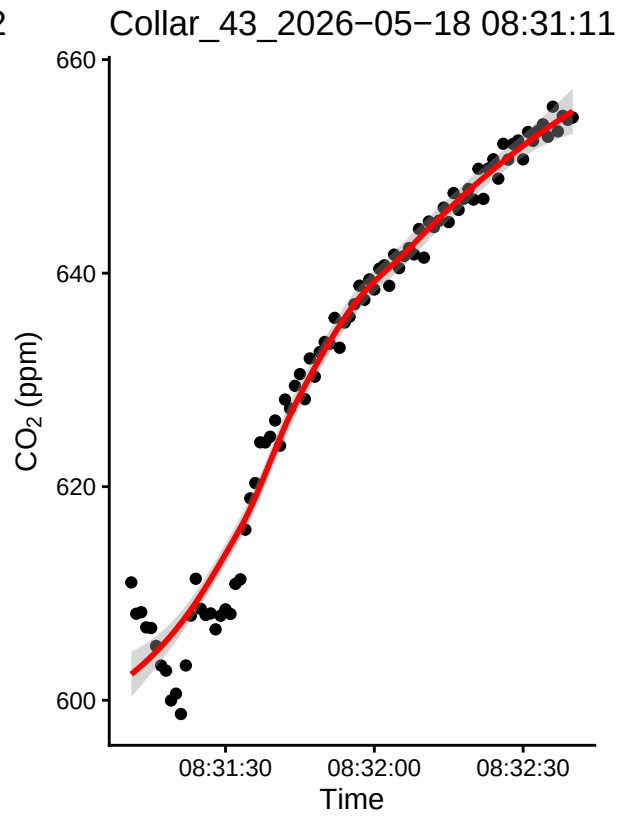
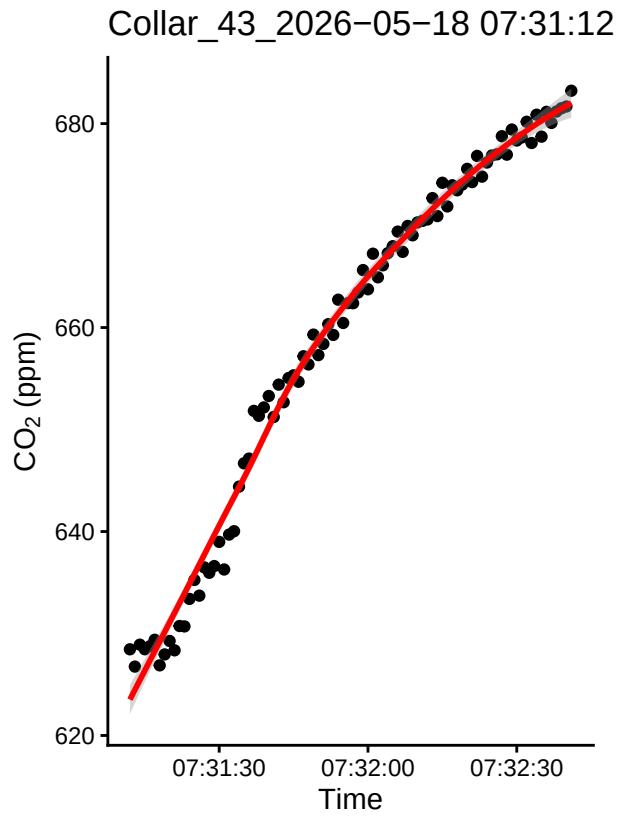


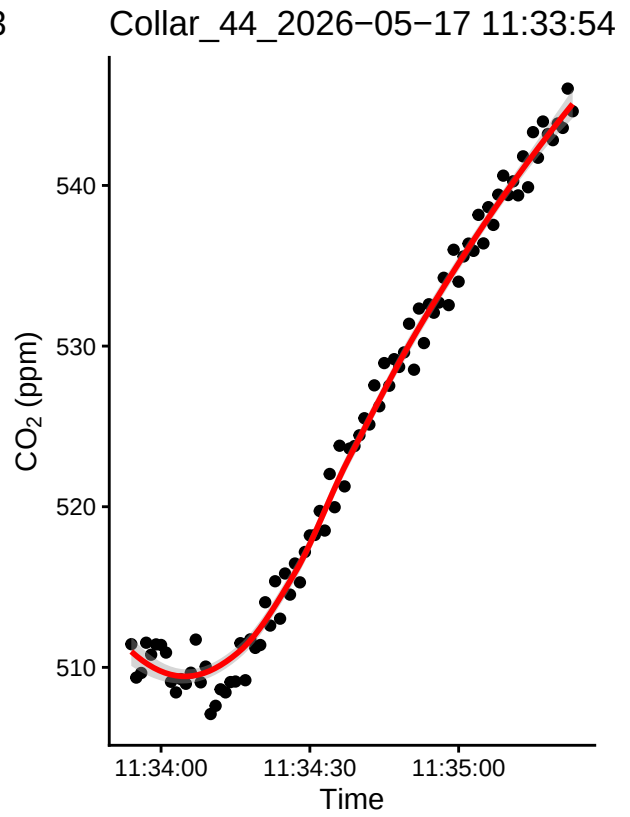
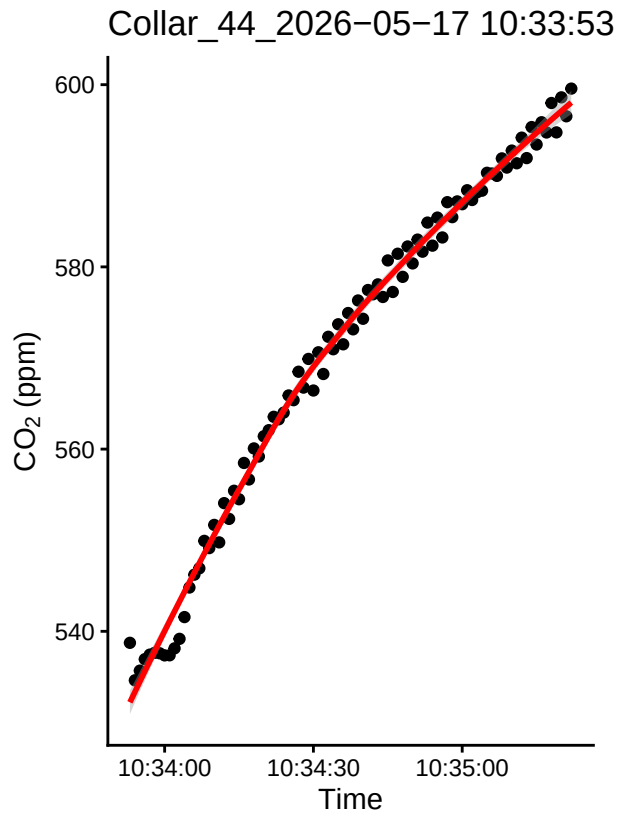


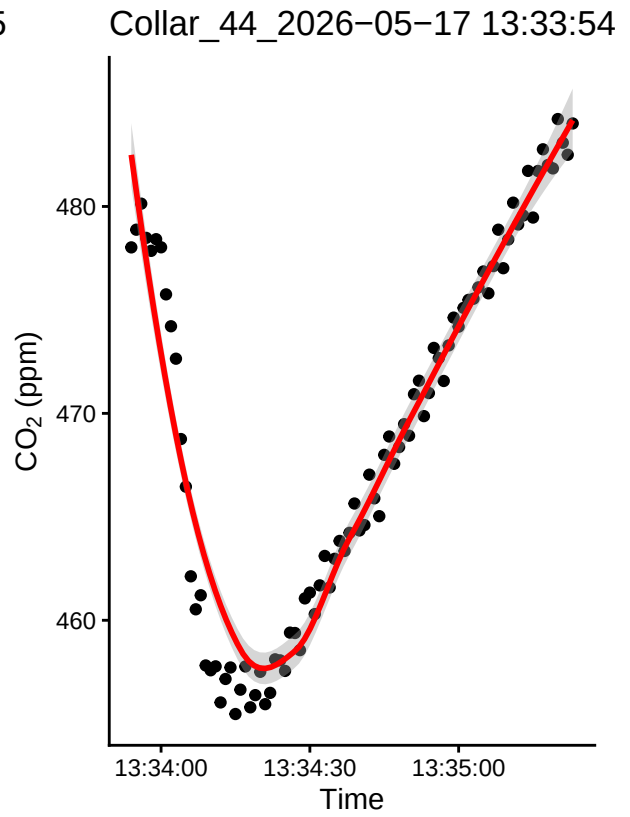
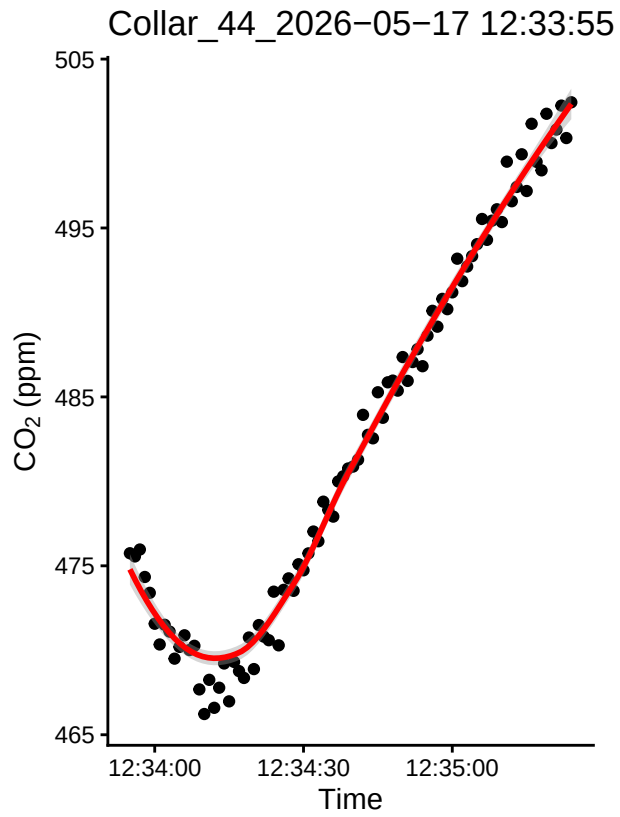


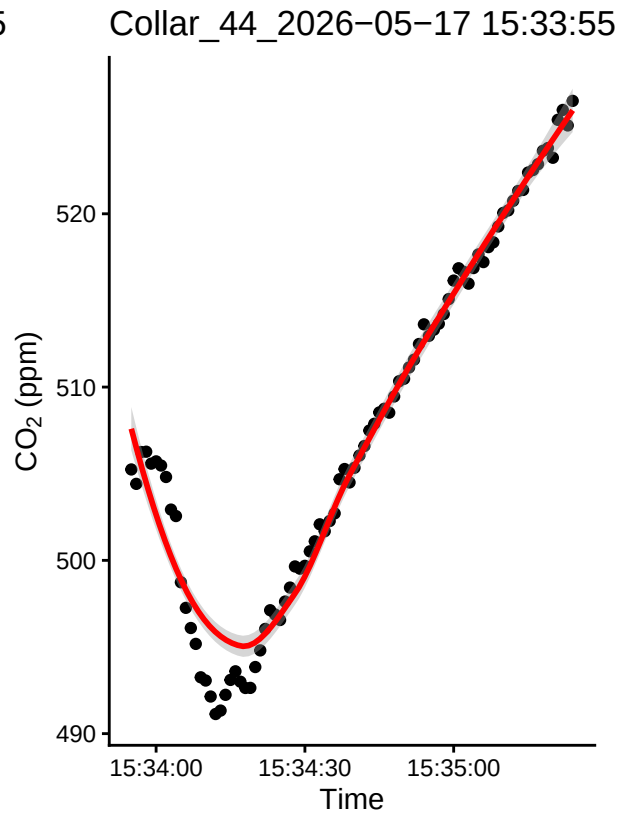
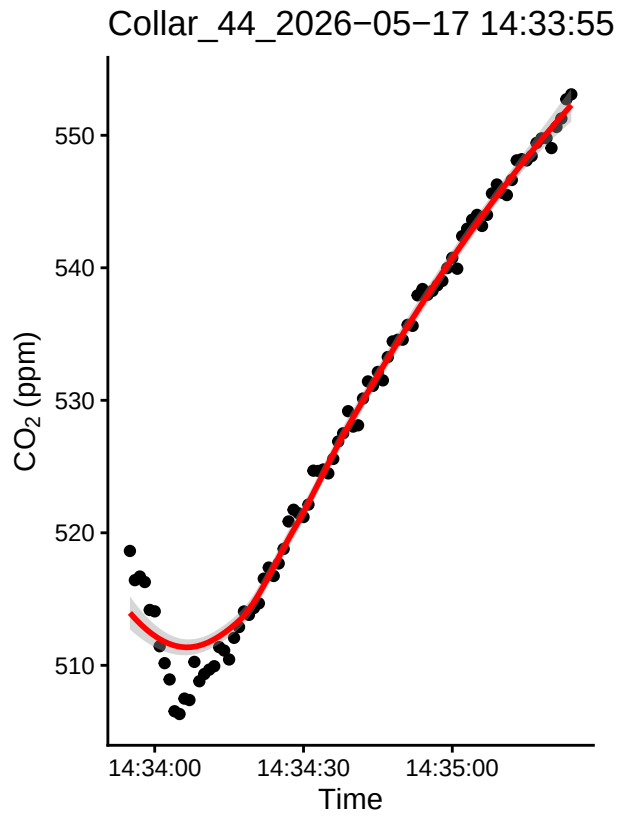


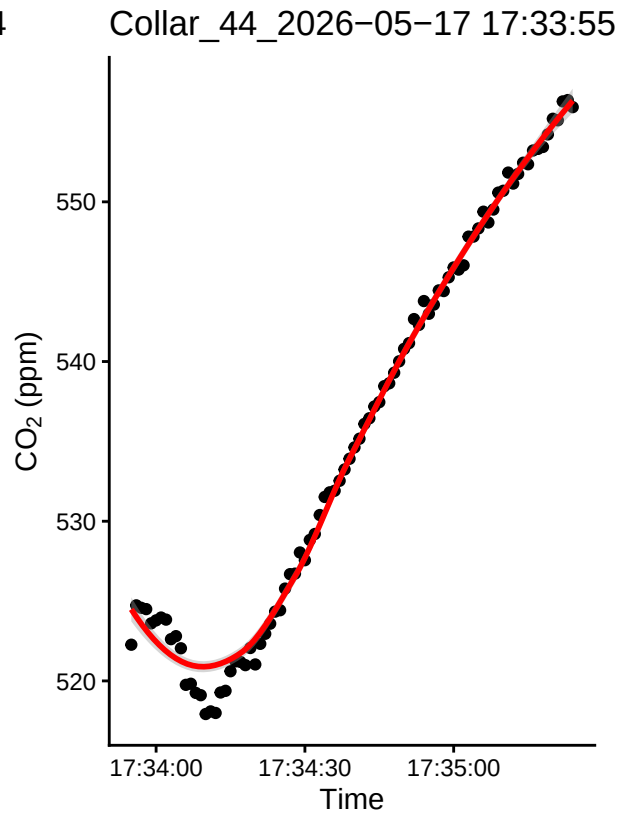
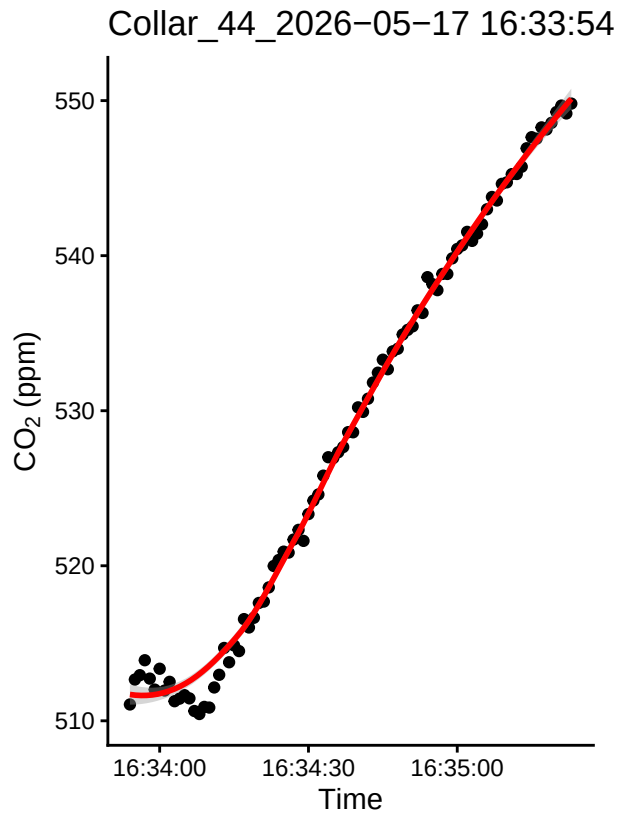


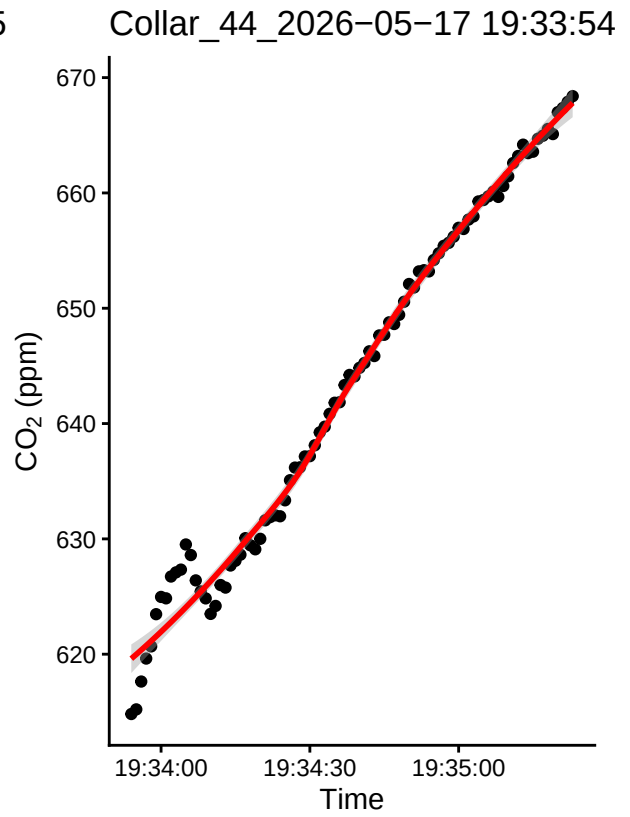
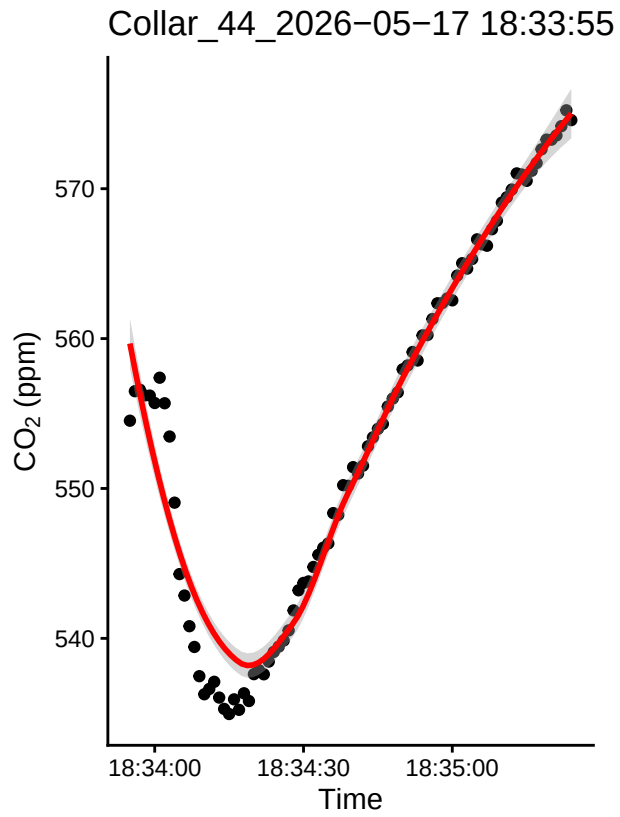


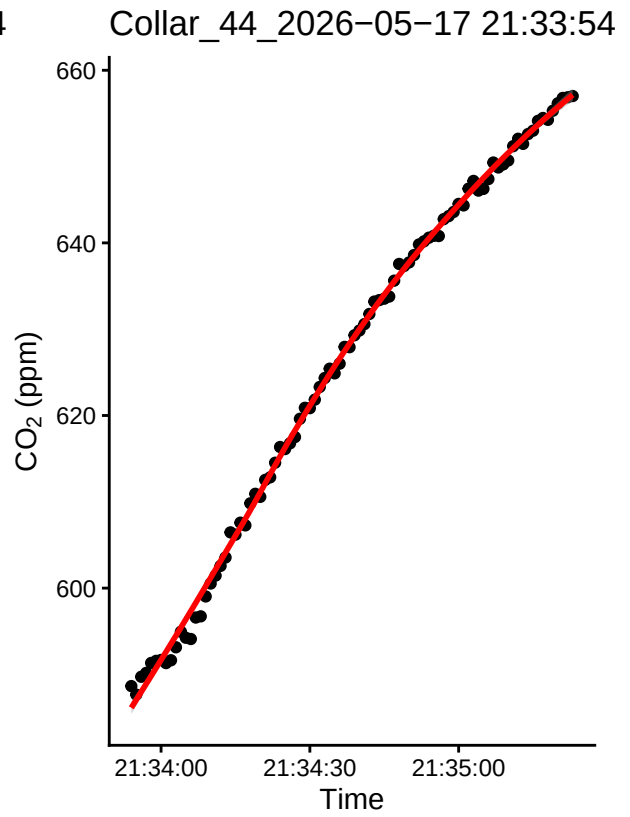
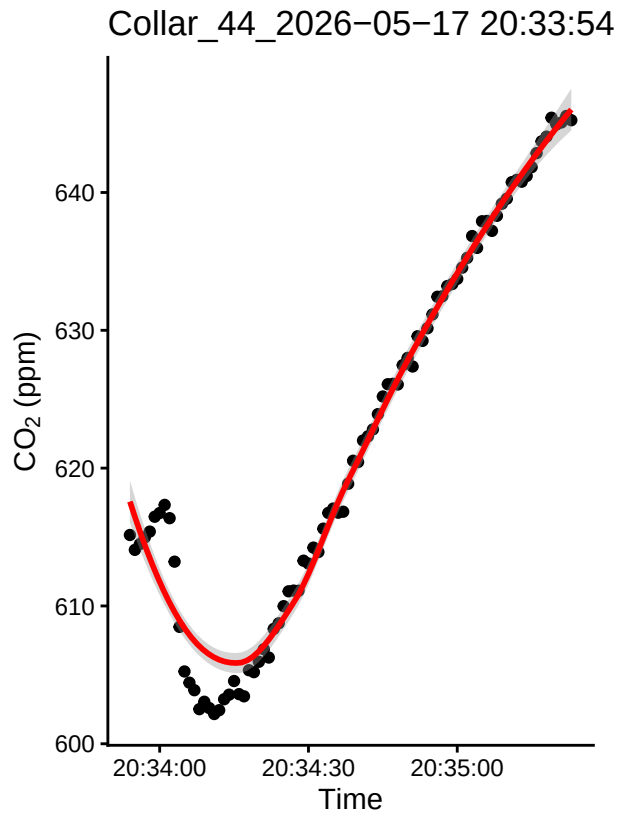


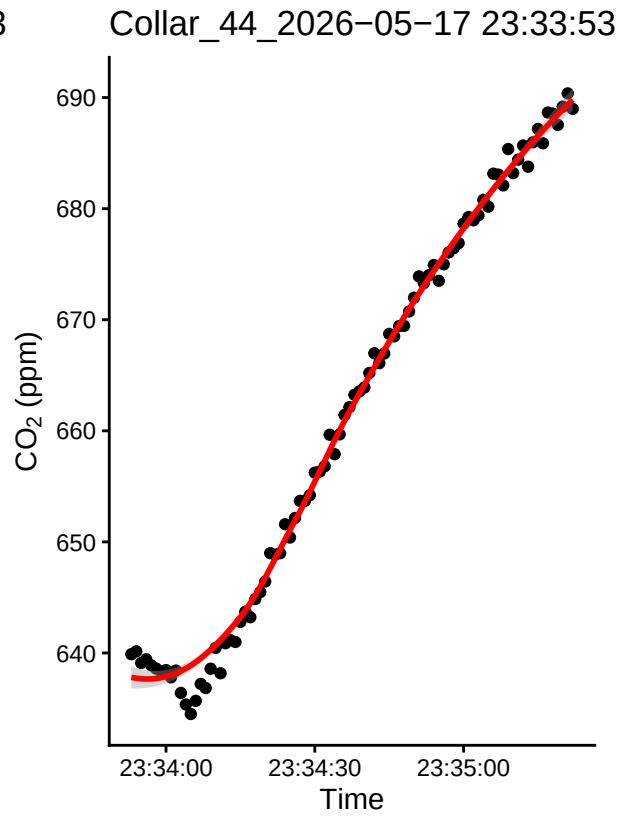
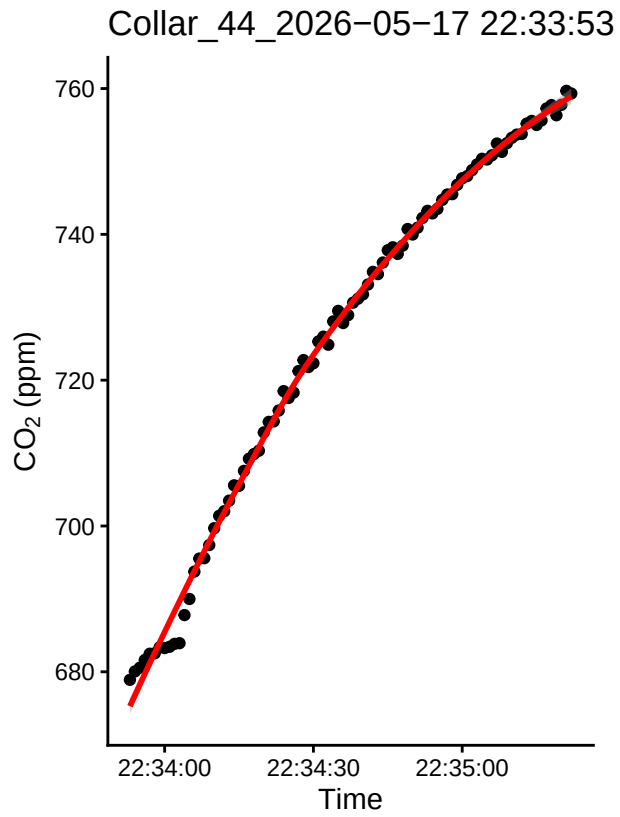


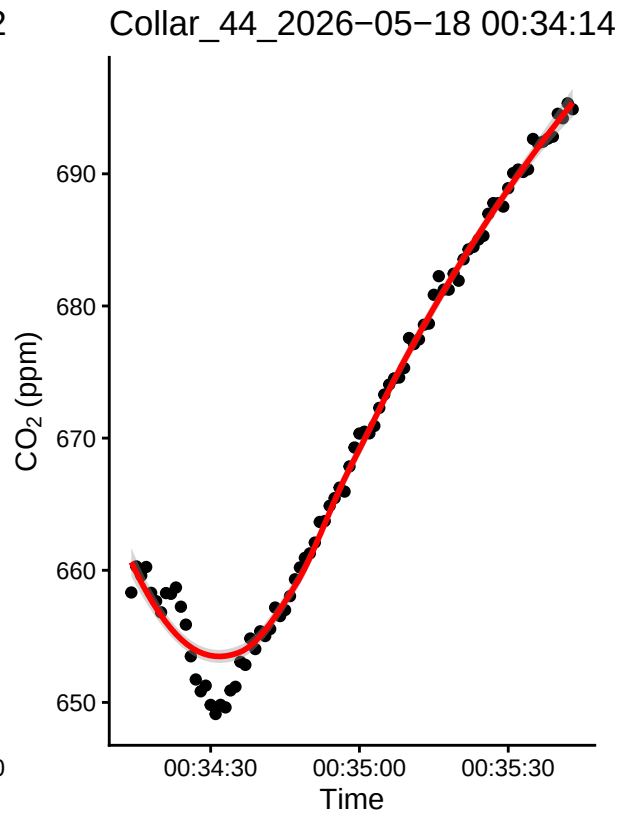
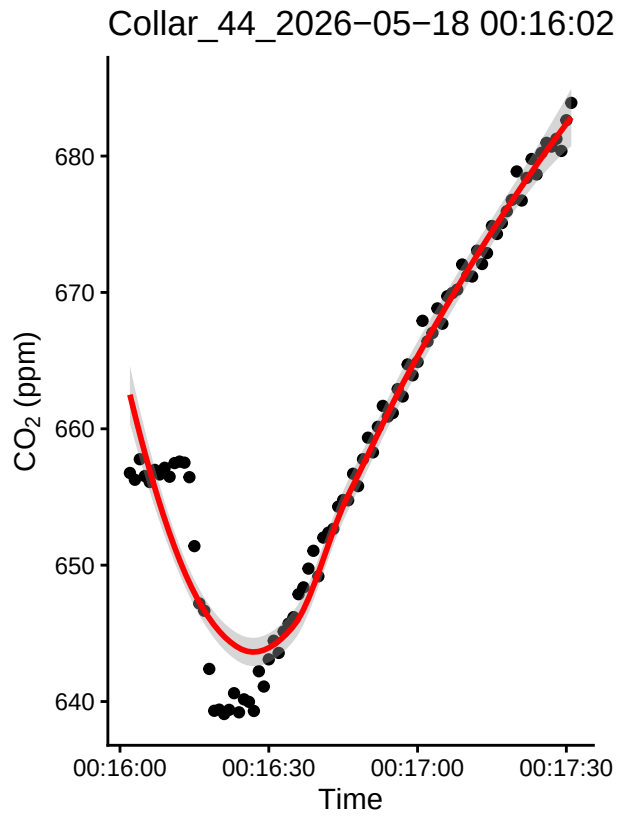


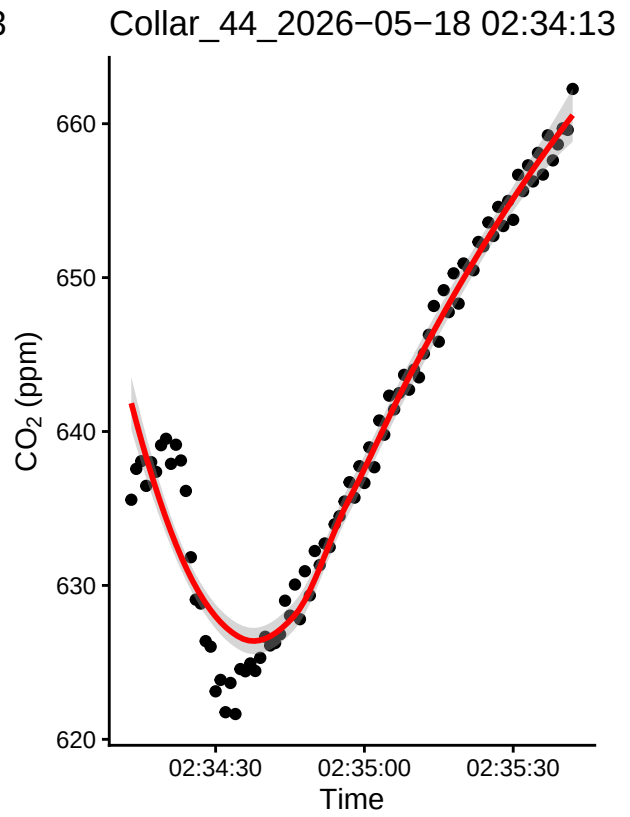
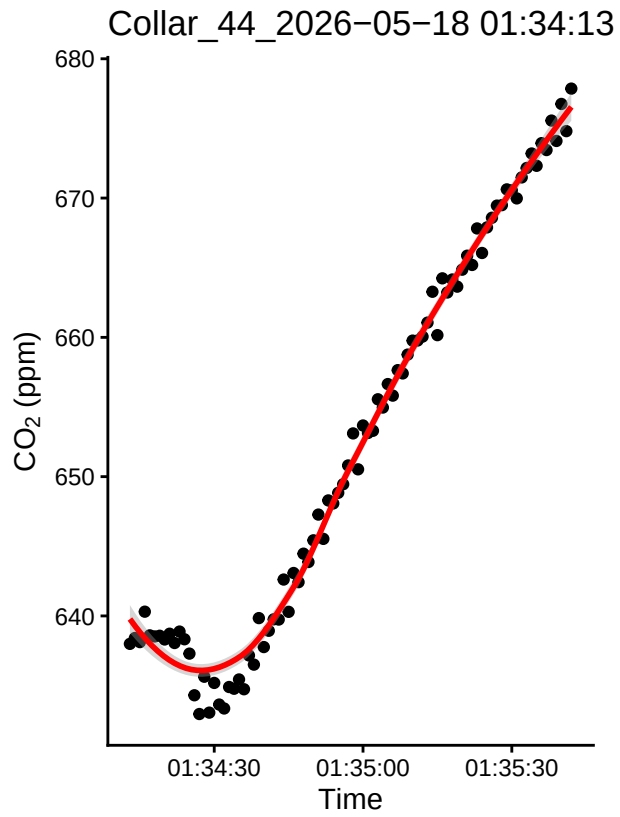


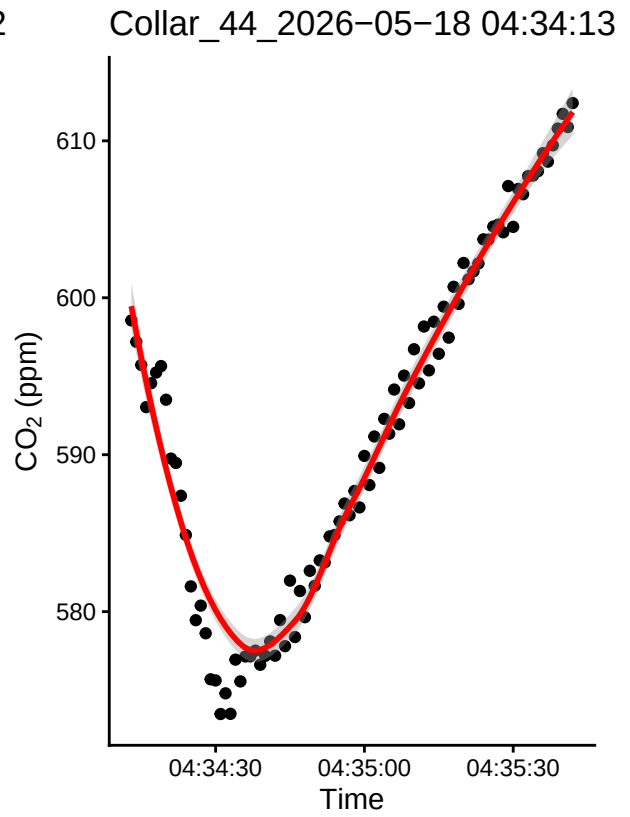
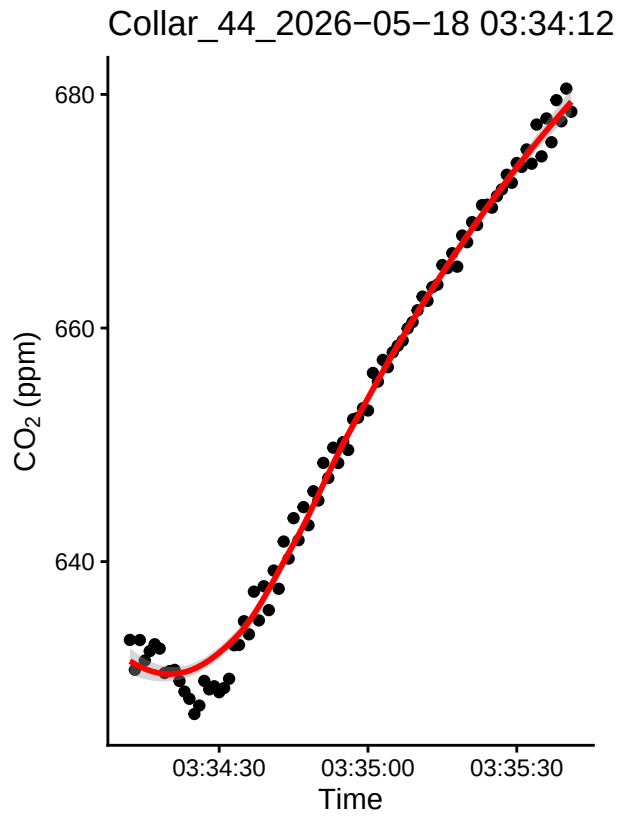


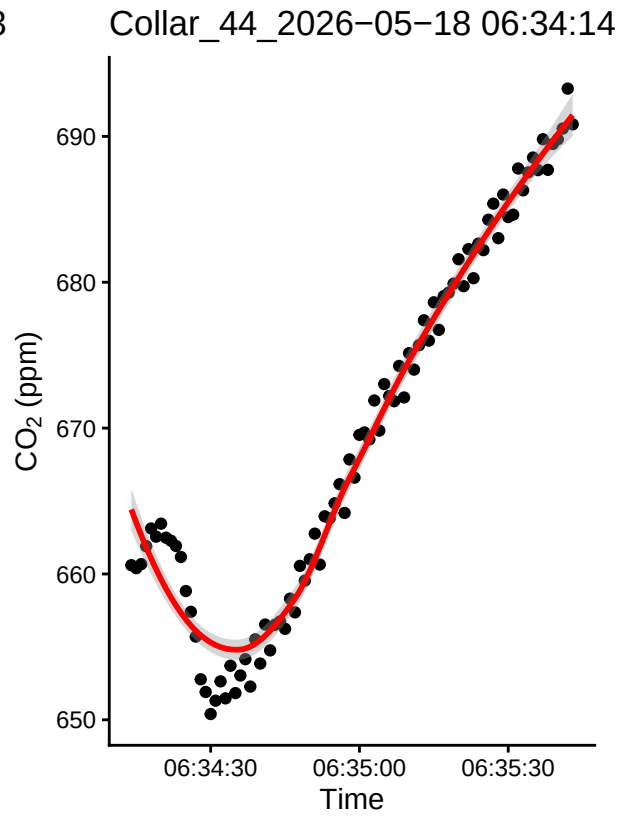
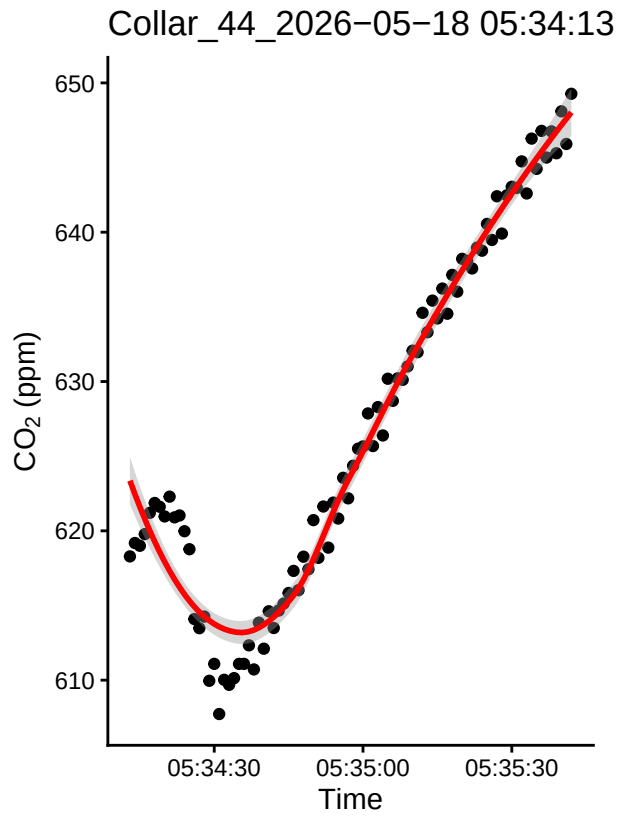


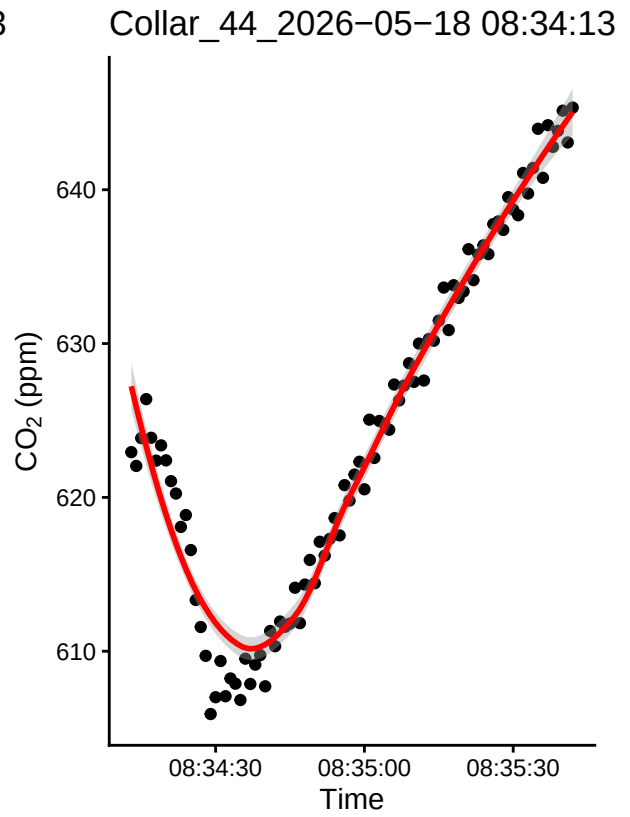
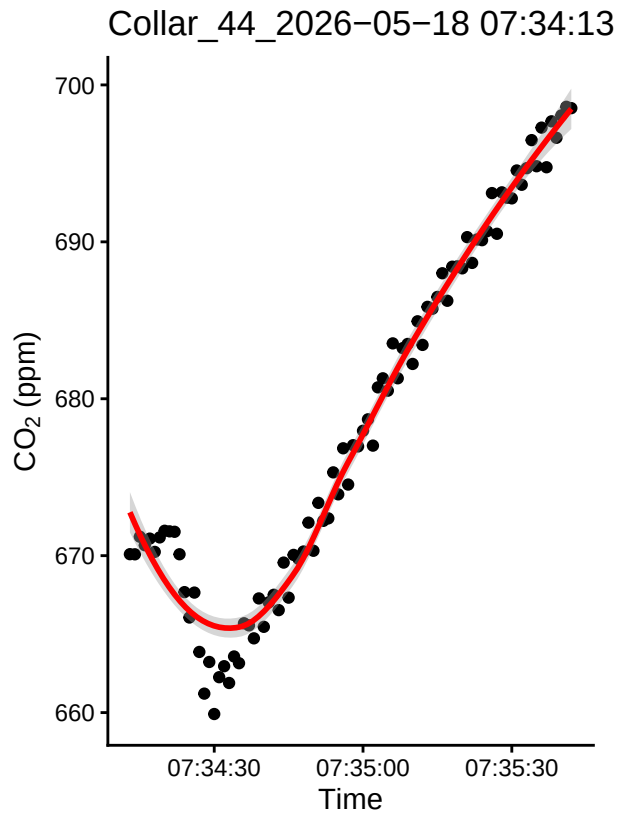


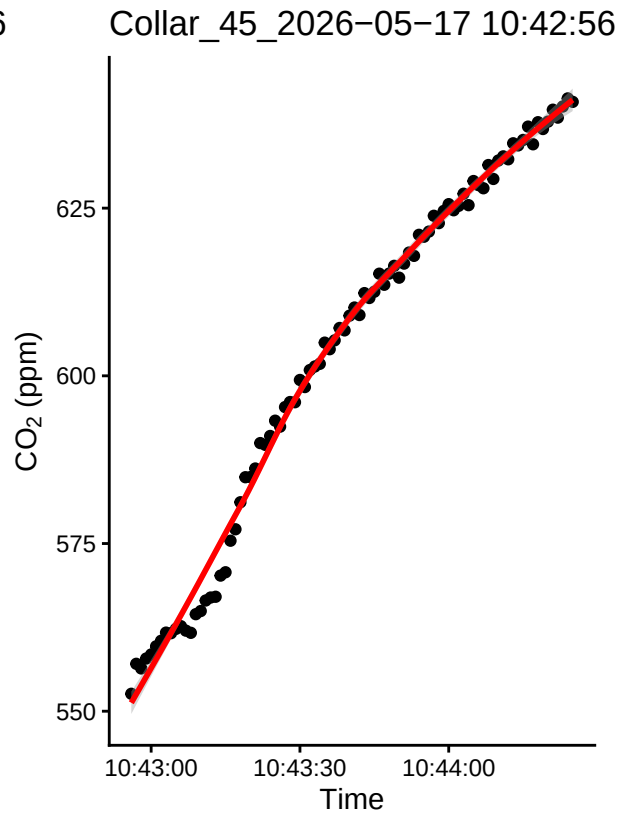
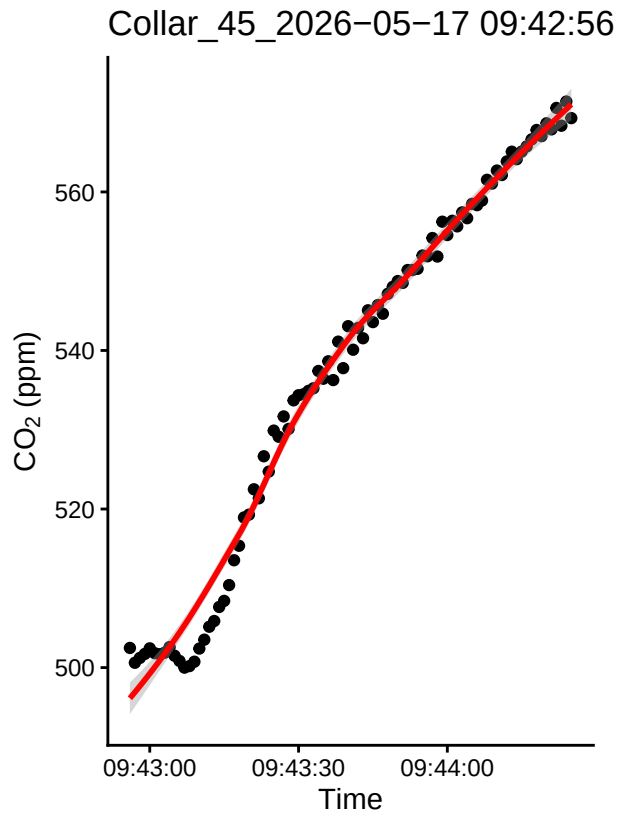


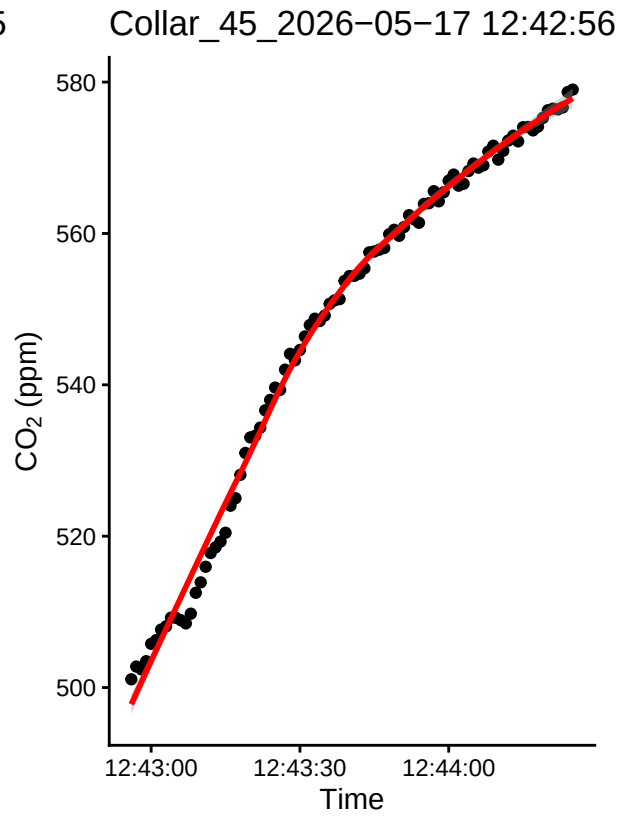
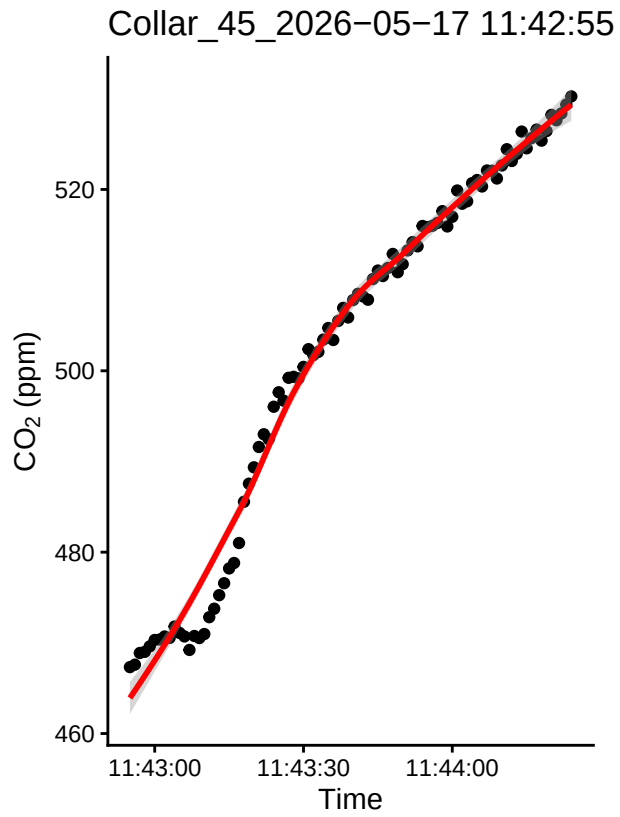


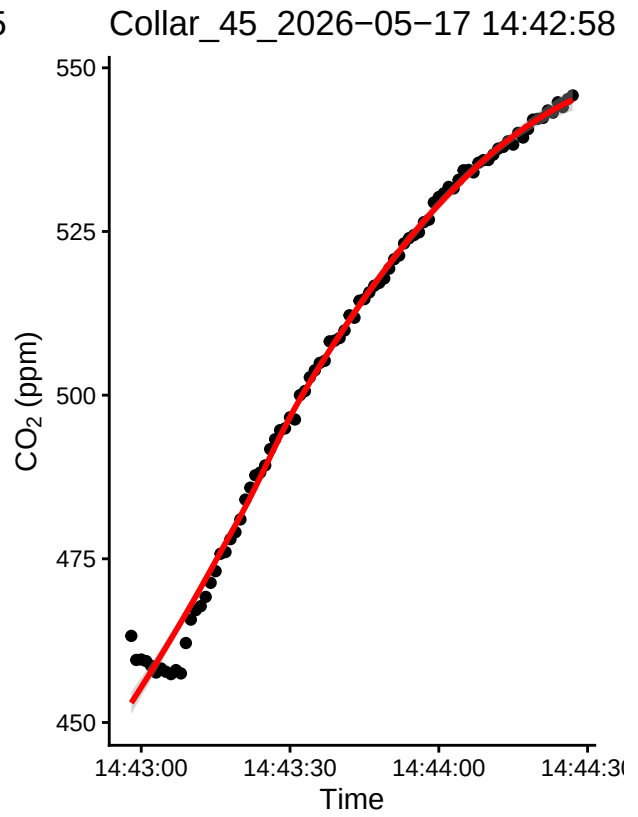
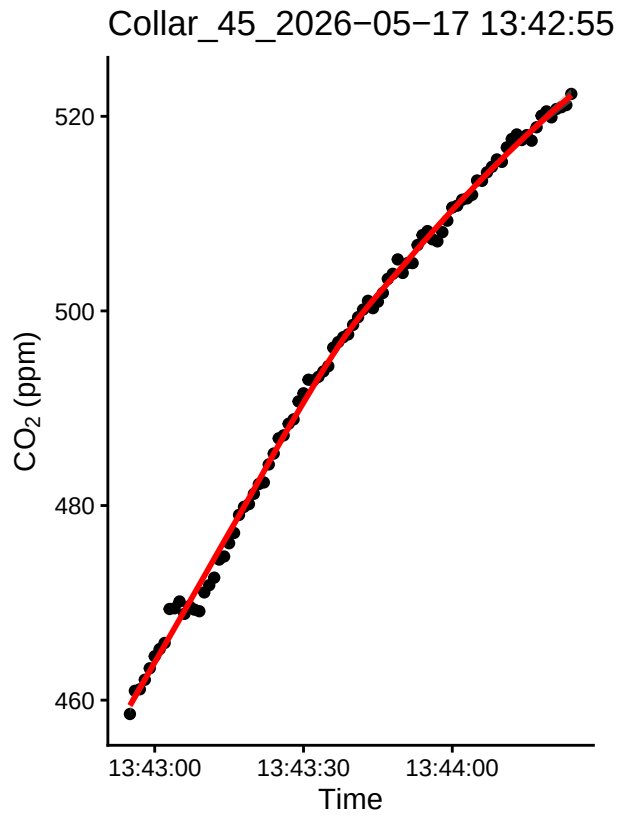


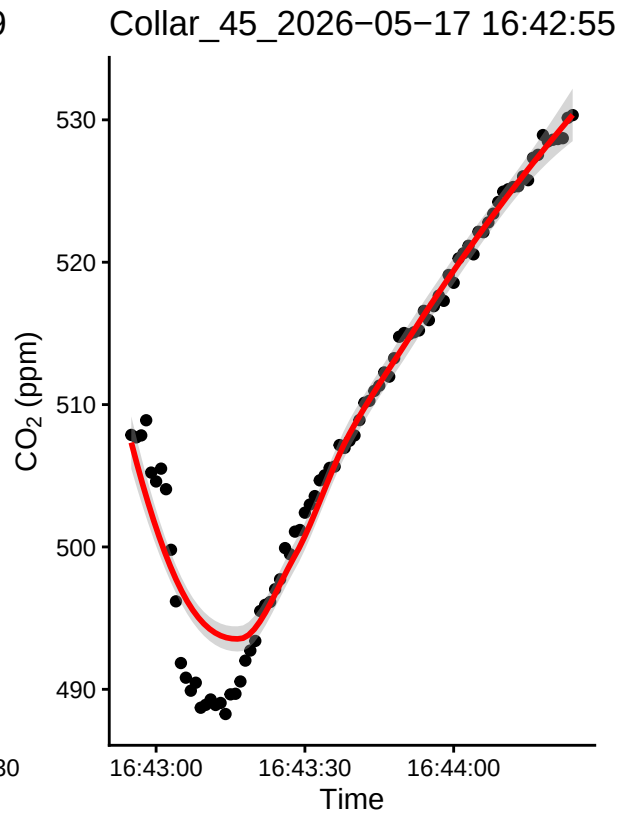
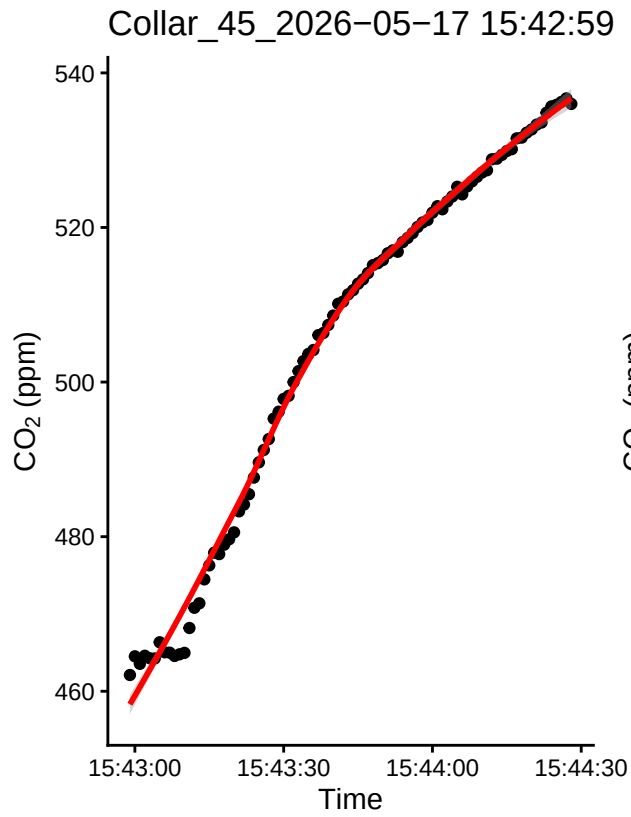


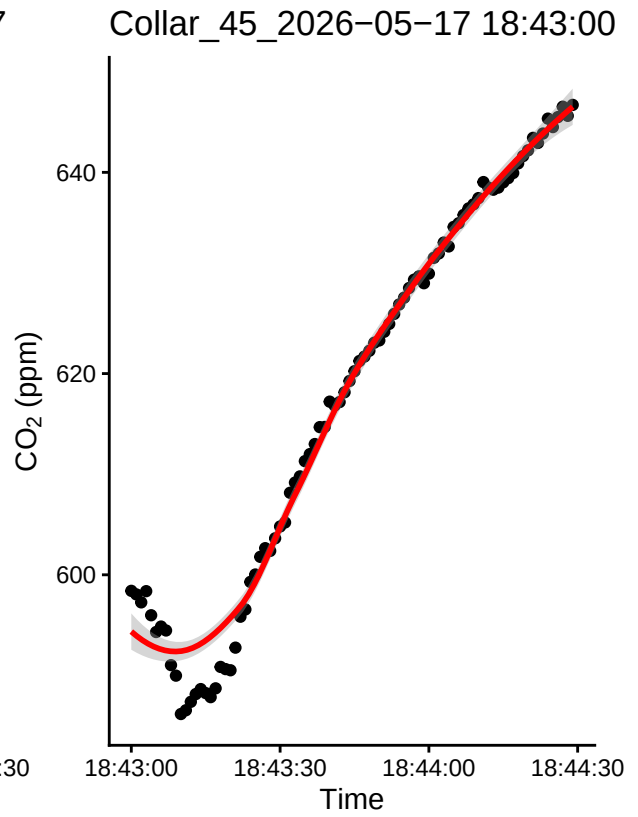
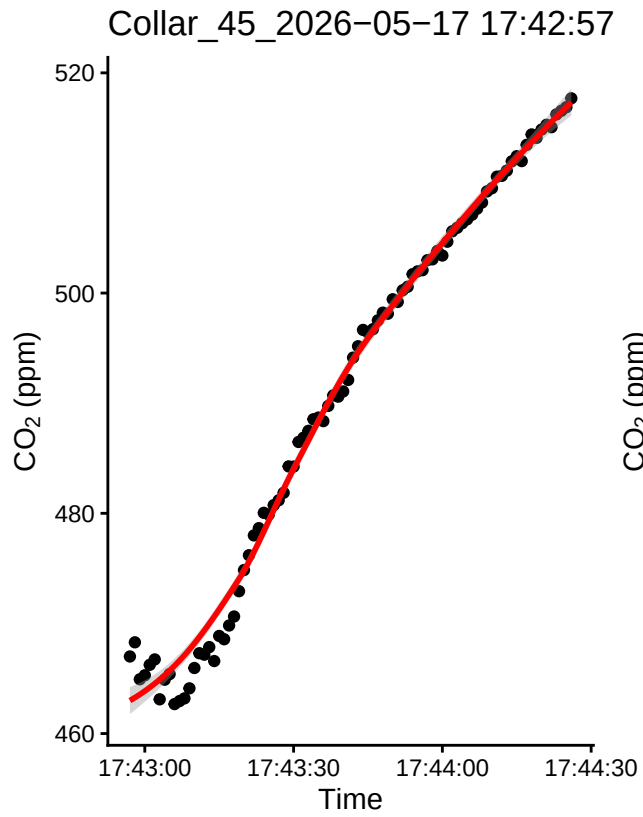


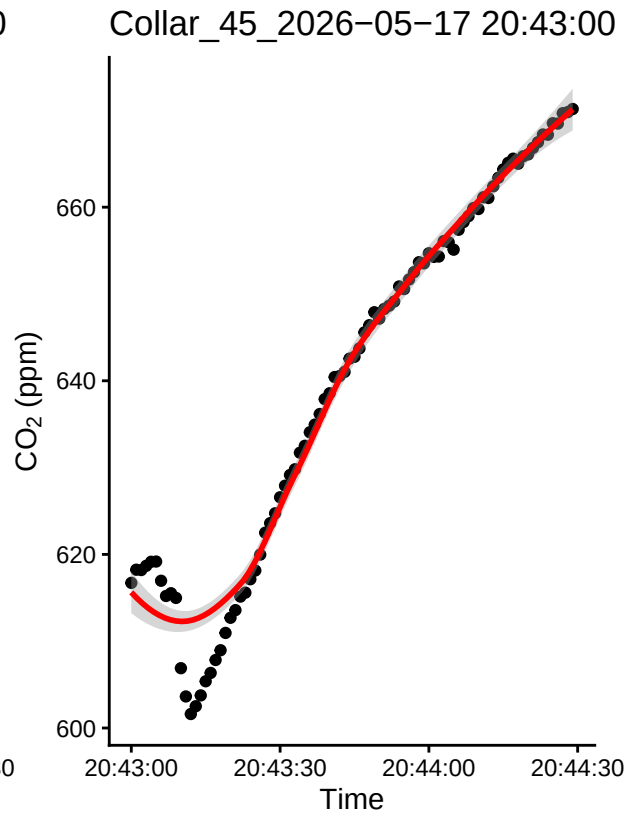
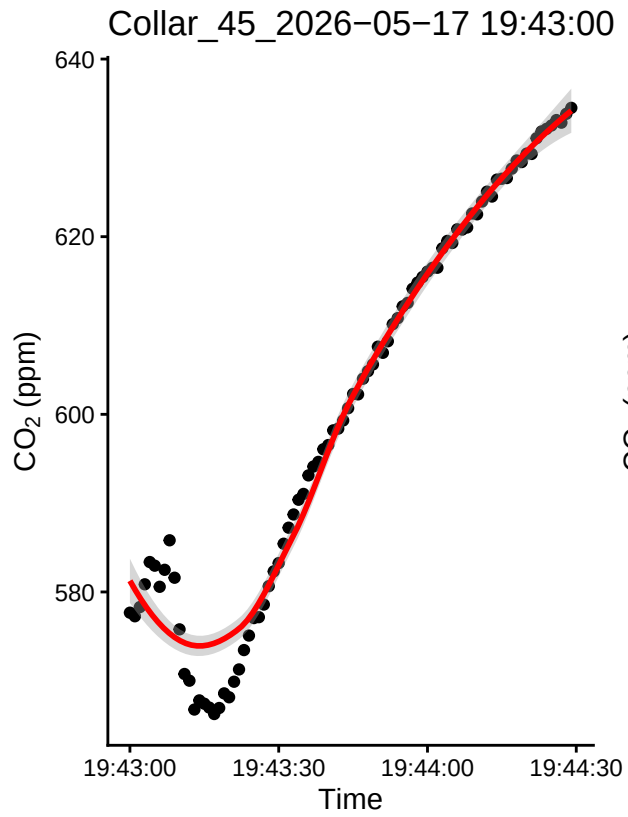


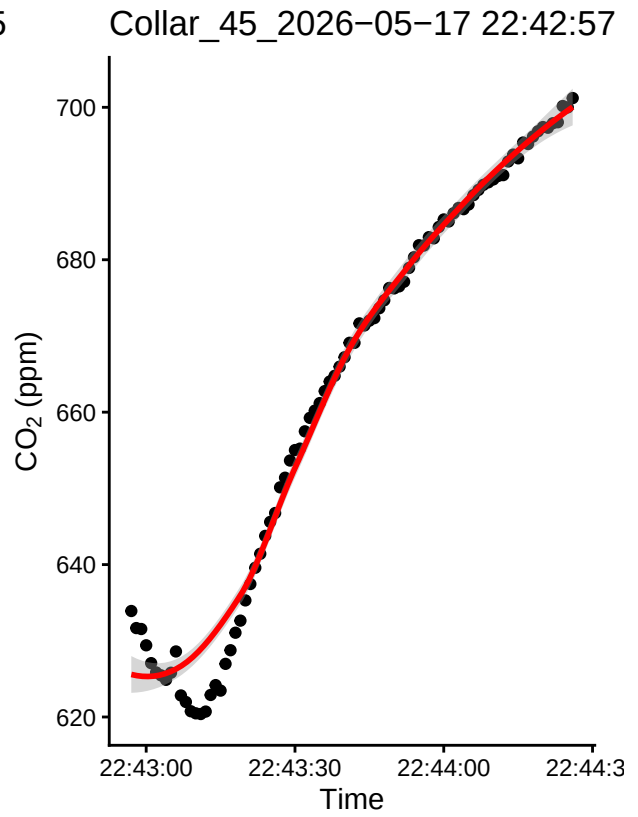
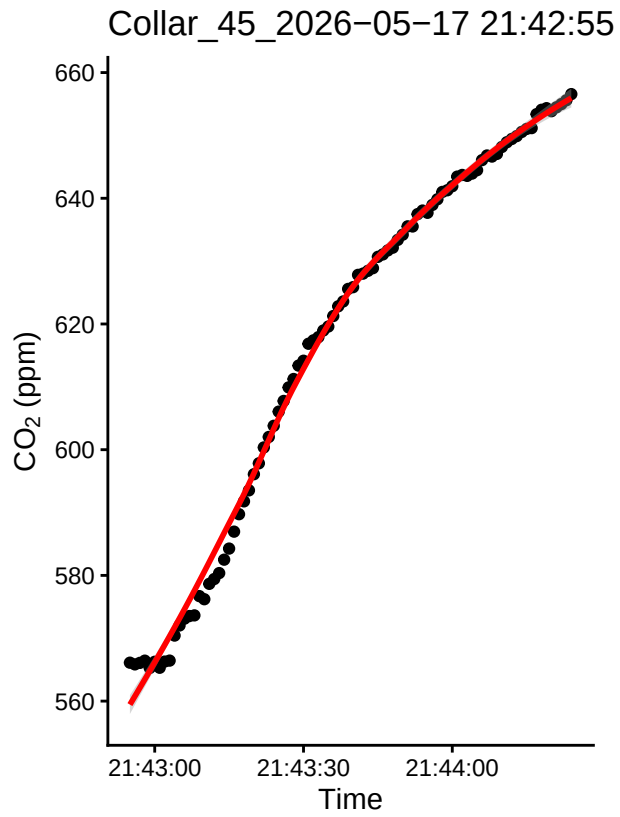


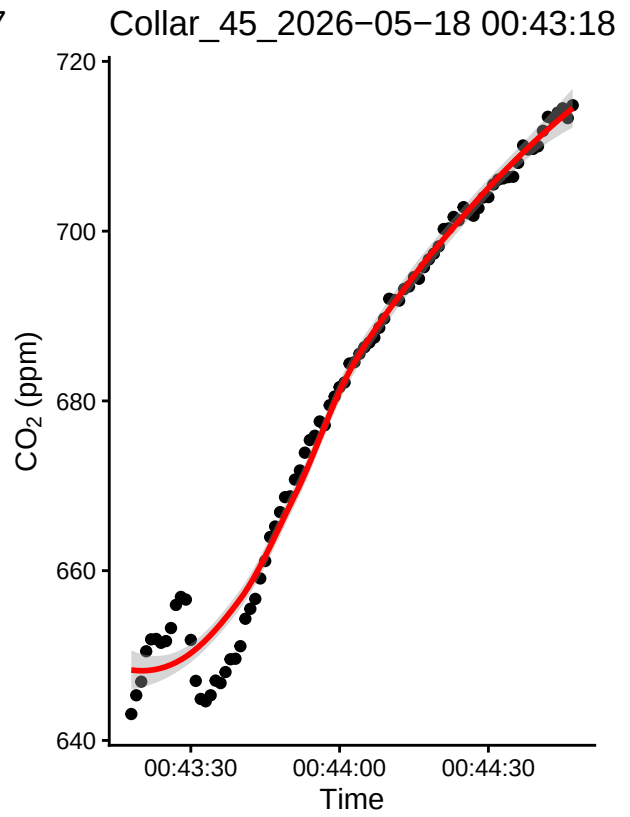
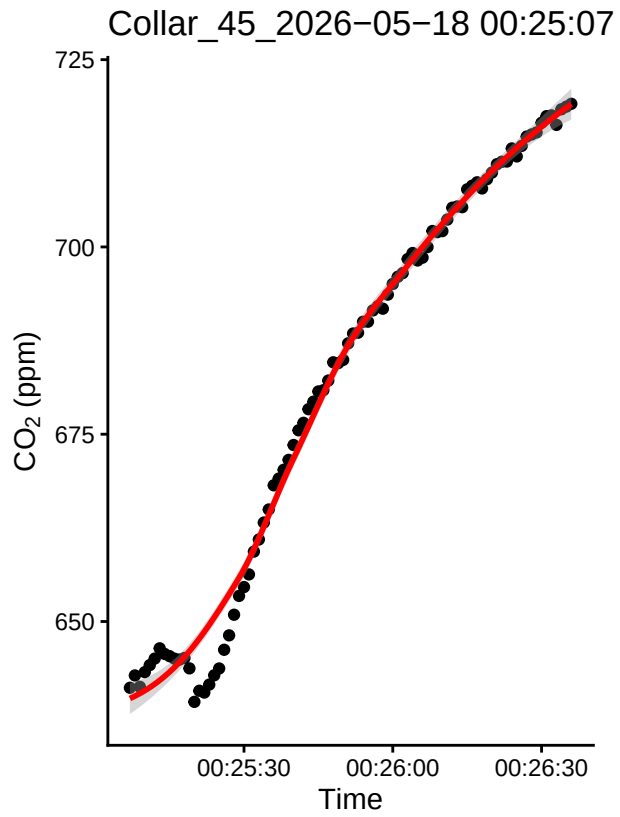


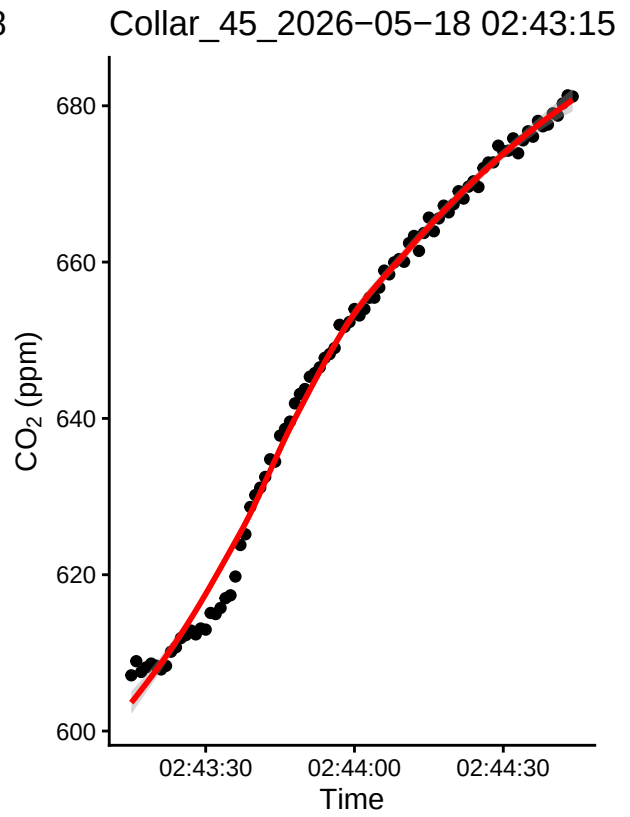
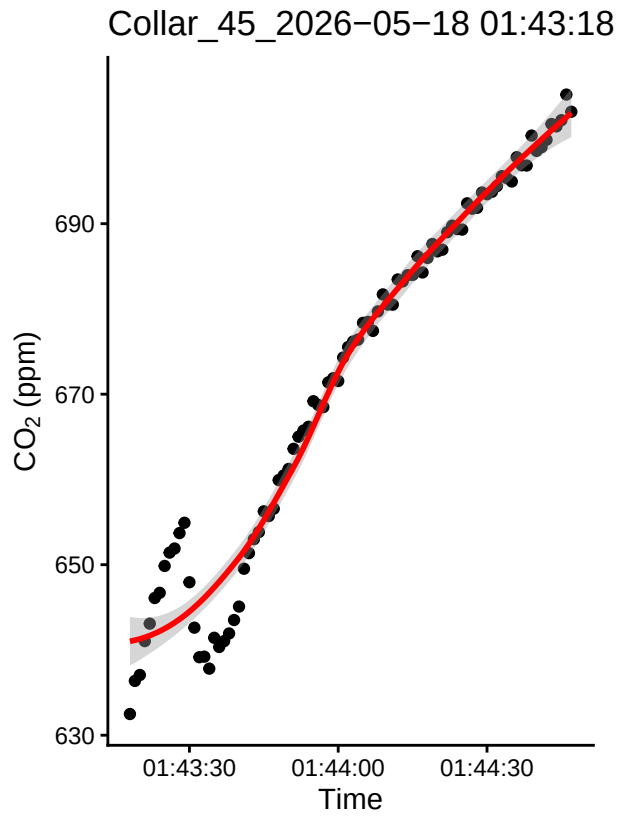


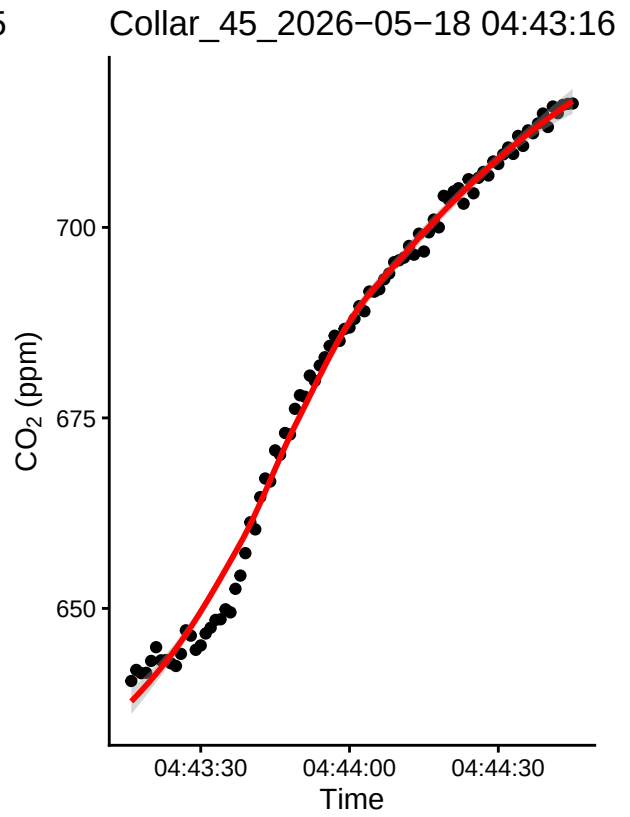
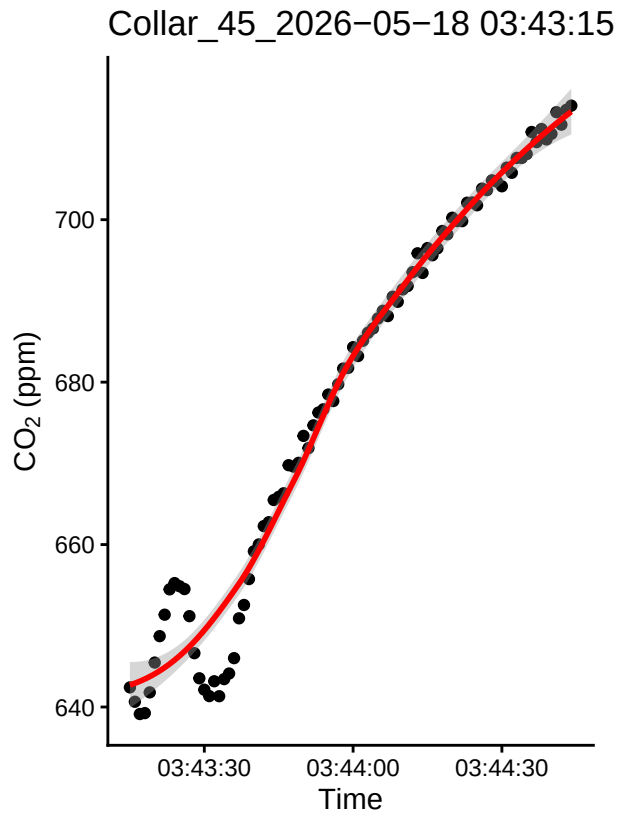


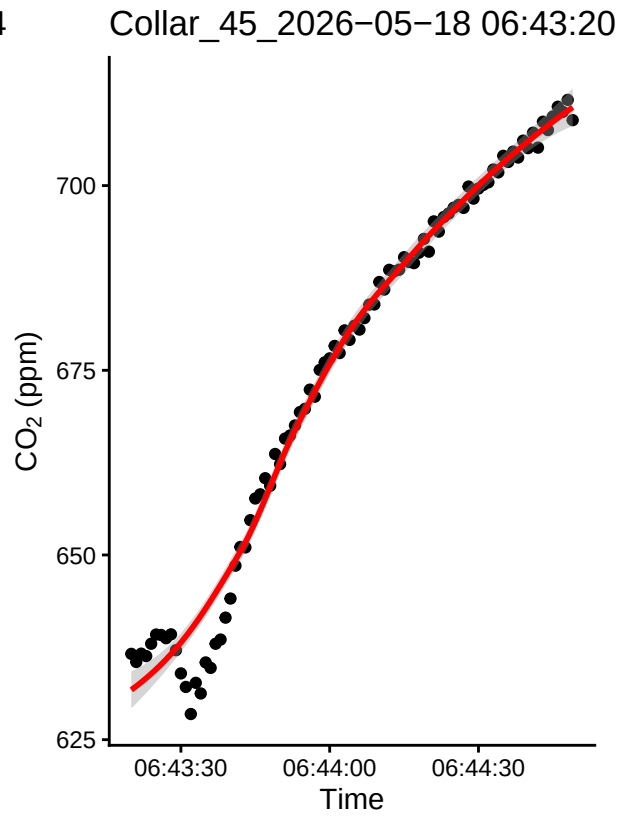
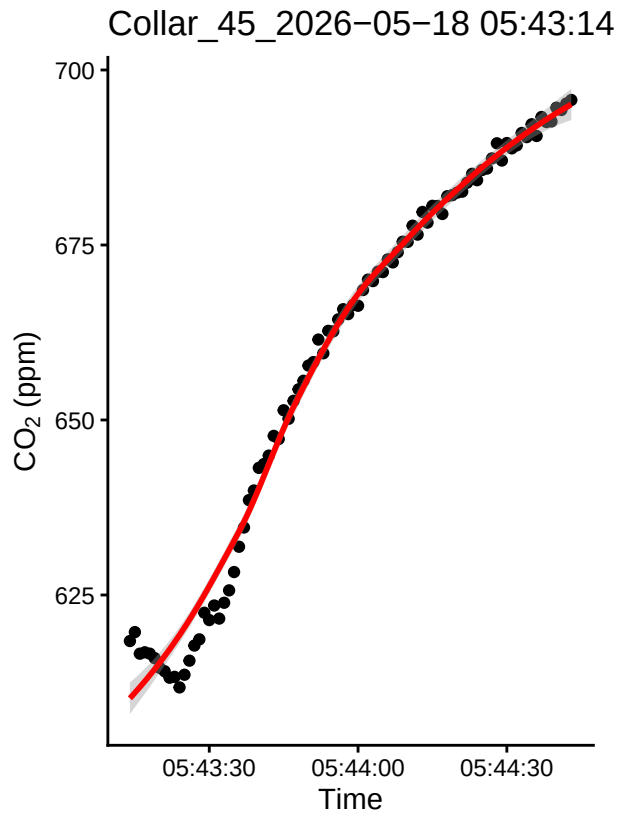


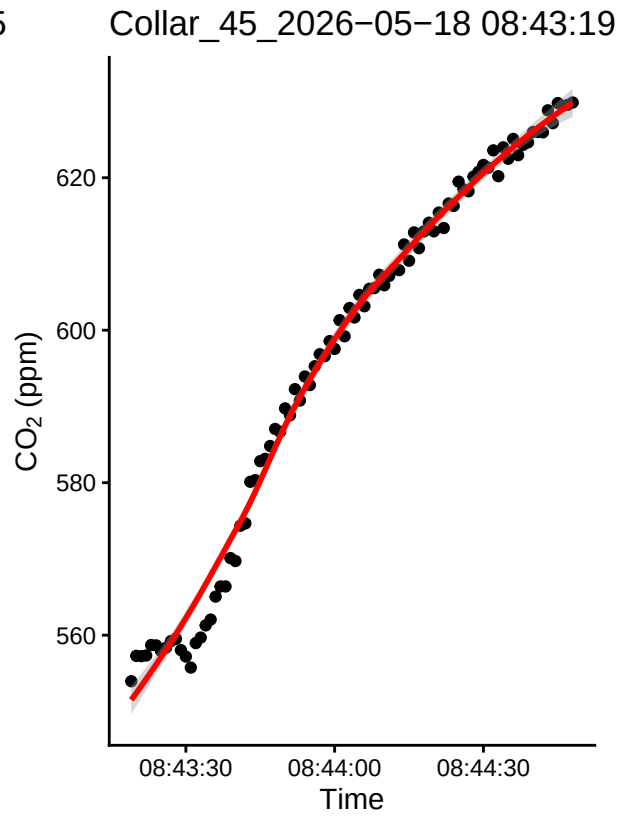
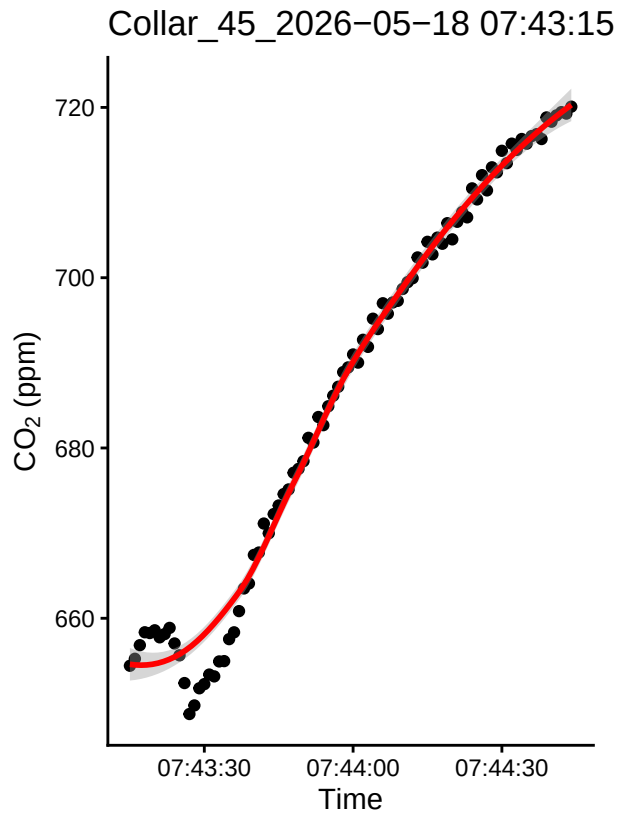




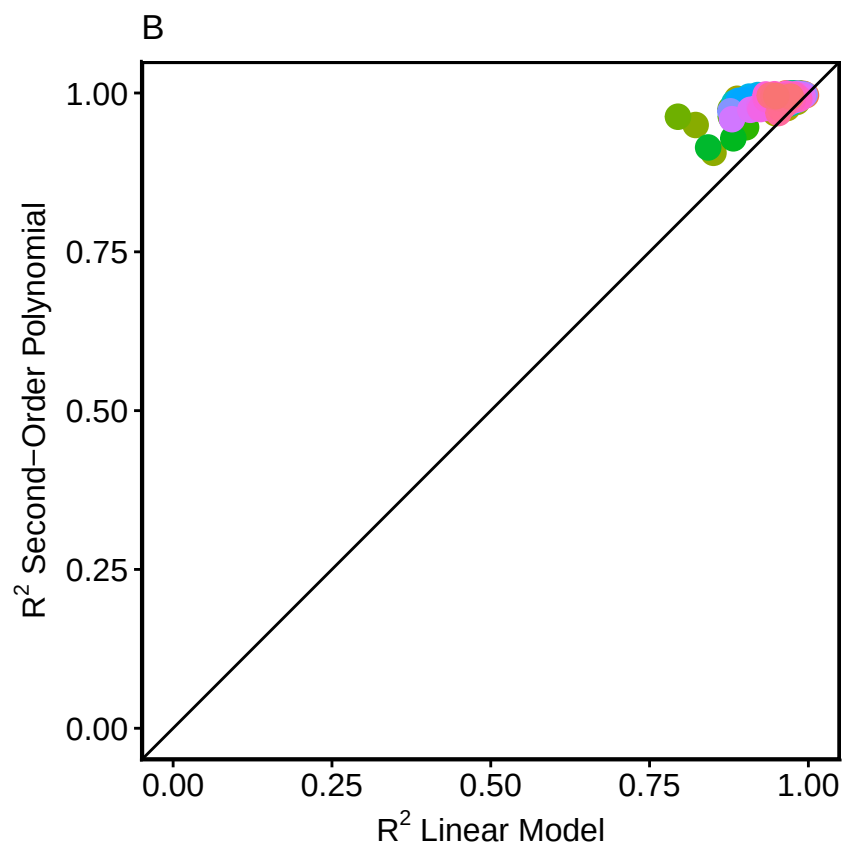
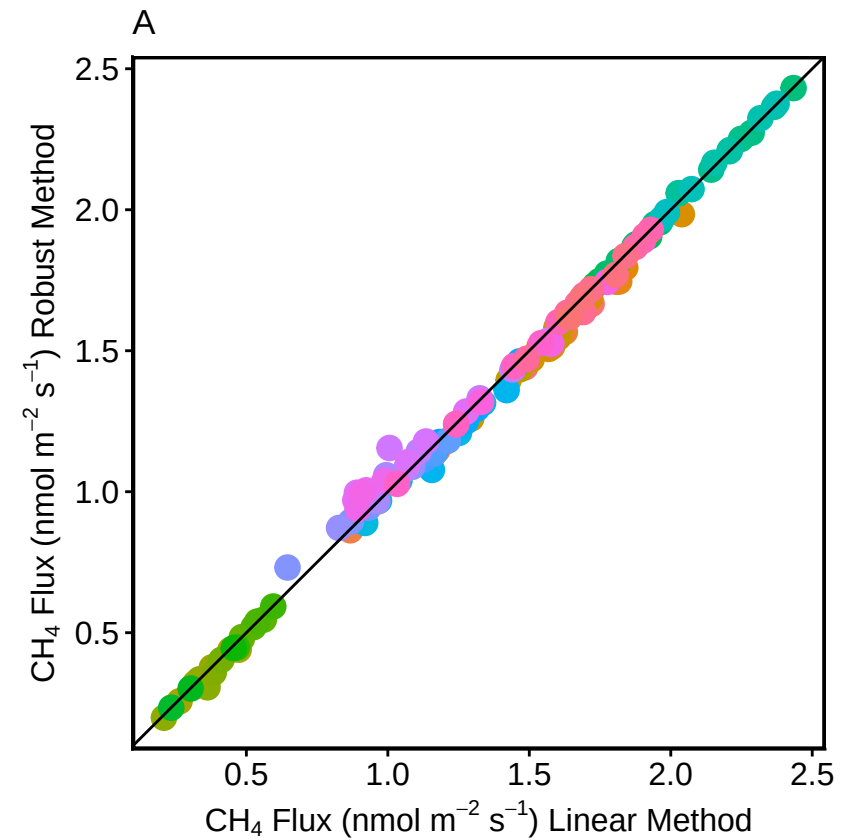




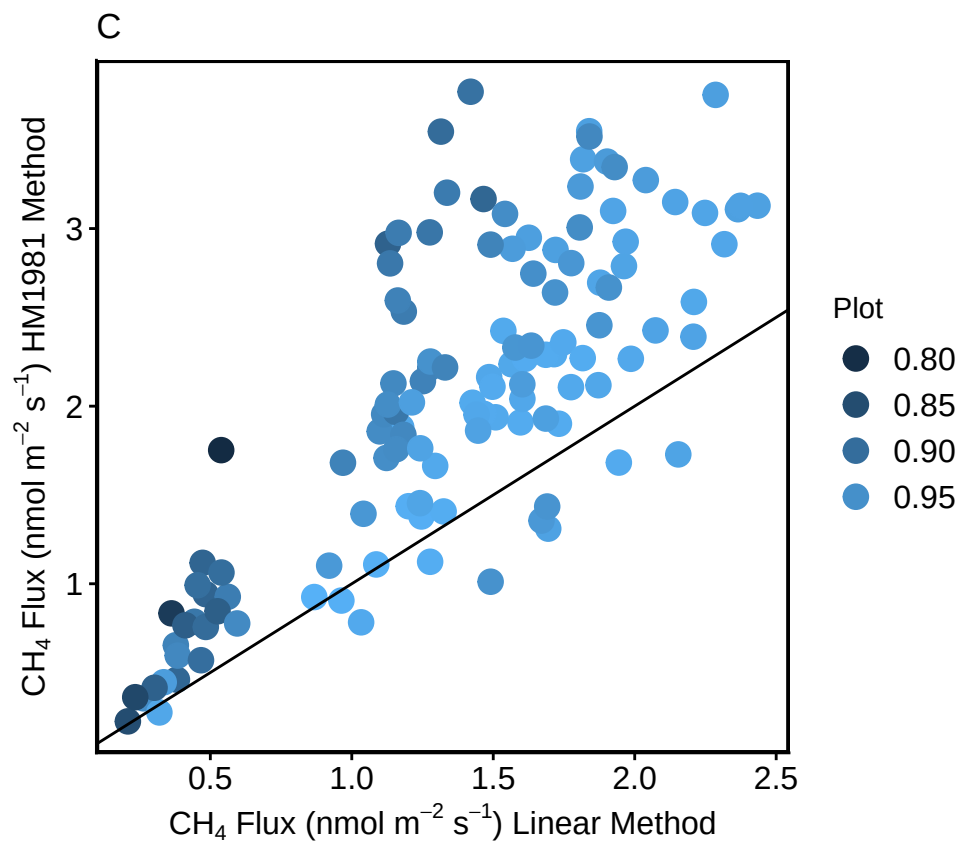




Do some visualization of QAQC plots to assess fluxes



```
## Warning: Removed 20 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Pull together dataframe and export for use elsewhere